

ABSTRACT ALGEBRA

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1. MONOIDS

Definition 1.0.1. Let S be a set. Let $n \geq 1$ be an integer (Definition 3.3.2). An n -ary operation/operator on S is a function (Definition A.1.6) $S^n \rightarrow S$ (Definition A.1.4).

For $n = 2$ and 3 , an n -ary operator on S is called a **binary operator** and **ternary operator** respectively.

One common way to denote a generic binary operator is as “multiplication” — given elements $a, b \in S$, one often write $a \cdot b$, or ab for short, as the image (Definition A.1.6) of the pair (a, b) via the binary operator $S \times S \rightarrow S$.

Definition 1.0.2. Let S be a set equipped with a binary operation (Definition 1.0.1) $\cdot : S \times S \rightarrow S$. An element $e \in S$ is called a **left identity element** if $e \cdot x = x$ for all $x \in S$. Similarly, an element $e \in S$ is called a **right identity element** if $x \cdot e = x$ for all $x \in S$. An element $e \in S$ is called an **identity element** (or **neutral element**) if it is both a left and a right identity element. If an identity element of $\cdot$ exists, then it is unique (Lemma 1.0.3).

Lemma 1.0.3. Let S be a set. Let $\cdot : S \times S \rightarrow S$ be a binary operator (Definition 1.0.1) on S . Assuming that there exists an identity element (Definition 1.0.2) of the binary operator $\cdot$, it is unique. More precisely, if e_1 and e_2 are two identity elements of $\cdot$, then $e_1 = e_2$.

Proof. Suppose that e_1 and E_2 are identity elements of $\cdot$. Since e_1 is in particular a left identity element of $\cdot$, we have that $e_1 \cdot e_2 = e_2$. Since e_2 is a right identity element of $\cdot$, we have that $e_1 \cdot e_2 = e_1$. Therefore, $e_1 = e_2$ as desired. $\square$

Definition 1.0.4. Let $\cdot : S \times S \rightarrow S$ be a binary operation (Definition 1.0.1) on a set S . It is said to be **associative** if

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in S.$$

Definition 1.0.5. Let $\cdot : S \times S \rightarrow S$ be a binary operation (Definition 1.0.1) on a set S such that there exists an identity element (Definition 1.0.2) $e \in S$. Let $s \in S$ be an element.

1. A **left inverse of s** is an element $a \in S$ such that $a \cdot s = e$.
2. A **right inverse of s** is an element $a \in S$ such that $s \cdot a = e$.
3. An **inverse of s** is an element $a \in S$ that is both a left and a right inverse of a .

In the case that $\cdot$ is associative (Definition 1.0.4), i.e. $(S, \cdot)$ is a monoid (Definition A.2.2), inverses of s are unique (Lemma 1.0.6) if they exist, so one may refer to an inverse as *the* inverse of s . In this case, the inverse of s is denoted by s^{-1} .

Lemma 1.0.6. Let $(S, \cdot)$ be a monoid (Definition A.2.2) with identity element (Definition 1.0.2) e . Let $s \in S$ be an element.

If an inverse (Definition 1.0.5) of s exists, then it is unique. More precisely, if a_1 and a_2 are inverses of s , then $a_1 = a_2$.

Proof. Suppose that a_1 and a_2 are inverses of s . By the associativity of $\cdot$, we have

$$(a_1 \cdot s) \cdot a_2 = a_1 \cdot (s \cdot a_2).$$

Since s is an inverse of a_1 and since e is the identity element, the left hand side expression above equals

$$e \cdot a_2 = a_2.$$

Similarly, the right hand side expression equals

$$a_1 \cdot e = a_1.$$

Therefore, $a_1 = a_2$ as desired. $\square$

Definition 1.0.7. Let $\cdot : S \times S \rightarrow S$ be a binary operation (Definition 1.0.1) on a set. It is said to be **commutative** if

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in S.$$

Definition 1.0.8 (Semigroup). A **semigroup** is a pair $(M, \cdot)$ where M is a set and $\cdot : M \times M \rightarrow M$ is an associative (Definition 1.0.4) binary operation (Definition 1.0.1), i.e. $\cdot$ satisfies the following:

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in M.$$

A semigroup is said to be **commutative** or **abelian** if $\cdot$ is also commutative (Definition A.2.3), i.e. $\cdot$ satisfies the following:

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in M.$$

Definition 1.0.9. Let $(M, \cdot)$ be a semigroup (Definition A.2.1) and let $a \in M$. For each positive integer $n \in \mathbb{Z}_{>0}$, the **n th power of a** , denoted by a^n , is the element of M defined recursively by

$$\begin{aligned} a^1 &:= a, \\ a^{n+1} &:= a \cdot a^n. \end{aligned}$$

(Lemma 1.0.10)

Lemma 1.0.10 (Powers of semigroup elements remain in the semigroup). Let $(M, \cdot)$ be a semigroup (Definition A.2.1) and let $a \in M$. For every positive integer $n \in \mathbb{Z}_{>0}$, the power (Definition 1.0.9) a^n belongs to M .

Proof. We proceed by induction on $n \in \mathbb{Z}_{>0}$. For $n = 1$, note that $a^1 = a \in M$ by definition. Suppose inductively that $a^k \in M$ for some $k \geq 1$. By the recursive definition of powers,

$$a^{k+1} := a \cdot a^k.$$

Since $a \in M$ and $a^k \in M$ by the inductive hypothesis, and M is closed under the binary operation $\cdot$ (as it is a semigroup), the product $a \cdot a^k$ belongs to M . Hence $a^{k+1} \in M$. By induction, $a^n \in M$ for all $n \in \mathbb{Z}_{>0}$. $\square$

Lemma 1.0.11. Let $(M, \cdot)$ be a semigroup (Definition A.2.1) and let $a \in M$. For all positive integers $m, n \in \mathbb{Z}_{>0}$,

$$a^m \cdot a^n = a^{m+n},$$

where a^k denotes the k th power of a (Definition 1.0.9).

Proof. Fix $n \in \mathbb{Z}_{>0}$ and induct on m . For the base case $m = 1$, we have by (Definition 1.0.9):

$$a^1 \cdot a^n = a \cdot a^n = a^{n+1} = a^{1+n},$$

so the identity holds. Inductively assume that $m \in \mathbb{Z}_{>0}$ satisfies $a^m \cdot a^n = a^{m+n}$ for all $n \in \mathbb{Z}_{>0}$. Then

$$a^{m+1} \cdot a^n = (a \cdot a^m) \cdot a^n$$

by the definition of powers. By the associativity of $\cdot$, this equals

$$a \cdot (a^m \cdot a^n) = a \cdot a^{m+n}$$

by the inductive hypothesis. Since $a \cdot a^{m+n} = a^{(m+n)+1} = a^{(m+1)+n}$ by definition of powers, the identity holds for $m+1$. This completes the induction. $\square$

Lemma 1.0.12 (Power of a product of commuting elements in a semigroup). Let $(S, \cdot)$ be a semigroup (Definition A.2.1), and let $g, h \in S$ be elements that commute, i.e., $gh = hg$. For any integer $n \in \mathbb{Z}_{>0}$,

$$(gh)^n = g^n h^n,$$

where g^n and h^n denote the n th powers (Definition 1.0.9) in S .

Proof. We prove $(gh)^n = g^n h^n$ by induction on $n \geq 1$.

For the base case $n = 1$, we have $(gh)^1 = gh = g^1 h^1$ by definition of powers.

Inductively assume that the identity holds for some $n \geq 1$. Then

$$\begin{aligned} (gh)^{n+1} &= (gh)^n (gh) \\ &= (g^n h^n)(gh) && \text{by the inductive hypothesis} \\ &= g^n (h^n g) h && \text{(associativity)} \\ &= g^n (gh^n) h && \text{(Lemma 1.0.13)} \\ &= (g^n g)(h^n h) \\ &= g^{n+1} h^{n+1}. \end{aligned}$$

This completes the induction for $n \geq 1$. $\square$

Lemma 1.0.13 (Commuting with powers of a commuting element). Let $(S, \cdot)$ be a semigroup (Definition A.2.1), and let $g, h \in S$ be elements that commute, i.e., $gh = hg$. For any integer $k \in \mathbb{Z}_{>0}$,

$$gh^k = h^k g,$$

where h^k denotes the k th power (Definition 1.0.9).

If $S = (M, \cdot)$ is a monoid (Definition A.2.2) (or a group (Definition 2.1.1)), the same statement holds for $k \in \mathbb{Z}_{\geq 0}$ (with $h^0 = e$). If $S = G$ is a group, it holds for all $k \in \mathbb{Z}$, where h^k denotes the integer power (Definition 2.1.4).

Proof. We first prove that $gh^k = h^k g$ by induction on $k \geq 1$ in the case that S is only assumed to be a semigroup.

For $k = 1$, we have $gh^1 = gh = hg = h^1g$ since g and h commute. Suppose inductively that $gh^k = h^k g$ for some $k \geq 1$. We have

$$\begin{aligned}
gh^{k+1} &= g(h^k h) \\
&= (gh^k)h \quad (\text{associativity}) \\
&= (h^k g)h \quad (\text{inductive hypothesis}) \\
&= h^k(gh) \\
&= h^k(hg) \quad (\text{since } g \text{ and } h \text{ commute}) \\
&= (h^k h)g \\
&= h^{k+1}g.
\end{aligned}$$

By induction, $gh^k = h^k g$ indeed holds for $k \geq 1$.

Assuming that S is a monoid, it now suffices to prove the case of $k = 0$. We have $gh^0 = ge = g = eg = h^0 g$ by definition (Definition 1.0.16).

Further assuming that S is a group, it now suffices to prove the case that $k < 0$. Let $m = -k > 0$. Since $h^k = h^{-m} = (h^{-1})^m$ (Definition 2.1.4), we have $gh^k = g(h^{-1})^m$. Since $m > 0$, the work shown above implies $g(h^{-1})^m = (h^{-1})^m g$, which in turn equals $h^k g$. $\square$

Lemma 1.0.14 (Powers commute iff elements commute). 1. If S be a semigroup (Definition A.2.1), and $g, h \in S$, then the following are equivalent:

- (a) g and h commute, i.e., $gh = hg$.
- (b) For all $m, n \in \mathbb{Z}_{>0}$, the powers g^m and h^n commute, i.e., $g^m h^n = h^n g^m$.
- 2. If M is a monoid (Definition A.2.2) and $g, h \in M$, then the following are equivalent:
 - (a) g and h commute.
 - (b) For all $m, n \in \mathbb{Z}_{\geq 0}$, the powers g^m and h^n commute, i.e., $g^m h^n = h^n g^m$.
- 3. If G is a group (Definition 2.1.1) and $g, h \in G$, then the following are equivalent:
 - (a) g and h commute.
 - (b) For all $m, n \in \mathbb{Z}$, the powers g^m and h^n commute, i.e., $g^m h^n = h^n g^m$.

Proof. Clearly in each case, the condition that the powers g^m and h^n commute imply that g and h commute because $m, n = 1$ are positive integers, and $g^1 = g$ and $h^1 = h$.

For a semigroup S , suppose that $g, h \in S$ are elements that commute, i.e. $gh = hg$. By Lemma 1.0.13, g commutes with h^n for all $n \in \mathbb{Z}_{>0}$. By Lemma 1.0.13 again, g^m commutes with h^n for all $m \in \mathbb{Z}_{>0}$.

Similarly, for a monoid M and commuting elements $g, h \in M$, we have that g^m commutes with h^n for all $m, n \in \mathbb{Z}_{\geq 0}$ and for a group G and commuting elements $g, h \in G$, we have that g^m commutes with h^n for all $m, n \in \mathbb{Z}$. $\square$

Definition 1.0.15 (Monoid). A **monoid** is a semigroup (Definition A.2.1) $(M, \cdot)$ such that there exists an identity element (Definition 1.0.2) $e \in M$ for $\cdot$, i.e. e has the property:

$$e \cdot a = a \cdot e = a \quad \text{for all } a \in M.$$

Definition 1.0.16. Let $(M, \cdot)$ be a monoid (Definition A.2.2) with identity element e , and let $a \in M$. For each $n \in \mathbb{Z}_{\geq 0}$, the *n th power of a* , denoted by a^n , is the element of M defined recursively by

$$\begin{aligned} a^0 &:= e, \\ a^{n+1} &:= a \cdot a^n. \end{aligned}$$

In the case that the monoid operation is written additively, the n th power of a is more usually written na or $n \cdot a$.

Lemma 1.0.17. Let $(M, \cdot)$ be a monoid (Definition A.2.2) with identity element e , and let $a \in M$. For all nonnegative integers $m, n \in \mathbb{Z}_{\geq 0}$,

$$a^m \cdot a^n = a^{m+n},$$

where a^k denotes the k th power of a (Definition 1.0.16).

Proof. Fix $n \in \mathbb{Z}_{\geq 0}$ and induct on m . For the base case $m = 0$, we have by (Definition 1.0.16):

$$a^0 \cdot a^n = e \cdot a^n = a^n = a^{0+n},$$

so the identity holds. Inductively assume that $m \in \mathbb{Z}_{\geq 0}$ satisfies $a^m \cdot a^n = a^{m+n}$. Then

$$a^{m+1} \cdot a^n = (a \cdot a^m) \cdot a^n$$

by the recursive definition of powers. By associativity of $\cdot$ from (Definition A.2.2), this equals

$$a \cdot (a^m \cdot a^n) = a \cdot a^{m+n}$$

by the inductive hypothesis. Since $a \cdot a^{m+n} = a^{(m+1)+n}$ by definition of powers, the identity holds for $m + 1$. This completes the induction. $\square$

Lemma 1.0.18. Let $(M, \cdot)$ be a monoid (Definition A.2.2) with identity element e . For all $n \in \mathbb{Z}_{\geq 0}$,

$$e^n = e,$$

where e^n denotes the n th power of e (Definition 1.0.16).

Proof. We prove this by induction. By the definition of powers in a monoid (Definition 1.0.16), $e^0 = e$. For $n \geq 0$, suppose inductively that $e^n = e$. We have $e^{n+1} = e \cdot e^n = e \cdot e = e$. Hence $e^n = e$ for all $n \in \mathbb{Z}_{\geq 0}$. $\square$

Lemma 1.0.19. Let $(M, \cdot)$ be a monoid (Definition A.2.2) with identity element e . Then e is its own inverse, i.e.,

$$e \cdot e = e \quad \text{and hence} \quad e^{-1} = e.$$

Proof. By definition of the identity element in a monoid (Definition A.2.2), we have $e \cdot e = e$. This means e satisfies the definition of being its own inverse (Definition 1.0.5): e is both a left inverse of e ($e \cdot e = e$) and a right inverse of e ($e \cdot e = e$). Therefore e is an inverse of itself. (If the inverse is unique in M , which holds when M is a monoid by (Lemma 1.0.6), then $e^{-1} = e$.) $\square$

Lemma 1.0.20 (Power of a product of commuting elements). Let $(M, \cdot)$ be a monoid (Definition A.2.2), and let $g, h \in M$ be elements that commute, i.e., $gh = hg$. For any integer $n \in \mathbb{Z}_{\geq 0}$,

$$(gh)^n = g^n h^n,$$

where g^n and h^n denote the n th powers (Definition 1.0.16) in M .

Proof. For $n > 0$, the statement follows directly from Lemma 1.0.12 applied to the underlying semigroup of M .

For $n = 0$, note that $(gh)^0 = e$ by definition (Definition 1.0.16), and $g^0 h^0 = e \cdot e = e$, so the identity holds.

For the group case with $n \in \mathbb{Z}$, if $n \geq 0$, the result follows from the monoid case applied to the underlying monoid of G . If $n < 0$, let $m = -n > 0$. Then

$$\begin{aligned} (gh)^n &= ((gh)^{-1})^m \\ &= (h^{-1}g^{-1})^m \\ &= (g^{-1}h^{-1})^m \quad \text{Lemma 2.1.7} \\ &= (g^{-1})^m (h^{-1})^m \quad (\text{by the monoid case}) \\ &= g^{-m} h^{-m} \\ &= g^n h^n, \end{aligned}$$

(Lemma 2.1.11) where we used $(gh)^{-1} = h^{-1}g^{-1}$ from a (Definition 2.1.1) and the fact that g^{-1} and h^{-1} commute when g and h commute. $\square$

2. BASIC GROUP THEORY

2.1. Definitions.

Definition 2.1.1 (Groups). A **group** is a monoid (Definition A.2.2) such that every element has an inverse (Lemma 1.0.6).

In other words, a group is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a binary operation (Definition 1.0.1), subject to the following conditions:

1. (Associativity) For all $g, h, k \in G$ one has

$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$

2. (Identity element) There exists an identity element (Definition 1.0.2), i.e. an element $e \in G$ such that for all $g \in G$,

$$e \cdot g = g \cdot e = g.$$

3. (Inverse elements) For all $g \in G$, there exists an inverse element (Definition 1.0.5) of g , i.e. an element $g^{-1} \in G$ such that

$$g \cdot g^{-1} = g^{-1} \cdot g = e.$$

A group $(G, \cdot)$ is often simply written as G , when the notation for the binary operation $\cdot$ is clear.

Assuming that the notation for the identity element of a group is not explicitly specified,

1. if the group operation is written multiplicatively, then one usually implicitly writes e or 1 as the identity of G , and
2. If the group operation is instead written additively, then one usually implicitly write 0 as the identity of G .

An **abelian group** or synonymously, a **commutative group**, is a group $(G, \cdot)$ whose binary operation $\cdot$ is **abelian** or **commutative** (Definition A.2.3), i.e. satisfies

$$g \cdot h = h \cdot g$$

for all $g, h \in G$. A group operation written additively, i.e. with $+$, is almost usually abelian. An abelian group's group operation may still often be written multiplicatively.

An abelian group is equivalent to a $\mathbb{Z}$ -module.

Definition 2.1.2. A **trivial group** is a group (Definition 2.1.1) consisting of exactly one element. That is, a trivial group is a group $(G, \cdot)$ such that $|G| = 1$. The unique element of a trivial group is necessarily the identity element (Definition 2.1.1) of the group.

A trivial group is always an abelian group (Definition 2.1.1) (since the unique element commutes with itself). When the fact that it is an abelian group is emphasized and its group structure is written additively, a trivial group is also called a **zero group** (or **zero object** in the category of abelian groups).

Up to isomorphism of groups, there is exactly one trivial group. In other words, the trivial group is the unique group of cardinality one up to isomorphism.

Remark 2.1.3. Given a set G and a binary operator $\cdot$ on it, the notation of $\cdot$ may often be omitted, especially when the binary operation is thought of as a “multiplication” (as opposed to, for example, “addition”); more precisely, one may write gh in place of $g \cdot h$ for $g, h \in G$.

Definition 2.1.4. Let G be a group (Definition 2.1.1) with identity element e and whose group operation is written multiplicatively, and let $g \in G$. For each integer $n \in \mathbb{Z}$, the **n th power of g** , denoted by g^n , is the element of G defined as follows:

- If $n \geq 0$, then g^n is the n th power (Definition 1.0.16) of g in the underlying monoid (Definition A.2.2) of G .
- If $n < 0$, then $g^n := (g^{-1})^{-n}$, where g^{-1} is the inverse (Definition 2.1.1) of g .

In the case that the group operation is written additively, the n th power of g is more usually written ng or $n \cdot g$, see (Definition 1.0.16).

Lemma 2.1.5. Let G be a group (Definition 2.1.1) with identity element e . For all $n \in \mathbb{Z}$,

$$e^n = e,$$

where e^n denotes the n th power of e (Definition 2.1.4).

Proof. By Lemma 1.0.18, the statement holds for all $n \in \mathbb{Z}_{\geq 0}$ since every group is a monoid. For $n < 0$, let $m = -n > 0$. Then by the definition of negative powers in a group (Definition 2.1.4), $e^n = (e^{-1})^m$. Since e is its own inverse (Lemma 1.0.19), we have $(e^{-1})^m = e^m = e$ by the nonnegative case. Thus, $e^n = e$ for all $n \in \mathbb{Z}$. $\square$

Lemma 2.1.6. Let G be a group with identity element e . An element $g \in G$ satisfies $g = e$ if and only if $g^{-1} = e$.

Proof. We show each direction separately.

Note that $e \cdot e = e$, so e is an inverse of e . By the identity element is its own inverse in a monoid (Lemma 1.0.19) (every group is a monoid), e is then the unique inverse of e , i.e. $e = e^{-1}$.

Conversely, suppose that $g^{-1} = e$. Then $g = (g^{-1})^{-1}$ by Corollary 2.1.16. Substituting $g^{-1} = e$, we get $g = e^{-1}$. From the above, $e^{-1} = e$, so $g = e$. $\square$

Lemma 2.1.7 (Powers commute iff elements commute). 1. If S be a semigroup (Definition A.2.1), and $g, h \in S$, then the following are equivalent:

- (a) g and h commute, i.e., $gh = hg$.
 - (b) For all $m, n \in \mathbb{Z}_{>0}$, the powers g^m and h^n commute, i.e., $g^m h^n = h^n g^m$.
2. If M is a monoid (Definition A.2.2) and $g, h \in M$, then the following are equivalent:
- (a) g and h commute.
 - (b) For all $m, n \in \mathbb{Z}_{\geq 0}$, the powers g^m and h^n commute, i.e., $g^m h^n = h^n g^m$.
3. If G is a group (Definition 2.1.1) and $g, h \in G$, then the following are equivalent:
- (a) g and h commute.
 - (b) For all $m, n \in \mathbb{Z}$, the powers g^m and h^n commute, i.e., $g^m h^n = h^n g^m$.

Proof. Clearly in each case, the condition that the powers g^m and h^n commute imply that g and h commute because $m, n = 1$ are positive integers, and $g^1 = g$ and $h^1 = h$.

For a semigroup S , suppose that $g, h \in S$ are elements that commute, i.e. $gh = hg$. By Lemma 1.0.13, g commutes with h^n for all $n \in \mathbb{Z}_{>0}$. By Lemma 1.0.13 again, g^m commutes with h^n for all $m \in \mathbb{Z}_{>0}$.

Similarly, for a monoid M and commuting elements $g, h \in M$, we have that g^m commutes with h^n for all $m, n \in \mathbb{Z}_{\geq 0}$ and for a group G and commuting elements $g, h \in G$, we have that g^m commutes with h^n for all $m, n \in \mathbb{Z}$. $\square$

Lemma 2.1.8 (Power of a product in a group with commuting elements). Let G be a group (Definition 2.1.1) and let $g, h \in G$ be elements that commute, i.e., $gh = hg$. For all integers $n \in \mathbb{Z}$,

$$(gh)^n = g^n h^n,$$

(Definition 2.1.4)

Proof. This follows directly from Lemma 1.0.20, which covers both the monoid case ($n \geq 0$) and the group case ($n \in \mathbb{Z}$) when the underlying monoid is a group.

For $n \in \mathbb{Z}_{\geq 0}$, this follows from Lemma 1.0.20. For $n < 0$, let $m = -n > 0$. We have

$$\begin{aligned}
(gh)^n &= ((gh)^{-1})^m \\
&= (h^{-1}g^{-1})^m \quad (\text{Lemma 2.1.11}) \\
&= (g^{-1}h^{-1})^m \quad (\text{Lemma 2.1.7}) \\
&= (g^{-1})^m(h^{-1})^m \quad (\text{Lemma 1.0.20}) \\
&= g^{-m}h^{-m} \\
&= g^n h^n.
\end{aligned}$$

□

Lemma 2.1.9. Let G be a group (Definition 2.1.1) and let $g \in G$. For all integers $m, n \in \mathbb{Z}$,

$$g^m g^n = g^{m+n},$$

where g^k denotes the k th power of g (Definition 2.1.4).

Proof. We proceed by case analysis on the signs of m and n . Recall from that (Definition 2.1.4) for $k \geq 0$, g^k is the power in the underlying monoid, while for $k < 0$, $g^k = (g^{-1})^{-k}$.

1. Suppose $m \geq 0$ and $n \geq 0$. Then $g^m g^n = g^{m+n}$ follows from additivity of powers in the underlying monoid by Lemma 1.0.17.
2. Suppose $m < 0$ and $n < 0$. Let $a = -m > 0$ and $b = -n > 0$. Then

$$g^m g^n = (g^{-1})^a (g^{-1})^b = (g^{-1})^{a+b}$$

by Lemma 1.0.17 applied to the element g^{-1} in the underlying monoid. Since $a + b = -(m + n)$, we have $(g^{-1})^{a+b} = g^{-(a+b)} = g^{m+n}$.

3. Suppose $m > 0$ and $n < 0$. Let $b = -n > 0$. We consider subcases using the trichotomy of order on $\mathbb{Z}$:

(a) Suppose $m \geq b$, so that $m - b \geq 0$. We compute:

$$\begin{aligned}
g^m g^n &= g^m (g^{-1})^b \\
&= g^{m-b} g^b (g^{-1})^b
\end{aligned}$$

because $m - b \geq 0$ and $b \geq 0$, so by Lemma 1.0.17 applied to the underlying monoid, $g^{m-b} g^b = g^{(m-b)+b} = g^m$. Now

$$g^b (g^{-1})^b = (gg^{-1})^b = e^b = e$$

by Lemma 1.0.20 applied to the commuting elements g and g^{-1} in the underlying monoid (note that g and g^{-1} commute in any group), and by w (Definition 2.1.1) we have $gg^{-1} = e$. Since e is the identity, $g^{m-b}e = g^{m-b}$. Since $m + n = m - b$, we have $g^{m+n} = g^{m-b}$.

(b) Suppose $m < b$, so that $b - m > 0$. We compute:

$$\begin{aligned}
g^m g^n &= g^m (g^{-1})^b \\
&= g^m (g^{-1})^m (g^{-1})^{b-m} \\
&= (gg^{-1})^m (g^{-1})^{b-m} \\
&= e^m (g^{-1})^{b-m} \\
&= (g^{-1})^{b-m},
\end{aligned}$$

where the second equality uses Lemma 1.0.17 applied to g^{-1} in the underlying monoid to write $(g^{-1})^b = (g^{-1})^m (g^{-1})^{b-m}$ (since $m \geq 0$, $b - m \geq 0$), the third equality uses that powers of g and g^{-1} commute in the monoid, the fourth uses $gg^{-1} = e$, and the fifth uses that e is the identity. Since $m + n = m - b < 0$, we have $g^{m+n} = (g^{-1})^{-(m-n)} = (g^{-1})^{b-m}$.

4. Suppose $m < 0$ and $n > 0$. The argument is symmetric to the previous case by swapping the roles of m and n and using commutativity of addition in $\mathbb{Z}$.

In all cases, $g^m g^n = g^{m+n}$ as desired. $\square$

Lemma 2.1.10. Let G be a group (Definition 2.1.1) and let $g \in G$. For all integers $n \in \mathbb{Z}$,

$$(g^n)^{-1} = g^{-n} = (g^{-1})^n,$$

where g^k denotes the k th power of g (Definition 2.1.4).

Proof. For the first equality, observe that by Lemma 2.1.9, we have $g^n g^{-n} = g^{n+(-n)} = g^0 = e$. Similarly, $g^{-n} g^n = g^{-n+n} = g^0 = e$. Therefore g^{-n} is the inverse of g^n , so $(g^n)^{-1} = g^{-n}$.

We show that $(g^{-1})^n = (g^n)^{-1}$ for all $n \in \mathbb{Z}$. Note that $g \cdot g^{-1} = e = g^{-1}g$, so g and g^{-1} commute. For $n \geq 0$, Lemma 1.0.20 then yields

$$g^n (g^{-1})^n = (gg^{-1})^n = e^n = e$$

(Lemma 2.1.5) Hence $(g^{-1})^n = (g^n)^{-1} = g^{-n}$. If $n < 0$, let $a = -n > 0$. Then

$$g^{-n} = g^a$$

and

$$(g^{-1})^n = ((g^{-1})^{-1})^a = g^a.$$

(Corollary 2.1.16) Therefore, $g^{-n} = (g^{-1})^n$ in this case as well.

Combining both equalities, $(g^n)^{-1} = g^{-n} = (g^{-1})^n$. $\square$

Lemma 2.1.11. Let G be a group (Definition 2.1.1) and let $g, h \in G$. Then

$$(gh)^{-1} = h^{-1}g^{-1}.$$

Proof. Since G is a group, it has an identity element e and every element has a two-sided inverse. We compute:

$$\begin{aligned}
 (gh)(h^{-1}g^{-1}) &= g(h(h^{-1}g^{-1})) \\
 &= g((hh^{-1})g^{-1}) \\
 &= g(eg^{-1}) \\
 &= gg^{-1} \\
 &= e.
 \end{aligned}$$

This shows that $h^{-1}g^{-1}$ is a right inverse of (gh) in G . By Lemma 2.1.12, $h^{-1}g^{-1}$ is a left inverse of (gh) in G as well, i.e. is an inverse (Definition 1.0.5) of (gh) . By the uniqueness of inverses in a group (Lemma 1.0.6), we have $(gh)^{-1} = h^{-1}g^{-1}$. $\square$

Lemma 2.1.12. Let G be a group (Definition 2.1.1) and let $g, h \in G$.

1. If h is a left inverse (Definition 1.0.5) of g (i.e., $hg = e$), then h is also a right inverse (Definition 1.0.5) of g (i.e., $gh = e$).
2. If h is a right inverse (Definition 1.0.5) of g (i.e., $gh = e$), then h is also a left inverse (Definition 1.0.5) of g (i.e., $hg = e$).

In particular, any one-sided inverse in a group is a two-sided inverse, and therefore equals the unique inverse g^{-1} (Lemma 1.0.6).

Proof. Let e denote the identity element of G . Since G is a group, every element has a two-sided inverse.

1. Suppose that $hg = e$. Since $h \in G$, h has a (two-sided) inverse $h^{-1} \in G$. Multiplying both sides of $hg = e$ on the left by h^{-1} gives

$$h^{-1}(hg) = h^{-1}e.$$

By associativity and the definition of inverse, the left side is $(h^{-1}h)g = eg = g$, and the right side is h^{-1} . Thus $g = h^{-1}$. Therefore h is the inverse of g , and in particular $gh = e$.

2. Suppose that $gh = e$. Since $g \in G$, g has a (two-sided) inverse $g^{-1} \in G$. Multiplying both sides of $gh = e$ on the left by g^{-1} gives

$$g^{-1}(gh) = g^{-1}e.$$

By associativity and the definition of inverse, the left side is $(g^{-1}g)h = eh = h$, and the right side is g^{-1} . Thus $h = g^{-1}$. Therefore h is the inverse of g , and in particular $hg = e$.

In both cases, the one-sided inverse h equals the unique two-sided inverse g^{-1} (Lemma 1.0.6). $\square$

Lemma 2.1.13. Let $(G, \cdot)$ be a group (Definition 2.1.1). For any $a, b, c \in G$,

1. $ab = ac$ if and only if $b = c$.
2. $ba = bc$ if and only if $a = c$.

Proof. 1. The “if” direction is trivial. To prove the “only if” direction, suppose that $a, b, c \in G$ satisfy $ab = ac$. Applying a^{-1} to both sides, we have

$$a^{-1} \cdot (ab) = a^{-1} \cdot (ac).$$

By the associativity of $\cdot$, this equality becomes

$$(a^{-1}a)b = (a^{-1} \cdot a)c,$$

which in turn becomes

$$eb = ec.$$

Since e is the identity element, we have $b = c$ as desired.

2. One obtains this through a symmetric argument.

□

Corollary 2.1.14. Let $(G, \cdot)$ be a group (Definition 2.1.1). For any $a, b \in G$,

1. if $ab = b$, then $a = e$.
2. if $ab = a$, then $b = e$.

Proof. 1. Let $a, b \in G$ satisfy $ab = b$. In particular, $ab = eb$. Apply Lemma 2.1.13.

2. One obtains this through a symmetric argument.

□

Corollary 2.1.15. Let $(G, \cdot)$ be a group (Definition 2.1.1). For any $a, b \in G$, if $ab = e$, then $b = a^{-1}$, and $a = b^{-1}$.

Proof. Let $a, b \in G$ satisfy $ab = e$. Since $e = a \cdot a^{-1}$, we have

$$a \cdot b = a \cdot a^{-1}.$$

Apply Lemma 2.1.13 to obtain that $b = a^{-1}$. Similarly, $e = b^{-1} \cdot b$, so

$$a \cdot b = b^{-1} \cdot b$$

and hence $a = b^{-1}$ by Lemma 2.1.13.

□

Corollary 2.1.16. Let $(G, \cdot)$ be a group (Definition 2.1.1). For any $a \in G$, $(a^{-1})^{-1} = a$.

Proof. Note that $(a^{-1}) \cdot a = e$. Applying Corollary 2.1.15, we have that $(a^{-1})^{-1} = a$. □

Definition 2.1.17 (Group homomorphism). Let $(G, \cdot)$ and $(H, *)$ be groups (Definition 2.1.1). A map $f : G \rightarrow H$ is called a **group homomorphism** if for all $g_1, g_2 \in G$ one has

$$f(g_1 \cdot g_2) = f(g_1) * f(g_2).$$

The collection of all groups with the group homomorphisms forms a locally small (Definition A.4.3) category (Definition A.4.1), called the **category of groups**.

If f is bijective (Definition A.1.3), then f is called a **group isomorphism**.

- Definition 2.1.18.** 1. The *category of groups* is the locally small (Definition A.4.3) category (Definition A.4.1) whose objects are groups (Definition 2.1.1) and whose morphisms are group homomorphisms (Definition 2.1.17). It is often denoted by notations such as **Grp**.
2. The *category of abelian groups* is the locally small (Definition A.4.3) category (Definition A.4.1) whose objects are abelian groups (Definition 2.1.1) and whose morphisms are group homomorphisms (Definition 2.1.17). It is often denoted by notations such as **Ab**. **Ab** is an abelian category.

Definition 2.1.19. Let $(G, \cdot)$ be a group (Definition 2.1.1). A subset (Definition A.1.2) $H \subseteq G$ is called a *subgroup of G* if

- the operations (Definition 1.0.1)

$$\cdot|_H : H \times H \rightarrow G, \quad (h_1, h_2) \mapsto h_1 \cdot h_2$$

$$^{-1}|_H : HH \rightarrow G, \quad h \mapsto h^{-1}$$

have images contained in H , and

- these operations along with the identity element of G make H into a group.

Equivalently (Proposition 2.1.20), H is a subgroup if:

- $H \neq \emptyset$,
- For all $h_1, h_2 \in H$, the product $h_1 \cdot h_2$ is an element of H ,
- For all $h \in H$, the inverse h^{-1} is an element of H .

We denote this relation by $H \leq G$, meaning “ H is a subgroup of G ”.

Proposition 2.1.20 (Subgroup criterion). Let G be a group (Definition 2.1.1) and let $H \subseteq G$ be a subset. Then H is a subgroup of G (Definition 2.1.19) if and only if the following conditions hold:

1. $H \neq \emptyset$,
2. H is closed under the group operation, i.e. for all $h_1, h_2 \in H$, the product $h_1 h_2$ is an element of H ,
3. H is closed under the inverse operator of G , i.e. for all $h \in H$, the inverse h^{-1} is an element of H .

Proof. Suppose that H is a subgroup of G , i.e. that the operations (Definition 1.0.1)

$$\cdot|_H : H \times H \rightarrow G, \quad (h_1, h_2) \mapsto h_1 \cdot h_2$$

$$\cdot^{-1}|_H : HH \rightarrow G, \quad h \mapsto h^{-1}$$

have images contained in H , and these operations along with the identity element of G make H into a group. By the group axioms, H has an identity element, so H is not the empty set. By the assumption that $\cdot|_H$ has image contained in H , we have that $h_1 h_2$ is an element of H for all $h_1, h_2 \in H$. Similarly, by the assumption that $^{-1}|_H : H \rightarrow G$ has image contained in H , we have that h^{-1} is an element of H for all $h \in H$.

Conversely, suppose that $H \neq \emptyset$, that H is closed under the group operation, and that H is closed under the inverse operator of G . We show that H is a subgroup of G . For any $h_1, h_2 \in H$, we have $h_1 \cdot |_H h_2 = h_1 \cdot h_2$, which is assumed to be in H , so the image of $\cdot|_H$ is contained in H . Similarly, for any $h \in H$, we have $h^{-1}|_H = h^{-1}$, which is assumed to be in H , so the image of $^{-1}|_H$ is contained in H .

Write $\cdot_H$ as shorthand for $\cdot|_H$. For any $h_1, h_2, h_3 \in H$, we have

$$(h_1 \cdot_H h_2) \cdot_H h_3 = (h_1 \cdot h_2) \cdot h_3 = h_1 \cdot (h_2 \cdot h_3) = h_1 \cdot_H (h_2 \cdot_H h_3)$$

so $\cdot_H$ is associative (Definition 1.0.4).

The identity element 1 of G is an identity element of $\cdot_H$ because

$$1 \cdot_H h = 1 \cdot h = h = h \cdot 1 = h \cdot_H 1$$

for all $h \in H$.

Lastly, for all $h \in H$, the element h^{-1} is an inverse of h with respect to the operation $\cdot_H$ on H because

$$h \cdot_H h^{-1} = h \cdot h^{-1} = 1$$

and

$$h^{-1} \cdot_H h = h^{-1} \cdot h = 1.$$

Thus, H is a subgroup of G as desired. $\square$

Proposition 2.1.21 (Subgroup criterion for finite groups). Let G be a finite group (Definition 2.1.1) and let $H \subseteq G$ be a subset (Definition A.1.2). Then H is a subgroup of G (Definition 2.1.19) if and only if the following conditions hold:

1. $H \neq \emptyset$,
2. H is closed under the group operation, i.e. For all $h_1, h_2 \in H$, the product $h_1 h_2$ is an element of H .

Proof. By Proposition 2.1.20, H is a subgroup of G if and only if the two conditions along with the following condition holds: that H is closed under the inverse operator of G , i.e. for all $h \in H$, the inverse h^{-1} is an element of H . Thus, it suffices to show that assuming that G is a finite group, and the two conditions hold of H , then H is also closed under the inverse operator of G .

For any $h \in H$, since G is finite (Definition 3.6.3), there exist some distinct positive integers a, b such that $h^a = h^b$ (Definition 2.1.4) — if there did not, G would have to have infinitely many distinct elements. Without loss of generality, say that $a > b$. In particular, $h^{a-b} = 1$ (Lemma 2.1.10) (Lemma 2.1.9), so $h^{a-b} \cdot h^{-1} = 1 \cdot h^{-1}$ and hence $h^{a-b-1} = h^{-1}$. Since $a > b$, the exponent $a - b - 1$ is at least 0. If it is 0, then $1 = h^{-1}$, so h^{-1} is an element of H . Otherwise, $a - b - 1$ is positive, in which case $h^{a-b-1} \in H$ (Lemma 1.0.10). Thus, h^{-1} is an element of H . (♠ TODO: formulate the principle of mathematical induction in terms of $P(1)$ and $P(n)$ implies $P(n + 1)$ for all positive integer n implies that $P(n)$ holds for all positive integers n) $\square$

Definition 2.1.22. Let G be a group and let $S \subseteq G$ be a subset. The **subgroup of G generated by S** is the smallest subgroup $\langle S \rangle$ of G containing S — more explicitly, it can be constructed as the intersection

$$\langle S \rangle = \bigcap_{S \subseteq H \leq G} H$$

of subgroups H of G that contain S .

All of the elements of S together are called **generators of $\langle S \rangle$** .

The subgroup $\langle S \rangle$ may also be called the **subgroup of G algebraically generated by S** for emphasis.

Definition 2.1.23. Let G be a group (Definition 2.1.1). It is said to be **finitely generated** if there exists some finite (Definition A.1.5) subset (Definition A.1.2) $S \subseteq G$ such that $\langle S \rangle = G$ (Definition 2.1.22).

Definition 2.1.24 (Conjugation). Let $(G, \cdot)$ be a group (Definition 2.1.1).

1. For $g, h \in G$, the **conjugate of h by g** is the element $ghg^{-1} \in G$. Some may choose an opposite convention, letting the conjugate of h by g refer to the element $g^{-1}hg$ instead.
2. For $g \in G$ and a subgroup (Definition 2.1.19) $H \leq G$, the **conjugate of H by g** is the group

$$gHg^{-1} := \{ghg^{-1} \in G : h \in H\}.$$

It is a subgroup of G .

Some may choose an opposite convention, letting the conjugate of H by g to refer to the group $g^{-1}Hg$ instead.

Definition 2.1.25 (Cosets). Let G be a group (Definition 2.1.1) and $H \leq G$ a subgroup (Definition 2.1.19). For each element $g \in G$, the **left coset of H in G determined by g** is

$$gH := \{gh \mid h \in H\},$$

and the **right coset of H in G determined by g** is

$$Hg := \{hg \mid h \in H\}.$$

Definition 2.1.26 (Coset Space / Quotient Set). (♠ TODO: AI generated, verify) Let G be a group (Definition 2.1.1) and $H \leq G$ a subgroup (Definition 2.1.19).

1. The **left coset space of G by H** (also called the **left quotient set of G by H**) is the set of all left cosets (Definition 2.1.25) of H in G :

$$G/H := \{gH \mid g \in G\}.$$

By definition, the cardinality (Definition A.1.13) of G/H is the index (Definition 2.1.27) of H in G , denoted $[G : H]$.

If $H \trianglelefteq G$ is a normal subgroup (Definition 2.1.36), then G/H carries a natural group structure and is called the quotient group (Definition 2.1.40) of G by H .

2. The **right coset space of G by H** (also called the **right quotient set of G by H**) is the set of all right cosets (Definition 2.1.25) of H in G :

$$H \backslash G := \{Hg \mid g \in G\}.$$

The cardinality of G/H is the index (Definition 2.1.27) of H in G , denoted $[G : H]$.

Definition 2.1.27. Let G be a group (Definition 2.1.1) and let $H \leq G$ be a subgroup (Definition 2.1.19). The *index of H in G* is the cardinality (Definition A.1.13) of the set

$$\{gH : g \in G\}$$

of left cosets (Definition 2.1.25) of H in G . This is equivalent to the cardinality of the set

$$\{Hg : g \in G\}$$

of right cosets (Definition 2.1.25) of H in G . (Proposition 2.1.32)

Definition 2.1.28. Let G be a group (Definition 2.1.1). The *order of G* , denoted by $|G|$, is the cardinality (Definition A.1.13) of the underlying set of G . If $|G|$ is finite (Definition A.1.5), then G is called a *finite group*. Otherwise, G is called an *infinite group*.

Lemma 2.1.29 (Cosets are equal iff product of elements is in subgroup). Let G be a group (Definition 2.1.1), let $H \leq G$ be a subgroup (Definition 2.1.19), and let $g_1, g_2 \in G$.

1. $g_1H = g_2H$ (Definition 2.1.25) if and only if $g_1^{-1}g_2 \in H$.
2. $Hg_1 = Hg_2$ (Definition 2.1.25) if and only if $g_1g_2^{-1} \in H$.

Proof. We first prove the statement for left cosets.

Suppose that $g_1H = g_2H$. Since $e \in H$, we have $g_1 = g_1e \in g_1H = g_2H$, so $g_1 = g_2h$ for some $h \in H$. Multiplying both sides on the left by g_1^{-1} and on the right by h^{-1} , we obtain $g_1^{-1}g_2 = h \in H$. Hence, $g_1^{-1}g_2 \in H$.

Conversely, suppose that $g_1^{-1}g_2 \in H$, and let $h' \in H$ be such that $g_1^{-1}g_2 = h'$. We show that $g_1H \subseteq g_2H$ and $g_2H \subseteq g_1H$. For any $x \in g_1H$, we have $x = g_1h$ for some $h \in H$. Writing $g_1 = g_2(h')^{-1}$, we obtain

$$x = g_2(h')^{-1}h.$$

Since $(h')^{-1}h \in H$, it follows that $x \in g_2H$. Therefore, $g_1H \subseteq g_2H$.

For the reverse inclusion, note that $g_2 = g_1h'$, so for any $y \in g_2H$, we have $y = g_2h$ for some $h \in H$. Writing this as

$$y = g_1h'h,$$

and since $h'h \in H$, we obtain $y \in g_1H$. Therefore, $g_2H \subseteq g_1H$.

It follows that $g_1H = g_2H$.

The statement for right cosets is analogous. Suppose that $Hg_1 = Hg_2$. Since $e \in H$, we have $g_1 = eg_1 \in Hg_1 = Hg_2$, so $g_1 = hg_2$ for some $h \in H$. Multiplying both sides on the left by h^{-1} and on the right by g_2^{-1} , we obtain $g_1g_2^{-1} = h \in H$. Conversely, suppose that $g_1g_2^{-1} \in H$, and let $h'' \in H$ be such that $g_1g_2^{-1} = h''$. Then $g_1 = h''g_2$, and for any $x \in Hg_1$, we have $x = hg_1 = hh''g_2 \in Hg_2$. For the reverse inclusion, $g_2 = (h'')^{-1}g_1$, so for any $y \in Hg_2$, we have $y = hg_2 = h(h'')^{-1}g_1 \in Hg_1$. Therefore, $Hg_1 = Hg_2$. $\square$

Lemma 2.1.30 (Cosets partition the group). Let G be a group (Definition 2.1.1) and let $H \leq G$ be a subgroup (Definition 2.1.19).

1. The collection of left cosets $\{gH : g \in G\}$ covers G , and any two distinct left cosets are disjoint. In other words, the left cosets form a partition (Definition A.1.11) of G .
2. The collection of right cosets $\{Hg : g \in G\}$ covers G , and any two distinct right cosets are disjoint. In other words, the right cosets form a partition (Definition A.1.11) of G .

Proof. We verify the conditions for the left cosets to form a partition of G ; the argument for right cosets is analogous.

Every element of G belongs to some left coset because for any $x \in G$, we have $x = xe \in xH$, where e is the identity element of G by (Definition 2.1.1).

Suppose that $g_1, g_2 \in G$ satisfy $g_1H \cap g_2H \neq \emptyset$. Then there exist $h_1, h_2 \in H$ such that $g_1h_1 = g_2h_2$. Multiplying both sides on the left by g_1^{-1} and on the right by h_2^{-1} (which exist since $H \leq G$), we obtain

$$g_1^{-1}g_2 = h_1h_2^{-1}.$$

Since $h_1h_2^{-1} \in H$, it follows that $g_1^{-1}g_2 \in H$, which implies that $g_1H = g_2H$ (Lemma 2.1.29).

Therefore, any two left cosets are either disjoint or equal. Since they also cover G , the left cosets form a partition of G . For right cosets, an intersecting pair $Hg_1 \cap Hg_2 \neq \emptyset$ yields $h_1g_1 = h_2g_2$ for some $h_1, h_2 \in H$, so $g_1g_2^{-1} = h_1^{-1}h_2 \in H$, which implies that $Hg_1 = Hg_2$ (Lemma 2.1.29). $\square$

Lemma 2.1.31 (All cosets have equal cardinality). Let G be a group (Definition 2.1.1) and let $H \leq G$ be a subgroup (Definition 2.1.19). For any $g \in G$, the map

$$\varphi_g : H \longrightarrow gH, \quad h \longmapsto gh$$

is a bijection (Definition A.1.3). In particular, $|gH| = |H|$ for all $g \in G$.

Proof. Fix $g \in G$ and define $\varphi_g : H \rightarrow gH$ by $\varphi_g(h) = gh$. We verify that φ_g is well-defined, surjective, and injective.

The map is well-defined because for any $h \in H$, the product gh lies in gH (Definition 2.1.25).

Given any $y \in gH$, there exists $h \in H$ such that $y = gh$ by the definition of the left coset gH , so $\varphi_g(h) = y$. Thus, φ_g is surjective.

Suppose $h_1, h_2 \in H$ satisfy $\varphi_g(h_1) = \varphi_g(h_2)$. Then $gh_1 = gh_2$, and by the left cancellation law (Lemma 2.1.13) in G we obtain $h_1 = h_2$. Thus, φ_g is injective.

Since φ_g is a bijection, the sets H and gH have the same cardinality (Definition A.1.13). $\square$

Proposition 2.1.32 (Left and right cosets of a subgroup have equal cardinality). (♠ TODO: AI generated, verify) Let G be a group (Definition 2.1.1) and let $H \leq G$ be a subgroup (Definition 2.1.19). The sets of left cosets (Definition 2.1.25) $G/H = \{gH : g \in G\}$ and right cosets $H \backslash G = \{Hg : g \in G\}$ have the same cardinality (Definition A.1.13).

Proof. (♠ TODO: AI generated, verify) Define a map $\phi : G/H \rightarrow H \backslash G$ by $\phi(gH) = Hg^{-1}$ for each $g \in G$. We first verify that ϕ is well-defined. Suppose that $g_1H = g_2H$. Then

$g_2^{-1}g_1 \in H$ (Lemma 2.1.29), so $(g_2^{-1}g_1)^{-1} = g_1^{-1}g_2 \in H$, which implies that $Hg_1^{-1} = Hg_2^{-1}$ (Lemma 2.1.29). Therefore $\phi(g_1H) = \phi(g_2H)$. Thus ϕ is well-defined.

We show that ϕ is injective (Definition 3.5.76). Suppose $\phi(g_1H) = \phi(g_2H)$. Then $Hg_1^{-1} = Hg_2^{-1}$, so $(g_1^{-1})(g_2^{-1})^{-1} = g_1^{-1}g_2 \in H$ (Lemma 2.1.29), which implies that $g_1H = g_2H$ (Lemma 2.1.29). Thus ϕ is injective.

We show that ϕ is surjective. Let $Hg \in H \backslash G$ be an arbitrary right coset. Then $\phi(g^{-1}H) = H(g^{-1})^{-1} = Hg$, so ϕ is surjective.

Since ϕ is a bijection (Definition A.1.3) between G/H and $H \backslash G$, the two sets have the same cardinality. $\square$

Theorem 2.1.33 (Lagrange's Theorem). Let G be a finite group (Definition 2.1.28) and let $H \leq G$ be a subgroup (Definition 2.1.19). Then the order (Definition 2.1.28) $|H|$ of H divides the order $|G|$ of G . More precisely,

$$|G| = |H| \cdot [G : H],$$

where $[G : H]$ is the index (Definition 2.1.27) of H in G .

Proof. Since G is finite, the left cosets $\{gH : g \in G\}$ form a finite collection. By Lemma 2.1.30, these cosets partition G . By Lemma 2.1.31, each coset gH has the same cardinality as H , namely $|H|$.

The index $[G : H]$ is by (Definition 2.1.27) the number of distinct left cosets. Since the cosets are pairwise disjoint and their union is G , we have

$$|G| = \sum_{gH \in G/H} |gH| = \sum_{gH \in G/H} |H| = [G : H] \cdot |H|,$$

where the sum runs over the distinct left cosets. This establishes the equality $|G| = |H| \cdot [G : H]$, from which it follows that $|H|$ divides $|G|$. $\square$

Proposition 2.1.34. Let G be a group (Definition 2.1.1) and let $g \in G$. Let $\langle g \rangle$ be the subgroup generated by (Definition 2.1.22) $\{g\}$, and let $\text{ord}(g)$ be the order of g .

1. If $\text{ord}(g)$ is finite, then $\langle g \rangle$ is a cyclic group of order (Definition 2.1.28) $\text{ord}(g)$. In particular,

$$|\langle g \rangle| = \text{ord}(g).$$

2. If $\text{ord}(g)$ is infinite, then $\langle g \rangle$ is an infinite cyclic group, and the map

$$\varphi : \mathbb{Z} \rightarrow \langle g \rangle, \quad n \mapsto g^n$$

is a bijection (Definition A.1.3). In particular, $|\langle g \rangle|$ is (countably) infinite (Definition A.1.5).

Proof. By Definition 2.1.22, the subgroup $\langle g \rangle$ consists of all finite products of g and g^{-1} , which simplifies to powers of g : $\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$ by (Definition 2.1.4). Define

$$\varphi : \mathbb{Z} \rightarrow \langle g \rangle, \quad n \mapsto g^n.$$

This map is surjective by construction, since every element of $\langle g \rangle$ is a power of g .

We analyze injectivity in the two cases.

1. Suppose that $\text{ord}(g) = m < \infty$. In other words, m is the smallest positive integer with $g^m = e$. We claim that $\varphi(n) = \varphi(k)$ if and only if $n \equiv k \pmod{m}$. Indeed, $\varphi(n) = \varphi(k)$ means $g^n = g^k$, which is equivalent to $g^{n-k} = e$ by the cancellation law in G . By the division algorithm, write $n - k = qm + r$ with $0 \leq r < m$. Then $e = g^{qm+r} = (g^m)^q g^r = g^r$. Since m is the smallest positive integer with $g^m = e$ and $0 \leq r < m$, we must have $g^{n-k} = e$ is equivalent to $r = 0$. In other words, $\varphi(n) = \varphi(k)$ if and only if $n - k = qm$, i.e. $n \equiv k \pmod{m}$.

It follows that φ induces a bijection from the set of residue classes $\{0, 1, \dots, m-1\}$ onto $\langle g \rangle$. Therefore $|\langle g \rangle| = m = \text{ord}(g)$.

2. Suppose that $\text{ord}(g)$ is infinite. In other words, no positive power of g equals e . If $\varphi(n) = \varphi(k)$ for some $n, k \in \mathbb{Z}$, then $g^n = g^k$, so $g^{n-k} = e$. If $n - k > 0$, this contradicts the assumption that no positive power of g is the identity. Similarly, if $n - k < 0$, then $g^{-(n-k)} = e$ with $-(n - k) > 0$, again a contradiction. Hence $n - k = 0$, so $n = k$. This shows φ is injective.

Since φ is both injective and surjective, it is a bijection, so $|\langle g \rangle|$ is countably infinite (Definition A.1.5).

□

Corollary 2.1.35. Let G be a finite group (Definition 2.1.28) and let $g \in G$. Then the order $\text{ord}(g)$ of g divides the order (Definition 2.1.28) $|G|$ of G .

Proof. Let $\langle g \rangle$ be the subgroup of G generated by g . By Proposition 2.1.34, we have $|\langle g \rangle| = \text{ord}(g)$. Since $\langle g \rangle \leq G$ and G is finite, Theorem 2.1.33 implies that $|\langle g \rangle|$ divides $|G|$. Therefore $\text{ord}(g)$ divides $|G|$. □

Definition 2.1.36. Let $(G, \cdot)$ be a group (Definition 2.1.1). A subgroup (Definition 2.1.19) $N \leq G$ is called a **normal subgroup of G** if N is closed under conjugation (Definition 2.1.24) by elements of G , i.e. for all $g \in G$ and $n \in N$, one has

$$gng^{-1} \in N.$$

Equivalently, N is normal in G if every left coset of N in G is also a right coset of N in G (Definition 2.1.55). We denote this relation by $N \trianglelefteq G$, meaning “ N is a normal subgroup of G ”.

Lemma 2.1.37. Let G be an abelian group (Definition 2.1.1). Every subgroup (Definition 2.1.19) of G is a normal subgroup (Definition 2.1.36).

Proof. Let $H \leq G$ be a subgroup. For any $g \in G$ and any $h \in H$, we compute the conjugate (Definition 2.1.24):

$$ghg^{-1} = hgg^{-1} = he = h \in H.$$

Thus, the subgroup H is closed under conjugation by elements of G , i.e. H is a normal subgroup. □

Definition 2.1.38. Let $\phi : G \rightarrow H$ be a group homomorphism (Definition 2.1.17).

1. The *kernel of ϕ* , denoted $\ker(\phi)$, is the set of elements in G that map to the identity element of H :

$$\ker(\phi) = \{g \in G \mid \phi(g) = e_H\}$$

where e_H is the identity element of H .

2. The *image of ϕ* , denoted $\text{im}(\phi)$ or $\phi(G)$, is the set of elements in H that are the image of some element in G :

$$\text{im}(\phi) = \{h \in H \mid \exists g \in G, \phi(g) = h\}$$

Proposition 2.1.39. Let $\phi : G \rightarrow H$ be a group homomorphism (Definition 2.1.17).

1. The kernel $\ker(\phi)$ (Definition 2.1.38) is a normal subgroup (Definition 2.1.36) of G .
2. The image (Definition 2.1.38) $\text{im}(\phi)$ is a subgroup (Definition 2.1.19) of H .
3. The homomorphism ϕ is injective (Definition A.1.3) if and only if $\ker(\phi) = \{e_G\}$, where e_G is the identity element of G .
4. The homomorphism ϕ is surjective (Definition A.1.3) if and only if $\text{im}(\phi) = H$.

Proof. Let e_G and e_H denote the identity elements of G and H , respectively.

1. We verify that $\ker(\phi)$ is a subgroup using the criterion from (Proposition 2.1.20). First, $\ker(\phi) \neq \emptyset$ because $\phi(e_G) = e_H$, so $e_G \in \ker(\phi)$ by (Definition 2.1.38); the equality $\phi(e_G) = e_H$ follows from (Definition 2.1.17). Second, if $g_1, g_2 \in \ker(\phi)$, then

$$\phi(g_1 g_2) = \phi(g_1) \phi(g_2) = e_H e_H = e_H,$$

so $g_1 g_2 \in \ker(\phi)$. Third, if $g \in \ker(\phi)$, then

$$\phi(g^{-1}) = \phi(g)^{-1} = e_H^{-1} = e_H,$$

so $g^{-1} \in \ker(\phi)$. Hence $\ker(\phi) \leq G$.

To show normality, let $x \in G$ and $k \in \ker(\phi)$. Compute:

$$\phi(x k x^{-1}) = \phi(x) \phi(k) \phi(x)^{-1} = \phi(x) e_H \phi(x)^{-1} = e_H,$$

so $x k x^{-1} \in \ker(\phi)$. Thus $\ker(\phi)$ is closed under conjugation by elements of G , making it a normal subgroup by (Definition 2.1.36).

2. We verify that $\text{im}(\phi)$ is a subgroup using the same criterion. First, $\text{im}(\phi) \neq \emptyset$ because $e_H = \phi(e_G) \in \text{im}(\phi)$. Second, if $h_1, h_2 \in \text{im}(\phi)$, there exist $g_1, g_2 \in G$ with $\phi(g_1) = h_1$ and $\phi(g_2) = h_2$, so

$$h_1 h_2 = \phi(g_1) \phi(g_2) = \phi(g_1 g_2) \in \text{im}(\phi).$$

Third, if $h \in \text{im}(\phi)$ with $\phi(g) = h$, then

$$h^{-1} = \phi(g)^{-1} = \phi(g^{-1}) \in \text{im}(\phi).$$

Hence $\text{im}(\phi) \leq H$.

3. Suppose first that ϕ is injective. If $g \in \ker(\phi)$, then $\phi(g) = e_H = \phi(e_G)$. By injectivity, $g = e_G$, so $\ker(\phi) = \{e_G\}$.

Conversely, suppose $\ker(\phi) = \{e_G\}$. If $\phi(g_1) = \phi(g_2)$ for some $g_1, g_2 \in G$, then

$$\phi(g_2^{-1} g_1) = \phi(g_2)^{-1} \phi(g_1) = e_H,$$

so $g_2^{-1} g_1 \in \ker(\phi) = \{e_G\}$. Hence, $g_2^{-1} g_1 = e_G$, which implies that $g_1 = g_2$. Therefore, ϕ is injective.

4. This follows directly from the definition of surjectivity and the image: ϕ is surjective means for every $h \in H$ there exists $g \in G$ with $\phi(g) = h$, which is equivalent to saying that every element of H lies in $\text{im}(\phi)$, i.e., $\text{im}(\phi) = H$.

□

Definition 2.1.40 (Quotient Group). Let G be a group (Definition 2.1.1) and $N \trianglelefteq G$ a normal subgroup (Definition 2.1.36). The *quotient group of G by N* , denoted by G/N , is the set of all left (or equivalently, right) cosets (Definition 2.1.25) of N in G ,

$$G/N := \{gN \mid g \in G\},$$

(Definition 2.1.26) endowed with the operation defined by

$$(gN)(hN) := (gh)N \quad \text{for all } g, h \in G.$$

This operation is well-defined (Proposition 2.1.41) because N is normal in G , and under this operation G/N is a group with identity element N and inverse given by $(gN)^{-1} = g^{-1}N$.

There is (Theorem 2.1.42) a canonical surjective homomorphism $G \rightarrow G/N$ given by $g \mapsto gN$.

Note that any abelian group (Definition 2.1.1) has a quotient by any subgroup (Definition 2.1.19) as all subgroups of an abelian group are normal (Lemma 2.1.37).

Proposition 2.1.41. Let G be a group (Definition 2.1.1) and N a normal subgroup (Definition 2.1.36) of G . The binary operation of the quotient group (Definition 2.1.40) G/N is well defined, i.e.

$$(gN)(hN) := (gh)N$$

is well defined, and G/N is indeed a group.

Proof. Let e denote the identity element of G . We first show that the operation on cosets is well-defined, then verify the group axioms.

To show that the operation $(gN)(hN) = (gh)N$ is well-defined, suppose that $g_1N = g_2N$ and $h_1N = h_2N$. Then $g_2^{-1}g_1 \in N$ (Lemma 2.1.29) and $h_2^{-1}h_1 \in N$ (Lemma 2.1.29). We need to show $(g_1h_1)N = (g_2h_2)N$, i.e., that $(g_2h_2)^{-1}(g_1h_1) \in N$. Compute:

$$(g_2h_2)^{-1}(g_1h_1) = h_2^{-1}g_2^{-1}g_1h_1.$$

Since N is normal, we have $h_2^{-1}(g_2^{-1}g_1)h_2 \in N$ (Definition 2.1.36). Therefore

$$h_2^{-1}g_2^{-1}g_1h_1 = (h_2^{-1}(g_2^{-1}g_1)h_2)(h_2^{-1}h_1).$$

Both factors lie in N : the first by normality and the second because $h_2^{-1}h_1 \in N$. Their product is therefore in N by closure of N under the group operation, which implies that $(g_1h_1)N = (g_2h_2)N$ (Lemma 2.1.29).

For any $g_1, g_2, g_3 \in G$, we have

$$((g_1N)(g_2N))(g_3N) = ((g_1g_2)N)(g_3N) = ((g_1g_2)g_3)N = (g_1(g_2g_3))N = (g_1N)((g_2N)(g_3N)),$$

where $(g_1g_2)g_3 = g_1(g_2g_3)$ by associativity of the group operation in G . Thus, the operation on G/N is associative (Definition 1.0.4).

The identity element (Definition 1.0.2) is the coset $eN = N$, because for any $gN \in G/N$ we have

$$(eN)(gN) = (eg)N = gN$$

and

$$(gN)(eN) = (ge)N = gN,$$

since e is the identity in G .

We show that $(g^{-1}N)$ is the inverse (Definition 1.0.5) of gN . We compute

$$(gN)(g^{-1}N) = (gg^{-1})N = eN = N$$

and

$$(g^{-1}N)(gN) = (g^{-1}g)N = eN = N,$$

so $(gN)^{-1} = g^{-1}N$.

This verifies that G/N satisfies all the group axioms with the well-defined operation. $\square$

Theorem 2.1.42. Let G be a group (Definition 2.1.1) and let N be a normal subgroup (Definition 2.1.36) of G . The map $\pi : G \rightarrow G/N$ (Definition 2.1.40) defined by $\pi(g) = gN$ is a surjective (Definition A.1.3) group homomorphism (Definition 2.1.17) with kernel (Definition 2.1.38) N . This map is called the *canonical projection* or *natural homomorphism*.

Proof. Let $\pi : G \rightarrow G/N$ be defined by $\pi(g) = gN$. We verify each property in turn.

1. For any $g_1, g_2 \in G$, we compute using the operation on G/N :

$$\pi(g_1g_2) = (g_1g_2)N = (g_1N)(g_2N) = \pi(g_1)\pi(g_2).$$

Thus, π is a homomorphism.

2. To see that π is surjective, note that we have $\pi(x) = xN$ for any coset $xN \in G/N$, so every element of G/N is in the image.
3. To see that the kernel of π is exactly N , recall that $eN = N$ is the identity element of G/N and, for any $g \in G$, that

$$\pi(g) = N \iff gN = N \iff g \in N$$

Therefore $\ker(\pi) = \{g \in G : \pi(g) = N\} = N$.

This establishes that π is a surjective homomorphism with kernel N . $\square$

Lemma 2.1.43 (Factorization of a Group Homomorphism through a Quotient). (♠ TODO: AI generated, verify) Let G and H be groups (Definition 2.1.1), let $\phi : G \rightarrow H$ be a group homomorphism (Definition 2.1.17), and let N be a normal subgroup (Definition 2.1.36) of G such that $N \subseteq \ker(\phi)$. Let $\pi : G \rightarrow G/N$ denote the canonical projection (Theorem 2.1.42).

Then there exists a unique group homomorphism (Definition 2.1.17) $\bar{\phi} : G/N \rightarrow H$ such that $\phi = \bar{\phi} \circ \pi$. Explicitly, $\bar{\phi}(gN) = \phi(g)$ for all $g \in G$. Moreover:

1. The homomorphism $\bar{\phi}$ is injective (Definition A.1.3) if and only if $N = \ker(\phi)$.
2. If $N = \ker(\phi)$, then $\bar{\phi}$ is a group homomorphism (Definition 2.1.17) that is a bijection onto $\text{im}(\phi)$.

Proof. (♠ TODO: AI generated, verify) Let e_H denote the identity element of H . Define $\bar{\phi} : G/N \rightarrow H$ by $\bar{\phi}(gN) = \phi(g)$.

We show that $\bar{\phi}$ is well-defined. Suppose that $g_1N = g_2N$ for some $g_1, g_2 \in G$. Then $g_2^{-1}g_1 \in N$ (Lemma 2.1.29). Since $N \subseteq \ker(\phi)$, we have $\phi(g_2^{-1}g_1) = e_H$. Because ϕ is a homomorphism (Definition 2.1.17), we obtain $\phi(g_2)^{-1}\phi(g_1) = e_H$. Multiplying both sides on the left by $\phi(g_2)$, we get $\phi(g_1) = \phi(g_2)$. Hence, $\bar{\phi}(g_1N) = \bar{\phi}(g_2N)$.

We show that $\bar{\phi}$ is a group homomorphism. For any $g_1N, g_2N \in G/N$, we have $(g_1N)(g_2N) = (g_1g_2)N$ (Proposition 2.1.41). Therefore,

$$\bar{\phi}((g_1N)(g_2N)) = \bar{\phi}((g_1g_2)N) = \phi(g_1g_2) = \phi(g_1)\phi(g_2) = \bar{\phi}(g_1N)\bar{\phi}(g_2N),$$

where $\phi(g_1g_2) = \phi(g_1)\phi(g_2)$ because ϕ is a homomorphism.

For any $g \in G$, we have $(\bar{\phi} \circ \pi)(g) = \bar{\phi}(gN) = \phi(g)$, so $\phi = \bar{\phi} \circ \pi$.

We show uniqueness. Suppose that $\psi : G/N \rightarrow H$ is another group homomorphism satisfying $\phi = \psi \circ \pi$. Then for any $gN \in G/N$, we have $\psi(gN) = \psi(\pi(g)) = \phi(g) = \bar{\phi}(gN)$, so $\psi = \bar{\phi}$.

We establish the injectivity criterion. The kernel of $\bar{\phi}$ is

$$\ker(\bar{\phi}) = \{gN \in G/N : \bar{\phi}(gN) = e_H\} = \{gN : \phi(g) = e_H\} = \{gN : g \in \ker(\phi)\}.$$

This equals the trivial coset N if and only if $\ker(\phi) \subseteq N$. Since we already have $N \subseteq \ker(\phi)$ by hypothesis, the injectivity of $\bar{\phi}$ is equivalent to $N = \ker(\phi)$ by Proposition 2.1.39(3).

When $N = \ker(\phi)$, the previous item implies that $\bar{\phi}$ is injective. The image of $\bar{\phi}$ equals $\text{im}(\phi)$ because for any $g \in G$, we have $\bar{\phi}(gN) = \phi(g)$, and conversely every element of $\text{im}(\phi)$ has the form $\phi(g) = \bar{\phi}(gN)$. Therefore, $\bar{\phi}$ is a bijection onto $\text{im}(\phi)$. $\square$

Lemma 2.1.44 (Fibers of a homomorphism are cosets of the kernel). Let G and H be groups (Definition 2.1.1), and let $\phi : G \rightarrow H$ be a group homomorphism (Definition 2.1.17). For any $g_0 \in G$ and any $h = \phi(g_0) \in \text{im}(\phi)$, the fiber of ϕ over h is the left coset of $\ker(\phi)$ determined by g_0 :

$$\phi^{-1}(h) = \{g \in G : \phi(g) = h\} = g_0 \cdot \ker(\phi).$$

Proof. (♠ TODO: AI generated, verify) Let $K = \ker(\phi)$ (Definition 2.1.38). We show set equality by proving both inclusions.

We first show that $g_0K \subseteq \phi^{-1}(h)$. Suppose that $g \in g_0K$. Then $g = g_0k$ for some $k \in K$. Because ϕ is a group homomorphism (Definition 2.1.17) and because $k \in K$ implies $\phi(k) = e_H$ (Definition 2.1.38), we have

$$\phi(g) = \phi(g_0k) = \phi(g_0)\phi(k) = h \cdot e_H = h.$$

Hence, $g \in \phi^{-1}(h)$.

Conversely, we show that $\phi^{-1}(h) \subseteq g_0K$. Suppose that $g \in G$ satisfies $\phi(g) = h$. Because ϕ is a homomorphism, we have

$$\phi(g_0^{-1}g) = \phi(g_0)^{-1}\phi(g) = h^{-1}h = e_H.$$

Therefore, $g_0^{-1}g \in K$, so $g = g_0(g_0^{-1}g) \in g_0K$.

Since both inclusions hold, $\phi^{-1}(h) = g_0K$ as desired. $\square$

Theorem 2.1.45 (First Isomorphism Theorem for Groups). Let G and H be groups (Definition 2.1.1), and let $\phi : G \rightarrow H$ be a group homomorphism (Definition 2.1.17). Let $\ker(\phi)$ denote the kernel of ϕ (Definition 2.1.38) and $\text{im}(\phi)$ denote the image of ϕ (Definition 2.1.38).

Then $\ker(\phi)$ is a normal subgroup (Definition 2.1.36) of G , and there exists a unique group homomorphism (Definition 2.1.17)

$$\bar{\phi} : G/\ker(\phi) \xrightarrow{\sim} \text{im}(\phi)$$

which is a bijection, such that $\bar{\phi}(g \cdot \ker(\phi)) = \phi(g)$ for all $g \in G$. Here $G/\ker(\phi)$ denotes the quotient group (Definition 2.1.40).

Proof. Let $K = \ker(\phi)$ and let e_H denote the identity element of H .

1. The kernel K is a normal subgroup of G by Proposition 2.1.39, so the quotient group (Definition 2.1.40) G/K is well-defined.
2. We define a map on objects

$$\bar{\phi} : G/K \longrightarrow \text{im}(\phi), \quad gK \longmapsto \phi(g).$$

To show that $\bar{\phi}$ is well-defined, suppose $g_1K = g_2K$ for some $g_1, g_2 \in G$. Then $g_2^{-1}g_1 \in K$, so $\phi(g_2^{-1}g_1) = e_H$ by Definition 2.1.38. Since ϕ is a group homomorphism (Definition 2.1.17), we compute

$$\phi(g_1) = \phi(g_2(g_2^{-1}g_1)) = \phi(g_2)\phi(g_2^{-1}g_1) = \phi(g_2)e_H = \phi(g_2).$$

Thus, $\bar{\phi}(g_1K) = \bar{\phi}(g_2K)$, so the map is independent of the choice of coset representative.

3. The map $\bar{\phi}$ is a group homomorphism because for any $g_1K, g_2K \in G/K$, we have $(g_1K)(g_2K) = (g_1g_2)K$ by Definition 2.1.40, and since ϕ is a homomorphism,

$$\bar{\phi}((g_1K)(g_2K)) = \bar{\phi}((g_1g_2)K) = \phi(g_1g_2) = \phi(g_1)\phi(g_2) = \bar{\phi}(g_1K) \cdot \bar{\phi}(g_2K).$$

(Proposition 2.1.41) (Definition 2.1.17)

4. We show that the map $\bar{\phi}$ is bijective. First, $\bar{\phi}$ is surjective onto $\text{im}(\phi)$ because for any $y \in \text{im}(\phi)$ there exists $g \in G$ such that $\phi(g) = y$ by Definition 2.1.38, and then $\bar{\phi}(gK) = \phi(g) = y$. Second, $\bar{\phi}$ is injective because if $\bar{\phi}(g_1K) = \bar{\phi}(g_2K)$ for some $g_1, g_2 \in G$, then $\phi(g_1) = \phi(g_2)$, and since ϕ is a homomorphism we have

$$\phi(g_2^{-1}g_1) = \phi(g_2)^{-1}\phi(g_1) = e_H.$$

Therefore $g_2^{-1}g_1 \in K$, which means $g_1K = g_2K$.

To show that $\bar{\phi}$ is unique, suppose $\psi : G/K \rightarrow \text{im}(\phi)$ is another homomorphism satisfying $\psi(gK) = \phi(g)$ for all $g \in G$. Then ψ agrees with $\bar{\phi}$ on every element of G/K , so $\psi = \bar{\phi}$.

This establishes the existence and uniqueness of the isomorphism $\bar{\phi} : G/\ker(\phi) \xrightarrow{\sim} \text{im}(\phi)$. $\square$

Theorem 2.1.46 (Second Isomorphism Theorem for Groups (Diamond Isomorphism Theorem)). Let G be a group (Definition 2.1.1). Let H be a subgroup (Definition 2.1.19) of G , and let N be a normal subgroup (Definition 2.1.36) of G .

Then $HN = \{hn : h \in H, n \in N\}$ is a subgroup (Definition 2.1.19) of G , $H \cap N$ is a normal subgroup (Definition 2.1.36) of H , and there is a group isomorphism (Definition 2.1.17)

$$\psi : H/(H \cap N) \xrightarrow{\sim} HN/N$$

(Definition 2.1.40) given by $\psi(h(H \cap N)) = hN$ for all $h \in H$.

Proof. Let e denote the identity element of G .

1. We show that the set $HN = \{hn : h \in H, n \in N\}$ is a subgroup of G . First, $e = ee \in HN$. Second, if $h_1n_1, h_2n_2 \in HN$, then $n_1h_2 = h_2(h_2^{-1}n_1h_2)$, so

$$h_1n_1h_2n_2 = h_1h_2(h_2^{-1}n_1h_2)n_2.$$

Since $h_2^{-1}n_1h_2 \in N$ by normality, the above lies in HN . Third, if $hn \in HN$, then we have $n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1})$, so

$$(hn)^{-1} = h^{-1}(hn^{-1}h^{-1}).$$

Since $hn^{-1}h^{-1} \in N$ by the normality of N in G , we have $(hn)^{-1} \in HN$. Therefore, HN is indeed a subgroup of G .

2. We show that the intersection $H \cap N$ is a normal subgroup of H . It is a subgroup since both H and N are subgroups. To show that $H \cap N$ is in fact a normal subgroup of H , let $h_0 \in H$ and $x \in H \cap N$. Then $h_0xh_0^{-1} \in N$ because $x \in N$ and $N \trianglelefteq G$, and also $h_0xh_0^{-1} \in H$ because $x \in H$ and H is a group. Hence $h_0xh_0^{-1} \in H \cap N$, so $H \cap N \trianglelefteq H$.
3. The quotient groups $H/(H \cap N)$ and HN/N are well-defined because $H \cap N \trianglelefteq H$ by part 2 and $N \trianglelefteq HN$ since $N \trianglelefteq G$ and HN is a subgroup of G containing N as a consequence of ??.
4. Define $\psi : H/(H \cap N) \rightarrow HN/N$ by $\psi(h(H \cap N)) = hN$. We show that it is a group isomorphism. To show that ψ is well defined, suppose that $h_1(H \cap N) = h_2(H \cap N)$ for some $h_1, h_2 \in H$. Then $h_2^{-1}h_1 \in H \cap N$, so in particular $h_2^{-1}h_1 \in N$, which implies that $h_1N = h_2N$ (Lemma 2.1.29). Hence $\psi(h_1(H \cap N)) = \psi(h_2(H \cap N))$.

The map ψ is a group homomorphism because for any $h_1, h_2 \in H$,

$$\psi((h_1(H \cap N))(h_2(H \cap N))) = \psi((h_1h_2)(H \cap N)) = h_1h_2N = (h_1N)(h_2N) = \psi(h_1(H \cap N)) \cdot \psi(h_2(H \cap N)).$$

(Definition 2.1.40)

We show that ψ is injective. Suppose that $\psi(h(H \cap N)) = N$, the identity in HN/N . Then $hN = N$, so $h \in N$. Since $h \in H$ as well, we have that $h \in H \cap N$, which means $h(H \cap N) = H \cap N$, the identity in $H/(H \cap N)$. Hence $\ker(\psi)$ is trivial, so ψ is injective by Proposition 2.1.39(3).

We show that ψ is surjective. Given any element $xN \in HN/N$, we have $x = hn$ for some $h \in H, n \in N$ so $xN = hnN = hN = \psi(h(H \cap N))$. Therefore, ψ is a group isomorphism $H/(H \cap N) \cong HN/N$.

□

Theorem 2.1.47 (Third Isomorphism Theorem for Groups (Factor Theorem)). Let G be a group (Definition 2.1.1). Let N and M be normal subgroups (Definition 2.1.36) of G with $M \subseteq N$.

Then $N/M = \{nM : n \in N\}$ is a normal subgroup (Definition 2.1.36) of the quotient group (Definition 2.1.40) G/M , and there is a group homomorphism (Definition 2.1.17)

$$\Phi : (G/M)/(N/M) \xrightarrow{\sim} G/N$$

which is a bijection, given by $\Phi((gM)(N/M)) = gN$ for all $g \in G$.

Proof. Let $\pi : G \rightarrow G/M$ be the canonical projection by Theorem 2.1.42.

1. We show that the set $N/M = \{nM : n \in N\}$ is a subgroup of G/M . Since $M \subseteq N$, the image $\pi(N)$ equals $\{nM : n \in N\} = N/M$. As N is a subgroup of G , its image under the homomorphism π is a subgroup by Proposition 2.1.39(2).

To see that N/M is normal in G/M , let $gM \in G/M$ and $nM \in N/M$. Compute the conjugate:

$$(gM)(nM)(gM)^{-1} = (gM)(nM)(g^{-1}M) = (gng^{-1})M.$$

Since $N \trianglelefteq G$, we have $gng^{-1} \in N$, so $(gng^{-1})M \in N/M$. Hence $N/M \trianglelefteq G/M$.

2. Define $\Phi : (G/M)/(N/M) \rightarrow G/N$ by $\Phi((gM)(N/M)) = gN$. To show that Φ is well-defined, suppose that $(g_1M)(N/M) = (g_2M)(N/M)$. In particular, $g_2^{-1}g_1M \in N/M$, so there exists $n \in N$ with $g_2^{-1}g_1M = nM$. This means $n^{-1}g_2^{-1}g_1 \in M \subseteq N$, hence $g_2^{-1}g_1 \in N$, which implies that $g_1N = g_2N$ (Lemma 2.1.29).
3. The map Φ is a group homomorphism because for any $g_1, g_2 \in G$,

$$\begin{aligned} \Phi(((g_1M)(N/M))((g_2M)(N/M))) &= \Phi((g_1g_2M)(N/M)) = g_1g_2N = (g_1N)(g_2N) \\ &= \Phi((g_1M)(N/M)) \cdot \Phi((g_2M)(N/M)). \end{aligned}$$

(Definition 2.1.40)

4. We show that the map Φ is injective. Suppose that $\Phi((gM)(N/M))$ equals N , the identity in G/N . In other words, $gN = N$, so $g \in N$. Thus, $gM \in N/M$, which means $(gM)(N/M) = N/M$, the identity in $(G/M)/(N/M)$. Therefore, $\ker(\Phi)$ is trivial, so Φ is injective by part 3 of Proposition 2.1.39.
5. The map Φ is surjective because given any $gN \in G/N$, we have $\Phi((gM)(N/M)) = gN$.

This establishes that Φ is a group isomorphism $(G/M)/(N/M) \cong G/N$. $\square$

2.1.1. Some basic examples of groups.

Definition 2.1.48. Let X be a set. The *symmetric group on X* or the *permutation group on X* , denoted by $\mathfrak{S}_X$, S_X , or Sym_X , is the group (Definition 2.1.1) whose elements are all bijections (Definition A.1.3) $\sigma : X \rightarrow X$, with the group operation being composition of maps:

$$(\sigma \circ \tau)(x) := \sigma(\tau(x)), \quad \sigma, \tau \in \mathfrak{S}_X, \quad x \in X.$$

The identity element is the identity map id_X (Definition A.1.7), and the inverse of σ is its inverse function σ^{-1} (Definition A.1.3).

Elements of the symmetric group are called *permutations*.

A symmetric group on a finite set (Definition A.1.5) of n elements is typically denoted by $\mathfrak{S}_n$, S_n , or Sym_n . Any two symmetric groups on sets of equal cardinality are isomorphic (Definition 2.1.17).

Definition 2.1.49. Let $\mathfrak{S}_n$ denote the symmetric group (Definition 2.1.48) on $\{1, 2, \dots, n\}$. For $\tau \in \mathfrak{S}_n$, the *parity of τ* is defined to be 0 if τ can be expressed as a product of an even number of transpositions, and 1 otherwise. The *signature of τ* or *sign of τ* is then denoted by $\text{sgn}(\tau)$, defined by

$$\text{sgn}(\tau) := (-1)^{\text{parity}(\tau)} \in \{1, -1\}.$$

Let X be a finite set and $\mathfrak{S}_X$ its symmetric group. The *signature or sign of an element of $\mathfrak{S}_X$* , denoted by $\text{sgn}(\sigma)$ for $\sigma \in \mathfrak{S}_X$, is defined by choosing any bijection $\phi : X \rightarrow \{1, 2, \dots, n\}$, setting $\tau := \phi \circ \sigma \in \mathfrak{S}_n$, and then

$$\text{sgn}(\sigma) := \text{sgn}(\tau).$$

This definition is independent of the choice of ϕ , and thus $\text{sgn} : \mathfrak{S}_X \rightarrow \{1, -1\}$ is a well-defined group homomorphism (Definition 2.1.17).

2.1.2. Normalizer of a subgroup and centralizer of a subset of a group.

Definition 2.1.50 (Normalizer). Let $(G, \cdot)$ be a group (Definition 2.1.1) and let $H \subseteq G$ be a subgroup (Definition 2.1.19). The *normalizer of H in G* is defined as

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

Lemma 2.1.51. Let $(G, \cdot)$ be a group (Definition 2.1.1) and let $H \subseteq G$ be a subgroup (Definition 2.1.19). The normalizer $N_G(H)$ (Definition 2.1.50) is a subgroup (Definition 2.1.19) of G .

Proof. We verify the subgroup axioms for $N_G(H)$ inside G . The identity element $e \in G$ satisfies $eHe^{-1} = eHe = H$, so $e \in N_G(H)$. For any $g_1, g_2 \in N_G(H)$, we have

$$(g_1g_2)H(g_1g_2)^{-1} = g_1(g_2Hg_2^{-1})g_1^{-1} = g_1Hg_1^{-1} = H,$$

so $g_1g_2 \in N_G(H)$. For any $g \in N_G(H)$, we have

$$g^{-1}Hg = g^{-1}(gHg^{-1})g = H,$$

so $g^{-1} \in N_G(H)$. Hence, $N_G(H)$ is a subgroup of G (Proposition 2.1.20), i.e. $N_G(H) \leq G$. $\square$

Proposition 2.1.52. Let $(G, \cdot)$ be a group (Definition 2.1.1) and let $H \subseteq G$ be a subgroup (Definition 2.1.19). The group H is a normal subgroup (Definition 2.1.36) of the normalizer $N_G(H)$ (Definition 2.1.50).

Proof. By Lemma 2.1.51, the set $N_G(H)$ is a subgroup (Definition 2.1.19) of G .

We first show that $H \subseteq N_G(H)$. For any $h \in H$, we have $hHh^{-1} = H$ because H is a subgroup (Definition 2.1.19) of G : the map $H \rightarrow H$ given by $x \mapsto h x h^{-1}$ is well-defined and has inverse $x \mapsto h^{-1} x h$. Hence, $h \in N_G(H)$ by definition of the normalizer (Definition 2.1.50).

We now show that $H \trianglelefteq N_G(H)$. For any $n \in N_G(H)$ and $h \in H$, we have $nHn^{-1} = H$ by definition of the normalizer (Definition 2.1.50), so in particular $nhn^{-1} \in H$. Therefore, H is a normal subgroup (Definition 2.1.36) of $N_G(H)$. $\square$

Definition 2.1.53 (Centralizer). Let $(G, \cdot)$ be a group (Definition 2.1.1) and let $S \subseteq G$ be a subset. The *centralizer of S in G* is defined as

$$C_G(S) = \{g \in G : gs = sg \text{ for all } s \in S\}.$$

Definition 2.1.54. Let $(G, \cdot)$ be a group (Definition 2.1.1). The *center of G* , denoted by $Z(G)$, is the subset (Definition A.1.2) of G defined by

$$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}.$$

The center $Z(G)$ is a normal subgroup (Definition 2.1.36) of G . Elements of $Z(G)$ are called *central elements of G* .

One says that G is *center free* if $Z(G) = \{1\}$.

2.1.3. Group actions.

Definition 2.1.55 (Left and Right Group Actions). Let $(G, \cdot)$ be a group (Definition 2.1.1) and let X be a set.

1. A *left group action of G on X* is a function

$$\varphi : G \times X \rightarrow X$$

such that for all $g, h \in G$ and all $x \in X$,

$$\begin{aligned}\varphi(e, x) &= x, \\ \varphi(gh, x) &= \varphi(g, \varphi(h, x)).\end{aligned}$$

For $g \in G$ and $x \in X$, the element $\varphi(g, x)$ is called the *image of x under the action of g* . We often write $g \cdot x$ for $\varphi(g, x)$.

If G acts on X on the left via a function $\varphi : G \times X \rightarrow X$, we write

$$G \curvearrowright X.$$

2. A *right group action of G on X* is a function

$$\psi : X \times G \rightarrow X$$

such that for all $g, h \in G$ and all $x \in X$,

$$\begin{aligned}\psi(x, e) &= x, \\ \psi(x, gh) &= \psi(\psi(x, g), h).\end{aligned}$$

For $x \in X$ and $g \in G$, the element $\psi(x, g)$ is called the *image of x under the right action of g* . We often write $x \cdot g$ for $\psi(x, g)$.

If G acts on X on the right via a function $\psi : X \times G \rightarrow X$, we write

$$X \curvearrowleft G.$$

Definition 2.1.56 (Orbit). Let $(G, \cdot)$ be a group (Definition 2.1.1) acting on (Definition 2.1.55) a set X , with action denoted by $g \cdot x$. For $x \in X$, the *orbit of x* under the action of G is

$$G \cdot x = \{g \cdot x : g \in G\}.$$

Definition 2.1.57 (Stabilizer). Let $(G, \cdot)$ be a group acting on a set X . For $x \in X$, the *stabilizer of x in G* is

$$\text{Stab}_G(x) = \{g \in G : g \cdot x = x\}.$$

Definition 2.1.58 (Fixed Point Set). Let $(G, \cdot)$ be a group acting on (Definition 2.1.55) a set X , say on the left. For $H \subseteq G$, the *fixed point set of H in X* or the *set of H -invariants in X* is defined by

$$X^H = \{x \in X : h \cdot x = x \text{ for all } h \in H\}.$$

Proposition 2.1.59. Let $(G, \cdot)$ be a group (Definition 2.1.1) acting (Definition 2.1.55) (on the left) on a set X . Define a binary relation (Definition A.1.8) $\sim$ on X by

$$x \sim y \iff \exists g \in G : g \cdot x = y.$$

Then $\sim$ is an equivalence relation (Definition A.1.9) on X , and the equivalence classes (Definition A.1.10) are exactly the orbits (Definition 2.1.56) $\{G \cdot x : x \in X\}$. In other words, the orbits partition (Definition A.1.11) X .

Proof. We first show that $\sim$ is an equivalence relation on X .

1. $\sim$ is reflexive because for every $x \in X$, we have $e \cdot x = x$ by the identity axiom of the group action (Definition 2.1.55), so $x \sim x$.
2. $\sim$ is symmetric in that if $x \sim y$, then there exists some $g \in G$ such that $g \cdot x = y$. Applying g^{-1} to both sides, we obtain

$$g^{-1} \cdot (g \cdot x) = g^{-1} \cdot y.$$

By the compatibility axiom of the group action (Definition 2.1.55), the left-hand side equals $(g^{-1}g) \cdot x = e \cdot x = x$. Hence $g^{-1} \cdot y = x$, so $y \sim x$.

3. $\sim$ is transitive in that if $x \sim y$ and $y \sim z$, then there exist $g, h \in G$ such that $g \cdot x = y$ and $h \cdot y = z$. Substituting the first equation into the second gives

$$z = h \cdot (g \cdot x) = (hg) \cdot x,$$

where the last equality follows from the compatibility axiom (Definition 2.1.55). Since $hg \in G$, we have $x \sim z$.

Next we show that the equivalence classes of $\sim$ coincide with the orbits. Fix $x \in X$. The equivalence class (Definition A.1.10) of x is

$$[x] = \{y \in X : x \sim y\} = \{y \in X : \exists g \in G, g \cdot x = y\}.$$

On the other hand, the orbit (Definition 2.1.56) of x is

$$G \cdot x = \{g \cdot x : g \in G\}.$$

These two sets are identical by definition. □

Proposition 2.1.60. Let $(G, \cdot)$ be a group (Definition 2.1.1) acting (Definition 2.1.55) (on the left) on a set X . For all $x \in X$, the stabilizer (Definition 2.1.57)

$$\text{Stab}_G(x) = \{g \in G : g \cdot x = x\}$$

is a subgroup (Definition 2.1.19) of G .

Proof. The stabilizer $\text{Stab}_G(x)$ is nonempty because $e \cdot x = x$ by the identity axiom of the group action (Definition 2.1.55), so $e \in \text{Stab}_G(x)$.

The stabilizer $\text{Stab}_G(x)$ is closed under the group operation because if $g_1, g_2 \in \text{Stab}_G(x)$, then $g_1 \cdot x = x$ and $g_2 \cdot x = x$, so

$$(g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x) = g_1 \cdot x = x,$$

where the first equality follows from the compatibility axiom (Definition 2.1.55). Hence $g_1 g_2 \in \text{Stab}_G(x)$.

The stabilizer $\text{Stab}_G(x)$ is closed under inverses because if $g \in \text{Stab}_G(x)$, then $g \cdot x = x$. Applying g^{-1} to both sides gives

$$g^{-1} \cdot (g \cdot x) = g^{-1} \cdot x.$$

The left-hand side equals $(g^{-1} g) \cdot x = e \cdot x = x$, so $g^{-1} \cdot x = x$. Hence $g^{-1} \in \text{Stab}_G(x)$. Therefore, $\text{Stab}_G(x)$ is a subgroup of G . $\square$

Proposition 2.1.61 (Orbit–Stabilizer bijection). Let $(G, \cdot)$ be a group (Definition 2.1.1) acting (Definition 2.1.55) (on the left) on a set X . For all $x \in X$, there is a bijection of sets

$$G / \text{Stab}_G(x) \longleftrightarrow G \cdot x, \quad g \text{Stab}_G(x) \mapsto g \cdot x.$$

(Definition 2.1.26) (Definition 2.1.25) In particular, if G is finite then

$$|G \cdot x| = [G : \text{Stab}_G(x)] = \frac{|G|}{|\text{Stab}_G(x)|}.$$

Proof. Let $H := \text{Stab}_G(x)$. We construct a bijection

$$\Phi : G/H \longrightarrow G \cdot x, \quad gH \mapsto g \cdot x.$$

We first show that Φ is well-defined. Suppose that $g_1 H = g_2 H$ for some $g_1, g_2 \in G$. Then $g_2^{-1} g_1 \in H$ (Lemma 2.1.29), so $(g_2^{-1} g_1) \cdot x = x$ by the definition of H (Definition 2.1.57). Applying g_2 to both sides gives

$$g_2 \cdot ((g_2^{-1} g_1) \cdot x) = g_2 \cdot x.$$

The left-hand side equals $(g_2 (g_2^{-1} g_1)) \cdot x = g_1 \cdot x$ by the compatibility axiom (Definition 2.1.55). Hence $g_1 \cdot x = g_2 \cdot x$, so $\Phi(g_1 H) = \Phi(g_2 H)$.

Next we show that Φ is surjective. Let $y \in G \cdot x$. By the definition of the orbit (Definition 2.1.56), there exists some $g \in G$ such that $y = g \cdot x$. Then $\Phi(gH) = g \cdot x = y$, so Φ is surjective.

Finally we show that Φ is injective. Suppose that $\Phi(g_1H) = \Phi(g_2H)$ for some $g_1, g_2 \in G$. Then $g_1 \cdot x = g_2 \cdot x$. Applying g_2^{-1} to both sides gives

$$g_2^{-1} \cdot (g_1 \cdot x) = g_2^{-1} \cdot (g_2 \cdot x).$$

The left-hand side equals $(g_2^{-1}g_1) \cdot x$ and the right-hand side equals $(g_2^{-1}g_2) \cdot x = e \cdot x = x$, both by the compatibility axiom (Definition 2.1.55). Therefore $g_2^{-1}g_1 \in H$, which implies that $g_1H = g_2H$ (Lemma 2.1.29). Hence Φ is injective.

The map Φ is therefore a bijection. In particular, if G is finite then the set of left cosets G/H has cardinality equal to the index (Definition 2.1.27) $[G : H]$, and by Lagrange's theorem (Theorem 2.1.33), this index equals $\frac{|G|}{|H|}$. Since Φ is a bijection (Definition A.1.3), we have

$$|G \cdot x| = [G : H] = \frac{|G|}{|\text{Stab}_G(x)|}.$$

□

Proposition 2.1.62. Let $(G, \cdot)$ be a group (Definition 2.1.1) acting (Definition 2.1.55) (on the left) on a set X . For any subgroup (Definition 2.1.19) $H \subseteq G$, we have $X^H \subseteq X^{H'}$ whenever $H' \subseteq H$ (Definition 2.1.58).

Proof. Let $x \in X^H$. By definition (Definition 2.1.58), this means that $h \cdot x = x$ for all $h \in H$. Since $H' \subseteq H$, every element $h' \in H'$ is also an element of H , so in particular $h' \cdot x = x$ for all $h' \in H'$. Hence $x \in X^{H'}$, which shows that $X^H \subseteq X^{H'}$. □

2.2. Constructions of Groups from others.

Definition 2.2.1 (Product of Groups). Let $\{G_i\}_{i \in I}$ be a family of groups indexed by a (possibly infinite (Definition A.1.5) but small (Definition A.1.1)) set I , each with group operation denoted multiplicatively and identity element e_i . The *(direct) product of the family $\{G_i\}_{i \in I}$* , denoted by $\prod_{i \in I} G_i$, is defined as the set of all tuples/functions

$$\prod_{i \in I} G_i := \{(g_i)_{i \in I} \mid g_i \in G_i \text{ for all } i \in I\},$$

equipped with the binary operation defined componentwise by

$$(g_i)_{i \in I} \cdot (h_i)_{i \in I} := (g_i h_i)_{i \in I}$$

for all $(g_i)_{i \in I}, (h_i)_{i \in I} \in \prod_{i \in I} G_i$. Then $\prod_{i \in I} G_i$ is a group with the identity element $(e_i)_{i \in I}$ and inverses given by

$$(g_i)_{i \in I}^{-1} = (g_i^{-1})_{i \in I}.$$

The product $\prod_{i=1}^n G_i$ is the product (Definition A.4.21) of the objects G_i in the category of groups (Definition 2.1.17). As a set, note that $\prod_{i \in I} G_i$ coincides with the product $\prod_{i \in I} G_i$ (Definition A.1.4) of the G_i as sets.

A self product of a group G (indexed by a small set I), is often denoted by G^I . A finite self product of a group G taken n times is often denoted by G^n . In case that G is abelian, these may be written as $G^{\oplus I}$ and $G^{\oplus n}$ respectively.

The product of finitely many groups $G_1, \dots, G_n$ is often denoted by $G_1 \times \dots \times G_n$.

2.3. Sylow theory.

Definition 2.3.1. Let G be a finite (Definition A.1.5) group (Definition 2.1.1) and p be a prime number. A subgroup (Definition 2.1.19) H of G is called a **p -subgroup** if the order of H is a power of p .

Definition 2.3.2. Let G be a finite (Definition A.1.5) group (Definition 2.1.1) of order $p^n m$, where p is a prime number and p does not divide m . A **Sylow p -subgroup of G** is a subgroup (Definition 2.1.19) of order p^n .

The set of all Sylow p -subgroups of a group G is denoted by $\text{Syl}_p(G)$. The number of Sylow p -subgroups is denoted by $n_p(G)$ or simply n_p .

Theorem 2.3.3. Let G be a finite (Definition A.1.5) group (Definition 2.1.1) and p be a prime number such that p divides the order of G .

1. (Sylow's first theorem) $\text{Syl}_p(G)$ (Definition 2.3.2) is non-empty; that is, G contains at least one subgroup of order p^n , where p^n is the highest power of p dividing $|G|$.
2. (Sylow's second theorem) If P_1 and P_2 are Sylow p -subgroups of G (Definition 2.3.2), then they are conjugate (Definition 2.1.24) in G . That is, there exists an element $g \in G$ such that $gP_1g^{-1} = P_2$.
3. (Sylow's third theorem) Let G be a finite group of order $p^n m$ where $p \nmid m$. The number of Sylow p -subgroups, n_p (Definition 2.3.2), satisfies the following conditions:

$$n_p \equiv 1 \pmod{p}$$

$$n_p \mid m$$

Furthermore, $n_p = [G : N_G(P)]$ for any Sylow p -subgroup P , where $N_G(P)$ is the normalizer (Definition 2.1.50) of P in G .

Corollary 2.3.4. Let G be a finite group and p a prime number. Every p -subgroup (Definition 2.3.1) of G is contained in some Sylow p -subgroup (Definition 2.3.2) of G .

Proof. (♠ TODO: AI generated, verify) Let Q be a p -subgroup of G . By Sylow's first theorem (Theorem 2.3.3), there exists at least one Sylow p -subgroup (Definition 2.3.2) P_0 of G .

The group Q acts on the set of left cosets $G/P_0 = \{gP_0 : g \in G\}$ by left multiplication: for each $q \in Q$ and $gP_0 \in G/P_0$, we define $q \cdot (gP_0) = qgP_0$.

The number of cosets is $|G/P_0| = [G : P_0] = m$, where $p \nmid m$ because $|P_0| = p^n$ is the highest power of p dividing $|G|$ by Definition 2.3.2.

By the orbit-stabilizer bijection (Proposition 2.1.61), each orbit of this action has cardinality equal to the index $[Q : \text{Stab}_Q(gP_0)]$, which divides $|Q|$ by Theorem 2.1.33. Since $|Q|$ is a power of p by Definition 2.3.1, each orbit has size either 1 or divisible by p . The orbits partition the set G/P_0 , so $m = \sum_i |\mathcal{O}_i|$. Because the total sum m is not divisible by p , there must exist at least one orbit whose size is not divisible by p . The only such divisor of a power of p is 1. Hence, there exists a fixed point $gP_0 \in G/P_0$ such that for all $q \in Q$, we have $qgP_0 = gP_0$.

This implies that for all $q \in Q$, we have $g^{-1}qg \in P_0$ by (Lemma 2.1.29). Therefore, $g^{-1}Qg \subseteq P_0$, which implies that $Q \subseteq gP_0g^{-1}$.

The subgroup $P = gP_0g^{-1}$ has the same order as P_0 , namely p^n . Hence, P is a Sylow p -subgroup of G by Definition 2.3.2. Therefore, Q is contained in the Sylow p -subgroup P . $\square$

2.4. Short exact sequences of general groups.

Definition 2.4.1. 1. A sequence of groups (Definition 2.1.1) and group homomorphisms (Definition 2.1.17)

$$\dots \xrightarrow{\phi_{i-1}} G_i \xrightarrow{\phi_i} G_{i+1} \xrightarrow{\phi_{i+1}} \dots$$

is called *exact at G_i* if $\text{im}(\phi_{i-1}) = \ker(\phi_i)$ (Definition 2.1.38). The sequence is called an *exact sequence* if it is exact at every group G_i .

2. A *short exact sequence of groups* is an exact sequence of the form

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \longrightarrow 1$$

where 1 denotes the trivial group. Exactness implies that ι is injective (Definition A.1.3), π is surjective (Definition A.1.3), and $\text{im}(\iota) = \ker(\pi)$.

Definition 2.4.2. Let $1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \longrightarrow 1$ be a short exact sequence of groups.

1. A *section* (or *splitting homomorphism*) of the short exact sequence is a group homomorphism (Definition 2.1.17) $s : H \rightarrow G$ such that $\pi \circ s = \text{id}_H$.
2. The short exact sequence $1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \longrightarrow 1$ is said to *split* if there exists a section $s : H \rightarrow G$.

2.5. Semidirect products.

Definition 2.5.1. Let G be a group.

1. An *automorphism of G* is an isomorphism (Definition 2.1.17) $\phi : G \rightarrow G$.
2. The *automorphism group of a group G* , denoted $\text{Aut}(G)$, is the set of all automorphisms of G equipped with the binary operation of function composition. For $\phi, \psi \in \text{Aut}(G)$, the composition $\phi \circ \psi$ is defined by $(\phi \circ \psi)(g) = \phi(\psi(g))$ for all $g \in G$.

The automorphism group of a group G is the automorphism group (Definition A.4.2) of G as an object of the category of groups (Definition 2.1.18).

Proposition 2.5.2. Let $\mathcal{C}$ be a locally small category (Definition A.4.3) and X be an object of $\mathcal{C}$. The automorphism group (Definition A.4.2) $\text{Aut}_{\mathcal{C}}(X)$ indeed forms a group (Definition 2.1.1) under composition. The identity element is the identity morphism id_X , and the inverse of an element f is its inverse morphism f^{-1} .

Definition 2.5.3. 1. Let N and H be groups (Definition 2.1.1), and let $\varphi : H \rightarrow \text{Aut}(N)$ be a homomorphism (Definition 2.1.17), where $\text{Aut}(N)$ is the group of automorphisms (Definition 2.5.1) of N . The *external semidirect product of N and H with respect to*

φ , denoted $N \rtimes_{\varphi} H$ or $N \rtimes H$ for brevity, is the group defined on the set $N \times H$ with the binary operation:

$$(n_1, h_1)(n_2, h_2) = (n_1\varphi(h_1)(n_2), h_1h_2)$$

for all $n_1, n_2 \in N$ and $h_1, h_2 \in H$.

We note that there is a short exact sequence

$$1 \rightarrow N \xrightarrow{n \mapsto (n, 1_G)} N \rtimes_{\varphi} H \xrightarrow{(n, h) \mapsto h} H \rightarrow 1$$

2. Let G be a group. G is the *internal semidirect product of a subgroup N and a subgroup H* if the following conditions hold:

- (a) N is a normal subgroup (Definition 2.1.36) of G .
- (b) $G = NH$, meaning every element $g \in G$ can be written as $g = nh$ for some $n \in N$ and $h \in H$.
- (c) $N \cap H = \{1\}$, where 1 is the identity element of G .

Theorem 2.5.4. Let G be a group (Definition 2.1.1) with subgroups (Definition 2.1.19) N and H . The group G is the internal semidirect product (Definition 2.5.3) of N and H if and only if G is isomorphic (Definition 2.1.17) to the external semidirect product (Definition 2.5.3) $N \rtimes_{\varphi} H$, where the homomorphism $\varphi : H \rightarrow \text{Aut}(N)$ is given by conjugation (Definition 2.1.24) within G :

$$\varphi(h)(n) = hnh^{-1}$$

for all $h \in H$ and $n \in N$. Under this isomorphism, the subgroups N and H of the internal product correspond to the subgroups $\{(n, 1_H) : n \in N\}$ and $\{(1_N, h) : h \in H\}$ of the external product, respectively.

Proof. (♠ TODO: AI generated, verify) We first define the conjugation action. If G is an internal semidirect product of N and H , then $N \trianglelefteq G$ by condition (1) of Definition 2.5.3, so for each $h \in H \subseteq G$ and $n \in N$, we have $hnh^{-1} \in N$. Define $\varphi : H \rightarrow \text{Aut}(N)$ by $\varphi(h)(n) = hnh^{-1}$. The map $\varphi(h)$ is a group automorphism of N because conjugation preserves the group operation. The assignment $h \mapsto \varphi(h)$ is a homomorphism because

$$\varphi(h_1h_2)(n) = (h_1h_2)n(h_1h_2)^{-1} = h_1(h_2nh_2^{-1})h_1^{-1} = \varphi(h_1)(\varphi(h_2)(n))$$

for all $h_1, h_2 \in H$ and $n \in N$.

Define $\Phi : N \rtimes_{\varphi} H \rightarrow G$ by $\Phi(n, h) = nh$. We verify that Φ is a group homomorphism. For $(n_1, h_1), (n_2, h_2) \in N \rtimes_{\varphi} H$, we have

$$\begin{aligned} \Phi((n_1, h_1)(n_2, h_2)) &= \Phi(n_1\varphi(h_1)(n_2), h_1h_2) \\ &= n_1\varphi(h_1)(n_2)h_1h_2 \\ &= n_1(h_1n_2h_1^{-1})h_1h_2 \\ &= n_1h_1n_2h_2 \\ &= \Phi(n_1, h_1) \cdot \Phi(n_2, h_2). \end{aligned}$$

We now prove the equivalence.

1. ($\Rightarrow$) Suppose that G is the internal semidirect product of N and H . We show that Φ is a group isomorphism.

For surjectivity, let $g \in G$. Since $G = NH$ by condition (2) of Definition 2.5.3, there exist $n \in N$ and $h \in H$ such that $g = nh = \Phi(n, h)$. Hence Φ is surjective.

For injectivity, suppose that $\Phi(n, h) = e_G$, where e_G denotes the identity element of G . Then $nh = e_G$, so $n = h^{-1}$. Since $n \in N$ and $h \in H$, we have $n \in N \cap H = \{e_G\}$ by condition (3) of Definition 2.5.3. Hence, $n = e_G$ and $h = e_H$. Therefore, Φ has trivial kernel, so Φ is injective by Proposition 2.1.39(3).

Under this isomorphism, the subgroup $N \times \{e_H\} = \{(n, e_H) : n \in N\}$ of $N \rtimes_\varphi H$ maps to $\{\Phi(n, e_H) : n \in N\} = \{ne_H : n \in N\} = N$, and the subgroup $\{e_N\} \times H = \{(e_N, h) : h \in H\}$ maps to H .

2. ($\Leftarrow$) Suppose that $\varphi(h)(n) = hnh^{-1}$ defines a homomorphism $H \rightarrow \text{Aut}(N)$ and that $\Phi : N \rtimes_\varphi H \rightarrow G$ is a group isomorphism. We show that G satisfies the three conditions of an internal semidirect product.

First, we verify that N is normal in G . In the external semidirect product $N \rtimes_\varphi H$, the subset $\{(n, e_H) : n \in N\}$ is a normal subgroup because for any $(a, b) \in N \rtimes_\varphi H$ and (n, e_H) with $n \in N$, we have

$$\begin{aligned} (a, b)(n, e_H)(a, b)^{-1} &= (a\varphi(b)(n), b) \cdot (\varphi(b^{-1})(a^{-1}), b^{-1}) \\ &= (a\varphi(b)(n)\varphi(b)(\varphi(b^{-1})(a^{-1})), bb^{-1}) \\ &= (a\varphi(b)(n)a^{-1}, e_H), \end{aligned}$$

which lies in $\{(n', e_H) : n' \in N\}$ because $a, \varphi(b)(n) \in N$ and N is a group. Since Φ maps this normal subgroup to $\{\Phi(n, e_H) : n \in N\} = \{n : n \in N\} = N$, the subgroup N is normal in G because isomorphisms preserve normality by (Definition 2.1.36).

Second, $G = NH$ because Φ is surjective: for every $g \in G$, there exists $(n, h) \in N \rtimes_\varphi H$ such that $g = \Phi(n, h) = nh$.

Third, we show that $N \cap H = \{e_G\}$. Suppose that $x \in N \cap H$. Then $\Phi(x, e_H) = xe_H = x$ and $\Phi(e_N, x) = e_Nx = x$, so $\Phi(x, e_H) = \Phi(e_N, x)$. Since Φ is injective, we have $(x, e_H) = (e_N, x)$, which implies $x = e_N = e_G$. Hence, $N \cap H = \{e_G\}$.

Therefore, G satisfies all three conditions of the internal semidirect product by Definition 2.5.3.

□

Theorem 2.5.5. Let N and H be groups (Definition 2.1.1). A group G is isomorphic (Definition 2.1.17) to a semidirect product (Definition 2.5.3) of N by H if and only if there exists a short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \longrightarrow 1$$

that splits, meaning there exists a section $s : H \rightarrow G$, i.e. a group homomorphism (Definition 2.1.17) such that $\pi \circ s = \text{id}_H$. In this case, $G \cong N \rtimes_\varphi H$ with $\varphi(h)(n) = s(h)ns(h)^{-1}$ (identifying N with its image $\iota(N)$).

Proof. (♠ TODO: AI generated, verify) We prove the two directions separately.

1. ($\Rightarrow$) Suppose that $G \cong N \rtimes_{\varphi} H$ for some homomorphism $\varphi : H \rightarrow \text{Aut}(N)$ by Definition 2.5.3. Define $\iota : N \rightarrow G$ by $\iota(n) = (n, e_H)$ and $\pi : G \rightarrow H$ by $\pi(n, h) = h$. The map ι is a group homomorphism because

$$\iota(n_1)\iota(n_2) = (n_1, e_H)(n_2, e_H) = (n_1\varphi(e_H)(n_2), e_H e_H) = (n_1 n_2, e_H) = \iota(n_1 n_2).$$

The map π is a group homomorphism because

$$\pi((n_1, h_1)(n_2, h_2)) = \pi(n_1\varphi(h_1)(n_2), h_1 h_2) = h_1 h_2 = \pi(n_1, h_1)\pi(n_2, h_2).$$

Define $s : H \rightarrow G$ by $s(h) = (e_N, h)$. This is a homomorphism because

$$s(h_1)s(h_2) = (e_N, h_1)(e_N, h_2) = (e_N\varphi(h_1)(e_N), h_1 h_2) = (e_N, h_1 h_2) = s(h_1 h_2).$$

Moreover, $\pi(s(h)) = \pi(e_N, h) = h$, so $\pi \circ s = \text{id}_H$. Hence s is a section by Definition 2.4.2.

Exactness at N : the kernel of ι is trivial because $\iota(n) = (n, e_H) = (e_N, e_H)$ if and only if $n = e_N$, so exactness holds at N by Proposition 2.1.39(3).

Exactness at G : we have $\text{im}(\iota) = \{(n, e_H) : n \in N\}$ and $\ker(\pi) = \{(n, h) : \pi(n, h) = e_H\} = \{(n, e_H) : n \in N\}$, so $\text{im}(\iota) = \ker(\pi)$.

Exactness at H : the map π is surjective because for every $h \in H$, we have $\pi(e_N, h) = h$. Hence, by Definition 2.4.1, the sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \longrightarrow 1$$

is short exact and splits.

2. ($\Leftarrow$) Suppose that $1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1$ is a short exact sequence of groups, and let $s : H \rightarrow G$ be a section by Definition 2.4.2. By Proposition 2.1.39(3), the map ι is injective, so we identify N with its image $\iota(N) = \ker(\pi)$ by Definition 2.4.1.

We first define the action of H on N . For each $h \in H$ and $n \in N$, let $\varphi(h)(n) = s(h)ns(h)^{-1}$. Since $N = \ker(\pi)$ is a normal subgroup of G by Proposition 2.1.39, the conjugate $s(h)ns(h)^{-1}$ lies in N , so $\varphi(h) : N \rightarrow N$ is well-defined. The map $\varphi(h)$ is an automorphism of N because conjugation by any group element preserves the group operation and has an inverse (conjugation by the inverse). The assignment $h \mapsto \varphi(h)$ is a homomorphism from H to $\text{Aut}(N)$ because

$$\varphi(h_1 h_2)(n) = s(h_1 h_2)ns(h_1 h_2)^{-1} = s(h_1)s(h_2)ns(h_2)^{-1}s(h_1)^{-1} = \varphi(h_1)(\varphi(h_2)(n))$$

for all $h_1, h_2 \in H$ and $n \in N$, where we used that s is a homomorphism.

Define $\Phi : N \rtimes_{\varphi} H \rightarrow G$ by $\Phi(n, h) = ns(h)$. We verify that Φ is a group homomorphism. For $(n_1, h_1), (n_2, h_2) \in N \rtimes_{\varphi} H$,

$$\begin{aligned} \Phi((n_1, h_1)(n_2, h_2)) &= \Phi(n_1\varphi(h_1)(n_2), h_1 h_2) \\ &= n_1\varphi(h_1)(n_2)s(h_1 h_2) \\ &= n_1s(h_1)n_2s(h_1)^{-1}s(h_1)s(h_2) \\ &= n_1s(h_1)n_2s(h_2) \\ &= \Phi(n_1, h_1)\Phi(n_2, h_2). \end{aligned}$$

We show that Φ is surjective. Let $g \in G$. Since $\pi : G \rightarrow H$ is surjective by exactness, let $h = \pi(g)$. Then $\pi(s(h)) = (\pi \circ s)(h) = h$, so $\pi(gs(h)^{-1}) = \pi(g)\pi(s(h))^{-1} = hh^{-1} = e_H$. Hence, $gs(h)^{-1} \in \ker(\pi) = N$. Write $gs(h)^{-1} = n$ for some $n \in N$. Then $g = ns(h) = \Phi(n, h)$, so Φ is surjective.

We show that Φ is injective by verifying it has trivial kernel. Suppose that $\Phi(n, h) = e_G$. Then $ns(h) = e_G$, so $n = s(h)^{-1}$. Applying π to both sides gives $\pi(n) = \pi(s(h))^{-1}$. Since $n \in N = \ker(\pi)$, we have $\pi(n) = e_H$. Also, $\pi(s(h)) = h$, so $e_H = h^{-1}$, which implies $h = e_H$. Hence, $s(h) = s(e_H) = e_G$ and $n = e_G^{-1} = e_N$. Therefore, $(n, h) = (e_N, e_H)$, so Φ has trivial kernel by Proposition 2.1.39(3). Hence, Φ is injective.

Therefore, $G \cong N \rtimes_{\varphi} H$.

□

Proposition 2.5.6. Let N and H be groups (Definition 2.1.1), and let $\varphi : H \rightarrow \text{Aut}(N)$ be the trivial homomorphism sending every element of H to the identity automorphism of N . The identity map on the set $N \times H$,

$$\text{id}_{N \times H} : N \times H \longrightarrow N \rtimes_{\varphi} H, \quad (n, h) \mapsto (n, h),$$

is a group isomorphism (Definition 2.1.17), where $N \times H$ carries the componentwise direct product (Definition 2.2.1) operation and $N \rtimes_{\varphi} H$ carries the semidirect product operation.

Proof. Since $\text{id}_{N \times H}$ is a bijection on the underlying sets, we only need to show that it preserves the group operations. The multiplication in $N \rtimes_{\varphi} H$ is given by

$$(n_1, h_1)(n_2, h_2) = (n_1\varphi(h_1)(n_2), h_1h_2).$$

Since φ is the trivial homomorphism, $\varphi(h)(n) = n$ for all $h \in H$ and $n \in N$. Hence the semidirect product multiplication simplifies to

$$(n_1, h_1)(n_2, h_2) = (n_1n_2, h_1h_2),$$

which is exactly the componentwise multiplication in the direct product $N \times H$.

Therefore, $\text{id}_{N \times H}$ is a group homomorphism. It is also clearly bijective (Definition A.1.3), so it is a group isomorphism (Definition 2.1.17). □

Proposition 2.5.7. Let N and H be finite groups (Definition 2.1.1), and let $\varphi : H \rightarrow \text{Aut}(N)$ be a homomorphism (Definition 2.1.17). The external semidirect product (Definition 2.5.3) $N \rtimes_{\varphi} H$ is a finite group of order

$$|N \rtimes_{\varphi} H| = |N| \cdot |H|.$$

Proof. (♠ TODO: write and reference that cardinality of cartesian product of sets is product of cardinalities) By Definition 2.5.3, the underlying set of $N \rtimes_{\varphi} H$ is the Cartesian product $N \times H$. Since N and H are finite groups, the cardinality of their Cartesian product is the product of their cardinalities:

$$|N \times H| = |N| \cdot |H|.$$

Therefore, $|N \rtimes_{\varphi} H| = |N| \cdot |H|$. □

Proposition 2.5.8. (♠ TODO: Recheck; this statement may be incorrect) Let N and H be groups (Definition 2.1.1), and let $\varphi : H \rightarrow \text{Aut}(N)$ be a homomorphism (Definition 2.1.17). Write the action of $h \in H$ on $n \in N$ as $h \cdot n = \varphi(h)(n)$. Let $Z(N)$ and $Z(H)$ denote the centers (Definition 2.1.54) of N and H , respectively, and let

$$N^H = \{n \in N : h \cdot n = n \text{ for all } h \in H\}$$

be the set of H -fixed points in N . The center (Definition 2.1.54) of the external semidirect product (Definition 2.5.3) $N \rtimes_{\varphi} H$ is

$$Z(N \rtimes_{\varphi} H) = (Z(N) \cap N^H) \times (Z(H) \cap \ker(\varphi)).$$

Proof. (♠ TODO: AI generated, verify) An element $(n, h) \in N \rtimes_{\varphi} H$ is central if and only if it commutes with every element of $N \rtimes_{\varphi} H$. Using the semidirect product multiplication $(x, k)(y, l) = (x \cdot k(y), kl)$, we compute for arbitrary $(x, k) \in N \rtimes_{\varphi} H$:

$$\begin{aligned} (n, h)(x, k) &= (n \cdot h(x), hk), \\ (x, k)(n, h) &= (x \cdot k(n), kh). \end{aligned}$$

These are equal if and only if both $hk = kh$ in H and $n \cdot h(x) = x \cdot k(n)$ in N . We analyze these conditions.

First, taking $(1_N, k)$ for arbitrary $k \in H$, we need $hk = kh$ (so $h \in Z(H)$) and $n \cdot h(1_N) = 1_N \cdot k(n)$, which simplifies to $n = k(n)$. Hence $n \in N^H$.

Next, taking $(x, 1_H)$ for arbitrary $x \in N$, we need $h1_H = 1_Hh$ (trivially satisfied) and $n \cdot h(x) = x \cdot 1_H(n) = xn$. Since we already have $n \in N^H$, the condition with (x, k) for arbitrary k becomes $n \cdot h(x) = xk(n) = xn$, which is the same as the condition from $(x, 1_H)$. Thus the only conditions are:

1. $h \in Z(H)$,
2. $n \in N^H$,
3. $nh(x) = xn$ for all $x \in N$.

Condition (3) can be rewritten as $\varphi(h)(x) = n^{-1}xn$ for all $x \in N$, i.e., $\varphi(h)$ is the inner automorphism given by conjugation by n^{-1} .

We now show that these three conditions are equivalent to $h \in Z(H) \cap \ker(\varphi)$ and $n \in Z(N) \cap N^H$.

($\Rightarrow$) Suppose (1), (2), and (3) hold. For any $k \in H$, since φ is a homomorphism from H to $\text{Aut}(N)$, we have $\varphi(k)\varphi(h) = \varphi(kh) = \varphi(hk) = \varphi(h)\varphi(k)$. Applying both sides to $x \in N$:

$$\varphi(k)(n^{-1}xn) = n^{-1}\varphi(k)(x)n,$$

where the last equality uses $\varphi(h) = \text{inn}(n^{-1})$ and the fact that $k(n) = n$ (since $n \in N^H$). The left side is $\varphi(k)(n^{-1})\varphi(k)(x)\varphi(k)(n) = n^{-1}\varphi(k)(x)n$ since $k(n) = n$. This gives no new information.

Instead, we argue directly. From (3), for every $x \in N$, $\varphi(h)(x) = n^{-1}xn$. Now take any $k \in H$ and apply $\varphi(k)$ to both sides of the equation in (3): $nh(x) = xn$. Since φ is a homomorphism,

$$\varphi(kh)(y) = \varphi(k)\varphi(h)(y)$$

for all $y \in N$. Using (3), $\varphi(k)(n^{-1}yn) = n^{-1}\varphi(k)(y)n$ by $k(n) = n$. On the other hand, $\varphi(kh)(y) = \varphi(hk)(y)$ since $h \in Z(H)$. This gives $\varphi(h)(\varphi(k)(y)) = n^{-1}\varphi(k)(y)n$, which is consistent with (3).

To see that $h \in \ker(\varphi)$ and $n \in Z(N)$, observe: for any $k \in H$ and $x \in N$, the condition $nh(x) = xk(n)$ must hold. With $k = 1_H$: $nh(x) = xn$. With arbitrary k and $x = 1_N$: $n = k(n)$, so $n \in N^H$. Now for any k , the condition reads $nh(x) = xk(n) = xn$ (since $n \in N^H$). This is identical to the $k = 1_H$ case, so no further constraint arises.

From (3), $\varphi(h)(x) = n^{-1}xn$. For any $k \in H$, since $h \in Z(H)$ and φ is a homomorphism, $\varphi(k)\varphi(h) = \varphi(h)\varphi(k)$. Applying to $y \in N$:

$$\varphi(k)(n^{-1}yn) = n^{-1}\varphi(k)(y)n.$$

The left side is $k(n^{-1})\varphi(k)(y)k(n) = n^{-1}\varphi(k)(y)n$ (since $k(n) = n$). So this holds identically for any choice of $h \in Z(H)$, $n \in N^H$, and $\varphi(h) = \text{inn}(n^{-1})$.

Now observe: if additionally we set $x = n$ in (3), then $nh(n) = nn = n^2$, so $h(n) = n$, i.e., n is fixed by $\varphi(h)$. This is automatic since $\text{inn}(n^{-1})(n) = n^{-1}nn = n$.

To conclude, we need to show that (3) implies both $h \in \ker(\varphi)$ and $n \in Z(N)$. Take any $x \in N$ and use the action homomorphism property: for all $k \in H$, $\varphi(k)\varphi(h) = \varphi(kh)$. Since $h \in Z(H)$, this equals $\varphi(hk) = \varphi(h)\varphi(k)$. Apply to x :

$$\varphi(k)(n^{-1}xn) = n^{-1}\varphi(k)(x)n.$$

As computed above, the left side is indeed $n^{-1}\varphi(k)(x)n$ by $k(n) = n$. This shows consistency but does not force $\varphi(h) = \text{id}_N$.

In fact, we need to check: can (n, h) be central with $\varphi(h) \neq \text{id}_N$? If so, the proposition's claim would be wrong. Let us verify by checking whether the conditions force $h \in \ker(\varphi)$.

The condition is that for all (x, k) : $nh(x) = xk(n)$. With $k = 1_H$: $nh(x) = xn$, so $\varphi(h)(x) = n^{-1}xn$. If this holds and $h \in Z(H)$, then for any $g \in H$:

$$\varphi(g)\varphi(h) = \varphi(gh) = \varphi(hg) = \varphi(h)\varphi(g).$$

So $\text{inn}(n^{-1})$ commutes with every element of the image of φ . This does not force $n \in Z(N)$ in general.

Wait — let me re-examine. The proposition claims equality, so either there is an additional assumption or my derivation missed a step. Let me check: for (n, h) to be central, we need it to commute with all elements. With $(1_N, g)$ for $g \in H$:

$$(n, h)(1_N, g) = (nh(1_N), hg) = (n, hg),$$

$$(1_N, g)(n, h) = (g(n), gh).$$

So we need $hg = gh$ (i.e., $h \in Z(H)$) and $n = g(n)$ for all $g \in H$ (i.e., $n \in N^H$).

With $(x, 1_H)$ for $x \in N$:

$$(n, h)(x, 1_H) = (nh(x), h),$$

$$(x, 1_H)(n, h) = (xn, h).$$

So we need $nh(x) = xn$ for all x , i.e., $\varphi(h)(x) = n^{-1}xn$.

Now combine: since $n \in N^H$, for any $g \in H$:

$$g(nh(x)) = g(xn).$$

Using the homomorphism property of φ and $n \in N^H$:

$$n \cdot gh(x) = g(x)n.$$

But also from (3), $gh(x) = g(n^{-1}xn) = g(n)^{-1}g(x)g(n) = n^{-1}g(x)n$. So:

$$n \cdot n^{-1}g(x)n = g(x)n,$$

which is trivially true. This gives no new constraint.

However, the conditions (3), $h \in Z(H)$, and $n \in N^H$ are exactly what's needed for centrality. The proposition claims this equals $(Z(N) \cap N^H) \times (Z(H) \cap \ker(\varphi))$. This would require $\text{inn}(n^{-1}) = \text{id}_N$, i.e., $n \in Z(N)$, and then $\varphi(h) = \text{id}_N$, i.e., $h \in \ker(\varphi)$. But the derivation shows that (n, h) is central iff $\varphi(h) = \text{inn}(n^{-1})$, not necessarily $\varphi(h) = \text{id}_N$ and $n \in Z(N)$.

Actually, wait. Let me re-derive more carefully. We need (n, h) to commute with (x, g) for ALL $g \in H$ simultaneously. The condition is:

$$nh(x) = xg(n) \quad \text{for all } x \in N, g \in H.$$

With $n \in N^H$, this becomes $nh(x) = xn$ for all x . But wait — we also need to check with (x, g) where both $x \neq 1_N$ and $g \neq 1_H$:

$$(n, h)(x, g) = (nh(x), hg),$$

$$(x, g)(n, h) = (xg(n), gh).$$

Equality requires $hg = gh$ and $nh(x) = xg(n)$. With $h \in Z(H)$ and $n \in N^H$, the second condition is $nh(x) = xn$. This must hold for all $x \in N$, which means $\varphi(h)(x) = n^{-1}xn$ for all x .

So the center consists of pairs (n, h) such that:

1. $h \in Z(H)$,
2. $n \in N^H$,
3. $\varphi(h) = \text{inn}(n^{-1})$.

The proposition claims this is $(Z(N) \cap N^H) \times (Z(H) \cap \ker(\varphi))$. These are equal if and only if condition (3) implies both $\varphi(h) = \text{id}_N$ and $n \in Z(N)$. But in general this is not true — one can have nontrivial inner automorphisms commuting with the image of φ .

Therefore, there may be an error in the proposition statement. The correct characterization should be:

$$Z(N \rtimes_{\varphi} H) = \{(n, h) : h \in Z(H), n \in N^H, \varphi(h) = \text{inn}(n^{-1})\}.$$

□

Proposition 2.5.9. Let N and H be abelian groups (Definition 2.1.1), and let $\varphi : H \rightarrow \text{Aut}(N)$ be a homomorphism (Definition 2.1.17). The external semidirect product (Definition 2.5.3) $N \rtimes_{\varphi} H$ is abelian if and only if φ is the trivial homomorphism (i.e., $\varphi(h) = \text{id}_N$ for all $h \in H$). In that case, $N \rtimes_{\varphi} H$ coincides with the direct product (Definition 2.2.1) $N \times H$ (Proposition 2.5.6).

Proof. The multiplication in $N \rtimes_{\varphi} H$ is $(n_1, h_1)(n_2, h_2) = (n_1 \cdot \varphi(h_1)(n_2), h_1 h_2)$.

Suppose that $N \rtimes_{\varphi} H$ is abelian. Taking arbitrary $n \in N$ and $h \in H$, commutativity of $(n, 1_H)$ and $(1_N, h)$ gives

$$(n, 1_H)(1_N, h) = (1_N, h)(n, 1_H).$$

so

$$(nh, h) = (\varphi(h)(n), h)$$

Comparing the first components, we get $\varphi(h)(n) = n$ for all $n \in N$ and $h \in H$. Hence $\varphi(h) = \text{id}_N$ for all h , so φ is the trivial homomorphism.

Conversely, suppose that φ is the trivial homomorphism. Then $\varphi(h)(n) = n$ for all $h \in H$ and $n \in N$, so the multiplication becomes

$$(n_1, h_1)(n_2, h_2) = (n_1 n_2, h_1 h_2).$$

Since N and H are both abelian, we have $(n_1 n_2, h_1 h_2) = (n_2 n_1, h_2 h_1) = (n_2, h_2)(n_1, h_1)$. Therefore, $N \rtimes_{\varphi} H$ is abelian. By Proposition 2.5.6, it coincides with the direct product $N \times H$. $\square$

2.5.1. Holomorph of a group.

Definition 2.5.10. Let G be a group (Definition 2.1.1). The *holomorph of G* , denoted $\text{Hol}(G)$, is the external semidirect product (Definition 2.5.3)

$$\text{Hol}(G) = G \rtimes \text{Aut}(G)$$

where $\text{Aut}(G)$ is the automorphism group (Definition 2.5.1) of G , and the action $\text{Aut}(G) \rightarrow \text{Aut}(G)$ is the natural evaluation action: for $\alpha \in \text{Aut}(G)$ and $g \in G$,

$$\alpha \cdot g = \alpha(g).$$

2.5.2.

Definition 2.5.11 (Category of group actions). (♠ TODO: AI generated, verify) (♠ TODO: I don't think this is standard terminology or notation for the category; saying the category of group actions suggests a category whose objects are sets equipped with a group action, not a group equipped with a group action.) The *category of group actions*, denoted $\mathbf{ActGrp}$, is defined as follows:

1. An object is a triple (N, H, ϕ) where N and H are groups (Definition 2.1.1) and $\phi : H \rightarrow \text{Aut}(N)$ is a homomorphism (Definition 2.1.17).
2. A morphism from (N, H, ϕ) to (M, K, ψ) is a pair (f, g) where $f : N \rightarrow M$ and $g : H \rightarrow K$ are homomorphisms (Definition 2.1.17) satisfying the compatibility condition

$$f(\phi(h)(n)) = \psi(g(h))(f(n))$$

for all $h \in H$ and $n \in N$.

Proposition 2.5.12 (Functoriality of the semidirect product). The external semidirect product (Definition 2.5.3) defines a functor

$$\rtimes : \mathbf{ActGrp} \longrightarrow \mathbf{Grp}$$

from the category of group actions (Definition 2.5.11) to the category of groups.

On objects, $\rtimes(N, H, \phi) = N \rtimes_{\phi} H$. On morphisms, given $(f, g) : (N, H, \phi) \rightarrow (M, K, \psi)$, the induced homomorphism

$$(f \rtimes g) : N \rtimes_{\phi} H \longrightarrow M \rtimes_{\psi} K$$

is defined by $(f \rtimes g)(n, h) = (f(n), g(h))$.

Proof. (♠ TODO: AI generated, verify) We first show that $f \rtimes g$ is a group homomorphism. For $(n_1, h_1), (n_2, h_2) \in N \rtimes_{\phi} H$, we have

$$\begin{aligned} (f \rtimes g)((n_1, h_1)(n_2, h_2)) &= (f \rtimes g)(n_1\phi(h_1)(n_2), h_1h_2) \\ &= (f(n_1\phi(h_1)(n_2)), g(h_1h_2)). \end{aligned}$$

Since f and g are group homomorphisms by Definition 2.5.11, we have $f(n_1\phi(h_1)(n_2)) = f(n_1)f(\phi(h_1)(n_2))$ and $g(h_1h_2) = g(h_1)g(h_2)$. The compatibility condition for morphisms in **ActGrp** gives $f(\phi(h_1)(n_2)) = \psi(g(h_1))(f(n_2))$. Hence,

$$(f \rtimes g)((n_1, h_1)(n_2, h_2)) = (f(n_1)\psi(g(h_1))(f(n_2)), g(h_1)g(h_2)).$$

On the other hand, by Definition 2.5.3,

$$(f \rtimes g)(n_1, h_1) \cdot (f \rtimes g)(n_2, h_2) = (f(n_1), g(h_1))(f(n_2), g(h_2)) = (f(n_1)\psi(g(h_1))(f(n_2)), g(h_1)g(h_2)).$$

Therefore, $f \rtimes g$ preserves the group operation.

Next we verify that functoriality holds for identities. For each object (N, H, ϕ) of **ActGrp**, the identity morphism is $(\text{id}_N, \text{id}_H)$. The induced homomorphism

$$(\text{id}_N \rtimes \text{id}_H)(n, h) = (\text{id}_N(n), \text{id}_H(h)) = (n, h),$$

which is the identity morphism on $N \rtimes_{\phi} H$.

Finally we verify that functoriality holds for composition. Let $(f, g) : (N, H, \phi) \rightarrow (M, K, \psi)$ and $(f', g') : (M, K, \psi) \rightarrow (L, J, \chi)$ be morphisms in **ActGrp**. For each $(n, h) \in N \rtimes_{\phi} H$,

$$(f' \rtimes g')((f \rtimes g)(n, h)) = (f' \rtimes g')(f(n), g(h)) = (f'(f(n)), g'(g(h))) = ((f' \circ f)(n), (g' \circ g)(h)).$$

On the other hand,

$$((f' \rtimes g') \circ (f \rtimes g))(n, h) = ((f' \circ f)(n), (g' \circ g)(h)),$$

so $(f' \rtimes g') \circ (f \rtimes g) = (f' \circ f) \rtimes (g' \circ g)$, which is the homomorphism induced by the composition of morphisms in **ActGrp**.

Hence, the external semidirect product defines a functor **ActGrp** $\rightarrow$ **Grp**. □

Corollary 2.5.13 (Surjection of acting groups induces surjection of semidirect products). Let $p : H_1 \twoheadrightarrow H_2$ be a surjective homomorphism (Definition 2.1.17). Let H_2 act on a group N via $\phi_2 : H_2 \rightarrow \text{Aut}(N)$, and let H_1 act on N via the pulled-back action $\phi_1 = \phi_2 \circ p$.

Then (id_N, p) is a morphism in the category of group actions (Definition 2.5.11), and the induced homomorphism

$$\text{id}_N \rtimes p : N \rtimes_{\phi_1} H_1 \longrightarrow N \rtimes_{\phi_2} H_2$$

given by $(\text{id}_N \rtimes p)(n, h_1) = (n, p(h_1))$ is a surjective homomorphism (Definition 2.1.17). Its kernel is $\{e_N\} \times \ker(p)$.

Proof. (♠ TODO: AI generated, verify) We first verify that (id_N, p) is a morphism in **ActGrp** by checking the compatibility condition from Definition 2.5.11. For each $h_1 \in H_1$ and $n \in N$, we have

$$\text{id}_N(\phi_1(h_1)(n)) = \phi_2(p(h_1))(n) = \phi_2(p(h_1))(\text{id}_N(n)),$$

where the first equality uses $\phi_1 = \phi_2 \circ p$ and the second uses that id_N is the identity map. Hence, (id_N, p) satisfies the compatibility condition.

By Proposition 2.5.12, the induced homomorphism $\text{id}_N \rtimes p : N \rtimes_{\phi_1} H_1 \rightarrow N \rtimes_{\phi_2} H_2$ is a group homomorphism.

We show that $\text{id}_N \rtimes p$ is surjective. Let $(n, h_2) \in N \rtimes_{\phi_2} H_2$. Since $p : H_1 \rightarrow H_2$ is surjective by hypothesis, there exists $h_1 \in H_1$ such that $p(h_1) = h_2$. Then $(\text{id}_N \rtimes p)(n, h_1) = (n, p(h_1)) = (n, h_2)$. Hence $\text{id}_N \rtimes p$ is surjective.

We now compute the kernel. An element (n, h_1) lies in the kernel if and only if

$$(\text{id}_N \rtimes p)(n, h_1) = (e_N, e_{H_2}),$$

which is equivalent to $n = e_N$ and $p(h_1) = e_{H_2}$. The condition $p(h_1) = e_{H_2}$ means precisely that $h_1 \in \ker(p)$. Therefore, the kernel of $\text{id}_N \rtimes p$ is

$$\{(n, h_1) : n = e_N \text{ and } h_1 \in \ker(p)\} = \{e_N\} \times \ker(p).$$

□

Corollary 2.5.14 (Inclusion of H -invariant subgroups induces inclusion of semidirect products). Let H be a group (Definition 2.1.1). Suppose H acts on a group N_2 via $\phi_2 : H \rightarrow \text{Aut}(N_2)$, and let $N_1 \leq N_2$ be a subgroup such that $\phi_2(h)(N_1) \subseteq N_1$ for all $h \in H$. Let $\phi_1 : H \rightarrow \text{Aut}(N_1)$ be the restricted action.

Then the inclusion $\iota : N_1 \hookrightarrow N_2$ is an H -equivariant homomorphism, and the induced homomorphism

$$\iota \rtimes \text{id}_H : N_1 \rtimes_{\phi_1} H \longrightarrow N_2 \rtimes_{\phi_2} H$$

given by $(\iota \rtimes \text{id}_H)(n, h) = (\iota(n), h)$ is an injective homomorphism (Definition 2.1.17).

Proof. (♠ TODO: AI generated, verify) We first verify that (ι, id_H) is a morphism in **ActGrp** by checking the compatibility condition from Definition 2.5.11. For each $h \in H$ and $n \in N_1$, we have

$$\iota(\phi_1(h)(n)) = \phi_2(h)(\iota(n)).$$

This holds because $\phi_1(h)$ is defined as the restriction of $\phi_2(h)$ to N_1 and ι is the inclusion map. Hence, (ι, id_H) satisfies the compatibility condition.

By Proposition 2.5.12, the induced homomorphism $\iota \rtimes \text{id}_H : N_1 \rtimes_{\phi_1} H \rightarrow N_2 \rtimes_{\phi_2} H$ is a group homomorphism.

We show that $\iota \rtimes \text{id}_H$ is injective by verifying it has trivial kernel. Suppose that $(\iota(n), h) = (e_{N_2}, e_H)$ for some $n \in N_1$ and $h \in H$. Then $h = e_H$ and $\iota(n) = e_{N_2}$. Since ι is the inclusion

of a subgroup, it is injective, so $\iota(n) = e_{N_2}$ implies $n = e_{N_1}$. Hence $(n, h) = (e_{N_1}, e_H)$, which is the identity element of $N_1 \rtimes_{\phi_1} H$. Therefore, $\iota \rtimes \text{id}_H$ has trivial kernel, so it is injective by Proposition 2.1.39(3). $\square$

Lemma 2.5.15. Let A and H be groups (Definition 2.1.1) with A abelian, and let $\varphi : H \rightarrow \text{Aut}(A)$ be a homomorphism (Definition 2.1.17), where $\text{Aut}(A)$ is the group of automorphisms (Definition 2.5.1) of A . Write the corresponding action of H on A by $H \times A \rightarrow A : (h, a) \mapsto h \cdot a$.

A subgroup $A' \leq A$ is invariant under the action of H , i.e. $h \cdot a' \in A'$ for all $h \in H$ and $a' \in A'$, if and only if A' , identified as $A' \rtimes \{1\}$, is a normal subgroup of $A \rtimes H$.

Proof. Let $A' \leq A$ be a subgroup. Given $a' \in A'$ and $(a, h) \in A \rtimes H$, Since A is abelian, one can compute that

$$(a, h) \cdot (a', 1) \cdot (a, h)^{-1} = (ha', 1).$$

Note that A' is invariant under the action of H if and only if $ha' \in A'$ for all $a' \in A'$ and $h \in H$, which in turn holds if and only if the above conjugate is an element of $A' \rtimes \{1\}$ for all $(a, h) \in A \rtimes H$ and $(a', 1) \in A' \rtimes \{1\}$, i.e. $A' \rtimes \{1\}$ is a normal subgroup of $A \rtimes H$. $\square$

2.5.3.

Theorem 2.5.16. Let A be a nontrivial abelian group and let G be a group acting on A via a homomorphism $\phi : G \rightarrow \text{Aut}(A)$. For each $g \in G$, denote the action of g on $a \in A$ by $g \cdot a$. Suppose that the following hold:

1. There exists a non-identity element $g_0 \in G \setminus \{1\}$ such that the fixed-point subgroup is trivial, i.e., $A^{g_0} = \{a \in A \mid g_0 \cdot a = a\} = \{0\}$.
2. For every non-identity element $g \in G \setminus \{1\}$, the map $(1 - g) : A \rightarrow A$ defined by $a \mapsto a - g \cdot a$ is not the zero map.

The semidirect product $A \rtimes G$ is center-free (Definition 2.1.54).

Proof. Let $(a_1, g_1) \in Z(A \rtimes G)$. By definition, (a_1, g_1) must commute with all elements $(a_2, g_2) \in A \rtimes G$. The equality $(a_1, g_1)(a_2, g_2) = (a_2, g_2)(a_1, g_1)$ is equivalent to

$$(a_1 + g_1 \cdot a_2, g_1 g_2) = (a_2 + g_2 \cdot a_1, g_2 g_1).$$

In particular, we must have

$$(A) \quad (1 - g_2) \cdot a_1 = (1 - g_1) \cdot a_2 \quad \text{for all } a_2 \in A, g_2 \in G.$$

We must show that $(a_1, g_1) = (0, 1)$.

Suppose for contradiction that $g_1 \neq 1$. Letting $g_2 = 1$ in (A), the left-hand side becomes 0, reducing the relation to:

$$0 = (1 - g_1) \cdot a_2 \quad \text{for all } a_2 \in A.$$

This implies that the map $(1 - g_1) : A \rightarrow A$ is identically the zero map, which contradicts directly contradicts Condition (2). Hence, $g_1 = 1$.

With $g_1 = 1$, the right-hand side of (A) becomes 0, so the relation simplifies to $(1 - g_2) \cdot a_1 = 0$ for all $g_2 \in G$. Letting g_2 be the element g_0 of Condition (1), the assumption that $A^{g_0} = \{0\}$ implies that $a_1 = 0$. Thus, $(a_1, g_1) = (0, 1)$ as desired. $\square$

3. BASIC RING THEORY

3.1. Definitions.

3.1.1. Rings, division rings, and fields.

Definition 3.1.1. A **ring** is a triple $(R, +, \cdot)$ where

1. $(R, +)$ is a commutative group (Definition 2.1.1), and
2. $(R, \cdot)$ is a monoid (Definition A.2.2).
3. $\cdot$ is distributive over $+$, i.e. for all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

Equivalently, a ring is a triple $(R, +, \cdot)$ where $+, \cdot : R \times R \rightarrow R$ are binary operations satisfying

1. $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ for all $a, b, c \in R$
2. There exists an element $0 \in R$ such that $a + 0 = a = 0 + a$ for all $a \in R$.
3. For every $a \in R$, there exists an element $-a \in R$ such that $a + (-a) = 0 = (-a) + a$ for all $a \in R$.
4. There exists an element $1 \in R$ such that $a \cdot 1 = a = 1 \cdot a$ for all $a \in R$.
5. For all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

The operation $+$ is often called **addition** and the operation $\cdot$ is often called **multiplication**. Accordingly, the identity element 0 of $+$ is often called the **additive identity** and the identity element 1 of $\cdot$ is often called the **multiplicative identity**.

While other notations may be used to denote addition and multiplication, if no notation is specified explicitly, then $+$ and $\cdot$ are universally used as default notations for addition and multiplication in a ring.

Some writers might not require a ring to have a multiplicative identity element, i.e. would define a ring so that $(R, +)$ is a commutative group, $(R, \cdot)$ is a semigroup, and $\cdot$ is distributive over $+$. Such writers would call the notion of ring in as above a **unital ring** to emphasize the existence of the multiplicative identity 1.

Lemma 3.1.2. Let R be a ring (Definition 3.1.1). For any $r \in R$, we have $0 \cdot r = 0 = r \cdot 0$.

Proof. For any $r \in R$, we have

$$0 \cdot r = (0 + 0) \cdot r = 0 \cdot r + 0 \cdot r.$$

By Corollary 2.1.14, we have that $0 = 0 \cdot r$. Similarly, $r \cdot 0 = 0$ as well. $\square$

Lemma 3.1.3. Let R be a ring (Definition 3.1.1). For any $r \in R$, we have $(-1) \cdot r = -r = r \cdot (-1)$.

Proof. For any $r \in R$, distributivity of $\cdot$ over $+$ yields

$$(1 + (-1)) \cdot r = 1 \cdot r + (-1) \cdot r.$$

On the one hand,

$$(1 + (-1)) \cdot r = 0 \cdot r = 0$$

by Lemma 3.1.2. On the other hand,

$$1 \cdot r + (-1) \cdot r = r + (-1) \cdot r,$$

so

$$0 = r + (-1) \cdot r.$$

By Corollary 2.1.15,

$$(-1) \cdot r = -r.$$

Similarly, recognizing that

$$((-1) + 1) \cdot r = (-1) \cdot r + 1 \cdot r,$$

one can formulate a similar argument to show that $r \cdot (-1) = -r$. $\square$

Corollary 3.1.4. Let R be a ring (Definition 3.1.1). We have

$$(-1) \cdot (-1) = 1.$$

Proof. By Lemma 3.1.3, we have

$$(-1) \cdot (-1) = -(-1).$$

By Corollary 2.1.16, $-(-1)$ equals 1. $\square$

Definition 3.1.5 (Opposite ring). Let $R = (R, +, \cdot, 0, 1)$ be a ring (Definition 3.1.1) with addition $+$, multiplication $\cdot$, additive identity 0, and multiplicative identity 1 (not necessarily commutative).

The *opposite ring of R* , denoted R^{op} , is the ring with the same underlying set R and the same addition $+$ and additive identity 0, but with multiplication defined by

$$r \star s := s \cdot r$$

for all $r, s \in R$.

That is, multiplication in R^{op} is the multiplication of R reversed in order.

If R is commutative (Definition 3.1.6), then R and R^{op} are naturally isomorphic to each other.

Definition 3.1.6. A *commutative (unital) ring* is a ring (Definition 3.1.1) $(R, +, \cdot)$ such that $\cdot$ is a commutative operation (Definition A.2.3), i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. “commutative” algebraists or number theorists), a *ring* refers to a commutative ring as above.

Definition 3.1.7 (Center of a ring). Let R be a (not necessarily commutative) ring (Definition 3.1.1). The *center of R* , denoted by $Z(R)$, is the subset of elements in R that commute with every element of R :

$$Z(R) := \{z \in R \mid zr = rz \text{ for all } r \in R\}.$$

The center $Z(R)$ is a commutative (Definition 3.1.6) subring (Definition 3.1.40) of R .

Elements of $Z(R)$ are called *central elements of R* .

Definition 3.1.8. Any set with a single element equipped with the only possible binary operators $+$ and $\cdot$ is a commutative ring (Definition 3.1.6) whose multiplicative and identity elements are both the sole element. Any ring with a single element is called the *zero ring* or the *trivial ring*.

Definition 3.1.9. Let $(R, +, \cdot)$ be a not-necessarily commutative ring (Definition 3.1.1).

1. An element $a \in R$ is a *left zero-divisor* if there exists a nonzero $x \in R$ such that $ax = 0$. Otherwise, a is called *left regular* or *left cancellable*.
2. An element $a \in R$ is a *right zero-divisor* if there exists a nonzero $x \in R$ such that $xa = 0$. Otherwise, a is called *right regular* or *right cancellable*.
3. An element $a \in R$ is a *zero-divisor* if it is a left zero-divisor or a right zero-divisor.
4. An element $a \in R$ is a *two-sided zero-divisor* if it is both a left zero-divisor and a right zero-divisor.
5. An element $a \in R$ is *regular*, *cancellable*, or a *non-zero-divisor* if it is both left and right regular.

A zero-divisor of any kind that is not itself 0 is said to be a *nonzero zero divisor* or a *nontrivial zero divisor* of its kind.

Definition 3.1.10. A non-zero ring (Definition 3.1.1) with no nontrivial zero divisors is called a *domain*. A $\dagger$ (Definition 3.1.10) that it also a commutative ring (Definition 3.1.6) is also called an *integral domain*.

Definition 3.1.11. Let $(R, +, \cdot)$ be a not-necessarily commutative ring (Definition 3.1.1). A *unit* or *invertible element of R* is an element $u \in R$ such that there exist an element $v \in R$ such that

$$uv = 1 = vu.$$

Such an element v is called the *multiplicative inverse of u* and is often denoted by u^{-1} . If it exists, then it is unique.

The set of units of R forms a group (Definition 2.1.1), often denoted by $R^\times$ or R^* , under the multiplication operation $\cdot$. It is called the *group of units* or *unit group* of R .

Definition 3.1.12. Let R be a ring (Definition 3.1.1). A element x of R is said to be *irreducible* if it is nonzero, it is not a unit (Definition 3.1.11), and if $x = ab$ implies that exactly one of a and b is a unit for any elements $a, b \in R$.

Definition 3.1.13. Let R be a ring (Definition 3.1.1) and let $a, b \in R$. We define the following notions of divisibility:

1. We say a **left divides** b (or a is a **left divisor of** b), written $a \mid_L b$, if there exists an element $x \in R$ such that $ax = b$.
2. We say a **right divides** b (or a is a **right divisor of** b), written $a \mid_R b$, if there exists an element $y \in R$ such that $ya = b$.
3. If R is commutative (Definition 3.1.6), these notions coincide, and we simply say a **divides** b or that a is a **divisor of** b , written $a \mid b$.

In terms of principal ideals (Definition 3.5.35), $a \mid_L b$ is equivalent to the containment of right ideals (Definition 3.5.34) $bR \subseteq aR$, and $a \mid_R b$ is equivalent to the containment of left ideals (Definition 3.5.34) $Rb \subseteq Ra$.

Definition 3.1.14. Let R be a ring (Definition 3.1.1). A **prime element of** R is an element $p \in R$ that is not zero, not a unit, and $p \mid ab$ implies that $p \mid a$ (Definition 3.1.13) or $p \mid b$.

If R is commutative (Definition 3.1.6), then $p \in R$ is prime if and only if the principal ideal (Definition 3.5.35) (p) generated by (Definition 3.5.35) p is a prime ideal (Definition 3.5.38).

Definition 3.1.15. Let $(R, +, \cdot)$ be a not-necessarily commutative ring (Definition 3.1.1). It is called a **division ring**, a **skew field**, or an **sfield**, if it is a nontrivial ring (Definition 3.1.8) in which every nonzero element $a \in R$ is a unit (Definition 3.1.11).

Definition 3.1.16 (Field). A **field** is commutative division (Definition 3.1.15) ring (Definition 3.1.6). In other words, a field is a commutative ring for which all nonzero elements have a multiplicative inverse (Definition 3.1.11).

Lemma 3.1.17. Let F be a field (Definition 3.1.16). Its unit group $F^\times$ (Definition 3.1.11) as a set equals $F - \{0\}$.

Proof. Since F is a field (Definition 3.1.16), it is a commutative division ring (Definition 3.1.15). By definition of a division ring, every nonzero element $a \in F - \{0\}$ is a unit (Definition 3.1.11), i.e., there exists $a^{-1} \in F$ such that $aa^{-1} = 1 = a^{-1}a$. Hence, $F - \{0\} \subseteq F^\times$.

Conversely, suppose that $u \in F^\times$. Then u is a unit, so $u \neq 0$ (Lemma 3.1.18) (since F is a field (Definition 3.1.16) and hence a nontrivial ring (Definition 3.1.8)). Therefore, $F^\times \subseteq F - \{0\}$, so $F^\times = F - \{0\}$ as desired. $\square$

Lemma 3.1.18. Let R be a nontrivial ring (Definition 3.1.8) and let $u \in R$ be a unit (Definition 3.1.11). Then $u \neq 0_R$.

Proof. Suppose for contradiction that $u = 0_R$. Since u is a unit, there exists $v \in R$ such that $uv = 1_R = vu$. But $uv = 0_R \cdot v = 0_R$, so $1_R = 0_R$, contradicting the assumption that R is a nontrivial ring. Hence $u \neq 0_R$. $\square$

Proposition 3.1.19. Let F be a field (Definition 3.1.16). Then the only ideals (Definition 3.5.34) of F are $\{0_F\}$ and F itself.

Proof. Let I be an ideal of F . If $I = \{0_F\}$, we are done. Otherwise, there exists some nonzero $a \in I$. Since F is a field, a is a unit (Definition 3.1.11), so $a^{-1} \cdot a = 1_F$. Because I is

an ideal, $a \in I$, and $a^{-1} \in F$, we have $1_F = a^{-1} \cdot a \in I$. Then for any $x \in F$, $x = x \cdot 1_F \in I$, so $I = F$. Hence the only ideals of F are $\{0_F\}$ and F . $\square$

Definition 3.1.20. Let R be a semiring, and let $a \in R$. For each nonnegative integer $n \in \mathbb{Z}_{\geq 0}$, the *n th power of a in R* , denoted by a^n , is defined as the n th power (Definition 1.0.16) of a with respect to the multiplicative monoid (Definition A.2.2) $(R, \cdot, 1_R)$. That is,

$$\begin{aligned} a^0 &:= 1_R, \\ a^{n+1} &:= a \cdot a^n. \end{aligned}$$

Definition 3.1.21. Let R be a ring (Definition 3.1.1), and let $a \in R$. For each integer $n \in \mathbb{Z}$, the *n th power of a in R* , denoted by a^n , is defined as follows:

- If $n \geq 0$, then a^n is the n th power (Definition 3.1.20) of a with respect to the underlying multiplicative semiring structure $(R, +, \cdot, 0_R, 1_R)$.
- If $n < 0$ and a is a unit of R (Definition 3.1.11), then $a^n := (a^{-1})^{-n}$, where a^{-1} is the multiplicative inverse (Definition 3.1.11) of a , and $(a^{-1})^{-n}$ is the power by the nonnegative integer $-n$ as in the first case.

A prominent example of a field would be the field of fractions of an integral domain.

Definition 3.1.22. Let R be an integral domain (Definition 3.1.10), and consider the set $R \times (R \setminus \{0\})$ as above. Define a relation $\sim$ on $R \times (R \setminus \{0\})$ by declaring that

$$(a, b) \sim (c, d) \quad \text{if and only if} \quad ad = bc,$$

for $a, c \in R$ and $b, d \in R \setminus \{0\}$. This relation is an equivalence relation. Its equivalence classes are denoted by $\frac{a}{b}$.

The set of equivalence classes

$$\left\{ \frac{a}{b} \mid a \in R, b \in R \setminus \{0\} \right\}$$

under the relation $\sim$ defined above is called the *field of fractions of R* , and is denoted by $\text{Frac}(R)$.

The operations on $\text{Frac}(R)$ are defined by

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= \frac{ad+bc}{bd}, \\ \frac{a}{b} \cdot \frac{c}{d} &= \frac{ac}{bd}, \end{aligned}$$

for $a, c \in R$ and $b, d \in R \setminus \{0\}$. With these operations, $\text{Frac}(R)$ is a field (Definition 3.1.16).

Equivalently, $\text{Frac}(R)$ may be defined as the localization of R (Definition 3.5.60) by the multiplicative subset $R \setminus \{0\}$ (Definition 3.5.59).

Definition 3.1.23. Let $(R, +, \cdot)$ and $(S, +, \cdot)$ be rings (Definition 3.1.1), not assumed to be commutative. A function $f : R \rightarrow S$ is called a *ring homomorphism* if for all $r_1, r_2 \in R$ the following properties hold:

1. $f(r_1 + r_2) = f(r_1) + f(r_2)$,
2. $f(r_1 r_2) = f(r_1) f(r_2)$,

3. $f(1_R) = 1_S$ where 1_R and 1_S denote the multiplicative identities in R and S , respectively.

A ring homomorphism is said to be a **ring isomorphism** if it is invertible as a map of sets.

An **R -ring** refers to a ring S equipped with a ring homomorphism $f : R \rightarrow S$.

We note that a ring homomorphism $f : R \rightarrow S$ yields a natural left R -module (Definition 3.5.1) structure on S and a natural right R -module structure on S respectively as follows for $r \in R$ and $s \in S$:

$$r \cdot s = f(r) \cdot s$$

$$s \cdot r = s \cdot f(r).$$

However, these left and right module structures need not yield a two-sided R -module structure.

Proposition 3.1.24. Let $\phi : R \rightarrow S$ be a ring homomorphism (Definition 3.1.23) between rings R and S . Then the kernel $\ker(\phi) = \{r \in R \mid \phi(r) = 0_S\}$ is a two-sided ideal (Definition 3.5.34) of R .

Proof. We verify the ideal axioms. Since $\phi(0_R) = 0_S$, we have $0_R \in \ker(\phi)$, so the kernel is nonempty.

Let $x, y \in \ker(\phi)$. Then $\phi(x) = 0_S$ and $\phi(y) = 0_S$. By additivity of ϕ , we have $\phi(x - y) = \phi(x) - \phi(y) = 0_S - 0_S = 0_S$, so $x - y \in \ker(\phi)$. Hence $\ker(\phi)$ is an additive subgroup of R .

Now let $r \in R$ and $x \in \ker(\phi)$. Then $\phi(rx) = \phi(r)\phi(x) = \phi(r) \cdot 0_S = 0_S$ and $\phi(xr) = \phi(x)\phi(r) = 0_S \cdot \phi(r) = 0_S$. Thus $rx \in \ker(\phi)$ and $xr \in \ker(\phi)$. Therefore $\ker(\phi)$ is a two-sided ideal of R . $\square$

Proposition 3.1.25. Let $\phi : R \rightarrow S$ be a ring homomorphism (Definition 3.1.23). Then ϕ is injective (Definition A.1.3) if and only if $\ker(\phi) = \{0_R\}$.

Proof. Suppose ϕ is injective. If $x \in \ker(\phi)$, then $\phi(x) = 0_S = \phi(0_R)$. By injectivity, $x = 0_R$, so $\ker(\phi) = \{0_R\}$.

Conversely, suppose $\ker(\phi) = \{0_R\}$. Let $x, y \in R$ such that $\phi(x) = \phi(y)$. Then $\phi(x - y) = \phi(x) - \phi(y) = 0_S$, so $x - y \in \ker(\phi) = \{0_R\}$. Hence $x - y = 0_R$, so $x = y$. Thus ϕ is injective. $\square$

Definition 3.1.26. 1. The **category of rings** is the locally small (Definition A.4.3) category (Definition A.4.1) whose objects are rings (Definition 3.1.1) R and whose morphisms $R \rightarrow S$ are ring homomorphisms (Definition 3.1.23). The category of rings over R is often denoted by notations such as **Ring**.

2. The **category of commutative rings** is the full subcategory (Definition A.4.10) of **Ring** consisting of the commutative rings (Definition 3.1.6). It is denoted by notations such as **CommRing** or **CRing**.

Definition 3.1.27. 1. Let R be a not necessarily commutative ring (Definition 3.1.1). The *category of rings over R* or the *category of R -rings* is the locally small (Definition A.4.3) category (Definition A.4.1) whose objects are R -rings (Definition 3.1.23) (S, φ) and whose morphisms $(S_1, \varphi_1) \rightarrow (S_2, \varphi_2)$ are given by ring homomorphisms $\psi : S_1 \rightarrow S_2$ such that $\varphi_2 = \psi \circ \varphi_1$.

$$\begin{array}{ccc} & R & \\ \varphi_1 \swarrow & & \searrow \varphi_2 \\ S_1 & \xrightarrow{\psi} & S_2 \end{array}$$

Equivalently, the category of R -rings is the category of objects under (Definition A.4.8) the object S in the category of rings (Definition 3.1.26). The category of rings over R is often denoted by notations such as **Ring _{R}** .

2. The category of commutative R -rings is the full subcategory of **Ring _{R}** consisting of the R -rings (A, φ) such that A is a commutative ring. It is denoted by notations such as **CommRing _{R}** or **CRing _{R}** .

Lemma 3.1.28. The category of rings (Definition 3.1.26) is equivalent (Definition A.4.7) to the category of $\mathbb{Z}$ -rings (Definition 3.1.27).

Proof. By ??, for every ring R there exists a unique ring homomorphism $\phi_R : \mathbb{Z} \rightarrow R$. This makes $\mathbb{Z}$ an initial object (Definition A.4.9) of **Ring**.

We construct an equivalence of categories $F : \mathbf{Ring} \rightarrow \mathbf{Ring}_{\mathbb{Z}}$. On objects, send each ring R to the pair (R, ϕ_R) , where $\phi_R : \mathbb{Z} \rightarrow R$ is the unique ring homomorphism. On morphisms, send each ring homomorphism $f : R \rightarrow S$ to itself, regarded as a morphism in **Ring _{$\mathbb{Z}$}** . To see that this is well-defined, note that for any ring homomorphism $f : R \rightarrow S$, the triangle

$$\begin{array}{ccc} & \mathbb{Z} & \\ \phi_R \swarrow & & \searrow \phi_S \\ R & \xrightarrow{f} & S \end{array}$$

commutes because both $f \circ \phi_R$ and ϕ_S are ring homomorphisms from $\mathbb{Z}$ to S , and by uniqueness in ?? they must coincide.

The functor F is clearly fully faithful (Definition A.4.26), since it acts as the identity on morphism sets. It is essentially surjective because every object of **Ring _{$\mathbb{Z}$}** has the form (R, ψ) for some ring homomorphism $\psi : \mathbb{Z} \rightarrow R$, and by uniqueness $\psi = \phi_R$, so $(R, \psi) = F(R)$.
(♠ TODO: statements that various homomorphisms in various categories are indeed closed under composition)

Therefore, F is an equivalence of categories (Definition A.4.7). □

Lemma 3.1.29. Let R be a not necessarily commutative ring (Definition 3.1.1). The category **Ring _{R}** (Definition 3.1.27) of rings over R has an initial object (Definition A.4.9), namely R equipped with the identity morphism.

Proof. Consider the object (R, id_R) in $\mathbf{Ring}_R$. Let (S, φ) be any other object of $\mathbf{Ring}_R$, where $\varphi : R \rightarrow S$ is a ring homomorphism. We show that there exists a unique morphism $f : (R, \text{id}_R) \rightarrow (S, \varphi)$ in $\mathbf{Ring}_R$.

A morphism $(R, \text{id}_R) \rightarrow (S, \varphi)$ is by definition a ring homomorphism $f : R \rightarrow S$ such that the following triangle commutes:

$$\begin{array}{ccc} & R & \\ \text{id}_R \swarrow & & \searrow \varphi \\ R & \xrightarrow{f} & S \end{array}$$

The commutativity condition requires $\varphi = f \circ \text{id}_R$, which simplifies to $f = \varphi$. Hence the only candidate is $f = \varphi$.

Since $\varphi : R \rightarrow S$ is already a ring homomorphism (as part of the object (S, φ)), it satisfies the morphism condition. Therefore, exactly one morphism exists from (R, id_R) to (S, φ) .

In conclusion, (R, id_R) is an initial object (Definition A.4.9) of $\mathbf{Ring}_R$. $\square$

Definition 3.1.30. Let R be a (not-necessarily commutative) ring with unity (Definition 3.1.1). An *R -algebra* is a ring A together with a ring homomorphism (Definition 3.1.23)

$$\varphi : R \rightarrow A$$

into the center $Z(A)$ (Definition 3.1.7) of A (so that $\varphi(r)$ commutes with every element of A for all $r \in R$), such that $\varphi(1_R) = 1_A$. The ring homomorphism φ is called the *structure map* of the algebra.

We note that older or more specialized texts might use less restrictive definitions of “algebra”, (e.g. just a ring homomorphism). One might call the above notion of algebra *central algebra* to distinguish it from the others.

Equivalently, an R -algebra consists of a ring A endowed with a two-sided R -module (Definition 3.5.1) structure for which the scalar multiplication satisfies

$$r \cdot (ab) = (r \cdot a)b = a(r \cdot b) \quad \text{for all } r \in R, a, b \in A.$$

In particular, any ring homomorphism between commutative rings (Definition 3.1.6) specifies an algebra structure.

Definition 3.1.31. 1. Let R be a not necessarily commutative ring (Definition 3.1.1). The *category of algebras over R* or the *category of R -algebras* is the locally small (Definition A.4.3) category (Definition A.4.1) whose objects are R -algebras (Definition 3.1.30) (A, φ) and whose morphisms $(A_1, \varphi_1) \rightarrow (A_2, \varphi_2)$ are given by ring homomorphisms $\psi : A_1 \rightarrow A_2$ such that $\varphi_2 = \psi \circ \varphi_1$.

$$\begin{array}{ccc} & R & \\ \varphi_1 \swarrow & & \searrow \varphi_2 \\ A_1 & \xrightarrow{\psi} & A_2 \end{array}$$

In other words, the category of R -algebras is the full subcategory (Definition A.4.10) of the category of R -rings (Definition 3.1.27) whose objects are the R -algebras.

The category of algebras over R is often denoted by notations such as $\mathbf{Alg}_R$ or $R\text{-}\mathbf{Alg}$.

2. The *category of commutative R -algebras* is the full subcategory of $\mathbf{Alg}_R$ consisting of those R -algebras (A, φ) for which A is a commutative ring. It is often denoted by $\mathbf{CAlg}_R$ or $\mathbf{CommAlg}_R$.

Definition 3.1.32. Let R be a commutative ring (Definition 3.1.6) and let (A, φ) be an R -algebra (Definition 3.1.30). The algebra A is called a *central R -algebra* if the structure map $\varphi : R \rightarrow Z(A)$ induces an isomorphism (Definition 3.1.23) between R and the center of A (Definition 3.1.7). Specifically, this requires two conditions:

1. The map φ is surjective onto $Z(A)$, meaning $Z(A) = \varphi(R)$.
2. The map φ is injective (faithful), so $R \cong \varphi(R)$.

In other words, A is central if $Z(A) \cong R$ via the structure map.

Definition 3.1.33 (Subalgebra of an algebra over a ring). Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let A be an R -algebra (Definition 3.1.30).

A subset $B \subseteq A$ is an *R -subalgebra* if:

- B is a subring of A (Definition 3.1.40),
- B is closed under scalar multiplication by R , i.e., for all $r \in R$ and $b \in B$, we have $r \cdot b \in B$.

In particular, B inherits the structure of an R -algebra as a subobject of A .

Definition 3.1.34 (Subalgebra generated by a subset). Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let A be an R -algebra (Definition 3.1.30). Given a subset $S \subseteq A$, the *subalgebra of A generated by S* , denoted by notations such as $\langle S \rangle = \langle S \rangle_R$, is the smallest R -subalgebra (Definition 3.1.33) of A that contains S .

Explicitly, the *subalgebra generated by S* is the intersection

$$\langle S \rangle_R = \bigcap_{T \subseteq S \text{ subalgebra}} T$$

of all R -subalgebras of A containing S . Equivalently, it consists of all R -linear combinations of finite products of elements from S and from the multiplicative identity element 1_A .

In case that R and A are both commutative (Definition 3.1.6), the subalgebra generated by S may be denoted by notations such as $R[S]$.

Definition 3.1.35 (Free associative algebra over a ring). Let R be a (not necessarily commutative) ring (Definition 3.1.1), and let $\{x_i\}_{i \in I}$ be a set of formal variables indexed by a set I .

The *free associative R -algebra on the variables $\{x_i\}_{i \in I}$* , denoted by notations such as $R\langle x_i \mid i \in I \rangle$ or $R\langle x_i \rangle_{i \in I}$, is defined as the R -algebra (Definition 3.1.30) whose underlying R -module is

freely generated by all finite words in the letters x_i (including the empty word, which acts as the multiplicative unit), with multiplication given by concatenation of words extended R -linearly.

Equivalently, $R\langle x_i \mid i \in I \rangle$ is the noncommutative polynomial ring generated by the x_i over R , satisfying no relations other than those required by the ring and algebra axioms.

In the case that I is a finite set, and writing $y_1, \dots, y_n$ for the variables x_i , it is customary to let $R\langle y_1, \dots, y_n \rangle$ denote the free associative R -algebra.

Definition 3.1.36 (Polynomial ring over a commutative ring). Let R be a commutative ring (Definition 3.1.6). For a set of variables $\{x_i\}_{i \in I}$, the *polynomial ring in variables $\{x_i\}$ over R* , denoted by notations such as $R[x_i \mid i \in I]$ or $R[x_i]_{i \in I}$, is defined as the commutative R -algebra (Definition 3.1.30) whose elements are finite R -linear combinations of monomials in the variables x_i , where the variables commute with each other and with elements of R .

That is, $R[x_i \mid i \in I]$ is the free commutative R -algebra generated by the set $\{x_i\}$.

In the case that I is a finite set, and writing $y_1, \dots, y_n$ for the variables x_i , it is customary to let $R[y_1, \dots, y_n]$ denote the polynomial ring.

Lemma 3.1.37. Let R be an integral domain (Definition 3.1.10). For any integer $n \geq 1$, the polynomial ring (Definition 3.1.36) $R[x_1, \dots, x_n]$ is an integral domain.

Proof. We first show that $R[x]$ is an integral domain. Let $f, g \in R[x]$ be nonzero polynomials. Write

$$f = a_0 + a_1x + \dots + a_dx^d \quad \text{and} \quad g = b_0 + b_1x + \dots + b_ex^e,$$

where $a_d \neq 0$ and $b_e \neq 0$ are the leading coefficients (Definition 3.1.38) of f and g , respectively, with $d = \deg(f)$ (Definition 3.1.38) and $e = \deg(g)$.

The product fg has as its coefficient of x^{d+e} the element a_db_e . Since R is an integral domain (Definition 3.1.10), it has no nontrivial zero divisors, so $a_db_e \neq 0$. Hence the coefficient of x^{d+e} in fg is nonzero, which implies that $fg \neq 0$. Therefore, $R[x]$ has no nontrivial zero divisors and is an integral domain.

For the general case with n variables, observe that $R[x_1, \dots, x_n] = (R[x_1, \dots, x_{n-1}])[x_n]$. By induction on n , if $R[x_1, \dots, x_{n-1}]$ is an integral domain, then adjoining the variable x_n yields another integral domain by the single-variable case established above. The base case $n = 1$ is exactly the argument above.

In all cases, $R[x_1, \dots, x_n]$ is an integral domain (Definition 3.1.10). □

Definition 3.1.38 (Degree and leading coefficient of a polynomial). Let R be a commutative ring (Definition 3.1.6) and let $f \in R[x]$ be a nonzero polynomial, so $f = a_0 + a_1x + \dots + a_nx^n$ with $a_i \in R$ and $a_n \neq 0$. The *degree* of f , denoted $\deg(f)$, is the largest integer $n \geq 0$ such that $a_n \neq 0$. The *leading coefficient* of f , denoted $\text{lc}(f)$ (or sometimes $\text{lead}(f)$), is the coefficient a_n .

If $f = 0$ is the zero polynomial, we define $\deg(0) = -\infty$ and the leading coefficient is undefined.

Definition 3.1.39. Let R be an integral domain (Definition 3.1.10). Let $n \geq 1$ be an integer. The *function field over R in n variables $x_1, \dots, x_n$* is the field (Definition 3.1.16) $R(x_1, \dots, x_n)$ defined as the field of fractions (Definition 3.1.22) of the polynomial ring (Definition 3.1.36) $R[x_1, \dots, x_n]$, which is an integral domain (Lemma 3.1.37).

$R(x_1, \dots, x_n)$ is also called the *field of rational numbers over R in n variables*. An element of $R(x_1, \dots, x_n)$ is called a *rational function over R in n variables*.

Definition 3.1.40 (Subring). Let R be a not necessarily commutative ring (Definition 3.1.1). A subset $S \subseteq R$ is called a *subring of R* if:

- S is itself a ring with the same operations as R (addition and multiplication inherited from R),
- the multiplicative identity 1_R of R is contained in S ,
- and the additive identity 0_R of R is contained in S .

Equivalently, $S \subseteq R$ is a subring if S is nonempty, closed under addition, multiplication, and additive inverses, and contains 1_R .

Definition 3.1.41. Let R be a not necessarily commutative ring (Definition 3.1.1). A *extension ring of R* is an R -algebra A such that the ring homomorphism $R \rightarrow A$ is an injection (Definition A.1.3). We also say that A/R is an *extension of rings*.

In this case, the image of $R \rightarrow A$ is a subring (Definition 3.1.40) of A isomorphic (Definition 3.1.23) to R ; we often identify R with this image to consider R as a subring of A .

3.2. Product rings.

Definition 3.2.1. Let R be a ring (Definition 3.1.1). An *idempotent element of R* is an element e such that $e^2 = e$.

Definition 3.2.2. Let R be a ring (Definition 3.1.1). A set $\{e_1, \dots, e_n\}$ in R is said to be *orthogonal* if $e_i e_j = 0$ for all $i \neq j$.

Definition 3.2.3. Let R be a ring (Definition 3.1.1). A set $\{e_1, \dots, e_n\}$ in R is said to consist of *orthogonal idempotents* if the set is orthogonal (Definition 3.2.2) and each e_i is idempotent (Definition 3.2.1).

Definition 3.2.4. Let R be a ring (Definition 3.1.1). A set $\{e_1, \dots, e_n\}$ in R is said to consist of *complete orthogonal idempotents* if the set is orthogonal idempotent and $\sum_i e_i = 1$.

Proposition 3.2.5. A ring (Definition 3.1.1) R is isomorphic to a product ring $\prod_i R_i$ if and only if R has a complete set of (Definition 3.2.4) central (Definition 3.1.7) orthogonal idempotents (Definition 3.2.3). More specifically, given a complete set $\{e_1, \dots, e_n\}$ of orthogonal idempotents of R , we have

$$R \cong \prod_i (e_i)R,$$

where $(e_i)R$ is the ideal of R generated by (e_i) which is in fact a ring (Definition 3.1.1) (albeit not a subring (Definition 3.1.40) of R), and given a decomposition

$$R \cong \prod_{i=1}^n R_i,$$

the set of elements $(e_1, 0, \dots, 0), (0, e_2, \dots, 0), \dots, (0, \dots, e_n)$ is a complete set of orthogonal idempotents.

Proof. Suppose that $R \cong \prod_{i=1}^n R_i$ for rings $R_1, \dots, R_n$. Let $\psi : R \rightarrow \prod_{i=1}^n R_i$ be a ring isomorphism. For each i , let $e'_i \in \prod_{j=1}^n R_j$ be the element with 1_{R_i} in the i -th coordinate and 0 in all other coordinates. Set $e_i = \psi^{-1}(e'_i)$. Since each e'_i is idempotent, orthogonal to e'_j for $j \neq i$, central, and $\sum e'_i = (1_{R_1}, \dots, 1_{R_n}) = 1_{\prod R_j}$, the same properties hold for e_i in R . Hence $\{e_1, \dots, e_n\}$ is a complete set of central (Definition 3.1.7) orthogonal idempotents.

Conversely, suppose that $\{e_1, \dots, e_n\}$ is a complete set of orthogonal idempotents (Definition 3.2.4) in R , with each e_i lying in the center (Definition 3.1.7) of R . For each i , let $R_i = e_i R = \{e_i r : r \in R\}$. Since e_i is central and idempotent, R_i is a ring with multiplicative identity e_i .

Define

$$\Phi : R \longrightarrow \prod_{i=1}^n R_i, \quad r \longmapsto (e_1 r, e_2 r, \dots, e_n r).$$

We first show that Φ is a ring homomorphism. For $r, s \in R$:

$$\Phi(r + s) = (e_1(r + s), \dots, e_n(r + s)) = (e_1 r + e_1 s, \dots, e_n r + e_n s) = \Phi(r) + \Phi(s).$$

For multiplication, using centrality of e_i :

$$\Phi(rs) = (e_1 rs, \dots, e_n rs) = (e_i^2 rs)_i = (e_i r e_i s)_i = (e_i r \cdot e_i s)_i = \Phi(r)\Phi(s).$$

Also $\Phi(1_R) = (e_1, \dots, e_n) = (1_{R_1}, \dots, 1_{R_n})$.

Next we show that Φ is injective (Definition 3.5.76). Suppose that $\Phi(r) = (0, \dots, 0)$. Then $e_i r = 0$ for all i . Since $\sum_{i=1}^n e_i = 1_R$, we have

$$r = 1_R \cdot r = \left(\sum_{i=1}^n e_i \right) r = \sum_{i=1}^n e_i r = 0.$$

(♠ TODO: reference how ring homomorphisms are injective iff they have trivial kernel) (♠ TODO: write definition of kernel of ring homomorphism) Therefore, Φ has trivial kernel, so Φ is injective.

Finally we show that Φ is surjective. Let $(x_1, \dots, x_n) \in \prod_{i=1}^n R_i$. For each i , write $x_i = e_i r_i$ for some $r_i \in R$. Set $r = \sum_{j=1}^n e_j r_j$. Then for any k :

$$e_k r = e_k \sum_{j=1}^n e_j r_j = \sum_{j=1}^n e_k e_j r_j.$$

By orthogonality, $e_k e_j = 0$ when $j \neq k$, and $e_k e_k = e_k^2 = e_k$. Hence

$$e_k r = e_k r_k = x_k.$$

Thus $\Phi(r) = (x_1, \dots, x_n)$, so Φ is surjective.

In conclusion, Φ is a ring isomorphism $R \cong \prod_{i=1}^n R_i$. $\square$

3.3. Arithmetic.

Definition 3.3.1 (Ordered ring). 1. An **ordered ring** is a ring (Definition 3.1.1) $(R, +, \cdot)$ equipped with a partial order $\leq$ such that

- (a) the additive group $(R, +)$ is an ordered abelian group, i.e. for all $a, b, c \in R$, if $a \leq b$, then $a + c \leq b + c$;
- (b) the order is compatible with multiplication: if $a \geq 0$ and $b \geq 0$, then $ab \geq 0$.

The set of **non-negative elements** of R is denoted by $R_{\geq 0} = \{a \in R : a \geq 0\}$.

If the partial order $\leq$ is a total order, then R is called a **totally ordered ring**. Equivalently, an ordered ring is totally ordered if and only if it satisfies **trichotomy**: for every $a \in R$, exactly one of $a > 0$, $a = 0$, or $a < 0$ holds.

- 2. Equivalently, an ordered ring may be defined as a ring $(R, +, \cdot)$ equipped with a strict partial order $<$ such that

- (a) the additive group $(R, +)$ is an ordered abelian group under $<$ (i.e. for all $a, b, c \in R$, if $a < b$, then $a + c < b + c$);
- (b) the order is compatible with multiplication in the following sense: for all $a, b \in R$, if $a > 0$ and $b > 0$, then $ab > 0$.

These two definitions are identified via the natural correspondence between a partial order $\leq$ and its associated strict partial order $<$ given by $a < b \iff a \leq b$ and $a \neq b$. When using the strict formulation, we adopt the notations that $a \leq b$ means $a < b$ or $a = b$, and $b > a$ (resp. $b \geq a$) means $a < b$ (resp. $a \leq b$). The set of **positive elements** is denoted by $R_{>0} = \{a \in R : a > 0\}$. For any $a, b \in R$, we have

$$a - b \in R_{>0} \iff a > b.$$

Definition 3.3.2. The **ring of integers**, universally denoted by $\mathbb{Z}$, is the unique (up to isomorphism (Definition 3.1.23)) ordered (Definition 3.3.1) integral domain (Definition 3.1.10) such that there exists a subset $P \subset \mathbb{Z}$ such that

- 1. P is nonempty, and
- 2. Every non-empty subset of P contains a least element.

Definition 3.3.3. The **field of rational numbers**, universally denoted by $\mathbb{Q}$, is the fraction field (Definition 3.1.22) of the integral domain (Definition 3.1.10) $\mathbb{Z}$ (Definition 3.3.2).

Definition 3.3.4. A **prime number** is a positive element of $\mathbb{Z}$ (Definition 3.3.2) that is an irreducible element (Definition 3.1.12) of $\mathbb{Z}$.

Explicitly, a prime number p is a positive integer that is not 1 and whose only positive integer divisors (Definition 3.1.13) are 1 and p

Equivalently, a prime number is a positive element of $\mathbb{Z}$ that is a prime element (Definition 3.1.14) of $\mathbb{Z}$.

A prime number is also called a *rational prime* to distinguish the notion of a prime number from other notions of "primes" that arise in arithmetic.

Definition 3.3.5. For each nonnegative integer $n \in \mathbb{Z}_{\geq 0}$, the *factorial of n* , denoted by $n!$, is the element of $\mathbb{Z}$ defined recursively by

$$\begin{aligned} 0! &:= 1, \\ n! &:= n \cdot (n-1)! \quad \text{for } n \geq 1. \end{aligned}$$

Equivalently, $n! = 1 \cdot 2 \cdots n$ for $n \geq 1$.

Definition 3.3.6. Let n, k be nonnegative integers with $0 \leq k \leq n$. The *binomial coefficient*, also called the *combinatorial choose function* and denoted by $\binom{n}{k}$, is defined as

$$\binom{n}{k} := \frac{n!}{k!(n-k)!}$$

where $n!$ is the factorial of n (Definition 3.3.5). If $k < 0$ or $k > n$, then $\binom{n}{k} := 0$.

Combinatorially, $\binom{n}{k}$ counts the number of subsets of cardinality k of a set with n elements.

Lemma 3.3.7 (Pascal's identity for binomial coefficients). For any nonnegative integer $n \in \mathbb{Z}_{\geq 0}$ and any integer $j \in \mathbb{Z}$, we have

$$\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}$$

where $\binom{n}{k}$ is the binomial coefficient (Definition 3.3.6) (with the convention $\binom{n}{k} = 0$ for $k < 0$ or $k > n$).

Proof. If $j \leq 0$, then $\binom{n}{j-1} = \binom{n}{j} = 0$ and $\binom{n+1}{j} = 0$, so the identity holds. If $j = 1$, then $\binom{n}{0} + \binom{n}{1} = 1 + n = n + 1 = \binom{n+1}{1}$. If $j = n + 1$, then $\binom{n}{n} + \binom{n}{n+1} = 1 + 0 = 1 = \binom{n+1}{n+1}$. If $j \geq n + 2$, then all three terms are 0. For $2 \leq j \leq n$, we compute directly using the factorial definition:

$$\begin{aligned} \binom{n}{j-1} + \binom{n}{j} &= \frac{n!}{(j-1)!(n-j+1)!} + \frac{n!}{j!(n-j)!} \\ &= \frac{n! \cdot j}{j!(n-j+1)!} + \frac{n! \cdot (n-j+1)}{j!(n-j+1)!} \\ &= \frac{n!(j+n-j+1)}{j!(n-j+1)!} \\ &= \frac{n!(n+1)}{j!(n+1-j)!} \\ &= \frac{(n+1)!}{j!(n+1-j)!} \\ &= \binom{n+1}{j}. \end{aligned}$$

(Definition 3.3.5) All cases are covered, so the identity holds for all $j \in \mathbb{Z}$. $\square$

Theorem 3.3.8 (Binomial theorem). Let R be a commutative ring (Definition 3.1.6), and let $a, b \in R$. For each nonnegative integer $n \in \mathbb{Z}_{\geq 0}$, we have

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

where $\binom{n}{k}$ is the binomial coefficient (Definition 3.3.6), and a^k, b^{n-k} are the k th and $(n-k)$ th powers (Definition 3.1.20) in the multiplicative monoid of R .

Proof. We proceed by induction on n .

For the base case $n = 0$, we have $(a + b)^0 = 1_R$ and $\binom{0}{0} a^0 b^0 = 1 \cdot 1_R \cdot 1_R = 1_R$, so the formula holds.

Inductively assume that the formula holds for some $n \geq 0$. We prove it for $n + 1$. Compute:

$$\begin{aligned} (a + b)^{n+1} &= (a + b) \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} a^{k+1} b^{n-k} + \sum_{k=0}^n \binom{n}{k} a^k b^{n+1-k}. \end{aligned}$$

In the first sum, shift the index by setting $j = k + 1$, so that it becomes $\sum_{j=1}^{n+1} \binom{n}{j-1} a^j b^{n+1-j}$. The second sum can be rewritten with the same index as $\sum_{j=0}^{n+1} \binom{n}{j} a^j b^{n+1-j}$. Combining, we obtain:

$$(a + b)^{n+1} = \binom{n}{0} a^0 b^{n+1} + \sum_{j=1}^n \left(\binom{n}{j-1} + \binom{n}{j} \right) a^j b^{n+1-j} + \binom{n}{n} a^{n+1} b^0.$$

The coefficient of $a^j b^{n+1-j}$ for $1 \leq j \leq n$ is $\binom{n}{j-1} + \binom{n}{j}$. By Lemma 3.3.7, this equals $\binom{n+1}{j}$. The boundary terms satisfy $\binom{n}{0} = 1 = \binom{n+1}{0}$ and $\binom{n}{n} = 1 = \binom{n+1}{n+1}$. Therefore:

$$(a + b)^{n+1} = \sum_{j=0}^{n+1} \binom{n+1}{j} a^j b^{n+1-j},$$

completing the induction. $\square$

3.4. Finite field.

Definition 3.4.1. Let R be a ring. Let $r \in R$, and let $n \in \mathbb{Z}_{\geq 0}$. The *n th power of r* , written r^n , is the element of R defined inductively as follows:

$$\begin{aligned} r^0 &= 1 \\ r^{n+1} &= r \cdot r^n. \end{aligned}$$

In other words, r^n is the n th power of r (Definition 1.0.16) as an element of the monoid of R under multiplication.

Definition 3.4.2. A *finite field* is a field (Definition 3.1.16) whose underlying set has finite (Definition A.1.5) cardinality (Definition A.1.13).

All finite fields have cardinality equal to the power (Definition 3.4.1) of a prime number (Definition 3.3.4). Moreover, for every $q = p^k$ for p a prime number and $k \geq 0$ an integer, any two fields of cardinality q are isomorphic as fields (Definition 4.1.1), and there exists a field of cardinality q (Theorem 3.4.3). One denotes by $\mathbb{F}_q$ a finite field of cardinality q .

Theorem 3.4.3. 1. All finite fields (Definition 3.4.2) have cardinality (Definition A.1.13) equal to the power (Definition 3.4.1) of a prime number (Definition 3.3.4).
 2. For every $q = p^k$ for p a prime number and $k \geq 0$ an integer,
 (a) any two fields of cardinality q are isomorphic as fields (Definition 4.1.1).
 (b) there is a field of cardinality exactly q .

Proof. (♠ TODO: AI generated, verify) We prove each part in turn.

1. Let F be a finite field. By Proposition 4.1.9, $\text{char}(F) = p$ for some prime number p , or $\text{char}(F) = 0$. The characteristic cannot be 0, because then the unique ring homomorphism $\mathbb{Z} \rightarrow F$ has trivial kernel by Definition 4.1.8, so its image is isomorphic to $\mathbb{Z}$, which is infinite — contradicting that F is finite. Hence $\text{char}(F) = p$.

The prime subfield of F is isomorphic to the finite field $\mathbb{F}_p$ by Definition 4.1.10. Viewing F as a vector space over its prime subfield $\mathbb{F}_p$ (which is valid by Lemma 4.1.4), we have that F is finite-dimensional because F is finite. Let $k = \dim_{\mathbb{F}_p} F$. By Definition 5.1.2, every element of F can be expressed uniquely as a linear combination of k basis vectors with coefficients in $\mathbb{F}_p$. Since each coefficient has p possible values, there are exactly p^k such linear combinations; hence $|F| = p^k$, which is the power of a prime number.

2. Fix a prime number p and an integer $k \geq 1$, and set $q = p^k$. We prove existence and uniqueness separately.

- (a) Suppose that F_1 and F_2 are two fields of cardinality q . By part (1), both have characteristic p , so their prime subfields are isomorphic to $\mathbb{F}_p$. Every element of F_i satisfies $x^q - x = 0$ in $F_i[x]$, because the multiplicative group $F_i^\times$ has order $q - 1$ and thus $a^{q-1} = 1$ for all nonzero $a \in F_i$ by Corollary 2.1.35; hence $a^q = a$ for all $a \in F_i$.

Each F_i is therefore the splitting field of the polynomial $x^q - x \in \mathbb{F}_p[x]$ over $\mathbb{F}_p$, since every element is a root and the field is generated by those roots. By Theorem 4.2.2, any two splitting fields of the same polynomial over the same base field are isomorphic; hence F_1 and F_2 are isomorphic as fields.

- (b) The polynomial $x^q - x \in \mathbb{F}_p[x]$ has a splitting field L over $\mathbb{F}_p$, by Theorem 4.2.2. We show that $|L| = q$. First, the formal derivative of $x^q - x$ in $\mathbb{F}_p[x]$ is $(x^q - x)' = qx^{q-1} - 1 = -1$, because $q = p^k \equiv 0 \pmod{p}$; hence $\gcd(x^q - x, (x^q - x)') = \gcd(x^q - x, -1) = 1$ in $\mathbb{F}_p[x]$. By ??, $x^q - x$ is a separable polynomial and therefore has exactly q distinct roots in L .

Let $S = \{\alpha \in L : \alpha^q - \alpha = 0\}$ be the set of these roots. We show that S is closed under field operations: for all $\alpha, \beta \in S$, we have $(\alpha + \beta)^q = \alpha^q + \beta^q = \alpha + \beta$ by the Freshman's dream identity in characteristic p , so $\alpha + \beta \in S$; similarly

$(\alpha\beta)^q = \alpha^q\beta^q = \alpha\beta$, so $\alpha\beta \in S$; and $(-\alpha)^q = -\alpha^q = -\alpha$, so $-\alpha \in S$. For nonzero $\alpha \in S$, we have $(\alpha^{-1})^q = (\alpha^q)^{-1} = \alpha^{-1}$, so $\alpha^{-1} \in S$. Hence S is a subfield of L . By definition of the splitting field, L is generated over $\mathbb{F}_p$ by the roots of $x^q - x$, which are exactly the elements of S ; hence $L = S$ and $|L| = q$.

This completes the proof of all three parts. $\square$

3.5. Modules of a ring.

Definition 3.5.1. Let R be a not-necessarily commutative ring (Definition 3.1.1).

1. A **left R -module** is an abelian group $(M, +)$ together with an operation $R \times M \rightarrow M$, denoted $(r, m) \mapsto rm$, such that for all $r, s \in R$ and $m, n \in M$:
 - $r(m + n) = rm + rn$,
 - $(r + s)m = rm + sm$,
 - $(rs)m = r(sm)$,
 - $1_R m = m$ where 1_R is the multiplicative identity of R .
2. A **right R -module** is defined similarly as an abelian group $(M, +)$ with an operation $M \times R \rightarrow M$, denoted $(m, r) \mapsto mr$, such that for all $r, s \in R$ and $m, n \in M$:
 - $(m + n)r = mr + nr$,
 - $m(r + s) = mr + ms$,
 - $m(rs) = (mr)s$,
 - $m1_R = m$.
3. Let R and S be (not necessarily commutative) rings (Definition 3.1.1).
 An **R - S -bimodule** (or an **R - S -module** or an (R, S) -module, etc.) is an abelian group (Definition 2.1.1) $(M, +)$ equipped with
 - (a) a left action of R :

$$R \times M \rightarrow M, \quad (r, m) \mapsto r \cdot m,$$

making M a left R -module (Definition 3.5.1),

- (b) a right action of S :

$$M \times S \rightarrow M, \quad (m, s) \mapsto m \cdot s,$$

making M a right S -module,

such that the left and right actions commute; that is, for all $r \in R$, $s \in S$, and $m \in M$,

$$r \cdot (m \cdot s) = (r \cdot m) \cdot s.$$

4. A **two-sided R -module** (or **R -bimodule**) is an R - R -bimodule.

If R is a commutative ring (Definition 3.1.6), then a left/right R -module can automatically be regarded as a two-sided R -module. As such, we simply talk about **R -modules** in this case.

Any abelian group is equivalent to a two-sided $\mathbb{Z}$ -module (Proposition 3.5.7). Moreover, any left R -module is equivalent to an $R - \mathbb{Z}$ -bimodule (Definition 3.5.1) (Proposition 3.5.9) and any right R -module is equivalent to an $\mathbb{Z} - R$ -bimodule (Definition 3.5.1) (Proposition 3.5.10). Given a left/right/two-sided R -module, its **natural bimodule structure** will refer to its structure as a $R - \mathbb{Z} / \mathbb{Z} - R / R - R$ bimodule. In this way, many definitions associated with

the notions of left/right/two-sided R -modules can be defined as special cases for definitions for R - S -bimodules.

Definition 3.5.2. Let R, S be not-necessarily commutative rings (Definition 3.1.1).

1. Let M be an R - S -bimodule whose abelian group (Definition 2.1.1) structure is given by the operator $+$. An R - S -submodule of M is a subgroup (Definition 2.1.19) $N \subseteq (M, +)$ if for all $r \in R$, $s \in S$, and $n \in N$, we have $rn \in N$ and $ns \in N$; in this case, N inherits an R - S bimodule structure from M .
2. If M is a left/right/two-sided R -module, then a left/right/two-sided R -submodule of M is a submodule of the natural bimodule structure (Definition 3.5.1) of M .

A submodule of M is a categorical subobject (Definition A.4.24) of M in the appropriate category of modules.

Proposition 3.5.3 (Submodule criterion). (♠ TODO: AI generated, verify) Let R, S be not-necessarily commutative rings (Definition 3.1.1) and let M be an R - S -bimodule (Definition 3.5.1). Let $N \subseteq M$ be a subset. Then N is an R - S -submodule of M (Definition 3.5.2) if and only if the following conditions hold:

- $N \neq \emptyset$,
- For all $n_1, n_2 \in N$, the sum $n_1 + n_2$ is an element of N ,
- For all $r \in R$ and $n \in N$, the product rn is an element of N ,
- For all $s \in S$ and $n \in N$, the product ns is an element of N .

In particular,

1. For a left R -module M , a subset $N \subseteq M$ is a left R -submodule if and only if it is nonempty, closed under addition, and closed under left multiplication by R
2. For a right R -module M , a subset $N \subseteq M$ is a right R -submodule if and only if it is nonempty, closed under addition, and closed under right multiplication by R
3. For a two-sided R -module M , a subset $N \subseteq M$ is a two-sided R -submodule if and only if it is nonempty, closed under addition, and closed under both left and right multiplication by R .

Proof. (♠ TODO: AI generated, verify) If N is an R - S -submodule of M , then the listed conditions hold by definition. Conversely, suppose that the listed conditions hold. We show that N is indeed an R - S -submodule of M .

First, we verify that N is a subgroup of the underlying abelian group of M . By the subgroup criterion (Proposition 2.1.20), it suffices to show that N is nonempty, closed under addition, and closed under additive inverses. The first two conditions are given. For closure under additive inverses, fix any $n \in N$. Since $(-1) \cdot n = -n$ (Lemma 3.5.8) and N is closed under left multiplication by elements of R , we have $-n = (-1) \cdot n \in N$. Therefore, N is a subgroup of M .

Next, N is closed under left multiplication by R and right multiplication by S by assumption. Hence, N is an R - S -submodule of M as desired. $\square$

Definition 3.5.4. Let R, S be (not-necessarily commutative) rings (Definition 3.1.1).

1. Let M and N be R - S -bimodules (Definition 3.5.1). A function $\varphi : M \rightarrow N$ is called an *R - S -bimodule homomorphism* or *R - S -linear* if it is a group homomorphism (Definition 2.1.17) of the underlying abelian groups of M and N and respects the scalar actions as follows: for all $m_1, m_2 \in M$, $r \in R$, and $s \in S$,

$$\begin{aligned}\varphi(r \cdot m_1) &= r \cdot \varphi(m_1), \\ \varphi(m_1 \cdot s) &= \varphi(m_1) \cdot s.\end{aligned}$$

2. Let M and N be left/right/two-sided R -modules (Definition 3.5.1). A function $\varphi : M \rightarrow N$ is called a *left/right/two-sided R -module homomorphism* if it is an bimodule homomorphism on the natural bimodule structures (Definition 3.5.1) of M and N . Such a function is also called *R -linear*.

Modules and homomorphisms of a fixed type (i.e. R - S -bimodules or left/right/two-sided R -modules) form a locally small (Definition A.4.3) category (Definition A.4.1).

A module homomorphism is an *isomorphism* if it is a bijective (Definition A.1.3) map. Equivalently, a module isomorphism is an isomorphism between the modules as objects in the appropriate module category (Definition 3.5.6)

Definition 3.5.5. (♠ TODO: AI generated, verify) Let $A_1, \dots, A_k, B$ be abelian groups (Definition 2.1.1) (written additively). A function

$$\Phi : A_1 \times \dots \times A_k \rightarrow B$$

is called a *multilinear map of abelian groups* (or simply a *multilinear map* when the context is clear) if for each $j = 1, \dots, k$ and for all fixed $a_i \in A_i$ with $i \neq j$, the map

$$A_j \rightarrow B, \quad a_j \mapsto \Phi(a_1, \dots, a_j, \dots, a_k)$$

is a group homomorphism (Definition 2.1.17) of abelian groups.

When $k = 2$, a multilinear map is called a *bilinear map of abelian groups* (or simply a *bilinear map*). Explicitly, a map $\Phi : A_1 \times A_2 \rightarrow B$ is bilinear if for all $a_1, a'_1 \in A_1$, $a_2, a'_2 \in A_2$:

$$\begin{aligned}\Phi(a_1 + a'_1, a_2) &= \Phi(a_1, a_2) + \Phi(a'_1, a_2), \\ \Phi(a_1, a_2 + a'_2) &= \Phi(a_1, a_2) + \Phi(a_1, a'_2).\end{aligned}$$

This is a special case of the definition of multilinear maps of modules over rings (Definition 5.10.1), obtained by taking all rings to be $\mathbb{Z}$ and all modules to be the underlying abelian groups (i.e., $\mathbb{Z}$ -modules).

Similarly, one may speak of *trilinear maps* ($k = 3$), *quadrilinear maps* ($k = 4$), etc.

Definition 3.5.6. Let R and S be (not necessarily commutative) rings (Definition 3.1.1).

1. The *category of (R, S) -bimodules* (or R - S -bimodules), denoted by notations such as ${}_R\mathbf{Mod}_S$, is the category whose objects are (R, S) -bimodules (Definition 3.5.1) and whose R - S -bimodule homomorphisms (Definition 3.5.4).

2. The *category of left R -modules*, denoted by notations such as ${}_R\mathbf{Mod}$ or $R - \mathbf{Mod}$, is the category ${}_R\mathbf{Mod}_{\mathbb{Z}}$, i.e. the category whose objects are left R -modules (Definition 3.5.1) and whose morphisms are left R -linear maps (Definition 3.5.4).
3. The *category of right R -modules*, denoted by notations such as $\mathbf{Mod}_R$ or $\mathbf{Mod} - R$, is the category ${}_{\mathbb{Z}}\mathbf{Mod}_R$, i.e. the category whose objects are right R -modules (Definition 3.5.1) and whose morphisms are right R -linear maps (Definition 3.5.4).

The category of bimodules can be canonically identified with module categories over tensor product rings (Definition 3.5.63):

- ${}_R\mathbf{Mod}_S$ is isomorphic to the category of left modules over the ring $R \otimes_{\mathbb{Z}} S^{\text{op}}$.
- ${}_R\mathbf{Mod}_S$ is isomorphic to the category of right modules over the ring $R^{\text{op}} \otimes_{\mathbb{Z}} S$.

Consequently, standard module-theoretic concepts (such as projective objects, injective objects, and flat objects) in ${}_R\mathbf{Mod}_S$ correspond exactly to the respective concepts in ${}_{R \otimes S^{\text{op}}}\mathbf{Mod}$.

Note that there are canonical isomorphisms of categories:

$${}_R\mathbf{Mod} \cong {}_R\mathbf{Mod}_{\mathbb{Z}} \quad \text{and} \quad \mathbf{Mod}_S \cong {}_{\mathbb{Z}}\mathbf{Mod}_S.$$

That is, left R -modules are exactly $(R, \mathbb{Z})$ -bimodules, and right S -modules are exactly $(\mathbb{Z}, S)$ -bimodules.

Proposition 3.5.7. (♠ TODO: AI generated, verify) The following categories are pairwise equivalent (Definition A.4.7): the category of abelian groups (Definition 2.1.18), the category of left $\mathbb{Z}$ -modules (Definition 3.5.6), the category of right $\mathbb{Z}$ -modules (Definition 3.5.6) $\mathbf{Mod}_{\mathbb{Z}}$, and the category of two-sided $\mathbb{Z}$ -modules (Definition 3.5.6) ${}_{\mathbb{Z}}\mathbf{Mod}_{\mathbb{Z}}$. More precisely, the equivalences can be constructed as follows:

1. Equivalence with left $\mathbb{Z}$ -modules. Let $F_{\ell}: \mathbf{Ab} \rightarrow {}_{\mathbb{Z}}\mathbf{Mod}$ be the functor that equips an abelian group $(A, +)$ with the canonical left $\mathbb{Z}$ -action

$$n \cdot m = \underbrace{m + \cdots + m}_{n \text{ times}} \quad (n > 0), \quad 0 \cdot m = 0_A, \quad (-n) \cdot m = -(n \cdot m) \quad (n > 0),$$

and acts as the identity on morphisms. Let $G_{\ell}: {}_{\mathbb{Z}}\mathbf{Mod} \rightarrow \mathbf{Ab}$ be the forgetful functor that sends a left $\mathbb{Z}$ -module to its underlying abelian group. Then:

- (a) $F_{\ell} \circ G_{\ell}$ is naturally isomorphic to $\text{id}_{{}_{\mathbb{Z}}\mathbf{Mod}}$, because for any left $\mathbb{Z}$ -module M the scalar multiplication by integers is uniquely determined by the additive structure of M : namely, $n \cdot m$ must equal $m + \cdots + m$ (n times) for $n > 0$, with $0 \cdot m = 0_M$ and $(-n) \cdot m = -(n \cdot m)$.
- (b) $G_{\ell} \circ F_{\ell} = \text{id}_{\mathbf{Ab}}$, because the canonical left $\mathbb{Z}$ -action on an abelian group uses only the addition and inverse operations already present.

Any group homomorphism between left $\mathbb{Z}$ -modules is automatically $\mathbb{Z}$ -linear because $f(n \cdot m) = n \cdot f(m)$ for all $n \in \mathbb{Z}$, by additivity of f . Hence, F_{ℓ} is fully faithful and essentially surjective, so it is an equivalence of categories (Definition A.4.7).

2. Equivalence with right $\mathbb{Z}$ -modules. Similarly, let $F_r: \mathbf{Ab} \rightarrow \mathbf{Mod}_{\mathbb{Z}}$ equip an abelian group $(A, +)$ with the canonical right $\mathbb{Z}$ -action

$$m \cdot n = \underbrace{m + \cdots + m}_{n \text{ times}} \quad (n > 0), \quad m \cdot 0 = 0_A, \quad m \cdot (-n) = -(m \cdot n) \quad (n > 0),$$

and let $G_r: \mathbf{Mod}_{\mathbb{Z}} \rightarrow \mathbf{Ab}$ be the forgetful functor. The same argument shows that F_r is an equivalence of categories with inverse G_r .

3. Equivalence with two-sided $\mathbb{Z}$ -modules. Let $F_{\ell_r}: \mathbf{Ab} \rightarrow {}_{\mathbb{Z}}\mathbf{Mod}_{\mathbb{Z}}$ equip an abelian group $(A, +)$ with the left and right $\mathbb{Z}$ -actions defined above (which coincide since $\mathbb{Z}$ is commutative), making $n \cdot m = m \cdot n$. Let $G_{\ell_r}: {}_{\mathbb{Z}}\mathbf{Mod}_{\mathbb{Z}} \rightarrow \mathbf{Ab}$ be the forgetful functor. Then:

- (a) $F_{\ell_r} \circ G_{\ell_r}$ is naturally isomorphic to $\text{id}_{{}_{\mathbb{Z}}\mathbf{Mod}_{\mathbb{Z}}}$, because any two-sided $\mathbb{Z}$ -module has scalar multiplication determined by the additive structure.
- (b) $G_{\ell_r} \circ F_{\ell_r} = \text{id}_{\mathbf{Ab}}$.

Any group homomorphism between two-sided $\mathbb{Z}$ -modules is automatically a bimodule map, so F_{ℓ_r} is fully faithful and essentially surjective.

Lemma 3.5.8 (Minus one times an element is its inverse in a $\mathbb{Z}$ -module). (♠ TODO: AI generated, verify) Let A be an abelian group and let it be equipped with the canonical $\mathbb{Z}$ -module structure (Definition 3.5.1) via the equivalence of categories between abelian groups and $\mathbb{Z}$ -modules (Proposition 3.5.7). For any $a \in A$, we have

$$(-1) \cdot a = -a,$$

where $-a$ denotes the inverse of a in the abelian group $(A, +)$.

Proof. (♠ TODO: AI generated, verify) By the definition of the canonical left $\mathbb{Z}$ -action on an abelian group (Proposition 3.5.7), for $n > 0$ we have $n \cdot a = \underbrace{a + \cdots + a}_{n \text{ times}}$, and $(-n) \cdot a = -(n \cdot a)$.

Taking $n = 1$, we get $(-1) \cdot a = -(1 \cdot a)$. Since $1 \cdot a = a$ by the module axiom $1_{\mathbb{Z}}a = a$, we obtain $(-1) \cdot a = -a$. $\square$

Proposition 3.5.9. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring (Definition 3.1.1). The category of left R -modules (Definition 3.5.6) ${}_R\mathbf{Mod}$ is equivalent (Definition A.4.7) to the category of R - $\mathbb{Z}$ -bimodules (Definition 3.5.6) ${}_R\mathbf{Mod}_{\mathbb{Z}}$.

We first construct a functor from left R -modules to R - $\mathbb{Z}$ -bimodules. Let $F: {}_R\mathbf{Mod} \rightarrow {}_R\mathbf{Mod}_{\mathbb{Z}}$ be the functor that equips a left R -module M with a right $\mathbb{Z}$ -action via

$$m \cdot n = \underbrace{m + \cdots + m}_{n \text{ times}} \quad (n > 0), \quad m \cdot 0 = 0_M, \quad m \cdot (-n) = -(m \cdot n) \quad (n > 0),$$

for $m \in M$ and $n \in \mathbb{Z}$. The left R -action and this right $\mathbb{Z}$ -action commute because the left action is additive:

$$r \cdot (m \cdot n) = r \cdot \underbrace{(m + \cdots + m)}_n = \underbrace{rm + \cdots + rm}_n = (rm) \cdot n.$$

Hence, M becomes an R - $\mathbb{Z}$ -bimodule. On morphisms, F acts as the identity: a left R -linear map is automatically $\mathbb{Z}$ -linear since it preserves addition and hence integer multiples.

Next we construct a functor from R - $\mathbb{Z}$ -bimodules to left R -modules. Let $G: {}_R\mathbf{Mod}_{\mathbb{Z}} \rightarrow {}_R\mathbf{Mod}$ be the forgetful functor that sends an R - $\mathbb{Z}$ -bimodule to its underlying left R -module (forgetting the right $\mathbb{Z}$ -action).

Finally we verify that F and G are mutually inverse equivalences:

1. $F \circ G$ is naturally isomorphic to id_{Mod_R} , because for any R - $\mathbb{Z}$ -bimodule M , the right $\mathbb{Z}$ -action is uniquely determined by the additive structure of M : in any abelian group (and hence in any left R -module), there is at most one way to define integer multiples, and the canonical action always works.
2. $G \circ F = \text{id}_{\text{Mod}_R}$, because the underlying left R -module of $F(M)$ is exactly M .

Any left R -linear map between bimodules is automatically a bimodule map: if $f: M \rightarrow N$ satisfies $f(rm) = rf(m)$, then it also satisfies $f(m \cdot n) = f(\sum m) = \sum f(m) = f(m) \cdot n$. Hence, F is fully faithful and essentially surjective.

Proposition 3.5.10. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring (Definition 3.1.1). The category of right R -modules (Definition 3.5.6) Mod_R is equivalent (Definition A.4.7) to the category of $\mathbb{Z}$ - R -bimodules (Definition 3.5.6) ${}_{\mathbb{Z}}\text{Mod}_R$.

We first construct a functor from right R -modules to $\mathbb{Z}$ - R -bimodules. Let $F: \text{Mod}_R \rightarrow {}_{\mathbb{Z}}\text{Mod}_R$ be the functor that equips a right R -module M with a left $\mathbb{Z}$ -action via

$$n \cdot m = \underbrace{m + \cdots + m}_{n \text{ times}} \quad (n > 0), \quad 0 \cdot m = 0_M, \quad (-n) \cdot m = -(n \cdot m) \quad (n > 0),$$

for $m \in M$ and $n \in \mathbb{Z}$. The left $\mathbb{Z}$ -action and the right R -action commute because the right action is additive:

$$(n \cdot m) \cdot r = (\underbrace{m + \cdots + m}_n) \cdot r = \underbrace{mr + \cdots + mr}_n = n \cdot (m \cdot r).$$

Hence, M becomes a $\mathbb{Z}$ - R -bimodule. On morphisms, F acts as the identity: a right R -linear map is automatically $\mathbb{Z}$ -linear since it preserves addition and hence integer multiples.

Next we construct a functor from $\mathbb{Z}$ - R -bimodules to right R -modules. Let $G: {}_{\mathbb{Z}}\text{Mod}_R \rightarrow \text{Mod}_R$ be the forgetful functor that sends a $\mathbb{Z}$ - R -bimodule to its underlying right R -module (forgetting the left $\mathbb{Z}$ -action).

Finally we verify that F and G are mutually inverse equivalences:

1. $F \circ G$ is naturally isomorphic to $\text{id}_{{}_{\mathbb{Z}}\text{Mod}_R}$, because for any $\mathbb{Z}$ - R -bimodule M , the left $\mathbb{Z}$ -action is uniquely determined by the additive structure of M .
2. $G \circ F = \text{id}_{\text{Mod}_R}$, because the underlying right R -module of $F(M)$ is exactly M .

Any right R -linear map between bimodules is automatically a bimodule map: if $f: M \rightarrow N$ satisfies $f(mr) = f(m)r$, then it also satisfies $f(n \cdot m) = n \cdot f(m)$ for all $n \in \mathbb{Z}$. Hence, F is fully faithful and essentially surjective.

3.5.1. Linear combinations, spanning, and linear independence for modules over rings.

Definition 3.5.11. Let R, S be (not-necessarily commutative) rings (Definition 3.1.1).

1. Let M be a R - S -bimodule (Definition 3.5.1). Given elements $x_1, \dots, x_n \in M$, $r_1, \dots, r_n \in R$, and $s_1, \dots, s_n \in S$, an element of the form

$$\sum_{i=1}^n r_i x_i s_i \in M$$

is called a *(R-S-)linear combination of $x_1, \dots, x_n$* . Given an arbitrary subset $\{m_i\}_{i \in I} \subseteq M$, a *(R-S-)linear combination of the m_i over R* refers to a linear combination of some finite subset of $\{m_i\}_{i \in I}$ over R .

- Let M be a left/right/two-sided R -module (Definition 3.5.1). A *linear combination of elements $\{m_i\}_{i \in I} \subseteq M$* refers to a linear combination of (a finite subset of) $\{m_i\}_{i \in I}$ for the natural bimodule structure (Definition 3.5.1) of M . In case that M is a left/right R -module, and given finitely many $m_1, \dots, m_n \in M$, a linear combination of these m_i is equivalently an element of the form

$$\begin{aligned} r_1x_1 + r_2x_2 + \dots + r_nx_n \\ x_1r_1 + x_2r_2 + \dots + x_nr_n \end{aligned}$$

for $r_1, \dots, r_n \in R$ respectively.

Definition 3.5.12 (Submodule generated by elements in an (R, S) -bimodule). Let R and S be (not necessarily commutative) rings (Definition 3.1.1).

- Let M be an (R, S) -bimodule (Definition 3.5.1).

Given a subset $X \subseteq M$, the *sub-bimodule of M generated by X* is the smallest (R, S) -sub-bimodule of M containing X . It is often denoted by notations such as $\langle X \rangle = \langle X \rangle_{R,S}$ and is more explicitly the intersection

$$\langle X \rangle_{R,S} = \bigcap_{X \subseteq T \subseteq M, T \text{ is a } (R,S)\text{-submodule of } M} T$$

of all (R, S) -submodules of M containing X .

Equivalently, $\langle X \rangle_{R,S}$ consists of all linear combinations (Definition 3.5.11) of X .

- If M is a left/right/two-sided R -module and given a subset $X \subseteq M$, the *submodule of M generated by X* is the submodule of the natural bimodule (Definition 3.5.1) of M generated by X . It is denoted by notations such as $\langle X \rangle = \langle X \rangle_R$.

Definition 3.5.13. Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be an R -module (Definition 3.5.1).

- If M is a left R -module (Definition 3.5.1), a finite subset $S = \{x_1, \dots, x_n\} \subseteq M$ is said to be *linearly independent over R* if, whenever $r_1, \dots, r_n \in R$ satisfy

$$r_1x_1 + r_2x_2 + \dots + r_nx_n = 0,$$

it follows that each $r_i = 0$.

- If M is a right R -module (Definition 3.5.1), a finite subset $S = \{x_1, \dots, x_n\} \subseteq M$ is said to be *linearly independent over R* if, whenever $r_1, \dots, r_n \in R$ satisfy

$$x_1r_1 + x_2r_2 + \dots + x_nr_n = 0,$$

it follows that each $r_i = 0$.

- If M is a two-sided R -module (Definition 3.5.1), a finite subset $S = \{x_1, \dots, x_n\} \subseteq M$ is said to be *linearly independent over R* if, whenever $r_1, \dots, r_n, s_1, \dots, s_n \in R$ satisfy

$$r_1x_1s_1 + r_2x_2s_2 + \dots + r_nx_ns_n = 0,$$

it follows that each $r_i s_i = 0$ in R . Equivalently, no nontrivial pair of coefficient actions on the x_i yields the zero element of M .

An arbitrary subset $S \subseteq M$ is **linearly independent over R** if every finite subset of S is linearly independent.

Definition 3.5.14. Let R and S be (not necessarily commutative) rings (Definition 3.1.1).

1. If M is an R - S -bimodule (Definition 3.5.1), then a subset $\{m_i\}_{i \in I} \subseteq M$ is said to **span M (as an R - S -bimodule)** if every element $m \in M$ is a linear combination (Definition 3.5.11) of $\{m_i\}_{i \in I}$.
2. If M is a left/right/two-sided R -module, then a subset $\{m_i\}_{i \in I} \subseteq M$ is said to **span M (as a left/right/two-sided R -module)** if $\{m_i\}_{i \in I}$ spans its natural bimodule structure (Definition 3.5.1).

In each case, such a set $\{m_i\}_{i \in I}$ is called a **generating set** or **spanning set of M over R** .

In each case, such a set S is called a **generating set** or **spanning set of M over R** .

Definition 3.5.15 (Finitely generated modules and bimodules). Let R and S be (not necessarily commutative) rings (Definition 3.1.1).

1. An R - S -bimodule M is **finitely generated**, **finite type** or **finite** (which is not to say that M is of finite cardinality (Definition A.1.13)) if it has a finite spanning set (Definition 3.5.14).
2. A left/right/two-sided R -module is **finitely generated**, **finite type**, or **finite** if has a finite spanning set (Definition 3.5.14), or equivalently if its natural bimodule structure (Definition 3.5.1) is finitely generated.

3.5.2. Categories of modules as abelian categories.

Definition 3.5.16 (Additive category). Let $\mathcal{A}$ be a locally small category (Definition A.4.3).

1. $\mathcal{A}$ is said to be a **preadditive category** if the following hold:
 - For any two objects A, B in $\mathcal{A}$, the set $\text{Hom}_{\mathcal{A}}(A, B)$ is an abelian group (Definition 2.1.1), and composition of morphisms is bilinear.
 - There is a zero object (Definition A.4.9) 0 in $\mathcal{A}$.
2. If $\mathcal{A}$ is preadditive, then it is called **additive** if it additionally satisfies the following:
 - For any two objects A, B in $\mathcal{A}$, there exists a product object $A \times B$ (Definition A.4.21), often written $A \oplus B$, called the **direct sum of A and B** . In fact, $A \oplus B$ is not only a product but also a coproduct (Definition 3.5.23) of A and B (Lemma A.5.8).

Given a finite collection $\{A_i\}_i$ of objects A_i in an additive category $\mathcal{A}$, we may more generally speak of the **direct sum** $\bigoplus_i A_i$; it has canonical injections from and projections to each A_i .

Definition 3.5.17 (Additive functor). Let $\mathcal{A}$ and $\mathcal{B}$ be pre-additive categories (Definition 3.5.16). A functor

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

is an **additive functor** if for every pair of objects $A, A' \in \mathcal{A}$, the induced map

$$F_{A,A'} : \text{Hom}_{\mathcal{A}}(A, A') \rightarrow \text{Hom}_{\mathcal{B}}(F(A), F(A'))$$

is a group homomorphism of abelian groups, or equivalently if it is enriched over the category $\mathbf{Ab}$ of abelian groups.

In the case that $\mathcal{A}$ and $\mathcal{B}$ are additive categories (Definition 3.5.16), this condition implies that F is compatible with the additive structures on $\mathcal{A}$ and $\mathcal{B}$. More precisely,

- F sends the zero object $0_{\mathcal{A}}$ of $\mathcal{A}$ to the zero object $0_{\mathcal{B}}$ of $\mathcal{B}$, i.e.,

$$F(0_{\mathcal{A}}) = 0_{\mathcal{B}}.$$

- F preserves finite direct sums: For any finite family of objects $\{A_i\}_{i=1}^n$ in $\mathcal{A}$,

$$F\left(\bigoplus_{i=1}^n A_i\right) \cong \bigoplus_{i=1}^n F(A_i)$$

via the canonical isomorphism induced by F applied to the canonical injections and projections.

Definition 3.5.18 (Abelian category). Let $\mathcal{A}$ be a category. The category $\mathcal{A}$ is an *abelian category* if:

- $\mathcal{A}$ is an additive category (Definition 3.5.16).
- Every morphism $f : A \rightarrow B$ has a kernel $\ker(f)$ and a cokernel $\operatorname{coker}(f)$ (Definition A.4.14).
- For every morphism $f : A \rightarrow B$, the canonical morphism $\operatorname{coim}(f) \rightarrow \operatorname{im}(f)$ is an isomorphism, where

$$\operatorname{coim}(f) = \operatorname{coker}(\ker(f) \rightarrow A), \quad \operatorname{im}(f) = \ker(B \rightarrow \operatorname{coker}(f)).$$

(♠ TODO: I think I need to re-check this definition) (♠ TODO: coimage)

.

It is also worth considering Grothendieck's additional axioms for abelian categories (Definition 3.5.19).

Definition 3.5.19 (Grothendieck's axioms for abelian categories (Ab1–Ab5)). Let $\mathcal{A}$ be an abelian category (Definition 3.5.18).

Grothendieck introduced the following hierarchy of additional axioms to express stronger completeness and exactness properties in $\mathcal{A}$ — we note that Ab1, Ab2, and Ab2* are already satisfied for any abelian category:

- **Ab1**: Every morphism in $\mathcal{A}$ has a kernel and a cokernel (Definition A.4.14).
- **Ab2**: Every monic (Definition A.4.23) in $\mathcal{A}$ is the kernel of its cokernel.
- **Ab2***: Every epi in $\mathcal{A}$ is the cokernel of its kernel.
- **AB3**: The category $\mathcal{A}$ is cocomplete (Definition A.4.16).
 - Since $\mathcal{A}$ is abelian (and hence admits equalizers as kernels (Definition 3.5.25)), this is equivalent to requiring that $\mathcal{A}$ has all small coproducts (Definition A.4.21) (direct sums).
- **AB4**: The category $\mathcal{A}$ satisfies AB3, and coproducts are exact (Definition 3.5.29) (when regarded as functors $\bigoplus_I : \mathcal{A}^I \rightarrow \mathcal{A}$).

- That is, the coproduct of a family of short exact sequences is a short exact sequence. Explicitly, for any family of short exact sequences $0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0$ indexed by a set I , the sequence

$$0 \rightarrow \bigoplus_{i \in I} A_i \rightarrow \bigoplus_{i \in I} B_i \rightarrow \bigoplus_{i \in I} C_i \rightarrow 0$$

is exact in $\mathcal{A}$.

- **AB5**: The category $\mathcal{A}$ satisfies AB3, and filtered colimits (Definition A.4.20) are exact (Definition 3.5.29) (when regarded as functors $\varinjlim_J : \mathcal{A}^J \rightarrow \mathcal{A}$).
 - Equivalently, for any filtered (Definition A.4.17) index category J and any directed system (Definition A.4.18) of short exact sequences $0 \rightarrow A_j \rightarrow B_j \rightarrow C_j \rightarrow 0$, the colimit sequence

$$0 \rightarrow \varinjlim A_j \rightarrow \varinjlim B_j \rightarrow \varinjlim C_j \rightarrow 0$$

is exact.

- Note: AB5 implies AB4. An abelian category satisfying AB5 and having a generator (Definition 3.5.20) is called a **Grothendieck category**.
- **AB6**: The category $\mathcal{A}$ satisfies AB3, and for any object X and any family of filtered subobjects $\{F_i\}_{i \in I}$ of X (where each F_i is a filter of subobjects), the intersection commutes with the limit:

$$\bigcap_{i \in I} (\varinjlim_{j \in F_i} U_{i,j}) = \varinjlim_{(j_i) \in \prod_{i \in I} F_i} (\bigcap_{i \in I} U_{i,j_i}).$$

(This axiom is less commonly cited but appears in Grothendieck's Tohoku paper).

- **AB3***: The category $\mathcal{A}$ is complete (Definition A.4.16) (i.e., has all small products).
- **AB4***: The category $\mathcal{A}$ satisfies AB3* and products are exact.
 - Note: This is rarely satisfied for module categories (e.g., it fails for Abelian groups), but it is satisfied for the category of sheaves on a space.
- **AB5***: The category $\mathcal{A}$ satisfies AB3* and filtered limits (inverse limits) are exact.

Notes:

- AB5 implies AB4, and AB4 implies AB3.
- AB5* implies AB4*, and AB4* implies AB3*.

Definition 3.5.20 (Generator of a category). Let $\mathcal{C}$ be a category (Definition A.4.1).

1. An object $G \in \mathcal{C}$ is called a **generator** (or **separator**) if for every pair of distinct morphisms $f, g : X \rightarrow Y$ in $\mathcal{C}$, there exists a morphism $h : G \rightarrow X$ such that

$$f \circ h \neq g \circ h.$$

In case that $\mathcal{C}$ is locally small (Definition A.4.3), this is equivalent to the condition that the representable functor

$$\mathrm{Hom}_{\mathcal{C}}(G, -) : \mathcal{C} \rightarrow \mathbf{Set}$$

is faithful (Definition A.4.26), which in turn is equivalent to the condition that for every object $X \in \mathcal{C}$, there exists an epimorphism

$$\bigoplus_{i \in I} G \twoheadrightarrow X$$

for some indexing set I , where $\bigoplus$ denotes the coproduct (Definition A.4.21) in $\mathcal{C}$.

2. A family $\{G_i\}_{i \in I}$ is called a **generating family** if for every pair of distinct morphisms $f, g : X \rightarrow Y$ in $\mathcal{C}$, there exists some index $i \in I$ and a morphism $h : G_i \rightarrow X$ such that

$$f \circ h \neq g \circ h.$$

In case $\mathcal{C}$ is locally small, this is equivalent to the condition that the collection of representable functors

$$\{\mathrm{Hom}_{\mathcal{C}}(G_i, -) : \mathcal{C} \rightarrow \mathbf{Set}\}_{i \in I}$$

is jointly faithful, which in turn is equivalent to the condition that for every object $X \in \mathcal{C}$, there exists a family of objects $\{G_i\}_{i \in J}$ from the generating set indexed by some set J , and an epimorphism

$$\bigoplus_{i \in J} G_i \twoheadrightarrow X.$$

Definition 3.5.21 (Hom of left/right/bi-modules). Let R, S, T be (not necessarily commutative) rings (Definition 3.1.1).

1. Let M and N be left R -modules (Definition 3.5.1). The **homomorphism group of left R -modules from M to N** is the abelian group

$$\mathrm{Hom}(M, N) = \mathrm{Hom}_R(M, N) = \mathrm{Hom}_{R\text{-}\mathbb{Z}}(M, N) := \{f : M \rightarrow N \mid f \text{ is a left } R\text{-module homomorphism}\}.$$

(Definition 3.5.4)

2. Let M and N be right R -modules (Definition 3.5.1). The **homomorphism group of right R -modules from M to N** is the abelian group

$$\mathrm{Hom}(M, N) = \mathrm{Hom}_R(M, N) = \mathrm{Hom}_{\mathbb{Z}\text{-}R}(M, N) := \{f : M \rightarrow N \mid f \text{ is a right } R\text{-module homomorphism}\}.$$

3. Let S be a (not necessarily commutative ring) and let M and N be R – S -bimodules (Definition 3.5.1). The **homomorphism group of R - S -bimodules from M to N** is the abelian group

$$\mathrm{Hom}(M, N) = \mathrm{Hom}_{R\text{-}S}(M, N) := \{f : M \rightarrow N \mid f \text{ is a } R\text{--}S\text{-bimodule homomorphism}\}$$

In each case, $\mathrm{Hom}(M, N)$ has a natural structure of an **abelian group** given by **pointwise addition**: for $f, g \in \mathrm{Hom}(M, N)$,

$$(f + g)(m) := f(m) + g(m),$$

and the zero morphism $\mathbf{0}$ given by $0(m) := 0_N$ acts as the identity element. The additive inverse $-f$ is defined by $(-f)(m) := -f(m)$. Moreover, depending on bi-module structures that M and N may be carrying, $\mathrm{Hom}(M, N)$ may itself carry additional module structures:

- In case that M is a $R - S$ -bimodule and N is a $R - T$ -bimodule, $\text{Hom}_R(M, N)$, the group of left R -module homomorphisms, is an $S - T$ -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(m \cdot s) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

- Dually, in case that M is a $S - R$ -bimodule and N is a $T - R$ -bimodule, $\text{Hom}_R(M, N)$, the group of right R -module homomorphisms, is an $S - T$ -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(s \cdot m) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

Some cases of interest may be when R , S , or T is in fact $\mathbb{Z}$ — these allow us to see module structures on $\text{Hom}(M, N)$ even when M and N are one-sided modules.

(♠ **TODO: state this as a theorem**) We furthermore note that $\text{Hom}_R(-, -)$ yields biadditive functors (Definition A.5.1)

$$\text{Hom}_R(-, -) : {}_R\mathbf{Mod}_S^{\text{op}} \times {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T$$

$$\text{Hom}_R(-, -) : {}_S\mathbf{Mod}_R^{\text{op}} \times {}_T\mathbf{Mod}_R \rightarrow {}_S\mathbf{Mod}_T.$$

(Definition A.4.12) (Definition 3.5.6) (Theorem 3.5.27)

Definition 3.5.22 (Product of Modules). Let R and S be (not necessarily commutative) rings (Definition 3.1.1), and let $\{M_i\}_{i \in I}$ be a (possibly infinite (Definition A.1.5) but small (Definition A.1.1)) family of (R, S) -bimodules (Definition 3.5.1).

left R -modules, of right R -modules, of two-sided R -modules (Definition 3.5.1), or of

The *(direct) product of the family $\{M_i\}_{i \in I}$* is defined, as a group (Definition 2.1.1), as the product of sets (Definition 2.2.1):

$$\prod_{i \in I} M_i := \{(m_i)_{i \in I} \mid m_i \in M_i \text{ for all } i \in I\}.$$

$\prod_{i \in I} M_i$ inherits a natural R - S module structure defined componentwise by the following rules for all $(m_i)_{i \in I}, (n_i)_{i \in I} \in \prod_{i \in I} M_i$ and all scalars $r \in R, s \in S$:

$$(m_i)_{i \in I} + (n_i)_{i \in I} := (m_i + n_i)_{i \in I}, \quad r \cdot (m_i)_{i \in I} \cdot s := (r \cdot m_i \cdot s)_{i \in I}.$$

The zero element of $\prod_{i \in I} M_i$ is the tuple $(0)_{i \in I}$, and additive inverses are given componentwise:

$$-(m_i)_{i \in I} := (-m_i)_{i \in I}.$$

Note that we can define the product of a family $\{M_i\}_{i \in I}$ of left/right/two-sided R -modules by taking the natural bimodule structure (Definition 3.5.1) of each module.

As usual (Definition 2.2.1), $\prod_{i \in I} M_i$ is the categorical product (Definition A.4.21) of the objects M_i in the appropriate category of modules (Definition 3.5.6). Moreover, the product of finitely many modules $M_1, \dots, M_n$ is often written as $M_1 \times \dots \times M_n$, which agrees with notation for the product of finitely many groups (Definition 2.2.1). We often write the self-product of a module M indexed by a (small) set I as M^I or by $M^{\oplus I}$. A finite self-product of a module M taken n times is often denoted by M^n or $M^{\oplus n}$; note that these all agree with the notations for abelian groups.

Definition 3.5.23 (Coproduct of Modules). Let R and S be (not necessarily commutative) rings (Definition 3.1.1), and let $\{M_i\}_{i \in I}$ be a (possibly infinite but small) family of (R, S) -bimodules.

The *coproduct (direct sum) of the family $\{M_i\}_{i \in I}$* , denoted by $\bigoplus_{i \in I} M_i$, is constructed as

$$\bigoplus_{i \in I} M_i := \left\{ (m_i)_{i \in I} \in \prod_{i \in I} M_i \mid m_i = 0 \text{ for all but finitely many } i \in I \right\}$$

(Definition 3.5.22) consisting of all tuples with only finitely many nonzero entries.

Addition and scalar multiplication in $\bigoplus_{i \in I} M_i$ are defined componentwise as in the direct product (Definition 3.5.22):

$$(m_i)_{i \in I} + (n_i)_{i \in I} := (m_i + n_i)_{i \in I}, \quad r \cdot (m_i)_{i \in I} \cdot s := (r \cdot m_i \cdot s)_{i \in I}, \quad r \in R, s \in S.$$

In all cases, the zero element is $(0)_{i \in I}$, and additive inverses are given by $-(m_i)_{i \in I} := (-m_i)_{i \in I}$.

Note that we can define the coproduct of a family $\{M_i\}_{i \in I}$ of left/right/two-sided R -modules by taking the natural bimodule structure (Definition 3.5.1) of each module.

Note that $\bigoplus_{i \in I} M_i$ is a submodule (Definition 3.5.2) of $\prod_{i \in I} M_i$. Moreover, $\bigoplus_{i \in I} M_i$ is the coproduct (Definition A.4.21) in the appropriate category of modules (Definition 3.5.6).

For finitely many modules $M_1, \dots, M_n$, the direct sum $\bigoplus_{j=1}^n M_j$, which may also be written as $M_1 \oplus \dots \oplus M_n$, is simply the usual Cartesian product $\prod_{j=1}^n M_j$ (Definition 3.5.22) of the modules, as every tuple automatically has only finitely many nonzero entries.

Definition 3.5.24. Let R, S be (not necessarily commutative) rings (Definition 3.1.1).

1. Let M be an R - S -bimodule (Definition 3.5.1). Let $N \subseteq M$ be a submodule of M (Definition 3.5.2).

The quotient group M/N (Definition 2.1.40), which is well defined as M is an abelian group (Definition 2.1.1) and hence N is a normal subgroup (Definition 2.1.36) (Proposition 2.1.52), has the structure of an R - S -bimodule — the (abelian) group structure is simply the group structure of M/N , whereas the R - S -bimodule structure is given as follows: for $m \in M$, $r \in R$, $s \in S$, we have

$$r \cdot (m + N) \cdot s = r \cdot m \cdot s + N.$$

This R - S -bimodule structure on M/N is called the *quotient R - S -bimodule of M by N* and is also denoted as M/N .

The canonical projection map

$$\pi : M \rightarrow M/N, \quad m \mapsto m + N,$$

is a surjective R -module homomorphism (Definition 3.5.4) with kernel (Definition 3.5.25) N .

2. Let M be a left/right/two-sided R -module. Let $N \subseteq M$ be a submodule of M . The *quotient R -module M/N* is the quotient of M by N for their respective natural bimodule structures (Definition 3.5.1).

The quotient M/N is the categorical quotient object (Definition A.4.25) of M by the subobject (Definition A.4.24) N in the appropriate category of modules

Definition 3.5.25. Let R, S be (not-necessarily commutative) rings with unity (Definition 3.1.1), and let M, N be R - S -bimodules (Definition 3.5.1). Let

$$\varphi : M \rightarrow N$$

be a homomorphism of R - S -bimodules (Definition 3.5.4). We define:

1. The **kernel of φ** is the submodule of M (Definition 3.5.2) given by

$$\ker(\varphi) := \{m \in M \mid \varphi(m) = 0\} \subseteq M.$$

2. The **image of φ** is the submodule of N given by

$$\operatorname{im}(\varphi) := \{\varphi(m) \mid m \in M\} \subseteq N.$$

3. The **cokernel of φ** is the quotient module of N (Definition 3.5.24) defined by

$$\operatorname{coker}(\varphi) := N / \operatorname{im}(\varphi).$$

4. The **coimage of φ** is the quotient module of M (Definition 3.5.24) defined by

$$\operatorname{coim}(\varphi) := M / \ker(\varphi).$$

It is not difficult to see that each of these are indeed R - S bimodules. In case M and N are left/right/two-sided R -modules, the **kernel, image, cokernel, and coimage** of a module homomorphism $\varphi : M \rightarrow N$ are respectively defined to be the kernel, image, cokernel, and coimage for the natural bimodule structures (Definition 3.5.1) of M and N .

The kernel, cokernel, image, and coimage of f are respectively the categorical kernel, cokernel (Definition A.4.14), image, and coimage (Definition A.4.15) (Lemma 3.5.26).

Lemma 3.5.26. Let R, S be (not-necessarily commutative) rings with unity (Definition 3.1.1), let M, N be R - S -bimodules (Definition 3.5.1), and let $f : M \rightarrow N$ be a module homomorphism (Definition 3.5.4). The kernel, cokernel, image, and coimage (Definition 3.5.25) of f are respectively the categorical kernel, cokernel (Definition A.4.14), image, and coimage (Definition A.4.15).

Proof. (♠ **TODO: AI generated, verify**) The category of R - S -bimodules is an abelian category, hence in particular a pointed category with the zero module as its zero object (Definition A.4.9). We verify each of the four claims.

1. The kernel $\ker(f) = \{m \in M : f(m) = 0\}$ is a submodule of M by definition (Definition 3.5.2). Let $i : \ker(f) \hookrightarrow M$ be the inclusion map. Then $f \circ i = 0$, the zero morphism $\ker(f) \rightarrow N$. Now let $g : Z \rightarrow M$ be any module homomorphism with $f \circ g = 0$. For every $z \in Z$, we have $f(g(z)) = 0$, so $g(z) \in \ker(f)$. Hence g factors uniquely through $\ker(f)$ via the map $u : Z \rightarrow \ker(f)$ defined by $u(z) = g(z)$. Therefore, $(\ker(f), i)$ is the categorical kernel (Definition A.4.14) of f .

2. The cokernel $\text{coker}(f) = N/\text{im}(f)$ is a quotient module by definition (Definition 3.5.24). Let $p : N \twoheadrightarrow N/\text{im}(f)$ be the canonical projection. Then $p \circ f = 0$ because $f(m) \in \text{im}(f)$ for all $m \in M$. Now let $g : N \rightarrow Z$ be any module homomorphism with $g \circ f = 0$. This means $\text{im}(f) \subseteq \ker(g)$, so by the universal property of quotient modules, g factors uniquely through $N/\text{im}(f)$ via $v : N/\text{im}(f) \rightarrow Z$ defined by $v(n + \text{im}(f)) = g(n)$. Therefore, $(\text{coker}(f), p)$ is the categorical cokernel (Definition A.4.14) of f .
3. The image $\text{im}(f) = f(M) \subseteq N$ is a submodule by definition. Let $m : \text{im}(f) \hookrightarrow N$ be the inclusion (a monomorphism in the category of modules). Let $e : M \twoheadrightarrow \text{im}(f)$ be defined by $e(m) = f(m)$ (an epimorphism because it is surjective onto $\text{im}(f)$). Then $f = m \circ e$. For any other factorization $f = m' \circ e'$ with $m' : I' \hookrightarrow N$ mono and $e' : M \twoheadrightarrow I'$ epi, we have an induced isomorphism $\varphi : \text{im}(f) \cong I'$ making the diagram commute. Therefore, $(\text{im}(f), m)$ is the categorical image (Definition A.4.15) of f .
4. The coimage $\text{coim}(f) = M/\ker(f)$ is a quotient module. Let $e : M \twoheadrightarrow M/\ker(f)$ be the canonical projection (an epimorphism). By the first isomorphism theorem for modules, (♠ TODO: AI generated, reference needed: first isomorphism theorem for modules) there exists an induced isomorphism $\bar{f} : M/\ker(f) \xrightarrow{\sim} \text{im}(f)$ such that $\bar{f}(m + \ker(f)) = f(m)$. Let m' be the composition of $\bar{f}$ with the inclusion $\text{im}(f) \hookrightarrow N$. Then $m' : M/\ker(f) \rightarrow N$ is a monomorphism (as a composition of an isomorphism and a monomorphism), and $f = m' \circ e$. This gives the categorical coimage (Definition A.4.15) factorization.

In all four cases, the module-theoretic constructions satisfy the universal properties of their categorical counterparts. $\square$

Theorem 3.5.27. Let R, S be (not necessarily commutative) rings (Definition 3.1.1). The category of (Definition 3.5.6) R - S -bimodules (Definition 3.5.1) is a Grothendieck (Definition 3.5.19) category and an $AB4^*$ (Definition 3.5.19) category.

Proof. We handwave details.

Given R - S -bimodules M and N , the set $\text{Hom}_{R\text{Mod}_S}(M, N)$ is an abelian group. Moreover, there is a 0-object, namely the zero module in $R\text{Mod}_S$. Therefore, $R\text{Mod}_S$ is preadditive (Definition 3.5.16). The direct sum of finitely many R - S -bimodules is their coproduct (Definition 3.5.23). Therefore, $R\text{Mod}_S$ is additive (Definition 3.5.16).

Given a morphism (Definition 3.5.4) $f : M \rightarrow N$ be R - S -bimodules, $\ker f$ and $\text{coker } f$ (Definition 3.5.25) are the categorical kernel and cokernel (Definition A.4.14) of f in $R\text{Mod}_S$ (Definition 3.5.6). Moreover, the monomorphisms (Definition A.4.23) in $R\text{Mod}_S$ are the injective module homomorphisms $f : M \rightarrow N$; such an f is the kernel of its cokernel. In other words, $R\text{Mod}_S$ satisfies $AB1$ and $AB2$ and hence is an abelian category (Definition 3.5.18).

Moreover, small coproducts (Definition A.4.21) exist in $R\text{Mod}_S$ (Definition 3.5.23), and it is easy to see that they are in fact exact, so $R\text{Mod}_S$ satisfies $AB3$ and $AB4$. To show that filtered colimits in $R\text{Mod}_S$ are exact, we first note that small (Definition A.4.22) colimits (Definition A.4.20) are right exact (Definition 3.5.29); for any small index category J and any system of short exact sequences $0 \rightarrow A_j \rightarrow B_j \rightarrow C_j \rightarrow 0$, the sequence $0 \rightarrow A \rightarrow$

$B \rightarrow C \rightarrow 0$ of diagrams is exact by ??, so applying the colimit functor yields a right exact sequence

$$\operatorname{colim}_{j \in J} A_j \rightarrow \operatorname{colim}_{j \in J} B_j \rightarrow \operatorname{colim}_{j \in J} C_j \rightarrow 0.$$

If J is additionally filtered (Definition A.4.17) so that the system is a directed (Definition A.4.18) one, then we show that the sequence

$$(B) \quad \varinjlim_{j \in J} A_j \rightarrow \varinjlim_{j \in J} B_j \rightarrow \varinjlim_{j \in J} C_j \rightarrow 0.$$

is left exact as well. Take some element¹ of the kernel of $\varinjlim_{j \in J} A_j \rightarrow \varinjlim_{j \in J} B_j$; such an element is represented by some element $a_j \in A_j$ for some $j \in J$. Since its image is zero in $\varinjlim_{j \in J} B_j$, it must be zero as an element of B_k for some $k \in J$. Since J is filtered, there exists some $k' \in J$ so that there are arrows $j \rightarrow k'$ and $k \rightarrow k'$ and so that the image of a_j in $B_{k'}$ is 0. The image of a_j in $A_{k'}$ is then 0 due to the assumption that

$$0 \rightarrow A_{k'} \rightarrow B_{k'} \rightarrow C_{k'} \rightarrow 0$$

is exact. Therefore, a_j is 0 in $\varinjlim_{j \in J} A_j$, so (B) is left exact as claimed and ${}_R\operatorname{Mod}_S$ is AB5.

Similarly as how we argued that ${}_R\operatorname{Mod}_S$ is AB3 and AB4, one can argue that ${}_R\operatorname{Mod}_S$ is AB3* and AB4*. Moreover, one can show that the R - S -bimodule $R \otimes_{\mathbb{Z}} S^{\operatorname{op}}$ (Definition 3.5.63) (Definition 3.1.5) is a generator for ${}_R\operatorname{Mod}_S$. $\square$

Definition 3.5.28. Let $\mathcal{A}$ be an additive category (Definition 3.5.16). A sequence

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

of morphisms in $\mathcal{A}$ is called a *short exact sequence* if the morphisms satisfy:

- $f : A \rightarrow B$ is a kernel of g (Definition A.4.14),
- $g : B \rightarrow C$ is a cokernel of f (Definition A.4.14),

In other words, the sequences $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C$ and $A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ are left and right exact respectively.

Definition 3.5.29. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.5.17) between abelian categories (Definition 3.5.18).

1. F is called *left exact* if it preserves all finite limits (Definition A.4.19); since F is an additive functor between abelian categories, this is equivalent to saying that F preserves kernels (Definition A.4.14)²

¹Recall ??

²To see this equivalence, note that

- finite products (Definition A.4.21) coincide with finite direct sums (Definition 3.5.16) in abelian categories,
- additive functors preserve finite direct sums by definition,
- kernels (Definition A.4.14) in a pointed category are equalizers, and all equalizers in abelian categories are kernels,
- if all finite products and equalizers exist in a general category, then all finite limits exist in the category (??)

Equivalently, for every left exact sequence in $\mathcal{A}$

$$0 \rightarrow A' \xrightarrow{f} A \xrightarrow{g} A''$$

the sequence

$$0 \rightarrow F(A') \xrightarrow{F(f)} F(A) \xrightarrow{F(g)} F(A'')$$

is exact at $F(A')$ and $F(A)$ (i.e., F preserves monomorphisms (Definition A.4.23) and exactness at the first two terms).

2. Dually, F is called **right exact** if it preserves all finite colimits (Definition A.4.19); since F is an additive functor between abelian categories, this is equivalent to saying that F preserves cokernels (Definition A.4.14).

Equivalently, for every right exact sequence in $\mathcal{A}$

$$A' \xrightarrow{f} A \xrightarrow{g} A'' \rightarrow 0,$$

the sequence

$$F(A') \xrightarrow{F(f)} F(A) \xrightarrow{F(g)} F(A'') \rightarrow 0$$

is exact at $F(A)$ and $F(A'')$ (i.e., F preserves epimorphisms (Definition A.4.23) and exactness at the last two terms).

3. F is called **exact** if it is both left and right exact.

Definition 3.5.30 (Natural module structures induced by opposite rings). Let R be a ring (not necessarily commutative) (Definition 3.1.1).

1. If M is a left R -module (Definition 3.5.1), then M naturally has a structure of a right R^{op} (Definition 3.1.5)-module defined by

$$m \cdot r := r \cdot m,$$

for all $m \in M$ and $r \in R$, where the multiplication on the right is the original left R -action on M but re-interpreted so elements of R^{op} act from the right.

2. Dually, if M is a right R -module, then M naturally has a structure of a left R^{op} -module defined by

$$r \cdot m := m \cdot r,$$

for all $m \in M$ and $r \in R$, where the multiplication on the left is the original right R -action on M but re-interpreted as a left action of R^{op} .

If R is a commutative ring, and given a left/right R -module M , recall that R^{op} is naturally isomorphic to R (Definition 3.1.5), so the right/left R^{op} -module structure on M is a right/left R -module structure. In fact, the left/right R -module and right/left R -module structures on M make M into an R - R -bimodule (Definition 3.5.1).

Remark 3.5.31. Given a general left/right R -module (Definition 3.5.1) M , the right/left R^{op} (Definition 3.1.5)-module structure does not necessarily make M into an R - R^{op} -bimodule (Definition 3.5.1).

3.5.3. Simple modules.

Definition 3.5.32. Let R be a not necessarily commutative ring (Definition 3.1.1). An R -module M is called **simple** if $M \neq 0$ and the only submodules of M (Definition 3.5.2) are $\{0\}$ and M itself.

Definition 3.5.33. A not necessarily commutative ring (Definition 3.1.1) R is called **simple** if $R \neq 0$ and the only two-sided ideals of R (Definition 3.5.34) are $\{0\}$ and R itself.

3.5.4. Ideals of a ring.

Definition 3.5.34. Let R be a (not necessarily commutative, possibly nonunital) ring (Definition 3.1.1). A **left ideal of R** is a subset $I \subseteq R$ such that

- $(I, +)$ is an additive subgroup (Definition 2.1.19) of $(R, +)$,
- $RI \subseteq I$, i.e., for all $r \in R$ and $x \in I$, one has $rx \in I$.

Similarly, a **right ideal of R** is a subset $I \subseteq R$ such that

- $(I, +)$ is an additive subgroup of $(R, +)$,
- $IR \subseteq I$, i.e., for all $r \in R$ and $x \in I$, one has $xr \in I$.

A **two-sided ideal** (or simply an **ideal**) of R is a subset $I \subseteq R$ which is both a left ideal and a right ideal of R . We denote by $I \trianglelefteq R$ the relation expressing that I is a two-sided ideal of R .

Equivalently, an left/right/two-sided ideal of R is a submodule (Definition 3.5.2) of R as an R -module (Definition 3.5.1).

A left/right/two-sided ideal is said to be **proper** if it is strictly contained in R .

Note that every left or right ideal of a commutative ring is a two-sided ideal.

Definition 3.5.35. Let R be a (not necessarily commutative) ring (Definition 3.1.1), and let $X \subseteq R$ be a subset (Definition A.1.2).

The **left ideal generated by X** is the smallest left ideal (Definition 3.5.34) of R containing X ; it equals the set of all finite sums of elements of the form rx with $r \in R$ and $x \in X$. It is often denoted as $\langle X \rangle_L$.

Similarly, the **right ideal generated by X** is the smallest right ideal of R containing X ; it equals the set of all finite sums of elements of the form xr with $x \in X$ and $r \in R$. It is often denoted as $\langle X \rangle_R$.

The **two-sided ideal generated by X** is the smallest two-sided ideal of R containing X ; it equals the set of all finite sums of elements of the form rxs with $r, s \in R$ and $x \in X$. It is often denoted as $\langle X \rangle$.

A left/right/two-sided ideal I of R is said to be **finitely generated** if there exists some finite subset X of R such that I equals the left/right/two-sided ideal generated by X . Moreover, I is said to be **principal** if there exists some subset X of R of cardinality 1 such that I

equals the left/right/two-sided ideal generated by X . An ideal generated by a finite set $\{x_1, \dots, x_n\} \subseteq R$ may be denoted as $\langle x_1, \dots, x_n \rangle$ (with the subscript of L or R added if the ideal is meant to be the left/right ideal generated by the set of these elements)

Definition 3.5.36. Let R be a ring, and let I, J be left ideals of R (Definition 3.5.34) (resp. right ideals, resp. two-sided ideals).

The *product ideal* IJ is the left (resp. right, resp. two-sided) ideal defined as the additive subgroup (Definition 2.1.19) of R generated by (Definition 3.5.35) all products xy where $x \in I$ and $y \in J$. In other words,

$$IJ = \left\{ \sum_{k=1}^n x_k y_k : n \geq 0, x_k \in I, y_k \in J \right\}.$$

More generally, we may speak of the product of finitely many ideals of R . In particular, given an ideal I and an integer $n \geq 0$, we may speak of I^n , the set of all finite sums of products $a_1 a_2 \dots a_n$ with each $a_i \in I$.

Definition 3.5.37. Let R be a not-necessarily commutative ring (Definition 3.1.1) and $I \subseteq R$ a two-sided ideal (Definition 3.5.34). The set

$$R/I = \{r + I : r \in R\}$$

is called the *quotient ring of R by I* , where $r + I = \{r + i : i \in I\}$ and the operations are defined by

$$\begin{aligned} (r + I) + (s + I) &:= (r + s) + I, \\ (r + I)(s + I) &:= rs + I. \end{aligned}$$

With these operations, R/I is a ring and the canonical projection $R \rightarrow R/I$, $r \mapsto r + I$ is a surjective ring homomorphism (Definition 3.1.23).

Definition 3.5.38. Let R be a (not necessarily commutative) ring (Definition 3.1.1). A proper two-sided ideal $P \subsetneq R$ (Definition 3.5.34) is called a *prime ideal* if the following equivalent conditions holds:

1. If I, J are left ideals and $IJ \subset P$ (Definition 3.5.36), then $I \subset P$ or $J \subset P$.
2. If I, J are right ideals and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
3. If I, J are two-sided ideals and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
4. If $x, y \in R$ with $xRy \subset P$, then $x \in P$ or $y \in P$.

A proper left/right/two-sided ideal $M \subsetneq R$ is called *maximal* if there exists no other left/right/two-sided ideal $J \trianglelefteq R$ such that $M \subsetneq J \subsetneq R$. Equivalently,

- a left/right ideal M of R is maximal if and only if the quotient module R/M (Definition 3.5.24) is a simple (Definition 3.5.32) left/right R -module.
- a two-sided ideal M of R is maximal if and only if the quotient ring R/M (Definition 3.5.37) is a simple ring (Definition 3.5.33).

Lemma 3.5.39. Let R be a commutative ring. An ideal $P \subseteq R$ is prime if and only if the following holds:

$$\text{For all } a, b \in R, \text{ if } ab \in P \text{ then } a \in P \text{ or } b \in P.$$

Proof. By Definition 3.5.38, a proper ideal P of the ring R is prime if for all two-sided ideals I, J of R , $IJ \subseteq P$ implies $I \subseteq P$ or $J \subseteq P$. Since R is commutative, every ideal is two-sided.

Suppose that P is a prime ideal in the sense of Definition 3.5.38. Let $a, b \in R$ with $ab \in P$. Let (a) and (b) denote the principal ideals generated by a and b , respectively. The product ideal $(a)(b)$ is generated by products of elements from (a) and (b) , which are of the form $(r_1a)(r_2b) = (r_1r_2)(ab)$ for $r_1, r_2 \in R$. Since $ab \in P$, every element of the form $(r_1r_2)(ab)$ is also in P . Thus, $(ab) \subseteq P$. Since P is prime, $(a) \subseteq P$ or $(b) \subseteq P$. In the first case, $a \in (a) \subseteq P$, so $a \in P$. In the second case, $b \in (b) \subseteq P$, so $b \in P$. Hence, $ab \in P$ implies $a \in P$ or $b \in P$.

Conversely, suppose that for all $a, b \in R$, if $ab \in P$ then $a \in P$ or $b \in P$. Let I, J be ideals of R with $IJ \subseteq P$. Suppose for contradiction that neither $I \subseteq P$ nor $J \subseteq P$. Then there exist $a \in I \setminus P$ and $b \in J \setminus P$. The product ab lies in IJ , which is a subset of P , so $ab \in P$. By hypothesis, either $a \in P$ or $b \in P$, a contradiction. Hence, $I \subseteq P$ or $J \subseteq P$.

Therefore, the two characterizations are equivalent for commutative rings. $\square$

Definition 3.5.40. Let R be a ring (Definition 3.1.1), not assumed commutative. The **Jacobson radical of R** , denoted by notations such as $J(R)$ or $\text{rad}(R)$, is the intersection of all maximal left ideals (Definition 3.5.38) of R :

$$J(R) = \bigcap \{M \subseteq R : M \text{ is a maximal left ideal of } R\}.$$

Equivalently, $J(R)$ is also the intersection of all maximal right ideals of R . Thus, $J(R)$ is a two-sided ideal of R .

3.5.5. Nakayama's lemma and consequences.

Lemma 3.5.41 (Determinant lemma for modules). Let R be a commutative ring (Definition 3.1.6), M a finitely generated right R -module, I an ideal of R , and $\varphi: M \rightarrow M$ an R -linear endomorphism such that $\varphi(M) \subseteq MI$. Let n be the number of generators of M .

Then there exist elements $a_1, \dots, a_{n-1} \in I$ and $a_n \in R$ such that

$$\varphi^n + a_1\varphi^{n-1} + a_2\varphi^{n-2} + \dots + a_{n-1}\varphi + (a_n - 1)\text{id}_M = 0$$

as an endomorphism of M .

In particular, if $I \subseteq J(R)$ (Definition 3.5.40), then φ is invertible.

Proof. (♠ TODO: AI generated, verify) Suppose that M is generated by n elements $m_1, \dots, m_n$ as a right R -module. Since $\varphi(M) \subseteq MI$, for each $j = 1, \dots, n$ there exist coefficients $b_{ji} \in I$ such that

$$\varphi(m_j) = \sum_{i=1}^n m_i b_{ji}.$$

Rearranging gives

$$\sum_{i=1}^n m_i(\delta_{ij}\varphi - b_{ji}) = 0 \quad \text{for each } j = 1, \dots, n,$$

where δ_{ij} is the Kronecker delta and we interpret $\delta_{ij}\varphi$ as scalar multiplication by φ .

Let $A = (\delta_{ij}\varphi - b_{ji}) \in M_n(\text{End}_R(M))$ be the matrix whose entries are endomorphisms of M . Since R is commutative and φ is R -linear, each entry $\delta_{ij}\varphi - b_{ji}$ commutes with every other entry: multiplication by scalars from I commute with each other because R is commutative, and φ commutes with right-multiplication by any element of R because φ is R -linear.

Let $C = \text{adj}(A) \in M_n(\text{End}_R(M))$ be the classical adjugate matrix (Proposition 5.1.10) of A . By the adjugate formula, $CA = (\det A)I_n$, where $\det(A)$ is computed in the commutative ring $\text{End}_R(M)^{\text{comm}}$ generated by these commuting endomorphisms.

Applying the relation CA to the column vector $(m_1, \dots, m_n)^\top$, each entry vanishes because $\sum_i (\delta_{ij}\varphi - b_{ji})m_j = 0$. Hence

$$\det(A) \cdot m_k = 0 \quad \text{for all } k = 1, \dots, n.$$

Since the m_k generate M , we have $\det(\delta_{ij}\varphi - b_{ji}) = 0$ as an endomorphism of M .

Expanding the determinant by the Leibniz formula (Definition 5.1.6) yields a polynomial in φ :

$$\varphi^n + c_1\varphi^{n-1} + \dots + c_{n-1}\varphi + c_n = 0,$$

where each coefficient c_k is a sum of products involving exactly k entries from the matrix (b_{ji}) , hence lies in $I^k \subseteq I$ for $k = 1, \dots, n$. Setting $a_k = c_k$ for $k = 1, \dots, n-1$ and noting that $c_n \in I$, we rewrite the relation as

$$\varphi^n + a_1\varphi^{n-1} + \dots + a_{n-1}\varphi + (a_n - 1)\text{id}_M = 0,$$

with $a_1, \dots, a_{n-1} \in I$ and $a_n = c_n + 1_R \in R$.

Now suppose that $I \subseteq J(R)$. Rewrite the equation as

$$\varphi \circ (\varphi^{n-1} + a_1\varphi^{n-2} + \dots + a_{n-1}\text{id}_M) = (1_R - c_n)\text{id}_M.$$

Since $c_n \in I \subseteq J(R)$, the element $1_R - c_n$ is a unit in R . Hence we can multiply by $(1_R - c_n)^{-1}$ to obtain

$$\varphi \circ (\varphi^{n-1} + a_1\varphi^{n-2} + \dots + a_{n-1}\text{id}_M) \cdot (1_R - c_n)^{-1} = \text{id}_M.$$

Thus φ has a right inverse. By the same polynomial relation, we also have

$$(\varphi^{n-1} + a_1\varphi^{n-2} + \dots + a_{n-1}\text{id}_M) \cdot (1_R - c_n)^{-1} \circ \varphi = \text{id}_M,$$

so φ also has a left inverse. Therefore, φ is invertible. □

Proposition 3.5.42 (Nonzero finitely generated module has a maximal submodule). (♠
TODO: AI generated, verify)

Let R be a ring (Definition 3.1.1) and let M be a nonzero finitely generated (Definition 3.5.15) right R -module. Then there exists a proper submodule (Definition 3.5.2) $L \subsetneq M$ that is maximal among proper submodules, i.e., no proper submodule of M strictly contains L .

Proposition 3.5.43 (Jacobson radical annihilates every simple right module). (♠ TODO: AI generated, verify)

Let R be a ring (Definition 3.1.1), not assumed commutative, and let S be a simple (Definition 3.5.32) right R -module. Then the Jacobson radical (Definition 3.5.40) $J(R)$ annihilates S , i.e., $s \cdot x = 0$ for all $s \in S$ and all $x \in J(R)$.

Lemma 3.5.44 (Nakayama's submodule lemma). (♠ TODO: AI generated, verify)

Let R be a ring (Definition 3.1.1), M a finitely generated (Definition 3.5.15) right R -module, N a submodule (Definition 3.5.2) of M , and I a left ideal of R with $I \subseteq J(R)$ (Definition 3.5.40).

If $N + IM = M$, then $N = M$.

Proof. (♠ TODO: AI generated, verify) Suppose for contradiction that $N \neq M$. Then the quotient module M/N is nonzero. Since M is finitely generated, every quotient of M is also finitely generated; in particular, M/N is a finitely generated right R -module.

We first show that $(M/N)I = M/N$. For each $\bar{x} \in M/N$, choose a representative $x \in M$. By hypothesis, $N + IM = M$, so $x = n + \sum_{k=1}^m y_k a_k$ for some $n \in N$, $y_k \in M$, and $a_k \in I$. Passing to the quotient, $\bar{x} = \sum_{k=1}^m \bar{y}_k a_k \in (M/N)I$. Hence, $(M/N)I = M/N$.

Since M/N is a nonzero finitely generated module, it has a maximal submodule by Proposition 3.5.42. Let L be a maximal submodule of M/N . The quotient $(M/N)/L$ is therefore a simple (Definition 3.5.32) right R -module.

We next show that $((M/N)/L)I = 0$. Since $I \subseteq J(R)$ and the Jacobson radical annihilates every simple right R -module by Proposition 3.5.43, we have $((M/N)/L)J(R) = 0$, which implies that $((M/N)/L)I = 0$.

On the other hand, since $(M/N)I = M/N$, the surjective R -linear map $M/N \rightarrow (M/N)/L$ implies that $((M/N)/L)I = (M/N)/L$. Therefore, $(M/N)/L = ((M/N)/L)I = 0$.

This contradicts the fact that a simple module is nonzero. Hence, $N = M$. □

Lemma 3.5.45 (Nakayama's lemma). (♠ TODO: AI generated, verify)

Let R be a ring (Definition 3.1.1), M a finitely generated (Definition 3.5.15) right R -module, and I a left ideal of R with $I \subseteq J(R)$ (Definition 3.5.40).

If $IM = M$, then $M = 0$.

Proof. (♠ TODO: AI generated, verify) Suppose that $IM = M$. Apply Nakayama's submodule lemma (Lemma 3.5.44) with $N = 0$. The hypothesis gives $0 + IM = M$, i.e., $IM = M$. Since M is finitely generated and $I \subseteq J(R)$, Nakayama's submodule lemma implies that $0 = M$. Therefore, $M = 0$. □

Proposition 3.5.46 (Nakayama's generator criterion). (♠ TODO: AI generated, verify)

Let R be a ring (Definition 3.1.1), M a finitely generated (Definition 3.5.15) right R -module, and let $m_1, \dots, m_n \in M$.

Suppose the images of $m_1, \dots, m_n$ in the quotient module $M/J(R)$ (Definition 3.5.40) M generate this quotient as a module over the quotient ring (Definition 3.5.37) $R/J(R)$. Then $m_1, \dots, m_n$ generate M as an R -module.

Proof. (♠ TODO: AI generated, verify) Suppose that the images $\bar{m}_1, \dots, \bar{m}_n$ of $m_1, \dots, m_n$ in $M/J(R)M$ generate this quotient as an $R/J(R)$ -module. Let N be the submodule of M generated by $m_1, \dots, m_n$.

We claim that $N + J(R)M = M$. Indeed, for any $x \in M$, its image $\bar{x} \in M/J(R)M$ can be written as $\bar{x} = \sum_{i=1}^n \bar{m}_i \cdot \bar{r}_i$ for some $\bar{r}_i \in R/J(R)$. Lifting the coefficients $r_i \in R$, we have

$$x - \sum_{i=1}^n m_i r_i \in J(R)M.$$

Hence $x \in N + J(R)M$. Therefore, $N + J(R)M = M$.

Since $J(R) \subseteq J(R)$ and M is finitely generated, apply Nakayama's submodule lemma (Lemma 3.5.44) with the ideal $I = J(R)$ to conclude that $N = M$. Therefore, $m_1, \dots, m_n$ generate M as an R -module. $\square$

Corollary 3.5.47 (Nakayama's lemma for local rings: vanishing). (♠ TODO: AI generated, verify)

Let $(R, \mathfrak{m})$ be a local ring (Definition 3.5.58) (not necessarily commutative) with unique maximal ideal $\mathfrak{m}$, and let M be a finitely generated (Definition 3.5.15) right R -module.

Then $M = 0$ if and only if $M/\mathfrak{m}M = 0$.

Proof. (♠ TODO: AI generated, verify) The forward direction is trivial: if $M = 0$, then $\mathfrak{m}M = 0$ and $M/\mathfrak{m}M = 0$.

For the converse, suppose that $M/\mathfrak{m}M = 0$. Since $(R, \mathfrak{m})$ is a local ring, $\mathfrak{m} = J(R)$. Apply Nakayama's lemma (Lemma 3.5.45) with $I = \mathfrak{m} = J(R)$. The hypothesis gives $\mathfrak{m}M = M$, and since M is finitely generated, Nakayama's lemma implies that $M = 0$. Therefore, $M = 0$. $\square$

Proposition 3.5.48 (Nakayama's lemma for local rings: minimal generating sets). (♠ TODO: AI generated, verify)

Let $(R, \mathfrak{m})$ be a local ring (Definition 3.5.58) (not necessarily commutative) with unique maximal ideal $\mathfrak{m}$ and residue field $k = R/\mathfrak{m}$. Let M be a finitely generated (Definition 3.5.15) right R -module.

1. The quotient $M/\mathfrak{m}M$ is a finite-dimensional right vector space (Definition 5.1.1) over k . If M can be generated by n elements as an R -module, then $\dim_k(M/\mathfrak{m}M) \leq n$.
2. Conversely, if $x_1 + \mathfrak{m}M, \dots, x_n + \mathfrak{m}M \in M/\mathfrak{m}M$ form a basis of $M/\mathfrak{m}M$ as a k -vector space, then $x_1, \dots, x_n$ generate M as an R -module.
3. The minimum number of generators of M as an R -module equals $\dim_k(M/\mathfrak{m}M)$. Moreover, any subset of M whose image in $M/\mathfrak{m}M$ is linearly independent over k can be extended to a minimal generating set of M .

Proof. (♠ TODO: AI generated, verify)

1. Since M is finitely generated as an R -module and $\mathfrak{m}$ is an ideal of R , the quotient $M/\mathfrak{m}M$ is annihilated by $\mathfrak{m}$. Hence $M/\mathfrak{m}M$ is a right module over $R/\mathfrak{m} = k$, i.e., a right vector space. Since M is finitely generated, the images of any finite generating set span $M/\mathfrak{m}M$, so it has finite dimension. If M can be generated by n elements, their images span $M/\mathfrak{m}M$, which implies $\dim_k(M/\mathfrak{m}M) \leq n$.
2. Suppose that $\bar{x}_1, \dots, \bar{x}_n$ form a basis of $M/\mathfrak{m}M$ over k . Let N be the submodule generated by lifts $x_1, \dots, x_n$. The images of $x_1, \dots, x_n$ in $M/\mathfrak{m}M$ generate this quotient, so $N + \mathfrak{m}M = M$. Since $(R, \mathfrak{m})$ is a local ring, $\mathfrak{m} = J(R)$. Apply Nakayama's submodule lemma (Lemma 3.5.44) with $I = \mathfrak{m}$ to conclude that $N = M$. Therefore, $x_1, \dots, x_n$ generate M .
3. Part (2) shows that a basis of $M/\mathfrak{m}M$ lifts to a generating set of the same size. Part (1) shows that any generating set has size at least $\dim_k(M/\mathfrak{m}M)$. Therefore, the minimum number of generators equals $\dim_k(M/\mathfrak{m}M)$.

For the extension statement, suppose $S \subseteq M$ is a subset whose image in $M/\mathfrak{m}M$ is linearly independent over k . Extend this to a basis of $M/\mathfrak{m}M$, and lift each new element to an element of M . By part (2), the resulting set generates M .

□

Lemma 3.5.49 (Nakayama's lemma with nilpotent ideal). (♠ TODO: AI generated, verify)

Let R be a ring (Definition 3.1.1), M a right R -module (not assumed finitely generated), and I a left ideal of R that is nilpotent (Definition 3.5.53).

If $IM = M$, then $M = 0$.

Proof. (♠ TODO: AI generated, verify) Suppose that $IM = M$. Since I is nilpotent, there exists some integer $k \geq 1$ such that $I^k = 0$.

We claim that $I^j M = M$ for all $j \geq 1$. For $j = 1$, this is the hypothesis. If $I^j M = M$, then

$$I^{j+1}M = I(I^j M) = IM = M.$$

In particular, $I^k M = M$. Since $I^k = 0$, we have $M = 0 \cdot M = 0$. Therefore, $M = 0$. □

Proposition 3.5.50 (Nakayama's surjection criterion). (♠ TODO: AI generated, verify)

Let A be a ring (Definition 3.1.1), $\mathfrak{a}$ a two-sided ideal (Definition 3.5.34) of A with $\mathfrak{a} \subseteq J(A)$ (Definition 3.5.40). Let M and N be left A -modules, with N finitely generated (Definition 3.5.15), and let $u: M \rightarrow N$ be an A -linear homomorphism.

If the induced homomorphism $\bar{u}: M/\mathfrak{a}M \rightarrow N/\mathfrak{a}N$ is surjective, then u is surjective. (cf. [?, Chapter 2, Exercise 10])

Proof. (♠ TODO: AI generated, verify) Suppose that $\bar{u}: M/\mathfrak{a}M \rightarrow N/\mathfrak{a}N$ is surjective. Let $K = \text{im}(u)$ be the image of u in N . We wish to show that $K = N$.

Consider the quotient module N/K . For any $\bar{n} \in N/\mathfrak{a}N$, by surjectivity of $\bar{u}$ there exists some $\bar{m} \in M/\mathfrak{a}M$ such that $\bar{u}(\bar{m}) = \bar{n}$. This means that for any representative $n \in N$ of $\bar{n}$, there exist $m \in M$ and $x \in \mathfrak{a}N$ such that $n - u(m) = x \in \mathfrak{a}N$. In other words, every element of N lies in $K + \mathfrak{a}N$.

Therefore $N = K + \mathfrak{a}N$, which implies $(K + \mathfrak{a}N)/K = N/K$, i.e., $\mathfrak{a}(N/K) = N/K$. Since N is finitely generated and $\mathfrak{a} \subseteq J(A)$, apply Nakayama's lemma (Lemma 3.5.45) to the finitely generated module N/K with ideal $\mathfrak{a}$ to conclude that $N/K = 0$. Therefore, $K = N$, which means u is surjective. $\square$

Proposition 3.5.51 (Nakayama's lemma for graded modules). (♠ TODO: AI generated, verify)

Let $R = \bigoplus_{d \geq 0} R_d$ be a *non-negatively graded ring* with R_0 a field, and let $M = \bigoplus_{d \geq 0} M_d$ be a *finitely generated graded right R -module*, meaning M is generated as an R -module by finitely many homogeneous elements. Let $R_{>0} = \bigoplus_{d > 0} R_d$.

1. If $R_{>0}M = M$, then $M = 0$. Equivalently, if $M/R_{>0}M = 0$, then $M = 0$.
2. The quotient $M/R_{>0}M$ is a finite-dimensional right vector space over R_0 . If $x_1 + R_{>0}M, \dots, x_n + R_{>0}M \in M/R_{>0}M$ form a basis as an R_0 -vector space, then homogeneous lifts $x_1, \dots, x_n \in M$ generate M as an R -module.
3. The minimum number of homogeneous generators of M equals $\dim_{R_0}(M/R_{>0}M)$.

Proof. (♠ TODO: AI generated, verify)

1. Suppose that $R_{>0}M = M$. Since M is finitely generated by homogeneous elements, let $d_{\max}$ be the maximum degree among all homogeneous generators of M . If $M \neq 0$, let $m \in M$ be a nonzero homogeneous element of maximal degree. By hypothesis, $m = \sum_i y_i r_i$ for some $y_i \in M$ and $r_i \in R_{>0}$. Since each r_i has positive degree, each term $y_i r_i$ has strictly greater degree than $\deg(y_i)$. But this contradicts the maximality of the degrees of the generators. Therefore, $M = 0$.
2. The quotient $M/R_{>0}M$ is annihilated by $R_{>0}$ because for any $\bar{x} \in M/R_{>0}M$ and any $r \in R_{>0}$, we have $\bar{x} \cdot r = 0$ since $x \cdot r \in R_{>0}M$. Hence $M/R_{>0}M$ is a right R_0 -module, i.e., a right vector space over the field R_0 . Since M is finitely generated as an R -module, its image generates $M/R_{>0}M$ as an R_0 -vector space, so it has finite dimension.

Now suppose that $\bar{x}_1, \dots, \bar{x}_n$ form a basis of $M/R_{>0}M$ over R_0 . Let N be the submodule generated by homogeneous lifts $x_1, \dots, x_n$. We have $N + R_{>0}M = M$, because every element of M maps to an R_0 -linear combination of $\bar{x}_i$'s modulo $R_{>0}M$. Apply part (1) to the quotient module M/N : we have $(M/N)/(R_{>0}(M/N)) = 0$, which implies $M/N = 0$. Therefore, $x_1, \dots, x_n$ generate M .

3. Part (2) shows that any basis of $M/R_{>0}M$ lifts to a generating set of the same size. Conversely, if $y_1, \dots, y_k$ are homogeneous generators of M , then their images span $M/R_{>0}M$ over R_0 . Hence $k \geq \dim_{R_0}(M/R_{>0}M)$. Therefore, the minimum number of homogeneous generators equals $\dim_{R_0}(M/R_{>0}M)$.

$\square$

Proposition 3.5.52 (Surjective endomorphism of a Noetherian module is an isomorphism).
(♠ TODO: AI generated, verify)

Let A be a ring (Definition 3.1.1) and let M be a left A -module.

1. If M is a Noetherian A -module (Definition 3.6.8) and $u: M \rightarrow M$ is a surjective A -linear endomorphism, then u is an isomorphism.
2. If M is a Artinian A -module (Definition 3.6.8) and $u: M \rightarrow M$ is an injective A -linear endomorphism, then u is an isomorphism.

(cf. [?, Chapter 6, Exercise 1])

Proof. (♠ TODO: AI generated, verify)

1. Suppose that M is Noetherian and $u: M \rightarrow M$ is surjective. We show that u is injective by showing that $\ker(u) = 0$. Consider the ascending chain of submodules

$$\ker(u) \subseteq \ker(u^2) \subseteq \ker(u^3) \subseteq \cdots,$$

where u^n denotes the n -fold composition of u with itself. Since $\ker(u^n)$ consists of all elements mapped to zero by u^n , and $u(\ker(u^{n+1})) \subseteq \ker(u^n)$, this chain is well-defined.

Because M is Noetherian, the ascending chain stabilizes: there exists some $N \geq 1$ such that $\ker(u^N) = \ker(u^{N+1})$. Since u is surjective, so is u^N . For any $x \in \ker(u)$, we have $u(x) = 0$. By surjectivity of u^N , there exists some $y \in M$ such that $u^N(y) = x$. Then $u^{N+1}(y) = u(x) = 0$, so $y \in \ker(u^{N+1}) = \ker(u^N)$. Hence $x = u^N(y) = 0$. Therefore, $\ker(u) = 0$, and u is injective.

2. Suppose that M is Artinian and $u: M \rightarrow M$ is injective. We show that u is surjective by showing that $\operatorname{im}(u) = M$. Consider the descending chain of submodules

$$M \supseteq \operatorname{im}(u) \supseteq \operatorname{im}(u^2) \supseteq \operatorname{im}(u^3) \supseteq \cdots.$$

Because M is Artinian, the descending chain stabilizes: there exists some $N \geq 1$ such that $\operatorname{im}(u^N) = \operatorname{im}(u^{N+1})$.

For any $x \in \operatorname{im}(u^N)$, since $\operatorname{im}(u^N) = \operatorname{im}(u^{N+1})$, there exists some $y \in M$ such that $x = u^{N+1}(y) = u(u^N(y))$. Hence $x \in \operatorname{im}(u|_{\operatorname{im}(u^N)})$. Because u is injective, the restriction $u|_{\operatorname{im}(u^N)}: \operatorname{im}(u^N) \rightarrow \operatorname{im}(u^{N+1}) = \operatorname{im}(u^N)$ is a bijection. In particular, it is surjective onto $\operatorname{im}(u^N)$.

Applying this argument iteratively, we see that u maps $\operatorname{im}(u^{N-1})$ onto $\operatorname{im}(u^N)$, and so on down to the fact that u maps M onto $\operatorname{im}(u) = M$. Therefore, u is surjective.

□

Definition 3.5.53. Let R be a not necessarily commutative ring (Definition 3.1.1). An ideal $I \subseteq R$ is *nilpotent* if there exists a positive integer $n \geq 1$ such that $I^n = \{0\}$ (Definition 3.5.36).

Definition 3.5.54. Let R be a not necessarily commutative ring (Definition 3.1.1).

A left/right R -module (Definition 3.5.1) M is *simple* (or *irreducible*) if $M \neq \{0\}$ and its only submodules are $\{0\}$ and M .

A left/right R -module is **semisimple** if it is isomorphic (Definition 3.1.23) to a direct sum (Definition 3.5.23) of simple left/right R -modules (Definition 3.5.1).

Definition 3.5.55. A ring (Definition 3.1.1) R is a **semisimple ring** if it is semisimple (Definition 3.5.54) as a left module (Definition 3.5.1) over itself.

By the Wedderburn-Artin theorem, this is equivalent to R being isomorphic to a finite direct product of matrix rings over division rings:

$$R \cong \prod_{i=1}^k M_{n_i}(D_i)$$

where each D_i is a division ring.

Proposition 3.5.56. For a ring (Definition 3.1.1) R , the following are equivalent:

1. R is a semisimple ring (Definition 3.5.55).
2. R is left Artinian (Definition 3.6.9) and its Jacobson radical (Definition 3.5.40) $J(R)$ is $\{0\}$.
3. Every left R -module (Definition 3.5.1) is semisimple.
4. Every left R -module is projective (Definition 3.5.77).
5. Every left R -module is injective.

Proposition 3.5.57. The property of being a semisimple ring (Definition 3.5.55) is left-right symmetric; a ring is left semisimple if and only if it is right semisimple.

Definition 3.5.58. Let R be a ring (Definition 3.1.1) with unity, not necessarily commutative. The ring R is called a **local ring** if it has a unique maximal left ideal (Definition 3.5.38). In this case, R also has a unique maximal right ideal, and these coincide with the Jacobson radical $J(R)$ of R (Definition 3.5.40).

The unique maximal left (and right) ideal of a local ring R may sometimes be denoted by $\mathfrak{m}_R$.

The following are each equivalent conditions for R to be a local ring:

1. $R/J(R)$ is a division ring (Definition 3.1.15)
2. $R = J(R) \cup R^\times$ (Definition 3.1.11)

Definition 3.5.59. Let R be a not necessarily commutative ring (Definition 3.1.1). A subset $S \subseteq R$ is called a **multiplicative subset** if

- $1 \in S$ (assuming R has unity),
- For all $s, t \in S$, one has $st \in S$.

Definition 3.5.60. Let R be a ring (Definition 3.1.1) and let $S \subseteq Z(R)$ be a multiplicative subset (Definition 3.5.59) contained in the center (Definition 3.1.7) of R . The **localization of R at S** , denoted by $S^{-1}R$, is the ring whose elements are equivalence classes of pairs $(r, s) \in R \times S$ under the equivalence relation

$$(r, s) \sim (r', s') \iff \exists u \in S \text{ such that } u(sr' - s'r) = 0.$$

Write $\frac{r}{s}$ for the equivalence class of (r, s) . Addition and multiplication on representatives are defined by

$$\begin{aligned}\frac{r}{s} + \frac{r'}{s'} &= \frac{rs' + r's}{ss'}, \\ \frac{r}{s} \cdot \frac{r'}{s'} &= \frac{rr'}{ss'}.\end{aligned}$$

The map $r \mapsto \frac{r}{1}$ defines a ring homomorphism (Definition 3.1.23); therefore, $S^{-1}R$ is naturally an R -algebra (Definition 3.1.30).

Given an element $f \in Z(R)$, the localization of R at $S = \{f^n : n \geq 0\}$ is denoted by R_f .

In the case that R is a commutative ring and if P is a prime ideal of R (Definition 3.5.38), then $R_P := S^{-1}R$ with $S = R \setminus P$ is called the *localization of R at P* . It is a local ring (Definition 3.5.58) whose maximal ideal (Definition 3.5.38) is given by

$$S^{-1}P = \left\{ \frac{p}{s} \in R_P : p \in P \right\}.$$

Definition 3.5.61. Let $(R, \mathfrak{m}_R)$ and $(S, \mathfrak{m}_S)$ be local rings (Definition 3.5.58), not necessarily commutative. A ring homomorphism (Definition 3.1.23) $\varphi : R \rightarrow S$ is called a *local morphism* (or *local homomorphism*) if $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$.

3.5.6. Tensor and Hom's of modules.

Definition 3.5.62 (Tensor product of bimodules). Let R, S, T be (not necessarily commutative) rings (Definition 3.1.1), let M be an R - S bimodule (Definition 3.5.1), and let N be an S - T bimodule. In the free abelian group $\mathbb{Z}[M \times N]$ generated by the Cartesian product $M \times N$ (Definition A.1.4), let U be the subgroup generated by (Definition 2.1.22) elements of the form

$$\begin{aligned}(m + m', n) - (m, n) - (m', n), \\ (m, n + n') - (m, n) - (m, n'), \\ (m \cdot s, n) - (m, s \cdot n),\end{aligned}$$

for all $m, m' \in M$, $n, n' \in N$, and $s \in S$. The *tensor product of M and N over S* is the quotient (Definition 2.1.40) abelian group

$$M \otimes_S N := \mathbb{Z}[M \times N]/U.$$

The image of an element of the form $(m, n) \in M \times N$ in $M \otimes_S N$ is denoted $m \otimes n$ and called a *pure tensor*. In general, the elements of $M \otimes_S N$ are finite sums

$$\sum_{i=1}^n m_i \otimes n_i \quad m_i \in M, n_i \in N$$

of pure tensors. Thus, the pure tensors satisfy the following relations:

$$\begin{aligned}(m + m') \otimes n &= m \otimes n + m' \otimes n \\ m \otimes (n + n') &= m \otimes n + m \otimes n' \\ (m \cdot s) \otimes n &= m \otimes (s \cdot n)\end{aligned}$$

This tensor product becomes naturally an R - T bimodule with left action and right action defined by

$$\begin{aligned} r \cdot (m \otimes n) &= (r \cdot m) \otimes n, \\ (m \otimes n) \cdot t &= m \otimes (n \cdot t), \end{aligned}$$

for all $r \in R$, $t \in T$, $m \in M$, and $n \in N$.

Inductively, given rings $R_0, \dots, R_k$ and R_{i-1} – R_i -bimodules M_i for $i = 1, \dots, k$, we may speak of the tensor product

$$M_0 \otimes_{R_1} M_1 \otimes_{R_2} \cdots \otimes_{R_{k-1}} M_k;$$

tensor products are associative(♠ **TODO:**), so parentheses are not strictly needed to notate them. Its *pure tensors* are elements of the form $m_0 \otimes m_1 \otimes \cdots \otimes m_k$ for $m_i \in M_i$, and its general elements are finite sums

$$\sum_{j=1}^n m_{0j} \otimes m_{1j} \otimes \cdots \otimes m_{kj} \quad m_{ij} \in M_i.$$

of pure tensors. It also has a natural R_0 – R_k -bimodule structure.

In general, $(M_0, \dots, M_k) \mapsto M_0 \otimes_{R_1} M_1 \otimes_{R_2} \cdots \otimes_{R_{k-1}} M_k$ defines a $(k+1)$ -ary additive functor (Definition A.5.1)

$${}_{R_0}\mathbf{Mod}_{R_1} \times \cdots \times {}_{R_{k-1}}\mathbf{Mod}_{R_k} \rightarrow {}_{R_0}\mathbf{Mod}_{R_k}$$

(Theorem 3.5.27).

Given a ring R and a two-sided R -module M , we may also speak of the *n -fold tensor product* $M^{\otimes n} = M^{\otimes_R n}$

Definition 3.5.63. Let k be a not necessarily commutative ring (Definition 3.1.1). Let R and S be k -rings (Definition 3.1.23) (not necessarily commutative). Assume that at least one of R or S is a k -algebra (Definition 3.1.30). The *tensor product ring* $R \otimes_k S$ is the k -module $R \otimes_k S$ (Definition 3.5.62) equipped with a multiplication defined on simple tensors by

$$(r_1 \otimes s_1) \cdot (r_2 \otimes s_2) = (r_1 r_2) \otimes (s_1 s_2)$$

and extended by linearity. This multiplication is well-defined and makes $R \otimes_k S$ into a k -ring under the ring homomorphism

$$k \rightarrow R \otimes_k S, \quad a \mapsto a \otimes 1 = 1 \otimes a.$$

The unit element is $1_R \otimes 1_S$.

In this ring, the subrings (Definition 3.1.40) $R \otimes 1$ and $1 \otimes S$ commute with each other; that is, for all $r \in R$ and $s \in S$,

$$(r \otimes 1) \cdot (1 \otimes s) = r \otimes s = (1 \otimes s) \cdot (r \otimes 1).$$

If R and S are both k -algebras, then $R \otimes_k S$ is also a k -algebra.

3.5.7. Restriction and extension of scalars.

Definition 3.5.64. Let R and S be associative rings with identity (Definition 3.1.1), and let $\varphi : R \rightarrow S$ be a unital ring homomorphism (Definition 3.1.23). Let T be a ring.

1. Let $(M, +)$ be an abelian group (Definition 2.1.1) equipped with either the structure of a $S - T$ -bimodule (Definition 3.5.1) $(M, +, \cdot_S)$ or the structure of a $T - S$ -bimodule $(M, +, \cdot_S)$.

The *restriction of scalars of M along φ* is the R -module structure on the same underlying abelian group $(M, +)$ defined as follows:

- If M is a $S - T$ -bimodule, the restriction of scalars of M along φ , often denoted by $\varphi_* M$ (or ${}_R M$), is the $R - T$ -bimodule whose R -action $\cdot_R : R \times M \rightarrow M$ is given by

$$r \cdot_R m := \varphi(r) \cdot_S m$$

for all $r \in R$ and $m \in M$.

- If M is a $T - S$ -bimodule, the restriction of scalars of M along φ , often denoted by $\varphi_* M$ (or M_R), is the $T - R$ -bimodule whose R -action $\cdot_R : M \times R \rightarrow M$ is given by

$$m \cdot_R r := m \cdot_S \varphi(r)$$

for all $r \in R$ and $m \in M$.

2. Let ${}_S \text{Mod}_T$ and ${}_T \text{Mod}_S$ be the categories of $S - T$ and $T - S$ -bimodules (Definition 3.5.6), respectively, and similarly for R .

The *restriction of scalars functor for modules* is the covariant functor (Definition A.4.5) induced by φ , defined for both left and right modules:

- For left S -modules, it is the functor $\varphi_* : {}_S \text{Mod}_T \rightarrow {}_R \text{Mod}_T$ defined as follows:
 - (a) On objects: For any left S -module M , $\varphi_*(M)$ is the left R -module obtained by restriction of scalars along φ .
 - (b) On morphisms: For any homomorphism (Definition 3.5.4) of left S -modules $h : M \rightarrow N$, the image $\varphi_*(h) : \varphi_*(M) \rightarrow \varphi_*(N)$ is the map h itself, viewed as a homomorphism of left R -modules.
- For right S -modules, it is the functor $\varphi_* : {}_T \text{Mod}_S \rightarrow {}_T \text{Mod}_R$ defined as follows:
 - (a) On objects: For any right S -module M , $\varphi_*(M)$ is the right R -module obtained by restriction of scalars along φ .
 - (b) On morphisms: For any homomorphism of right S -modules $h : M \rightarrow N$, the image $\varphi_*(h) : \varphi_*(M) \rightarrow \varphi_*(N)$ is the map h itself, viewed as a homomorphism of right R -modules.

In either context, if φ is an inclusion map (making R a subring of S), this functor is often called the *forgetful functor*.

3. Let Ring_S (or S/Ring) denote the category of S -rings (Definition 3.1.27). Let Ring_R be defined similarly.

The *restriction of scalars functor for rings*, denoted by $\varphi_* : \text{Ring}_S \rightarrow \text{Ring}_R$ is the functor defined as follows:

- On objects: Let (A, ψ) be an S -ring. Then $\varphi_*(A)$ is the R -ring $(A, \psi \circ \varphi)$, where the structure map is the composition $R \xrightarrow{\varphi} S \xrightarrow{\psi} A$.

- On morphisms: For any morphism of S -rings $h : (A, \psi_A) \rightarrow (B, \psi_B)$, the image $\varphi_*(h)$ is the map h itself, which satisfies $h \circ (\psi_A \circ \varphi) = \psi_B \circ \varphi$ and is thus a morphism of R -rings.

This functor simply pre-composes the structure map with φ , effectively "forgetting" the factorization through S .

The restriction of scalars functor restricts to a functor $\varphi_* : \mathbf{Alg}_S \rightarrow \mathbf{Alg}_R$ (Definition 3.1.30).

Definition 3.5.65. Let R and S be rings (Definition 3.1.1) (not necessarily commutative), and let $f : R \rightarrow S$ be a ring homomorphism (Definition 3.1.23). This homomorphism gives S the structure of an (R, R) -bimodule (Definition 3.5.1) via restriction of scalars (Definition 3.5.64). Let T be a ring (Definition 3.1.1).

1. The *extension of scalars* (or *base change* or *induction*) along f is defined separately for modules:
 - For an $(R - T)$ -bimodule M , the extension of scalars of M along f is the $(S - T)$ -bimodule $S \otimes_R M$ (Definition 3.5.62) where S is viewed as an (S, R) -bimodule. In particular, the action of $s' \in S$ on a simple tensor (Definition 3.5.62) $s \otimes m$ is given by $s' \cdot (s \otimes m) = (s's) \otimes m$.
 - For a $(T - R)$ -bimodule M , the extension of scalars of M along f is the $(T - S)$ -bimodule $M \otimes_R S$ where S is viewed as an (R, S) -bimodule. In particular, the action of $s' \in S$ on a simple tensor $m \otimes s$ is given by $(m \otimes s) \cdot s' = m \otimes (ss')$.
2. The *base change functor* (or *extension of scalars functor* or *induction functor*), denoted by f^* , $S \otimes_R -$, $- \otimes_R S$, or Ind_R^S , is given by:
 - For left R -modules:

$$f^* : {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T, \quad M \mapsto S \otimes_R M.$$

(Definition 3.5.6) This is the left adjoint (Definition A.4.11) to the restriction of scalars (Definition 3.5.64) functor $f_* : {}_S\mathbf{Mod}_T \rightarrow {}_R\mathbf{Mod}_T$.

- For right R -modules:

$$f^* : {}_T\mathbf{Mod}_R \rightarrow {}_T\mathbf{Mod}_S, \quad M \mapsto M \otimes_R S.$$

This is the left adjoint to the restriction of scalars functor $f_* : {}_T\mathbf{Mod}_S \rightarrow {}_T\mathbf{Mod}_R$.

3. Let A be an R -ring (Definition 3.1.23), i.e. a ring equipped with a ring homomorphism $R \rightarrow A$. Assume that S or A is an R -algebra (Definition 3.1.30).

The *base change of the algebra A along f* is the S -ring defined as

$$A_S := S \otimes_R A$$

(Definition 3.5.63) equipped with the natural homomorphism $S \rightarrow S \otimes_R A$ given by $s \mapsto s \otimes 1_A$. As a ring, the multiplication in A_S is determined by $(s_1 \otimes a_1)(s_2 \otimes a_2) = (s_1 s_2) \otimes (a_1 a_2)$ for $s_1, s_2 \in S$ and $a_1, a_2 \in A$.

4. Then the base change construction induces functors in the following situations:
 - (a) If S is only an R -ring, then base change induces a functor $f^* : \mathbf{Alg}_R \rightarrow \mathbf{Ring}_S$.
 - (b) If S is an R -algebra, then base change induces a functor $f^* : \mathbf{Ring}_R \rightarrow \mathbf{Ring}_S$ which restricts to a functor $f^* : \mathbf{Alg}_R \rightarrow \mathbf{Alg}_S$.

In either case, the base change functor is defined as follows:

- On objects: For any R -algebra (A, φ) , $f^*(A)$ is the S -algebra $S \otimes_R A$ defined above.
- On morphisms: For any homomorphism of R -algebras $h : A \rightarrow B$, the image $f^*(h)$ is the map $\text{id}_S \otimes h : S \otimes_R A \rightarrow S \otimes_R B$, defined by $s \otimes a \mapsto s \otimes h(a)$.

(♠ TODO: comment on adjunction)

Definition 3.5.66. Let R and S be rings (Definition 3.1.1) (not necessarily commutative), and let $f : R \rightarrow S$ be a ring homomorphism (Definition 3.1.23). The homomorphism f endows S with two bimodule structures — that of an (R, S) -bimodule and of an (S, R) -bimodule via restriction of scalars (Definition 3.5.64). Let T be a ring.

1. The *co-extension of scalars* (or simply *co-extension*) along f is defined separately for modules:
 - For an $R - T$ -bimodule M , the co-extension of scalars of M along f is the $S - T$ -bimodule $\text{Hom}_R(S, M)$ (Definition 3.5.21) of left R -module homomorphisms from the $R - S$ -bimodule S to M .
 - For a $T - R$ -bimodule M , the co-extension of scalars of M along f is the $T - S$ -bimodule $\text{Hom}_R(S, M)$ of right R -module homomorphisms from the $S - R$ -bimodule S to M .
2. The *co-extension of scalars functor* (or *coinduction functor*), denoted by $f^!$ or CoInd_R^S , is the functor given by:
 - For left R -modules:

$$f^! : {}_R\text{Mod}_T \rightarrow {}_S\text{Mod}_T, \quad M \mapsto \text{Hom}_R(S, M).$$

This functor is the right adjoint (Definition A.4.11) to the restriction of scalars functor $f_* : {}_S\text{Mod} \rightarrow {}_R\text{Mod}$.

- For right R -modules:

$$f^! : {}_T\text{Mod}_R \rightarrow {}_T\text{Mod}_S, \quad M \mapsto \text{Hom}_R(S, M).$$

This functor is the right adjoint to the restriction of scalars functor $f_* : \text{Mod}_S \rightarrow \text{Mod}_R$.

Theorem 3.5.67 (Extension-Restriction and Restriction-Coextension adjunction for modules). Let R and S be rings (Definition 3.1.1) (not necessarily commutative), and let $f : R \rightarrow S$ be a ring homomorphism (Definition 3.1.23). Let T be a ring.

1. **Extension-Restriction adjunction:**

- (a) **Restriction on the Left:** The extension of scalars functor (Definition 3.5.65)

$$f^* = S \otimes_R - : {}_R\text{Mod}_T \rightarrow {}_S\text{Mod}_T$$

(Definition 3.5.6) is left adjoint (Definition A.4.11) to the restriction of scalars functor (Definition 3.5.64)

$$f_* : {}_S\text{Mod}_T \rightarrow {}_R\text{Mod}_T.$$

Explicitly, for any $R - T$ -bimodule M and any $S - T$ -bimodule N , there is a natural isomorphism:

$$\text{Hom}_{S-T}(S \otimes_R M, N) \cong \text{Hom}_{R-T}(M, f_* N).$$

(b) **Restriction on the Right:** The extension of scalars functor

$$f^* = - \otimes_R S : {}_T\text{Mod}_R \rightarrow {}_T\text{Mod}_S$$

is left adjoint to the restriction of scalars functor

$$f_* : {}_T\text{Mod}_S \rightarrow {}_T\text{Mod}_R.$$

Explicitly, for any $T - R$ -bimodule M and any $T - S$ -bimodule N , there is a natural isomorphism:

$$\text{Hom}_{T-S}(M \otimes_R S, N) \cong \text{Hom}_{T-R}(M, f_* N).$$

2. Restriction-Coextension adjunction:

(a) **Restriction on the Left:** The co-extension of scalars functor (Definition 3.5.66)

$$f^! = \text{Hom}_R(S, -) : {}_R\text{Mod}_T \rightarrow {}_S\text{Mod}_T$$

is right adjoint (Definition A.4.11) to the restriction of scalars functor

$$f_* : {}_S\text{Mod}_T \rightarrow {}_R\text{Mod}_T.$$

Explicitly, for any $S - T$ -bimodule N and any $R - T$ -bimodule M , there is a natural isomorphism:

$$\text{Hom}_{R-T}(f_* N, M) \cong \text{Hom}_{S-T}(N, \text{Hom}_R(S, M)).$$

(b) **Restriction on the Right:** The co-extension of scalars functor

$$f^! = \text{Hom}_R(S, -) : {}_T\text{Mod}_R \rightarrow {}_T\text{Mod}_S$$

is right adjoint to the restriction of scalars functor

$$f_* : {}_T\text{Mod}_S \rightarrow {}_T\text{Mod}_R.$$

Explicitly, for any $T - S$ -bimodule N and any $T - R$ -bimodule M , there is a natural isomorphism:

$$\text{Hom}_{T-R}(f_* N, M) \cong \text{Hom}_{T-S}(N, \text{Hom}_R(S, M))$$

where here $\text{Hom}_R(S, M)$ uses the left R -module structure on S and the right R -module structure on M .

Theorem 3.5.68 (Adjunction for Rings). (♠ TODO:) Let $f : R \rightarrow S$ be a homomorphism of rings. Let Ring_R and Ring_S denote the categories of R -rings (Definition 3.1.27) and S -rings, respectively.

There is an adjunction between these categories:

1. The *restriction of scalars functor*, $f_* : \text{Ring}_S \rightarrow \text{Ring}_R$, sends an S -ring $(B, \psi : S \rightarrow B)$ to the R -ring $(B, \psi \circ f)$.
2. The *extension of scalars functor*, $f^* : \text{Ring}_R \rightarrow \text{Ring}_S$, sends an R -ring $(A, \varphi : R \rightarrow A)$ to the *pushout* $S \amalg_R A$ in the category of rings (the free product of S and A with amalgamation over R).

Then f^* is left adjoint to f_* . For any R -ring A and any S -ring B , there is a natural bijection:

$$\mathrm{Hom}_{\mathrm{Ring}_S}(S \amalg_R A, B) \cong \mathrm{Hom}_{\mathrm{Ring}_R}(A, f_* B).$$

Special Case (Algebras): If R is commutative and we restrict to the full subcategory Alg_R of R -algebras (where the structure map maps to the center), and similarly for S (assuming S is an R -algebra), then the extension functor is given by the tensor product $S \otimes_R A$, and the adjunction becomes:

$$\mathrm{Hom}_{\mathrm{Alg}_S}(S \otimes_R A, B) \cong \mathrm{Hom}_{\mathrm{Alg}_R}(A, f_* B).$$

Proposition 3.5.69 (Universal Property of the Tensor Product of Bimodules). Let R, S, T be (not necessarily commutative) rings (Definition 3.1.1). Let M be an R - S bimodule (Definition 3.5.1) and let N be an S - T bimodule. Let P be an R - T bimodule. Then for every R - T bilinear map (Definition 5.10.1)

$$\beta : M \times N \rightarrow P,$$

that is, a map satisfying

$$\begin{aligned} \beta(m + m', n) &= \beta(m, n) + \beta(m', n), \\ \beta(m, n + n') &= \beta(m, n) + \beta(m, n'), \\ \beta(r \cdot m, n) &= r \cdot \beta(m, n), \\ \beta(m, n \cdot t) &= \beta(m, n) \cdot t, \\ \beta(m \cdot s, n) &= \beta(m, s \cdot n), \end{aligned}$$

for all $m, m' \in M$, $n, n' \in N$, $r \in R$, $s \in S$, $t \in T$, there exists a unique R - T bimodule homomorphism

$$\tilde{\beta} : M \otimes_S N \rightarrow P$$

such that $\tilde{\beta}(m \otimes n) = \beta(m, n)$ for all $m \in M$, $n \in N$.

Theorem 3.5.70 (Tensor-Hom Adjunction for Bimodules).

1. Let R, S, T be (not necessarily commutative) rings (Definition 3.1.1). Let M be an R - S bimodule (Definition 3.5.1), let N be an S - T bimodule, and let P be an R - T bimodule. Then there is a natural isomorphism of abelian groups

$$\mathrm{Hom}_{R-T}(M \otimes_S N, P) \cong \mathrm{Hom}_{S-T}(N, \mathrm{Hom}_R(M, P))$$

(Definition 3.5.62) (Definition 3.5.21). Note that Hom_{R-T} is the abelian group of R - T bimodule homomorphisms, Hom_{S-T} is the abelian group of S - T bimodule homomorphisms, and $\mathrm{Hom}_R(M, P)$, which is the abelian group of left R -linear homomorphisms $M \rightarrow P$, is endowed with the structure of an S - T bimodule via

$$\begin{aligned} (s \cdot f)(m) &= f(m \cdot s), \\ (f \cdot t)(m) &= f(m) \cdot t, \end{aligned}$$

for all $s \in S$, $t \in T$, $f \in \mathrm{Hom}_R(M, P)$, $m \in M$. Intuitively, this expresses that $M \otimes_S -$ is left adjoint (Definition A.4.11) to $\mathrm{Hom}_R(M, -)$ in the category of bimodules.

2. Let R, S, T be (not necessarily commutative) rings (Definition 3.1.1). Let M be an R - S bimodule (Definition 3.5.1), let N be an S - T bimodule, and let P be an R - T bimodule. Then there is a natural isomorphism of abelian groups

$$\mathrm{Hom}_{R-T}(M \otimes_S N, P) \cong \mathrm{Hom}_{R-S}(M, \mathrm{Hom}_T(N, P))$$

(Definition 3.5.62) (Definition 3.5.21).

Note that Hom_{R-T} is the abelian group of R - T bimodule homomorphisms, Hom_{R-S} is the abelian group of R - S bimodule homomorphisms, and $\mathrm{Hom}_T(N, P)$, which is the abelian group of right T -linear homomorphisms $N \rightarrow P$, is endowed with the structure of an R - S bimodule via

$$\begin{aligned} (r \cdot f)(n) &= r \cdot f(n), \\ (f \cdot s)(n) &= f(n \cdot s), \end{aligned}$$

for all $r \in R$, $s \in S$, $f \in \mathrm{Hom}_T(N, P)$, and $n \in N$.

Intuitively, this expresses that $-\otimes N$ is left adjoint (Definition A.4.11) to $\mathrm{Hom}_T(N, -)$ in the category of bimodules.

Proposition 3.5.71. Let $\mathcal{A}, \mathcal{B}$ be additive categories (Definition 3.5.16) and let $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{A}$ be adjoint (Definition A.4.11) functors (Definition A.4.5), say with $F \dashv G$ (i.e. F is left adjoint to G).

1. Assuming that the adjunction isomorphisms

$$\alpha = \alpha_{X,Y} : \mathrm{Hom}_{\mathcal{B}}(F(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{A}}(X, G(Y))$$

are isomorphisms of abelian groups for each pair of objects $X, Y \in \mathcal{A}$, the functors F and G are additive (Definition 3.5.17).

2. If $\mathcal{A}$ and $\mathcal{B}$ are abelian (Definition 3.5.18), then the left adjoint functor F is right exact (Definition 3.5.29) and the right adjoint functor G is left exact (Definition 3.5.29).

Proof. Given a morphism $f : X' \rightarrow X$ in $\mathcal{A}$, the fact that the $\alpha_{X,Y}$ are natural isomorphisms with respect to the first input means that we have the commuting diagrams

$$(C) \quad \begin{array}{ccc} \mathrm{Hom}_{\mathcal{B}}(F(X), Y) & \xrightarrow{\alpha_{X,Y}} & \mathrm{Hom}_{\mathcal{A}}(X, G(Y)) \\ \downarrow -\circ F(f) & & \downarrow -\circ f \\ \mathrm{Hom}_{\mathcal{B}}(F(X'), Y) & \xrightarrow{\alpha_{X',Y}} & \mathrm{Hom}_{\mathcal{A}}(X', G(Y)). \end{array}$$

1. We show that F is additive; that G is additive can be obtained dually. To do so, we need to show that for all objects $A_1, A_2 \in \mathcal{A}$, the induced map

$$F_{A_1, A_2} : \mathrm{Hom}_{\mathcal{A}}(A_1, A_2) \rightarrow \mathrm{Hom}_{\mathcal{B}}(F(A_1), F(A_2))$$

is a group homomorphism of abelian groups. In other words, given morphisms $f, g : A_1 \rightarrow A_2$, we need to show that $F(f + g) = F(f) + F(g)$.

Write $\eta : \mathrm{Id}_{\mathcal{A}} \rightarrow GF$ for the unit natural transformation; in particular, given each object $A \in \mathcal{A}$, we write $\eta_A : A \rightarrow GF(A)$ for the unit morphism (Definition A.4.11).

For each $f : A_1 \rightarrow A_2$, we claim that

$$(D) \quad \alpha(F(f)) = \eta_{A_2} \circ f.$$

To see this, we use (C) in the case of $X = A_2$, $Y = F(A_2)$, and $X' = A_1$. Start with $\text{id}_{\mathcal{D}} \in \text{Hom}_{\mathcal{D}}(F(A_2), F(A_2))$ in the upper left corner. Traversing to the right then down sends $\text{id}_{\mathcal{D}}$ to the unit map η_{A_2} via α and then to $\eta_{A_2} \circ f$. Traversing down then to the right sends $\text{id}_{\mathcal{D}}$ to $F(f)$ then to $\alpha(F(f))$. Therefore, (D) holds.

In particular,

$$\alpha(F(f + g)) = \eta_B \circ (f + g).$$

Since $f + g : A_1 \rightarrow A_2$ and $\eta_{A_2} : A_2 \rightarrow GF(A_2)$ are morphisms in the abelian category $\mathcal{B}$, where composition of morphisms is distributive over addition, we have

$$\eta_{A_2} \circ (f + g) = (\eta_{A_2} \circ f) + (\eta_{A_2} \circ g).$$

Applying (D), we have

$$\alpha(F(f + g)) = \alpha(F(f)) + \alpha(F(g)).$$

Since α is assumed to be an isomorphism of groups, we conclude

$$F(f + g) = F(f) + F(g)$$

2. We show that F is right exact; that G is left exact can be obtained dually. It suffices to show that F preserves cokernels (Definition A.4.14), i.e. that if $f : A_1 \rightarrow A_2$ is a morphism in $\mathcal{A}$, then $F(\text{coker } f) \cong \text{coker } F(f)$. Let $q : A_2 \rightarrow Q$ be the cokernel of f .

$$A_1 \xrightarrow{f} A_2 \xrightarrow{q} Q$$

We show that $F(q) : F(A_2) \rightarrow F(Q)$ is the cokernel of $F(f)$, i.e. that $F(q)$ possesses the universal property of $\text{coker } F(f)$.

$$F(A_1) \xrightarrow{F(f)} F(A_2) \xrightarrow{F(q)} F(Q)$$

Equivalently, we show that $F(\text{coker } f)$ possesses the universal property of $\text{coker } F(f)$. Since F is additive, note that

$$F(q) \circ F(f) = F(q \circ f) = F(0) = 0.$$

Now suppose that $g : F(A_2) \rightarrow Z$ is a morphism such that $g \circ F(f) = 0$. We need to show that g factors uniquely through $F(q)$.

Apply the commutativity of (C) in the case that $X = A_2$, $Y = Z$, and $X' = A_1$; this yields

$$\alpha(g \circ F(f)) = \alpha(g) \circ f.$$

Since $g \circ F(f) = 0$, we have $\alpha(g) \circ f = 0$ in $\mathcal{A}$. By the universal property of $\text{coker } f$, the map $\alpha(g)$ factors uniquely through $\text{coker } f$, i.e. there is a unique map $\psi : Q \rightarrow G(Z)$ such that $\psi \circ q = \alpha(g)$.

$$\begin{array}{ccccc} A_1 & \xrightarrow{f} & A_2 & \xrightarrow{q} & Q \\ & & \searrow \alpha(g) & & \downarrow \exists! \psi \\ & & & & G(Z) \end{array}$$

Applying α^{-1} , we have

$$\alpha^{-1}(\psi \circ q) = g.$$

By the naturality of the adjunction isomorphism in the first variable again, we obtain

$$g = \alpha^{-1}(\psi \circ q) = \alpha^{-1}(\psi) \circ F(q).$$

In particular,

$$\begin{array}{ccccc}
 F(A_1) & \xrightarrow{F(f)} & F(A_2) & \xrightarrow{F(q)} & F(Q) \\
 & & \searrow g & & \downarrow \alpha^{-1}(\psi) \\
 & & & & Z
 \end{array}$$

commutes. In fact, $\alpha^{-1}(\psi)$ is the unique morphism making the above commute because α is a bijection and ψ is unique. Therefore, $F(q) : F(A_2) \rightarrow F(Q)$ is indeed the cokernel of $F(f)$ as desired.

□

Proposition 3.5.72. Let R, S, T be (not necessarily commutative) rings (Definition 3.1.1). Recall that categories of modules are abelian (Definition 3.5.18) (Theorem 3.5.27).

1. Let M be an R - S -bimodule (Definition 3.5.1). The functor $M \otimes_S - : {}_S\mathbf{Mod}_T \rightarrow {}_R\mathbf{Mod}_T$ (Definition 3.5.62) is a right exact functor (Definition 3.5.29).
2. Let N be an S - T -bimodule. The functor $- \otimes_S N : {}_R\mathbf{Mod}_S \rightarrow {}_R\mathbf{Mod}_T$ is a right exact functor (Definition 3.5.29).
3. Let M be an R - S -bimodule. The functor

$$\mathrm{Hom}(M, -) : {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T$$

(Definition 3.5.21) is a left exact functor (Definition 3.5.29).

Now let M be an S - R -bimodule. The functor

$$\mathrm{Hom}(M, -) : {}_T\mathbf{Mod}_R \rightarrow {}_S\mathbf{Mod}_T$$

is a left exact functor (Definition 3.5.29).

4. Let N be an R - T -bimodule. The functor

$$\mathrm{Hom}(-, N) : {}_R\mathbf{Mod}_S^{\mathrm{op}} \rightarrow {}_S\mathbf{Mod}_T$$

(Definition A.4.12) is a left exact functor (Definition 3.5.29).

Now let N be an T - R -bimodule. The functor

$$\mathrm{Hom}(-, N) : {}_S\mathbf{Mod}_R^{\mathrm{op}} \rightarrow {}_S\mathbf{Mod}_T$$

is a left exact functor (Definition 3.5.29).

Proof. Note that the tensor-hom adjunction (Theorem 3.5.70) yields natural isomorphisms of abelian groups. Proposition 3.5.71 implies the additivity of the above functors and implies the right exactness of the tensor functors and the left exactness of $\mathrm{Hom}(M, -)$.

Also recall ??.

□

Definition 3.5.73 (Flat module over a ring). Let R be a (not necessarily commutative) ring (Definition 3.1.1).

1. Let M be a left R -module. The module M is said to be *flat (with respect to the left R -module structure)* if the functor

$$- \otimes_R M : \mathbf{Mod}_R \rightarrow \mathbf{Ab}$$

(♠ TODO: comment on how I might as well be talking about other more specific kinds of modules) (Definition 3.5.62) from the category of right R -modules to abelian groups is exact; that is, for every exact sequence of right R -modules

$$0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0,$$

the induced sequence

$$0 \rightarrow N' \otimes_R M \rightarrow N \otimes_R M \rightarrow N'' \otimes_R M \rightarrow 0$$

is exact.

Equivalently, M is flat if $\text{Tor}_1^R(-, M) = 0$ (??).

2. Let M be a right R -module. The module M is said to be *flat (with respect to the right R -module structure)* if the functor

$$M \otimes_R - : {}_R\text{Mod} \rightarrow \mathbf{Ab}$$

from the category of left R -modules to abelian groups is exact; that is, for every exact sequence of right R -modules

$$0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0,$$

the induced sequence

$$0 \rightarrow M \otimes_R N' \rightarrow M \otimes_R N \rightarrow M \otimes_R N'' \rightarrow 0$$

is exact.

Equivalently, M is flat if $\text{Tor}_1^R(M, -) = 0$ (??).

Equivalently in either case, M is flat if it is flat with respect to the biadditive functor

$$- \otimes_R - : \mathbf{Mod}_R \times {}_R\mathbf{Mod} \rightarrow \mathbf{Ab}.$$

(Definition A.5.2)

Definition 3.5.74 (Faithfully flat module). Let R be a (not necessarily commutative) ring (Definition 3.5.75), and let M be a left/right R -module (Definition 3.5.1).

We say that M is a *faithfully flat left/right R -module* if:

1. M is flat (Definition 3.5.73) as a left/right R -module, i.e. the appropriate functor $- \otimes_R M : \mathbf{Mod}_R \rightarrow \mathbf{Ab}$ (Definition 3.5.62) or $M \otimes_R - : {}_R\mathbf{Mod} \rightarrow \mathbf{Ab}$ is exact.
2. The functor $- \otimes_R M : \mathbf{Mod}_R \rightarrow \mathbf{Ab}$ or $M \otimes_R - : {}_R\mathbf{Mod} \rightarrow \mathbf{Ab}$ is faithful (Definition A.4.26). Since the flatness of M means that the tensor functors are exact (Definition 3.5.29), the faithfulness of these functors is equivalent to saying that for any left/right R -module N , if $M \otimes_R N = 0$ or $N \otimes_R M = 0$ as appropriate, then $n = 0$.

Equivalently, the functor $M \otimes_R -$ or $- \otimes_R M$ as appropriate reflects (Definition A.4.27) exactness (Definition 3.5.28) and injectivity (Definition A.1.3), making M a generator (Definition 3.5.20) in the category of left/right R -modules.

Definition 3.5.75 (Flat ring homomorphism). (♠ TODO: flat module) Let R and S be (not necessarily commutative) rings (Definition 3.1.1), and let $\varphi : R \rightarrow S$ be a ring homomorphism (Definition 3.1.23).

1. We may say that S is *flat as a left module* or that φ is *flat (as a left module)* if $- \otimes_R S : \text{Mod}_R \rightarrow \text{Mod}_S$ (Definition 3.5.62) is exact (Definition 3.5.29).
2. We may say that S is *flat as a right module* or that φ is *flat (as a right module)* if $S \otimes_R - : {}_R\text{Mod}_R \rightarrow {}_S\text{Mod}$ (Definition 3.5.62) is exact (Definition 3.5.29).

These two notions of flatness are different in general. However, if φ equips S with the structure of an R -algebra (Definition 3.1.30), then these two notions of flatness coincide. In this case, we may speak of the notion of *flatness of R -algebras* and say that S is *flat over R* .

If S is faithfully flat (Definition 3.5.74) as a left/right R -module, then we may say that φ is *faithfully flat (as a left/right R -module)*.

Definition 3.5.76 (Injective and Projective objects in a general category). Let $\mathcal{C}$ be a category (Definition A.4.1)

- An object $I \in \mathcal{C}$ is called *injective* if it satisfies the following extension property: for every monomorphism (Definition A.4.23) $m : A \rightarrow B$ in $\mathcal{C}$ and every morphism $f : A \rightarrow I$, there exists a morphism $\tilde{f} : B \rightarrow I$ such that the diagram

$$\begin{array}{ccc} & I & \\ & \uparrow f & \\ A & \xrightarrow{m} & B \end{array} \quad \begin{array}{c} \nwarrow \tilde{f} \\ \end{array}$$

commutes, i.e., $\tilde{f} \circ m = f$.

- Dually, an object $P \in \mathcal{C}$ is called *projective* if it satisfies the following lifting property: for every epimorphism (Definition A.4.23) $e : X \rightarrow Y$ in $\mathcal{C}$ and every morphism $g : P \rightarrow Y$, there exists a morphism $\tilde{g} : P \rightarrow X$ such that the diagram

$$\begin{array}{ccc} & P & \\ & \downarrow g & \\ X & \xrightarrow{e} & Y \end{array} \quad \begin{array}{c} \nwarrow \tilde{g} \\ \end{array}$$

commutes, i.e., $e \circ \tilde{g} = g$.

Definition 3.5.77. Let R and S be (not necessarily commutative) rings (Definition 3.1.1). A *projective R - S -bimodule* is an (R, S) -bimodule (Definition 3.5.1) P that satisfies any of the following equivalent conditions:

1. The functor

$$\text{Hom}_{{}_R\text{Mod}_S}(P, -) : {}_R\text{Mod}_S \rightarrow \text{Ab}$$

is an exact functor (Definition 3.5.29) between the abelian categories ${}_R\text{Mod}_S$ (Definition 3.5.6) and Ab (Definition 2.1.18).

2. P is a direct summand of a free (R, S) -bimodule. (A free (R, S) -bimodule is a direct sum of copies of the tensor product $R \otimes_{\mathbb{Z}} S^{\text{op}}$, equipped with the natural left R -action and right S -action).
3. P is a projective object (Definition 3.5.76) in the category ${}_R\text{Mod}_S$. That is, for every surjective homomorphism of (R, S) -bimodules $f : M \rightarrow N$ and every homomorphism $g : P \rightarrow N$, there exists a homomorphism $h : P \rightarrow M$ such that $f \circ h = g$.

Note that P is a projective R - S -bimodule if and only if it is a projective left module over the ring $R \otimes_{\mathbb{Z}} S^{\text{op}}$ (Definition 3.5.63) (Definition 3.1.5) (see ??).

Being a projective bimodule is a strictly stronger condition than being projective as a left or right module.

- A bimodule ${}_R P_S$ may be projective as a left R -module (i.e., projective in ${}_R \mathbf{Mod}$) without being a projective bimodule.
- Similarly, it may be projective as a right S -module (i.e., projective in $\mathbf{Mod}_S$) without being a projective bimodule.
- A bimodule that is projective on both sides is sometimes called *biprojective*, but this does not imply it is a projective object in ${}_R \mathbf{Mod}_S$. For example, let $R = S = \mathbb{Q}[x]$. The bimodule $P = \mathbb{Q}[x]$ (with the natural action on both sides) is free (hence projective) as both a left R -module and a right R -module. However, it is *not* a projective (R, R) -bimodule because it is not a direct summand of the free bimodule $R \otimes_{\mathbb{Z}} R^{\text{op}}$.

Proposition 3.5.78. Let R and S be rings (Definition 3.1.1), and let $f : R \rightarrow S$ be a ring homomorphism (Definition 3.1.23). Let T be a ring.

The base change functor (Definition 3.5.65) (extension of scalars) for bimodules,

$$f^* = S \otimes_R - : {}_R \mathbf{Mod}_T \rightarrow {}_S \mathbf{Mod}_T,$$

(Definition 3.5.6) preserves projective objects. That is, if M is a projective R - T -bimodule, then $S \otimes_R M$ is a projective S - T -bimodule.

Similarly, the functor $f^* = - \otimes_R S : {}_T \mathbf{Mod}_R \rightarrow {}_T \mathbf{Mod}_S$ sends projective T - R -bimodules to projective T - S -bimodules.

3.6. Properties of rings and ring homomorphisms.

3.6.1. Finitely generated, finitely presented, and finite algebras.

Definition 3.6.1. Let R be a (not-necessarily commutative) ring (Definition 3.1.1) and let A be an R -algebra (Definition 3.1.30). We say that A is a *finitely generated R -algebra* or synonymously that the ring homomorphism (Definition 3.1.23) $R \rightarrow A$ is *of finite type* if there exists a finite subset $\{a_1, \dots, a_n\} \subseteq A$ such that R -subalgebra of A generated by (Definition 3.1.34) $\{a_1, \dots, a_n\}$ is equal to A . Equivalently, every element of A can be expressed, using a finite composition of ring operations and R -linear combinations, from the generators $\{a_1, \dots, a_n\}$.

Definition 3.6.2 (Finitely presented algebra over a ring). Let R be a (not necessarily commutative) ring (Definition 3.1.1). An R -algebra (Definition 3.1.30) A is said to be *finitely presented* if there exists an integer $n \geq 0$ and a surjective R -algebra homomorphism

$$\varphi : R\langle x_1, \dots, x_n \rangle \twoheadrightarrow A$$

where $R\langle x_1, \dots, x_n \rangle$ is the free R -algebra on n generators (Definition 3.1.35), such that the kernel $\ker(\varphi)$ is a finitely generated two-sided ideal (Definition 3.5.34) of $R\langle x_1, \dots, x_n \rangle$.

In other words, A admits a presentation as

$$A \cong R\langle x_1, \dots, x_n \rangle / I,$$

where I is a finitely generated (Definition 3.5.35) two-sided ideal.

If R and A are commutative rings, this recovers the usual definition of a finitely presented commutative R -algebra by replacing $R\langle x_1, \dots, x_n \rangle$ with the polynomial ring $R[x_1, \dots, x_n]$ and I a finitely generated ideal.

Definition 3.6.3. Let $B \rightarrow A$ be a B -algebra (Definition 3.1.30). We say that A is a *finite B -algebra* if A is finitely generated (Definition 3.6.1) as a B -module.

3.6.2. Integral elements and morphisms of rings.

Definition 3.6.4 (Integral element over a ring). Let R be a commutative ring with unity (Definition 3.1.6).

1. Let A be an R -algebra (Definition 3.1.30). An element $a \in A$ is called *integral over R* if there exists a monic polynomial

$$p(x) = x^n + r_{n-1}x^{n-1} + \dots + r_1x + r_0$$

with coefficients $r_i \in R$ such that

$$p(a) = a^n + r_{n-1}a^{n-1} + \dots + r_1a + r_0 = 0 \quad \text{in } A.$$

2. Let A be an extension ring (Definition 3.1.41) of R . The ring extension A/R is called an *integral extension* if every element of A is integral over R .
3. Let A be an extension ring (Definition 3.1.41) of R . The *integral closure of R in A* , sometimes denoted by $\tilde{A}$, is the subring

$$\tilde{A} = \{a \in A : a \text{ is integral over } R\}.$$

We say that R is integrally closed in A if $\tilde{A}$ coincides with A (considered as a subring of R (Definition 3.1.40)).

4. Let R be an integral domain with field of fractions $K = \text{Frac}(R)$. We say that R is *integrally closed* if it is integrally closed as a subring of K .

3.6.3. PID's and UFD's.

Definition 3.6.5 (Principal ideal ring/domain (PID)). Let R be a commutative unital ring (Definition 3.1.6). Then R is a *principal ideal ring (PIR)* if every ideal of R is principal (Definition 3.5.35). If R is additionally an , (Definition 3.1.10) then R is said to be a *principal ideal domain (PID)*

Definition 3.6.6 (Unique factorization domain (UFD) Factorial ring). An integral domain (Definition 3.1.10) R is called a *unique factorization domain (UFD)* or *factorial ring* if every nonzero nonunit element (Definition 3.1.11) of R can be factored as a product of irreducible elements (Definition 3.1.12) uniquely up to order and units (Definition 3.1.11).

3.6.4. Artinian and Noetherian rings.

Definition 3.6.7. Let $(S, \leq)$ be a partially ordered set.

1. S satisfies the **Ascending Chain Condition (ACC)** if every sequence $s_1 \leq s_2 \leq s_3 \leq \dots$ of elements in S eventually stabilizes, i.e., there exists $n \in \mathbb{N}$ such that $s_n = s_{n+1} = s_{n+2} = \dots$.
2. S satisfies the **Descending Chain Condition (DCC)** if every sequence $s_1 \geq s_2 \geq s_3 \geq \dots$ of elements in S eventually stabilizes, i.e., there exists $n \in \mathbb{N}$ such that $s_n = s_{n+1} = s_{n+2} = \dots$.

Definition 3.6.8. Let R be a ring (Definition 3.1.1) and M be a left or right R -module (Definition 3.5.1).

1. M is a **Noetherian module** if the set of its submodules (Definition 3.5.2), ordered by inclusion, satisfies the ACC (Definition 3.6.7).
2. M is an **Artinian module** if the set of its submodules, ordered by inclusion, satisfies the DCC (Definition 3.6.7).

Definition 3.6.9. A ring (Definition 3.1.1) R is a **left Artinian ring** if it is Artinian as a left module (Definition 3.5.1) over itself. That is, R satisfies the DCC (Definition 3.6.7) on left ideals (Definition 3.5.34). A **right Artinian ring** is defined dually as satisfying the DCC on right ideals (Definition 3.5.34). A ring is **Artinian** if it is both left and right Artinian.

Proposition 3.6.10. Let R be a left/right Artinian ring (Definition 3.6.9). Then:

1. R is also a left/right Noetherian ring (Hopkins-Levitzki Theorem).
2. The Jacobson radical (Definition 3.5.40) $J(R)$ is a nilpotent ideal (Definition 3.5.53).
3. $R/J(R)$ is a semisimple ring (Definition 3.5.55).

Theorem 3.6.11. A ring (Definition 3.1.1) R is left Artinian (Definition 3.6.9) if and only if it is left Noetherian and has Krull dimension zero.

4. FIELD THEORY

4.1. Field extensions and degrees.

Definition 4.1.1. Let K and L be fields (Definition 3.1.16).

1. A map $\phi : K \rightarrow L$ is a **field homomorphism** if it satisfies the following axioms for all $x, y \in K$:
 - (a) $\phi(x + y) = \phi(x) + \phi(y)$ (additivity);
 - (b) $\phi(xy) = \phi(x)\phi(y)$ (multiplicativity);
 - (c) $\phi(1_K) = 1_L$ (unitality).

Equivalently, a field homomorphism is simply a ring homomorphism (Definition 3.1.23) between fields.

If such a map exists, we often say that K **embeds into** L and that the map is an **embedding of K into L** ; this terminology is justified because field homomorphisms are injective (Definition A.1.3) as set maps (Proposition 4.1.2).

2. A field homomorphism $\phi : K \rightarrow L$ is an *isomorphism* if it is bijective (Definition A.1.3).
3. The set of all homomorphisms from a field K to a field L is denoted by $\text{Hom}(K, L)$.
4. Assuming that K and L have a common subfield F , an *F-embedding of K into L* is an embedding $K \rightarrow L$ that acts as the identity map (Definition A.1.7) on F . The set of F -embeddings of K into L is denoted by $\text{Hom}_F(K, L)$.

If K and L are fields such that K embeds into L , then that means that there is a subfield (Definition 4.1.3) of L that is isomorphic to K .

Proposition 4.1.2. Let K and L be fields (Definition 3.1.16) and let $\phi : K \rightarrow L$ be a field homomorphism (Definition 4.1.1). Then ϕ is injective (Definition A.1.3).

Proof. Since ϕ is a field homomorphism, it is in particular a ring homomorphism (Definition 3.1.23).

The kernel $\ker(\phi)$ is an ideal of K (Proposition 3.1.24). The only ideals of K are $\{0_K\}$ and K itself (Proposition 3.1.19). Since $\phi(1_K) = 1_L \neq 0_L$, we have $1_K \notin \ker(\phi)$, so $\ker(\phi) \neq K$. Therefore, $\ker(\phi) = \{0_K\}$. Hence, ϕ is injective (Definition A.1.3) (Proposition 3.1.25). $\square$

Definition 4.1.3 (Field Extension). Let K be a field and let L be a field such that $K \subseteq L$ and the operations of K are the restrictions of those of L . Then L is called a *extension field (or just an extension) of K* . The notation L/K is often used synonymously; we say that L/K is a *field extension*. Moreover, K is said to be a *subfield of L* .

Lemma 4.1.4. Let L/K be a field extension (Definition 4.1.3). Then L is a vector space (Definition 5.1.1) over K .

Proof. By (Definition 4.1.3), the field L contains K as a subfield, so the addition and multiplication operations of K are restrictions of those of L . We verify that L , equipped with its own addition and with scalar multiplication defined by the restriction of multiplication to $K \times L$, satisfies the vector space axioms over K .

Since $(L, +)$ is an abelian group (being a field), it serves as the additive group of the vector space. Scalar multiplication $K \times L \rightarrow L$ given by $(\lambda, x) \mapsto \lambda x$ satisfies the required axioms: for all $\lambda, \mu \in K$ and $x, y \in L$,

$$\begin{aligned}\lambda(x + y) &= \lambda x + \lambda y, \\ (\lambda + \mu)x &= \lambda x + \mu x, \\ \lambda(\mu x) &= (\lambda\mu)x,\end{aligned}$$

and

$$1_K \cdot x = x,$$

all of which follow from the distributivity, associativity, and identity axioms of the field L . Hence L is a vector space over K by (Definition 5.1.1). $\square$

Definition 4.1.5 (Degree of an Extension, Finite Extension). Let L/K be a field extension (Definition 4.1.3). The *degree of the extension* is $[L : K] := \dim_K(L)$, i.e. the dimension (Definition 5.1.2) of L as a K -vector space (Lemma 4.1.4).

Definition 4.1.6. Let L/K be a field extension (Definition 4.1.3).

It is said to be a *finite field extension* and L is said to be a *finite extension of K* if $[L : K] < \infty$.

A finite field extension of degree 2, 3, 4, 5, 6, etc. is called *quadratic*, *cubic*, *quartic*, *quintic*, *sextic*, etc.

Lemma 4.1.7 (Tower Law). Let $K \subseteq F \subseteq L$ be a tower of field extensions (Definition 4.1.3). Then

$$[L : K] = [L : F] \cdot [F : K],$$

where the product is interpreted in the extended sense: if either $[L : F]$ or $[F : K]$ is infinite, then $[L : K]$ is infinite.

Proof. By (Lemma 4.1.4), we may regard L as a vector space over F and F as a vector space over K . We consider the finite case first; the infinite cases follow by a limiting argument.

Suppose both $[L : F] = n < \infty$ and $[F : K] = m < \infty$. Let $\{x_1, \dots, x_n\}$ be a basis of L over F , and let $\{y_1, \dots, y_m\}$ be a basis of F over K , by (Definition 4.1.5). We claim that the set

$$B = \{x_i y_j : 1 \leq i \leq n, 1 \leq j \leq m\}$$

is a basis of L over K .

First, we show spanning. Given any $\ell \in L$, we can write $\ell = \sum_{i=1}^n a_i x_i$ for some $a_1, \dots, a_n \in F$, since the x_i span L over F . Each coefficient $a_i \in F$ can be written as $a_i = \sum_{j=1}^m b_{ij} y_j$ for some $b_{ij} \in K$, since the y_j span F over K . Substituting, we obtain

$$\ell = \sum_{i=1}^n \left(\sum_{j=1}^m b_{ij} y_j \right) x_i = \sum_{i=1}^n \sum_{j=1}^m b_{ij} (x_i y_j),$$

so B spans L over K .

Next, we show linear independence. Suppose $\sum_{i,j} c_{ij} x_i y_j = 0$ for some scalars $c_{ij} \in K$. Rearranging the sum:

$$\sum_{i=1}^n \left(\sum_{j=1}^m c_{ij} y_j \right) x_i = 0.$$

Since $\{x_1, \dots, x_n\}$ is linearly independent over F , we must have $\sum_{j=1}^m c_{ij} y_j = 0$ in F for each i . But $\{y_1, \dots, y_m\}$ is linearly independent over K , so $c_{ij} = 0$ for all j . Hence B is linearly independent.

Since B has exactly nm elements, we conclude $[L : K] = nm = [L : F] \cdot [F : K]$.

If either $[L : F]$ or $[F : K]$ is infinite, then the set B constructed above is infinite and still linearly independent over K , so $[L : K]$ must also be infinite. This completes the proof in all cases. $\square$

Definition 4.1.8. Let R be a ring (Definition 3.1.1). There exists a unique ring homomorphism (Definition 3.1.23) $\psi : \mathbb{Z} \rightarrow R$ defined by mapping $n \mapsto n \cdot 1_R$. The non-negative generator of the kernel of this map, $\ker(\psi) = \langle p \rangle \subseteq \mathbb{Z}$, is called the *characteristic of R* , denoted by $\text{char}(R)$.

Proposition 4.1.9. Let K be a field (Definition 3.1.16). Then the characteristic (Definition 4.1.8) $\text{char}(K)$ is either 0 or a prime number p .

Proof. Let $\phi : \mathbb{Z} \rightarrow K$ be the unique ring homomorphism sending $1_{\mathbb{Z}}$ to 1_K . The kernel of ϕ is an ideal of $\mathbb{Z}$, so by (Definition 3.6.5), there exists a unique nonnegative integer n such that $\ker(\phi) = (n)$. By (Definition 4.1.8), this integer n is precisely $\text{char}(K)$.

We claim that n is either 0 or a prime number. Suppose for contradiction that $n > 1$ is composite, so $n = ab$ for some integers a, b with $1 < a, b < n$. Then $ab \in \ker(\phi)$, which means $\phi(a)\phi(b) = \phi(ab) = 0_K$. Since K is a field, it has no zero divisors by (Definition 3.1.16), so either $\phi(a) = 0_K$ or $\phi(b) = 0_K$. This implies either $a \in \ker(\phi)$ or $b \in \ker(\phi)$, contradicting the minimality of n as the smallest positive generator of $\ker(\phi)$. Therefore n cannot be composite.

It follows that $\text{char}(K)$ is either 0 or a prime number. □

Definition 4.1.10. Let K be a field. The *prime subfield of K* is the intersection of all subfields (Definition 4.1.3) of K .

- If $\text{char}(K) = p > 0$, the prime subfield is isomorphic to the finite field (Definition 3.4.2) $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$.
- If $\text{char}(K) = 0$, the prime subfield is isomorphic to the field of rational numbers (Definition 3.3.3) $\mathbb{Q}$.

Definition 4.1.11 (Algebraic Element, Algebraic Extension). Let L/K be a field extension (Definition 4.1.3) and let $x \in L$.

- If there exists a nonzero polynomial $f(t) \in K[t]$ such that $f(x) = 0$, then x is called an *algebraic element over K* . There exists a unique such monic irreducible polynomial $f(t)$, which is called the *minimal polynomial of x over K* .
- Otherwise, x is called a *transcendental element over K* .

If every $x \in L$ is algebraic over K , then L/K is called an *algebraic extension*.

Definition 4.1.12. A field (Definition 3.1.16) F is said to be *algebraically closed* if every nonconstant polynomial $f(x) \in F[x]$ has a root in F , i.e., for every such $f(x)$ there exists $a \in F$ with $f(a) = 0$.

Definition 4.1.13. Let K be a field (Definition 3.1.16). A field extension (Definition 4.1.3) L/K is called an *algebraic closure of K* if the following two conditions hold:

1. Every element $a \in L$ is algebraic over K (Definition 4.1.11).
2. The field L is algebraically closed (Definition 4.1.12).

One often writes an algebraic closure of K by $\overline{K}$.

Algebraic closures of a fixed field are not unique, but they are unique up to isomorphism.

Definition 4.1.14 (Field Generated by Elements). Let L/K be a field extension (Definition 4.1.3) and let $S \subseteq L$ be a subset. The *field generated by S over K* is the smallest

subfield of L that contains K and S . Equivalently, it is the intersection of all subfields of L containing $K \cup S$. This field is denoted by $K(S)$.

In the special case where $S = \{x_1, \dots, x_n\}$ is finite, one writes $K(x_1, \dots, x_n)$. If $n = 1$, then $K(x)$ is called a *simple field extension*.

Definition 4.1.15 (Finitely generated field extension). Let E/F be a field extension (Definition 4.1.3). We say that E/F is *finitely generated* if there exist finitely many elements $\alpha_1, \dots, \alpha_n \in E$ such that $E = F(\alpha_1, \dots, \alpha_n)$ (Definition 4.1.14), where $F(\alpha_1, \dots, \alpha_n)$ denotes the smallest subfield of E containing F and $\{\alpha_1, \dots, \alpha_n\}$.

4.2. Splitting fields and normal extensions.

Definition 4.2.1 (Normal Extension). Let K be a field and let $S \subseteq K[t]$ be a set of polynomials.

- A field $L \supseteq K$ is a *splitting field of S over K* if
 - every polynomial in S splits completely into linear factors over L , and
 - L is generated over K (Definition 4.1.14) by the roots of the polynomials in S , namely

$$L = K(\{\alpha \mid f(\alpha) = 0, f \in S\}).$$

- An algebraic extension L/K is called a *normal extension* if L is the splitting field of a family of polynomials in $K[t]$.

Theorem 4.2.2 (Existence and Uniqueness of Splitting Fields). Let K be a field (Definition 3.1.16) and let $f \in K[t]$ be a non-constant polynomial.

1. **(Existence)** There exists a splitting field (Definition 4.2.1) of f over K .
2. **(Uniqueness up to isomorphism)** Let L_1 and L_2 be two splitting fields (Definition 4.2.1) of f over K . Then there exists a K -isomorphism $L_1 \xrightarrow{\sim} L_2$.

More generally, the same existence and uniqueness statements hold for any non-empty set $S \subseteq K[t]$ of polynomials.

Lemma 4.2.3 (Normal Extension via Embeddings). Let L/K be a finite extension (Definition 4.1.6) and let $\overline{K}$ be an algebraic closure (Definition 4.1.13) of K . Then L/K is a normal extension (Definition 4.2.1) if and only if every K -embedding (Definition 4.1.1) $\sigma: L \rightarrow \overline{K}$ satisfies $\sigma(L) = L$.

4.3. Separable extensions.

Definition 4.3.1 (Separable Element, Separable Extension). (♠ TODO: separable polynomial) Let L/K be a field extension (Definition 4.1.3) and let $x \in L$ be algebraic over K (Definition 4.1.11) with minimal polynomial (Definition 4.1.11) $m_{x,K}(t) \in K[t]$.

- The element x is *separable over K* if $m_{x,K}(t)$ has distinct roots in a splitting field (Definition 4.2.1).
- The element x is *inseparable over K* otherwise.

An algebraic extension (Definition 4.1.11) L/K is called **separable extension** if every element $x \in L$ is separable over K .

See also Definition 4.5.6, which defines separable field extensions in greater generality.

Proposition 4.3.2. Let L/K be a finite extension (Definition 4.1.6) of fields (Definition 3.1.16) of degree (Definition 4.1.5) n , and let $\bar{K}$ be an algebraic closure (Definition 4.1.13) of K . Then L/K is a separable extension (Definition 4.3.1) if and only if there are exactly n distinct K -embeddings (Definition 4.1.1) of L into $\bar{K}$.

Definition 4.3.3. A field F is said to be **separably closed** if every nonconstant separable polynomial $f(x) \in F[x]$ has a root in F , i.e., for every such $f(x)$ there exists $a \in F$ with $f(a) = 0$ (equivalently, every separable polynomial factors completely into linear factors over F).

Definition 4.3.4. Let K be a field (Definition 3.1.16). A field extension (Definition 4.1.3) L/K is called a **separable closure of K** if the following hold:

1. The extension L/K is separable algebraic (Definition 4.3.1) (i.e., every $a \in L$ is separable over K).
2. The field L is separably closed (Definition 4.3.3).

One often writes a separable closure of K by K^{sep} . Some also write separable closures of K by $\bar{K}$, which may conflict with the common notation for algebraic closures of K .

Definition 4.3.5. A **field** k is said to be a **perfect field** if it satisfies one of the following equivalent conditions: (♠ TODO: irred polynomial, root)

1. Every irreducible polynomial over k has distinct roots in an algebraic closure (Definition 4.1.11) of k (i.e., every algebraic extension (Definition 4.1.11) of k is separable (Definition 4.3.1)).
2. The characteristic of k is 0, or the characteristic of k is $p > 0$ and the Frobenius endomorphism $\Phi : k \rightarrow k$, defined by $\Phi(x) = x^p$, is an automorphism.

Proposition 4.3.6. Let k be a field (Definition 3.1.16). If k is algebraically closed (Definition 4.1.12), of characteristic (Definition 4.1.8) 0, or a finite field (Definition 3.4.2), then k is a perfect field (Definition 4.3.5).

Proof. (♠ TODO: AI generated, verify) We show that k satisfies one of the equivalent conditions for being a perfect field given in Definition 4.3.5. We treat the three hypotheses separately.

1. Suppose that k is algebraically closed by Definition 4.1.12. Every irreducible polynomial over k is linear, because any nonconstant polynomial has a root in k and therefore factors as a product of linear terms. A linear polynomial has exactly one root, which is simple; hence every irreducible polynomial over k has distinct roots in any algebraic closure of k . Thus k is perfect by condition (1) of Definition 4.3.5.
2. Suppose that $\text{char}(k) = 0$. Then k is perfect directly by condition (2) of Definition 4.3.5, which states that a field of characteristic 0 is perfect.

- Suppose that k is a finite field by Definition 3.4.2. By Proposition 4.1.9, $\text{char}(k) = p$ for some prime number p . The Frobenius endomorphism $\Phi : k \rightarrow k$, given by $\Phi(x) = x^p$, is a ring endomorphism by ???. Since Φ is a homomorphism of fields, it is injective by Proposition 4.1.2. Because k is finite, an injective map from k to itself is also surjective; hence Φ is an automorphism. Therefore k is perfect by condition (2) of Definition 4.3.5.

In all three cases, k is a perfect field. □

4.4. Galois theory.

Definition 4.4.1 (Field Automorphism). Let L be a field. A *field automorphism of L* is a bijective ring homomorphism $L \rightarrow L$.

Definition 4.4.2 (Automorphism Group of a Field Extension). Let L/K be a field extension (Definition 4.1.3).

- An automorphism (Definition 4.4.1) $\sigma : L \rightarrow L$ is called a *K -automorphism* if σ is a field automorphism of L that fixes every element of K , i.e. $\sigma(a) = a$ for all $a \in K$.
- The set of all such K -automorphisms of L forms a group (Definition 2.1.1) under composition, called the *automorphism group of the extension L/K* . It is usually denoted by $\text{Aut}(L/K)$ or $\text{Aut}_K(L)$.

Definition 4.4.3 (Galois Extension). An extension (Definition 4.1.3) L/K is called a *Galois extension* if it is both a normal extension (Definition 4.2.1) and a separable algebraic extension (Definition 4.3.1). Its *Galois group*, usually denoted by $\text{Gal}(L/K)$, is defined to be the automorphism group $\text{Aut}(L/K)$ (Definition 4.4.2).

Proposition 4.4.4 (Equivalent Conditions for a Finite Galois Extension). Let L/K be a finite extension (Definition 4.1.6). The following are equivalent:

- L/K is both a normal extension (Definition 4.2.1) and a separable extension (Definition 4.3.1) (i.e., a Galois extension (Definition 4.4.3)).
- $|\text{Aut}(L/K)| = [L : K]$.
- There exist $\alpha \in L$ and $\beta \in K$ such that $L = K(\alpha)$ and β is the product of the distinct K -embeddings (Definition 4.1.1) of α into an algebraic closure (Definition 4.1.13) of K .

Theorem 4.4.5 (Fundamental Theorem of Galois Theory). Let L/K be a finite (Definition 4.1.6) Galois extension of fields (Definition 4.4.3) with Galois group $G = \text{Gal}(L/K)$. Then there is an inclusion-reversing bijection between the set of subgroups $H \leq G$ and the set of intermediate fields $K \subseteq F \subseteq L$, given by (♠ TODO: L^H)

$$H \longmapsto L^H := \{x \in L \mid \sigma(x) = x \text{ for all } \sigma \in H\},$$

$$F \longmapsto \text{Gal}(L/F).$$

This correspondence has the following properties:

- (i) L^H is a subfield of L containing K for each subgroup $H \leq G$.
- (ii) $\text{Gal}(L/F)$ is a subgroup of G for each intermediate field F .
- (iii) $H_1 \subseteq H_2 \iff L^{H_2} \subseteq L^{H_1}$.
- (iv) $[L^H : K] = |G : H|$, $[L : L^H] = |H|$.
- (v) F/K is Galois if and only if $\text{Gal}(L/F) \trianglelefteq G$,
in which case $\text{Gal}(F/K) \cong G / \text{Gal}(L/F)$.

Corollary 4.4.6 (Normality of the whole extension). If L/K is a finite (Definition 4.1.6) Galois extension of fields (Definition 4.4.3) with Galois group G , then $K = L^G$. (♠ TODO: L^G)

Theorem 4.4.7 (Primitive element theorem). Let L/K be a finite (Definition 4.1.6) separable extension (Definition 4.3.1). Then there exists an element $\alpha \in L$ such that

$$L = K(\alpha).$$

(Definition 4.1.14) In other words, every finite separable extension is simple (Definition 4.1.14).

Theorem 4.4.8 (Infinite Galois Theory). Let L/K be a (possibly infinite) Galois extension with Galois group $G = \text{Gal}(L/K)$. Endow G with the Krull topology, i.e. the unique compact, Hausdorff, totally disconnected topology whose neighborhood basis at the identity is given by the open subgroups $\text{Gal}(L/F)$, where F runs through the finite Galois intermediate extensions $K \subseteq F \subseteq L$.

Then there is an inclusion-reversing bijection between: - the closed subgroups $H \leq G$, and - the intermediate fields $K \subseteq F \subseteq L$,

given by

$$\begin{aligned} H &\longmapsto L^H := \{x \in L \mid \sigma(x) = x \text{ for all } \sigma \in H\}, \\ F &\longmapsto \text{Gal}(L/F). \end{aligned}$$

This correspondence satisfies:

- (i) $H \leq G$ closed $\iff F = L^H$ for some intermediate field F .
- (ii) F/K is finite Galois $\iff \text{Gal}(L/F)$ is open in G .
- (iii) F/K is Galois $\iff \text{Gal}(L/F) \trianglelefteq G$,
in which case $\text{Gal}(F/K) \cong G / \text{Gal}(L/F)$.

Proposition 4.4.9 (Krull topology characterization). For a (possibly infinite) Galois extension L/K , the Galois group $G = \text{Gal}(L/K)$ is a profinite group, i.e. a compact, Hausdorff, totally disconnected topological group isomorphic to an inverse limit of finite groups.

4.5. Transcendental extensions.

Definition 4.5.1 (Algebraically independent elements). Let E/F be a field extension (Definition 4.1.3), and let $S = \{x_i\}_{i \in I}$ be a subset of E indexed by some set I . We say that S is *algebraically independent over F* if for every finite subset $\{x_{i_1}, \dots, x_{i_n}\} \subseteq S$ and every

non-zero polynomial $P \in F[X_1, \dots, X_n]$, we have $P(x_{i_1}, \dots, x_{i_n}) \neq 0$. Equivalently, S is algebraically independent over F if the natural ring homomorphism (Definition 3.1.23)

$$F[X_i : i \in I] \rightarrow E$$

sending $X_i \mapsto x_i$ is injective, where $F[X_i : i \in I]$ denotes the polynomial ring (Definition 3.1.36) over F in the indeterminates $\{X_i\}_{i \in I}$.

Definition 4.5.2 (Transcendence basis). Let E/F be a field extension (Definition 4.1.3). A subset $T \subseteq E$ is called a **transcendence basis** of E over F if:

1. T is algebraically independent over F , i.e., there is no non-trivial polynomial relation with coefficients in F among finitely many elements of T , and
2. E is algebraic (Definition 4.1.11) over $F(T)$, where $F(T)$ denotes the field generated by (Definition 4.1.14) F and T .

Definition 4.5.3 (Transcendence degree). Let E/F be a field extension (Definition 4.1.3). The **transcendence degree of E over F** , denoted $\text{tr.deg}_F(E)$ or $\text{trdeg}_F(E)$, is defined as the cardinality (Definition A.1.13) of any transcendence basis (Definition 4.5.2) of E over F . By a classical result, all transcendence bases of E over F have the same cardinality, so the transcendence degree is well-defined.

If E/F is algebraic (Definition 4.1.11), then $\text{tr.deg}_F(E) = 0$. If no transcendence basis is finite, then $\text{tr.deg}_F(E)$ is an infinite cardinal.

Definition 4.5.4 (Function field). Let k be a field (Definition 3.1.16). A field K is called a **function field over k** (or a **function field in n variables over k**) if K is a finitely generated field extension (Definition 4.1.15) of k with transcendence degree (Definition 4.5.3) n over k , where $n \geq 1$. Equivalently, the function field in n variables over k is isomorphic to the fraction field (Definition 3.1.22) of the polynomial ring (Definition 3.1.36) $k[x_1, \dots, x_n]$ where $x_1, \dots, x_n$ are indeterminate variables. Accordingly, the function field in n variables over k is often denoted as $k(x_1, \dots, x_n)$.

When $n = 1$, we say that K is a **function field of one variable over k** or an **algebraic function field over k** and denote it by $k(x)$.

Definition 4.5.5 (Separating transcendence basis). Let E/F be a field extension (Definition 4.1.3), and let $T \subseteq E$ be a transcendence basis (Definition 4.5.2) of E over F . We say that T is a **separating transcendence basis** if the algebraic extension $E/F(T)$ (Definition 4.1.14) is separable (Definition 4.5.6), i.e., every element of E is separable (Definition 4.3.1) over $F(T)$.

Definition 4.5.6 (Separable field extension). Let E/F be a field extension (Definition 4.1.3). In general (allowing transcendental extensions), E/F is called **separable** if there exists a separating transcendence basis (Definition 4.5.5) for E over F . Equivalently, E/F is separable if every finitely generated (Definition 4.1.15) intermediate field $F \subseteq K \subseteq E$ has a separating transcendence basis over F . In particular, a field extension E/F is algebraic and separable in the above sense if and only if it is a separable algebraic extension in the sense of Definition 4.3.1.

5. BASIC LINEAR ALGEBRA

5.1. Definitions.

Definition 5.1.1 (Vector space over a field). Let $(k, +, \cdot)$ be a field (Definition 3.1.16). A *vector space over k* or a *k -vector space* is a triple $(V, +, \cdot)$ ³ where

1. $(V, +)$ is an abelian group, and
2. $\cdot$ is a map $k \times V \rightarrow V$, called *scalar multiplication*

such that the following axioms hold for all $a, b \in k$ and all $u, v \in V$:

1. (Compatibility with field multiplication)

$$(ab) \cdot v = a \cdot (b \cdot v).$$

2. (Identity scalar)

$$1 \cdot v = v.$$

3. (Distributivity over vector addition)

$$a \cdot (u + v) = a \cdot u + a \cdot v.$$

4. (Distributivity over scalar addition)

$$(a + b) \cdot v = a \cdot v + b \cdot v.$$

Definition 5.1.2. Let F be a field (Definition 3.1.16), and let V be an F -vector space (Definition 5.1.1). A subset $B \subseteq V$ is called a *basis of V* if: (i) B is linearly independent (Definition 3.5.13) over F , and (ii) B spans (Definition 3.5.14) V .

If B is a basis, we define the *dimension of V over F* (or *rank of V over F*), denoted by

$$\dim_F(V),$$

to be the cardinality of B . Any two bases of V have the same cardinality; hence this value is uniquely determined by V and F .

This is a special case of the fact that any two linearly independent (Definition 3.5.13) generating sets of a free F -module (Definition 3.5.1) have the same cardinality when F is a field.

Definition 5.1.3. Let F be a field (Definition 3.1.16), and let V and W be F -vector spaces (Definition 5.1.1). A function $T : V \rightarrow W$ is called a *(homo)morphism of vector spaces over F* , or an *F -linear map*, if for all $u, v \in V$ and all $a, b \in F$, we have

$$T(au + bv) = aT(u) + bT(v).$$

The set of all such morphisms from V to W is denoted by

$$\text{Hom}_F(V, W).$$

³Note that $+$ and $\cdot$ are abuse of notation here as these are already used for the addition and multiplication of $\cdot$.

Definition 5.1.4. Let R be a (not necessarily commutative) ring (Definition 3.1.1). Depending on the module (Definition 3.5.1) structure of M , we define its dual module as follows:

1. If M is a left R -module (Definition 3.5.1), then the *(right) dual module of M* is

$$M^* = M^\vee := \text{Hom}_R(M, R) = \text{Hom}_{R-\mathbb{Z}}(M, R).$$

(Definition 3.5.21) Note that it is a right R -module, as M is a $R - \mathbb{Z}$ -bimodule and R is an $R - R$ -bimodule.

2. If M is a right R -module (Definition 3.5.1), then the *(left) dual module of M* is

$${}^*M = {}^\vee M := \text{Hom}_R(M, R) = \text{Hom}_{\mathbb{Z}-R}(M, R).$$

(Definition 3.5.21) Note that it is a left R -module, as M is a $\mathbb{Z} - R$ -bimodule and R is an $R - R$ -bimodule.

3. If M is a two-sided R -module, then the *dual of M* usually refers to either the right or the left dual as above.

If a convention for the rightness or the leftness of modules is fixed, then the appropriate dual module is often denoted by $M^\vee$.

In any case, the functor $M \mapsto M^\vee$ is a contravariant functor (Definition A.4.5) from the appropriate category of modules (Definition 3.5.6) to itself. More explicitly, the dual module constructions yield functors

$$\begin{aligned} (-)^\vee : {}_R \text{Mod}^{\text{op}} &\rightarrow \text{Mod}_R \\ (-)^\vee : \text{Mod}_R^{\text{op}} &\rightarrow {}_R \text{Mod}. \end{aligned}$$

(Definition A.4.12) (Definition 3.5.6)

If R is a field (Definition 3.1.16) F and V is an F -vector space (Definition 5.1.1), then the dual module

$$V^* = V^\vee := \text{Hom}_F(V, F)$$

is called the *dual vector space of V* .

Definition 5.1.5. Let R be a (not necessarily commutative) ring (Definition 3.1.1), and let $m, n \in \mathbb{N}$. The set of all $m \times n$ arrays with entries in R is denoted by

$$M_{m,n}(R) = \{(a_{ij}) \mid a_{ij} \in R \text{ for all } i, j\}.$$

Elements of $M_{m,n}(R)$ are called *matrices over R* of size $m \times n$. Given a matrix $A \in M_{m,n}(R)$ and $1 \leq i \leq m, 1 \leq j \leq n$, the value a_{ij} is called the *(i, j)-th-entry of A* and may be denoted by $A_{i,j}$.

In the case that $m = n$, the set of $n \times n$ matrices over R is often denoted by $M_n(R)$. The elements of $M_n(R)$ are called *square matrices over R* .

Matrix addition $A + B$ and multiplication AB are defined by the usual componentwise and summation formulas. More precisely, if $A, B \in M_{m,n}(R)$, then

$$(A + B)_{ij} = a_{ij} + b_{ij},$$

and if $A \in M_{m,n}(R)$ and $B \in M_{n,p}(R)$, then

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}, \quad 1 \leq i \leq m, 1 \leq j \leq p.$$

Definition 5.1.6. Let R be a (not necessarily commutative) ring (Definition 3.1.1), and let $n \geq 1$ be an integer. Let $A = (a_{ij}) \in M_n(R)$ be an $n \times n$ matrix (Definition 5.1.5) over R .

The *determinant of A* is the element $\det(A) \in R$ given by the Leibniz formula

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{n,\sigma(n)},$$

where S_n (Definition 2.1.48) is the symmetric group on $\{1, \dots, n\}$ (Definition 2.1.48), and $\operatorname{sgn}(\sigma)$ is the *sign of the permutation σ* (♠ TODO: AI generated, reference needed: definition of sign of a permutation).

For $n = 0$, we set $\det(A) = 1_R$.

5.1.1. Determinants over commutative rings.

Proposition 5.1.7. Let R be a commutative ring (Definition 3.1.6), and let $A = (a_{ij}) \in M_n(R)$ be an $n \times n$ matrix. Then

$$\det(A^\top) = \det(A),$$

where $A^\top$ (Definition 5.4.6) is the transpose of A (Definition 5.4.6).

Proposition 5.1.8. Let R be a commutative ring (Definition 3.1.6), and let $A, B \in M_n(R)$ be two $n \times n$ matrices. Then

$$\det(AB) = \det(A) \det(B).$$

Proposition 5.1.9. Let R be a commutative ring (Definition 3.1.6), and let I_n be the $n \times n$ identity matrix. Then

$$\det(I_n) = 1_R.$$

Proposition 5.1.10. Let R be a commutative ring (Definition 3.1.6), and let $A = (a_{ij}) \in M_n(R)$ be an $n \times n$ matrix. Let $\operatorname{adj}(A) \in M_n(R)$ be the *classical adjugate of A* (♠ TODO: AI generated, reference needed: definition of classical adjugate matrix), whose (i, j) -entry is $(-1)^{i+j}$ times the determinant of the submatrix obtained by deleting row j and column i from A . Then

$$\operatorname{adj}(A) \cdot A = A \cdot \operatorname{adj}(A) = \det(A) I_n.$$

Proposition 5.1.11. Let R be a commutative ring (Definition 3.1.6), and let $A \in M_n(R)$ be an $n \times n$ matrix. Then A is invertible in $M_n(R)$ if and only if $\det(A)$ is a unit in R . In that case,

$$A^{-1} = \det(A)^{-1} \operatorname{adj}(A).$$

Proposition 5.1.12. Let R be a commutative ring (Definition 3.1.6), and let $n \geq 1$.

The determinant function $\det: M_n(R) \rightarrow R$ has the following properties:

1. It is multilinear in the rows of the matrix, i.e. if one row of $A \in M_n(R)$ is replaced by a sum of two vectors or multiplied by a scalar from R , then $\det(A)$ behaves linearly with respect to that operation on that row while all other rows are held fixed.
2. It is alternating in the rows, i.e. if any two rows of A are equal, then $\det(A) = 0$. Equivalently, swapping two rows of A multiplies the determinant by -1_R .
3. The same multilinearity and alternation properties hold for columns: this follows from the fact that $\det(A^\top) = \det(A)$.

Proposition 5.1.13 (Cramer's rule). Let R be a commutative ring (Definition 3.1.6), and let $A = (a_{ij}) \in M_n(R)$ be an invertible $n \times n$ matrix with inverse given by $\det(A)^{-1} \operatorname{adj}(A)$. Let $b \in R^{n \times 1}$ be a column vector, and let $x \in R^{n \times 1}$ be the unique solution to $Ax = b$.

For each $k = 1, \dots, n$, the k -th component of x is

$$x_k = \frac{\det(A_k)}{\det(A)},$$

where A_k is the matrix obtained from A by replacing its k -th column with the vector b .

5.2. Dimension theory and rank.

Definition 5.2.1. Let F be a field (Definition 3.1.16). Let V and W be F -vector spaces (Definition 5.1.1), and let $T : V \rightarrow W$ be a linear map (Definition 5.1.3).

The *kernel of T* is the subspace of V defined by

$$\ker(T) = \{v \in V : T(v) = 0_W\}.$$

The *image of T* (or *range of T*) is the subspace of W defined by

$$\operatorname{im}(T) = \{T(v) : v \in V\}.$$

Note that since every F -vector space is an F -module (Definition 3.5.1) and every linear map is a module homomorphism (Definition 3.5.4), the kernel and image here are special cases of Definition 3.5.25. (♠ TODO: AI generated, verify)

Proposition 5.2.2. Let F be a field (Definition 3.1.16). Let V and W be F -vector spaces (Definition 5.1.1), and let $T : V \rightarrow W$ be a linear map (Definition 5.1.3).

Then $\ker(T)$ is a subspace (Definition 3.5.2) of V , and $\operatorname{im}(T)$ is a subspace of W .

This is the vector-space special case of Definition 3.5.25, where kernel and image of any module homomorphism are submodules. (♠ TODO: AI generated, verify)

Proof. (♠ TODO: AI generated, verify) Let 0_V and 0_W denote the zero vectors in V and W , respectively.

We show that $\ker(T)$ is a subspace of V . By Proposition 3.5.3, it suffices to verify nonemptiness, closure under addition, and closure under scalar multiplication. First, $T(0_V) = 0_W$ by Definition 5.1.3, so $0_V \in \ker(T)$ and $\ker(T) \neq \emptyset$. Second, if $v_1, v_2 \in \ker(T)$, then

$$T(v_1 + v_2) = T(v_1) + T(v_2) = 0_W + 0_W = 0_W,$$

so $v_1 + v_2 \in \ker(T)$. Third, if $c \in F$ and $v \in \ker(T)$, then

$$T(cv) = cT(v) = c \cdot 0_W = 0_W,$$

so $cv \in \ker(T)$. Hence, $\ker(T)$ is a subspace of V .

We show that $\text{im}(T)$ is a subspace of W using the same criterion. First, $T(0_V) = 0_W$, so $0_W \in \text{im}(T)$ and $\text{im}(T) \neq \emptyset$. Second, if $w_1, w_2 \in \text{im}(T)$, there exist $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$, so

$$w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2) \in \text{im}(T).$$

Third, if $c \in F$ and $w \in \text{im}(T)$ with $T(v) = w$, then

$$cw = cT(v) = T(cv) \in \text{im}(T).$$

Hence, $\text{im}(T)$ is a subspace of W . □

Definition 5.2.3. Let F be a field (Definition 3.1.16). Let V and W be F -vector spaces (Definition 5.1.1), and let $T : V \rightarrow W$ be a linear map (Definition 5.1.3).

The **rank of T** is the dimension (Definition 5.1.2) of its image (Definition 5.2.1):

$$\text{rank}(T) = \dim_F(\text{im}(T)).$$

The **nullity of T** is the dimension of its kernel (Definition 5.2.1):

$$\text{nullity}(T) = \dim_F(\ker(T)).$$

Let A be an $m \times n$ matrix over F (Definition 5.1.5). Identifying A with the linear map $L_A : F^n \rightarrow F^m$ given by left multiplication, the **rank of A** is defined as $\text{rank}(L_A)$.

The rank and nullity of a linear map are the vector-space special case of the dimension of the image and kernel of a Definition 3.5.4, where $\text{im}(T)$ and $\ker(T)$ carry natural module structures. (♠ TODO: AI generated, verify)

Theorem 5.2.4 (Rank–Nullity Theorem). Let F be a field (Definition 3.1.16). Let V and W be F -vector spaces (Definition 5.1.1), with $\dim_F(V) < \infty$. Let $T : V \rightarrow W$ be a linear map (Definition 5.1.3).

Then

$$\dim_F(\ker(T)) + \dim_F(\text{im}(T)) = \dim_F(V),$$

or equivalently, $\text{nullity}(T) + \text{rank}(T) = \dim_F(V)$.

This is the vector-space special case of the additivity of dimension in short exact sequences: if $0 \rightarrow \ker(T) \rightarrow V \rightarrow \text{im}(T) \rightarrow 0$ is a Definition 3.5.28, then $\dim(V) = \dim(\ker(T)) + \dim(\text{im}(T))$. (♠ TODO: AI generated, verify)

Proof. (♠ TODO: AI generated, verify) By Proposition 5.2.2, $\ker(T)$ is a subspace of V . Let $k = \dim_F(\ker(T))$, and let $\{v_1, \dots, v_k\}$ be a basis for $\ker(T)$.

By Lemma 5.2.5, we can extend $\{v_1, \dots, v_k\}$ to a basis $\{v_1, \dots, v_k, u_1, \dots, u_r\}$ for V , where $r = \dim_F(V) - k$.

We claim that $\{T(u_1), \dots, T(u_r)\}$ is a basis for $\text{im}(T)$.

We first show that these vectors span $\text{im}(T)$. Let $w \in \text{im}(T)$. Then there exists $v \in V$ such that $T(v) = w$. Write v in terms of the basis for V :

$$v = c_1v_1 + \dots + c_kv_k + d_1u_1 + \dots + d_ru_r$$

for some scalars $c_1, \dots, c_k, d_1, \dots, d_r \in F$. Applying T and using linearity gives

$$w = T(v) = c_1T(v_1) + \dots + c_kT(v_k) + d_1T(u_1) + \dots + d_rT(u_r).$$

Since each $v_i \in \ker(T)$, we have $T(v_i) = 0_W$, so the first k terms vanish. Hence

$$w = d_1T(u_1) + \dots + d_rT(u_r),$$

which shows that $\{T(u_1), \dots, T(u_r)\}$ spans $\text{im}(T)$.

Next we show linear independence. Suppose that $d_1T(u_1) + \dots + d_rT(u_r) = 0_W$ for some scalars $d_1, \dots, d_r \in F$. By linearity,

$$T(d_1u_1 + \dots + d_ru_r) = 0_W,$$

so $d_1u_1 + \dots + d_ru_r \in \ker(T)$. Since this vector lies in $\ker(T)$, it can be written as a linear combination of the basis vectors of $\ker(T)$:

$$d_1u_1 + \dots + d_ru_r = c_1v_1 + \dots + c_kv_k$$

for some scalars $c_1, \dots, c_k \in F$. Rearranging gives

$$-c_1v_1 - \dots - c_kv_k + d_1u_1 + \dots + d_ru_r = 0_V.$$

Since $\{v_1, \dots, v_k, u_1, \dots, u_r\}$ is a basis for V , it is linearly independent, so all coefficients must be zero. In particular, $d_1 = \dots = d_r = 0$. Hence, $\{T(u_1), \dots, T(u_r)\}$ is linearly independent by (Definition 5.1.2).

Therefore, $\{T(u_1), \dots, T(u_r)\}$ is a basis for $\text{im}(T)$ of size r . This means

$$\dim_F(\text{im}(T)) = r = \dim_F(V) - k = \dim_F(V) - \dim_F(\ker(T)).$$

Rearranging yields the desired equality. □

Lemma 5.2.5. (♠ TODO: AI generated, verify) Let F be a field (Definition 3.1.16). Let V be an F -vector space (Definition 5.1.1) with $\dim_F(V) < \infty$, and let W be a subspace of V . Every basis of W can be extended to a basis of V .

5.3. Direct sums and quotient spaces.

Definition 5.3.1. Let F be a field (Definition 3.1.16), and let V be an F -vector space (Definition 5.1.1).

(Internal direct sum.) Let $W_1, \dots, W_n$ be subspaces of V . Their sum $W_1 + \dots + W_n = \{w_1 + \dots + w_n : w_i \in W_i\}$ is called the *internal direct sum* of $W_1, \dots, W_n$ if every $v \in W_1 + \dots + W_n$ has a unique expression $v = w_1 + \dots + w_n$ with $w_i \in W_i$. In this case we write

$$W_1 \oplus \dots \oplus W_n.$$

(External direct sum.) Let $V_1, \dots, V_n$ be F -vector spaces. Their *external direct sum* is the Cartesian product $V_1 \times \dots \times V_n$ equipped with component-wise addition and scalar multiplication, denoted

$$V_1 \oplus \dots \oplus V_n.$$

The external direct sum of vector spaces is the special case of Definition 3.5.22 for F -modules. For finite families, it coincides with the coproduct of modules (Definition 3.5.23). (♠ TODO: AI generated, verify)

Theorem 5.3.2. Let F be a field (Definition 3.1.16). Let V be an F -vector space (Definition 5.1.1), and let $W_1, \dots, W_n$ be subspaces of V whose internal direct sum $W = W_1 \oplus \dots \oplus W_n$ exists.

The map $\Phi : W_1 \oplus \dots \oplus W_n \rightarrow W$ defined by

$$\Phi(w_1, \dots, w_n) = w_1 + \dots + w_n$$

is an isomorphism of F -vector spaces (Definition 5.1.3) between the external and internal direct sums.

This is the vector-space special case of the canonical isomorphism between an internal direct sum of submodules and Definition 3.5.23. (♠ TODO: AI generated, verify)

Proof. (♠ TODO: AI generated, verify) By Definition 5.3.1, the external direct sum $W_1 \oplus \dots \oplus W_n$ is the Cartesian product equipped with component-wise addition and scalar multiplication.

We first show that Φ is a linear map. For $(w_1, \dots, w_n), (w'_1, \dots, w'_n) \in W_1 \oplus \dots \oplus W_n$ and $c \in F$, we have

$$\Phi(w_1 + w'_1, \dots, w_n + w'_n) = (w_1 + w'_1) + \dots + (w_n + w'_n) = (w_1 + \dots + w_n) + (w'_1 + \dots + w'_n) = \Phi(w_1, \dots, w_n) + \Phi(w'_1, \dots, w'_n)$$

and

$$\Phi(cw_1, \dots, cw_n) = cw_1 + \dots + cw_n = c(w_1 + \dots + w_n) = c\Phi(w_1, \dots, w_n).$$

Hence Φ is linear by Definition 5.1.3.

We show that Φ is surjective. Let $w \in W$. By the definition of the internal direct sum (Definition 5.3.1), every element of $W = W_1 + \dots + W_n$ can be written as a sum $w_1 + \dots + w_n$ with $w_i \in W_i$. Hence, $\Phi(w_1, \dots, w_n) = w$, so Φ is surjective.

We show that Φ is injective by verifying it has trivial kernel. Suppose that $\Phi(w_1, \dots, w_n) = 0_V$. Then $w_1 + \dots + w_n = 0_V$ with $w_i \in W_i$. By the uniqueness of the expression in an internal direct sum (Definition 5.3.1), this implies that each $w_i = 0_{W_i}$. Hence, $(w_1, \dots, w_n) = (0_{W_1}, \dots, 0_{W_n})$, which is the zero vector of the external direct sum. Therefore, Φ has trivial kernel, so it is injective by Proposition 2.1.39(3) (applying the group-theoretic statement to the underlying abelian groups of vector spaces).

Hence, Φ is a bijective linear map, i.e., an isomorphism of F -vector spaces. $\square$

Definition 5.3.3. Let F be a field (Definition 3.1.16), let V be an F -vector space (Definition 5.1.1), and let W be a subspace of V .

The *quotient vector space* of V by W is the set of cosets

$$V/W = \{v + W : v \in V\}$$

equipped with addition and scalar multiplication defined by

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W, \quad c \cdot (v + W) = (c \cdot v) + W.$$

The map $\pi : V \rightarrow V/W$ defined by $\pi(v) = v + W$ is called the *canonical projection onto V/W* .

This is the special case for vector spaces of Definition 3.5.24, where the quotient of a module by a submodule carries a natural module structure. (♠ TODO: AI generated, verify)

Proposition 5.3.4. Let F be a field (Definition 3.1.16).

(Direct sum.) If $V_1, \dots, V_n$ are finite-dimensional F -vector spaces, then

$$\dim_F(V_1 \oplus \dots \oplus V_n) = \sum_{i=1}^n \dim_F(V_i).$$

(Quotient.) If V is a finite-dimensional F -vector space and W is a subspace, then

$$\dim_F(V/W) = \dim_F(V) - \dim_F(W).$$

These are the vector-space special cases of the additivity of length in Definition 3.5.28 of finite-length objects. (♠ TODO: AI generated, verify)

Proof. (♠ TODO: AI generated, verify) We prove each part separately.

1. (Direct sum.) Let $d_i = \dim_F(V_i)$ for each $i = 1, \dots, n$, and let $\{e_{i,1}, \dots, e_{i,d_i}\}$ be a basis for V_i . We claim that the set of all vectors

$$\{(v_1, \dots, v_n) : v_i \in \{e_{i,1}, \dots, e_{i,d_i}\} \text{ for each } i\}$$

forms a basis for $V_1 \oplus \dots \oplus V_n$. The total number of such vectors is $\sum_{i=1}^n d_i$.

We first show spanning. Let $(v_1, \dots, v_n) \in V_1 \oplus \dots \oplus V_n$. For each i , write $v_i = c_{i,1}e_{i,1} + \dots + c_{i,d_i}e_{i,d_i}$ for some scalars $c_{i,j} \in F$. By component-wise addition and scalar multiplication in the external direct sum (Definition 5.3.1), we have

$$(v_1, \dots, v_n) = \sum_{j=1}^{d_1} c_{1,j}(e_{1,j}, 0, \dots, 0) + \dots + \sum_{j=1}^{d_n} c_{n,j}(0, \dots, 0, e_{n,j}),$$

so the set spans $V_1 \oplus \dots \oplus V_n$.

Next we show linear independence. Suppose that a linear combination of these vectors equals $(0, \dots, 0)$. By component-wise operations in the external direct sum, each component must separately equal zero. Since $\{e_{i,1}, \dots, e_{i,d_i}\}$ is linearly independent for each i , all coefficients in the i -th component must be zero. Hence, all coefficients are zero, so the set is linearly independent by (Definition 5.1.2).

Therefore, the dimension of $V_1 \oplus \dots \oplus V_n$ equals the number of basis vectors, which is $\sum_{i=1}^n d_i$.

2. (Quotient.) Let $\pi : V \rightarrow V/W$ be the canonical projection defined by $\pi(v) = v + W$ (Definition 5.3.3). The map π is linear because

$$\pi(v_1 + v_2) = (v_1 + v_2) + W = (v_1 + W) + (v_2 + W) = \pi(v_1) + \pi(v_2),$$

and $\pi(cv) = cv + W = c(v + W) = c\pi(v)$ for all $v, v_1, v_2 \in V$ and $c \in F$.

The kernel of π is precisely W , because $\pi(v) = 0_{V/W} = W$ if and only if $v + W = W$, which holds if and only if $v \in W$. Hence, $\ker(\pi) = W$.

The map π is surjective by definition of the quotient space: every element of V/W has the form $v + W = \pi(v)$ for some $v \in V$. Therefore, $\text{im}(\pi) = V/W$.

Applying the rank-nullity theorem (Theorem 5.2.4) to π gives

$$\dim_F(\ker(\pi)) + \dim_F(\text{im}(\pi)) = \dim_F(V).$$

Substituting $\ker(\pi) = W$ and $\text{im}(\pi) = V/W$ yields the desired formula.

□

5.4. Matrix representations.

Definition 5.4.1. (♠ TODO: is this for column vectors? There should be also a thing for row vectors) Let F be a field (Definition 3.1.16). Let V and W be finite-dimensional F -vector spaces (Definition 5.1.1), with ordered bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$, respectively.

For any linear map (Definition 5.1.3) $T : V \rightarrow W$, the *matrix representation of T with respect to $\mathcal{B}$ and $\mathcal{C}$* is the unique $m \times n$ matrix (Definition 5.1.5) over F , denoted

$$[T]_{\mathcal{B}}^{\mathcal{C}},$$

whose (i, j) -entry is the coefficient of w_i in the expansion of $T(v_j)$ in the basis $\mathcal{C}$. That is,

$$T(v_j) = \sum_{i=1}^m [T]_{\mathcal{B}}^{\mathcal{C}}(i, j) w_i.$$

The matrix representation of a linear map with respect to bases is the special case for free F -modules of the matrix of a Definition 3.5.4 between free modules of finite rank with chosen bases. (♠ TODO: AI generated, verify)

Definition 5.4.2. Let R be a (not necessarily commutative) ring (Definition 3.1.1). Two matrices $A, B \in M_n(R)$ (Definition 5.1.5) are called *similar over R* (or *conjugate over R*) if there exists an invertible matrix (Definition 5.4.15) $P \in \text{GL}_n(R)$ such that

$$B = P^{-1}AP.$$

If M is a free R -module of finite rank and $T : M \rightarrow M$ is an R -linear (Definition 3.5.4) endomorphism, then the matrices representing (Definition 5.4.17) T with respect to two different ordered bases of M are similar. Conversely, any two similar matrices arise as matrix representations of the same endomorphism with respect to different bases (Definition 5.4.3).

Definition 5.4.3. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n .

(♠ TODO: make a version of this applicable for row vectors)

Let $\mathcal{E} = (e_1, \dots, e_n)$ and $\mathcal{F} = (f_1, \dots, f_n)$ be two ordered bases of M . The *change-of-basis matrix from $\mathcal{F}$ to $\mathcal{E}$* (or the *transition matrix from $\mathcal{F}$ to $\mathcal{E}$*) is the matrix $P \in M_n(R)$ (Definition 5.1.5) whose j -th column consists of the coordinates of f_j in the basis $\mathcal{E}$:

$$P = (P(i, j))_{1 \leq i, j \leq n} \quad \text{where} \quad f_j = \sum_{i=1}^n e_i P(i, j)$$

if M is a right R -module, and

$$P = (P(i, j))_{1 \leq i, j \leq n} \quad \text{where} \quad f_j = \sum_{i=1}^n P(i, j) e_i$$

if M is a left R -module.

In either case, P is invertible (Definition 5.4.15) and its inverse P^{-1} is the change-of-basis matrix from $\mathcal{E}$ to $\mathcal{F}$.

(♠ TODO: coordinate column/row vector with respect to a basis) For any element $x \in M$, write $[x]_{\mathcal{E}}$ and $[x]_{\mathcal{F}}$ for its coordinate column vectors (Definition 5.4.10) with respect to $\mathcal{E}$ and $\mathcal{F}$, respectively. The change-of-basis matrix relates these coordinate vectors by

$$[x]_{\mathcal{E}} = P [x]_{\mathcal{F}}$$

for all $x \in M$ (Lemma 5.4.11). Consequently, for any R -linear map $T : M \rightarrow M$, the matrix representations (Definition 5.4.17) of T with respect to $\mathcal{E}$ and $\mathcal{F}$ are related by

$$[T]_{\mathcal{E}}^{\mathcal{E}} = P [T]_{\mathcal{F}}^{\mathcal{F}} P^{-1}.$$

Definition 5.4.4. Let F be a field (Definition 3.1.16). Two $n \times n$ matrices (Definition 5.1.5) A and B over F are called *similar* (over F) if there exists an invertible matrix P over F such that

$$B = P^{-1} A P.$$

Let V be a finite-dimensional F -vector space, and let $T : V \rightarrow V$ be a linear endomorphism. If $\mathcal{B}$ and $\mathcal{B}'$ are two ordered bases of V , the matrices $[T]_{\mathcal{B}}^{\mathcal{B}}$ and $[T]_{\mathcal{B}'}^{\mathcal{B}'}$ are similar. The invertible matrix $P = [I]_{\mathcal{B}'}^{\mathcal{B}}$ is called the *change-of-basis matrix from $\mathcal{B}'$ to $\mathcal{B}$* . Similarity of matrices corresponds to the fact that two matrix representations of the same endomorphism of a free module with respect to different bases are related by conjugation; this is a special case for F -modules of the change-of-basis behavior for Definition 3.5.4s. (♠ TODO: AI generated, verify)

Proposition 5.4.5. Let F be a field (Definition 3.1.16). Let A and $B = P^{-1} A P$ be two similar $n \times n$ matrices (Definition 5.1.5) over F .

Then A and B share the same determinant (Definition 5.1.6), trace, characteristic polynomial, and rank (Definition 5.2.3).

These invariants are well-defined on the underlying endomorphism of a free module, independent of the choice of basis; they are special cases of invariants of Definition 3.5.4s between free modules of finite rank. (♠ TODO: AI generated, verify)

Proof. (♠ TODO: AI generated, verify) Let $L_A : F^n \rightarrow F^n$ and $L_B : F^n \rightarrow F^n$ denote the linear maps defined by left multiplication by A and B , respectively. The matrix P defines an invertible linear map $L_P : F^n \rightarrow F^n$. Similarity means that for every vector $v \in F^n$, we have

$$L_B(v) = P^{-1}APv = L_{P^{-1}}(L_A(L_P(v))).$$

In other words, $L_B = L_{P^{-1}} \circ L_A \circ L_P$.

We prove each invariance separately.

1. (Determinant.) By the multiplicativity of the determinant (Proposition 5.1.8), we have

$$\det(B) = \det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P).$$

Since F is a field, multiplication in F is commutative, so $\det(P^{-1}) \det(P) = \det(P)^{-1} \det(P) = 1$. Hence, $\det(B) = \det(A)$.

2. (Trace.) The trace of a matrix is the sum of its diagonal entries. For $B = P^{-1}AP$, we have

$$\text{tr}(B)_{ii} = \sum_{j=1}^n (P^{-1})_{ij} A_{jk} P_{ki},$$

where we use the standard matrix multiplication formula. Summing over i gives

$$\text{tr}(B) = \sum_{i,j,k} (P^{-1})_{ij} A_{jk} P_{ki}.$$

By commutativity of addition in F , we may rearrange the summation order:

$$\text{tr}(B) = \sum_{j,k} A_{jk} \left(\sum_i P_{ki} (P^{-1})_{ij} \right).$$

The inner sum is the (k, j) -entry of $PP^{-1} = I_n$, which equals δ_{kj} , the Kronecker delta. Hence, only terms with $j = k$ survive:

$$\text{tr}(B) = \sum_j A_{jj} \delta_{jj} = \sum_j A_{jj} = \text{tr}(A).$$

3. (Characteristic polynomial.) For each $t \in F$, we have

$$tI_n - B = tI_n - P^{-1}AP = P^{-1}(tI_n - A)P.$$

Taking determinants of both sides and applying part (1), we get

$$\det(tI_n - B) = \det(P^{-1}) \det(tI_n - A) \det(P) = \det(tI_n - A).$$

Hence, $p_B(t) = p_A(t)$ by ??.

4. (Rank.) The rank of a matrix equals the dimension of the image of its associated linear map (Definition 5.2.3). Since $L_B = L_{P^{-1}} \circ L_A \circ L_P$ and both L_P and $L_{P^{-1}}$ are isomorphisms (because P is invertible), the image of L_B equals $L_{P^{-1}}(\text{im}(L_A))$. The map $L_{P^{-1}}$ restricted to $\text{im}(L_A)$ is injective, so it preserves dimension. Hence,

$$\dim_F(\text{im}(L_B)) = \dim_F(\text{im}(L_A)).$$

Therefore, $\text{rank}(B) = \text{rank}(A)$.

□

Definition 5.4.6. Let R be a ring (Definition 3.1.1).

For any $m \times n$ matrix (Definition 5.1.5) A over R , the *transpose of A* is the $n \times m$ matrix $A^\top$ defined by $A^\top(i, j) = A(j, i)$ for all $1 \leq i \leq n$, $1 \leq j \leq m$.

Definition 5.4.7. (♠ TODO: AI generated, verify) Let F be a field (Definition 3.1.16).

Let V and W be F -vector spaces, and let $T : V \rightarrow W$ be a linear map (Definition 5.1.3). The *dual map of T* is the linear map

$$T^* : W^* \rightarrow V^*,$$

defined by $(T^*(\phi))(v) = \phi(T(v))$ for all $\phi \in W^*$ and $v \in V$, where V^* denotes the dual space (Definition 5.1.4) of V .

The dual map T^* is the vector-space special case of the dual homomorphism induced by a Definition 5.1.4 contravariant functor on modules.

Proposition 5.4.8. (♠ TODO: AI generated, verify) Let F be a field (Definition 3.1.16).

Let V and W be finite-dimensional F -vector spaces with ordered bases $\mathcal{B}$ and $\mathcal{C}$, respectively. Let $T : V \rightarrow W$ be a linear map (Definition 5.1.3), and let $T^* : W^* \rightarrow V^*$ be the dual map of T (Definition 5.4.7).

Let $\mathcal{B}^*$ and $\mathcal{C}^*$ denote the dual bases. Then

$$[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*} = [T]_{\mathcal{B}}^{\mathcal{C}\top},$$

where $A \mapsto A^\top$ is the transpose of a matrix (Definition 5.4.6).

5.4.1. Row/column vectors and coordinate representations.

Definition 5.4.9. Let R be a ring (Definition 3.1.1) (not necessarily commutative). Let $n \in \mathbb{N}$.

1. A *row vector of length n over R* is an element of $M_{1,n}(R)$ (Definition 5.1.5), i.e., a $1 \times n$ matrix over R . We denote the set of row vectors of length n over R by $R^{1 \times n}$ or $M_{1,n}(R)$.
2. A *column vector of length n over R* is an element of $M_{n,1}(R)$, i.e., an $n \times 1$ matrix over R . We denote the set of column vectors of length n over R by $R^{n \times 1}$ or $M_{n,1}(R)$.

- When R is a commutative ring (Definition 3.1.6), the set of column vectors $R^{n \times 1}$ carries a natural structure of a left R -module (Definition 3.5.1) (and also a right R -module) via componentwise addition and scalar multiplication. Similarly, the set of row vectors $R^{1 \times n}$ carries a natural structure of a right R -module (and also a left R -module) via componentwise addition and scalar multiplication.
- The transpose (Definition 5.4.6) operation gives a bijection between row vectors and column vectors:

$$(\cdot)^\top : R^{1 \times n} \rightarrow R^{n \times 1}, \quad (a_1, \dots, a_n) \mapsto \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}.$$

Its inverse is the transpose map $(\cdot)^\top : R^{n \times 1} \rightarrow R^{1 \times n}$.

Definition 5.4.10. Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n with an ordered basis $\mathcal{E} = (e_1, \dots, e_n)$.

- Suppose M is a right R -module. For any $x \in M$, there exist unique scalars $a_1, \dots, a_n \in R$ such that

$$x = \sum_{i=1}^n e_i a_i.$$

The *coordinate column vector of x with respect to $\mathcal{E}$* is the column vector

$$[x]_{\mathcal{E}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in R^{n \times 1}.$$

The *coordinate row vector of x with respect to $\mathcal{E}$* is the row vector

$$[x]_{\mathcal{E}}^r = (a_1 \ \cdots \ a_n) \in R^{1 \times n}.$$

- Suppose M is a left R -module. For any $x \in M$, there exist unique scalars $a_1, \dots, a_n \in R$ such that

$$x = \sum_{i=1}^n a_i e_i.$$

The *coordinate column vector of x with respect to $\mathcal{E}$* is the column vector

$$[x]_{\mathcal{E}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in R^{n \times 1}.$$

The *coordinate row vector of x with respect to $\mathcal{E}$* is the row vector

$$[x]_{\mathcal{E}}^r = (a_1 \ \cdots \ a_n) \in R^{1 \times n}.$$

In both cases, the map $x \mapsto [x]_{\mathcal{E}}$ is an isomorphism of (left/right) R -modules $M \cong R^{n \times 1}$, and the map $x \mapsto [x]_{\mathcal{E}}^r$ is an isomorphism $M \cong R^{1 \times n}$ (with the appropriate module structure on the space of row or column vectors (Definition 5.4.9)).

Lemma 5.4.11. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n .

(♠ TODO: Do a version of this for row vectors) (♠ TODO: coordinate column vectors with respect to bases) Let $\mathcal{E} = (e_1, \dots, e_n)$ and $\mathcal{F} = (f_1, \dots, f_n)$ be two ordered bases of M , and let $P \in M_n(R)$ (Definition 5.1.5) be the change-of-basis matrix from $\mathcal{F}$ to $\mathcal{E}$ (Definition 5.4.3). For any $x \in M$, denote by $[x]_{\mathcal{E}}, [x]_{\mathcal{F}} \in R^{n \times 1}$ (Definition 5.4.9) its coordinate column vectors with respect to $\mathcal{E}$ and $\mathcal{F}$, respectively.

1. If M is a right R -module, then for all $x \in M$,

$$[x]_{\mathcal{E}} = P [x]_{\mathcal{F}}.$$

2. If M is a left R -module, then for all $x \in M$,

$$[x]_{\mathcal{E}} = P [x]_{\mathcal{F}}.$$

In particular, the formula $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$ holds for both left and right modules with the same column-vector convention.

Proof. Write $x = \sum_{j=1}^n f_j a_j$ if M is a right R -module, or $x = \sum_{j=1}^n a_j f_j$ if M is a left R -module, so that $[x]_{\mathcal{F}} = (a_1, \dots, a_n)^{\top}$.

1. Suppose that M is a right R -module. By definition of P , we have $f_j = \sum_{i=1}^n e_i P(i, j)$. Substituting,

$$x = \sum_{j=1}^n \left(\sum_{i=1}^n e_i P(i, j) \right) a_j = \sum_{i=1}^n e_i \left(\sum_{j=1}^n P(i, j) a_j \right).$$

The coefficient of e_i in this expression is $\sum_{j=1}^n P(i, j) a_j$, which is precisely the i -th entry of the column vector $P[x]_{\mathcal{F}}$. Hence, $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$.

2. Suppose that M is a left R -module. By definition of P , we have $f_j = \sum_{i=1}^n P(i, j) e_i$. Substituting,

$$x = \sum_{j=1}^n a_j \left(\sum_{i=1}^n P(i, j) e_i \right) = \sum_{i=1}^n \left(\sum_{j=1}^n a_j P(i, j) \right) e_i.$$

The coefficient of e_i in this expression is $\sum_{j=1}^n a_j P(i, j)$, which is the i -th entry of $P[x]_{\mathcal{F}}$ (since $P(i, j)$ is the (i, j) -entry of P and a_j is the j -th entry of $[x]_{\mathcal{F}}$). Hence, $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$.

□

5.4.2. Endomorphisms and matrix representations.

Definition 5.4.12. Let $\mathcal{C}$ be a category (Definition A.4.1) and let X be an object of $\mathcal{C}$.

An *endomorphism of X* is a morphism $f : X \rightarrow X$ in $\mathcal{C}$.

Definition 5.4.13. Let $\mathcal{C}$ be a locally small category (Definition A.4.3) and let X be an object of $\mathcal{C}$.

The *endomorphism semigroup of X* , denoted $\text{End}_{\mathcal{C}}(X)$, is the set of all endomorphisms of X (Definition 5.4.12). It is indeed a semigroup (Definition A.2.1) (Proposition 5.4.14) under the composition of morphisms in $\mathcal{C}$.

Proposition 5.4.14. Let $\mathcal{C}$ be a locally small category (Definition A.4.3) and let X be an object of $\mathcal{C}$.

The endomorphism monoid (Definition 5.4.13) $\text{End}_{\mathcal{C}}(X)$ is indeed a semigroup (Definition A.2.2) under composition of morphisms.

Proof. Since $\mathcal{C}$ is locally small, the collection of endomorphisms $X \rightarrow X$ in $\mathcal{C}$ is a set. Composition of morphisms in $\mathcal{C}$ restricts to a binary operation $\circ : \text{End}_{\mathcal{C}}(X) \times \text{End}_{\mathcal{C}}(X) \rightarrow \text{End}_{\mathcal{C}}(X)$ because the composition of two endomorphisms $f, g : X \rightarrow X$ is again a morphism $f \circ g : X \rightarrow X$. Composition of morphisms in any category is associative, so $\text{End}_{\mathcal{C}}(X)$ is a semigroup (Definition A.2.1). Moreover, the identity morphism id_X is the identity element for the binary operation $\circ$, so $\text{End}_{\mathcal{C}}(X)$ is indeed a monoid. $\square$

5.4.3. Invertible matrices and the general linear group.

Definition 5.4.15. Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let n be a positive integer.

A matrix $A \in M_n(R)$ (Definition 5.1.5) is called *invertible* (or *nonsingular*) if there exists a matrix $B \in M_n(R)$ such that

$$AB = BA = I_n,$$

where I_n is the identity matrix. Such a matrix B is necessarily unique and is called the *inverse of A* , denoted A^{-1} .

The set of all invertible $n \times n$ matrices over R forms a group (Definition 2.1.1) under matrix multiplication (Proposition 5.4.16), called the general linear group $\text{GL}_n(R)$; the inverse of an invertible matrix A in the above sense is thus the inverse (Definition 1.0.5) of A in the group $\text{GL}_n(R)$.

Proposition 5.4.16. Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let n be a positive integer. The set $\text{GL}_n(R)$ of invertible $n \times n$ matrices (Definition 5.4.15) over R , equipped with matrix multiplication, forms a group (Definition 2.1.1).

Proof. We verify the group axioms.

1. We first verify closure under matrix multiplication. If $A, B \in \text{GL}_n(R)$, then AB is invertible with inverse $B^{-1}A^{-1}$, since

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$

and similarly $(B^{-1}A^{-1})(AB) = I_n$. Hence $AB \in \text{GL}_n(R)$.

2. Next, we verify associativity. Matrix multiplication is associative: for any $A, B, C \in M_n(R)$, one has $(AB)C = A(BC)$. This holds in particular for elements of $\text{GL}_n(R)$.

3. We now verify the existence of an identity element. The identity matrix I_n is invertible (since $I_n I_n = I_n$), so $I_n \in \text{GL}_n(R)$. Moreover, $I_n A = A I_n = A$ for all $A \in \text{GL}_n(R)$.
4. Finally, we verify the existence of inverses. By definition, every $A \in \text{GL}_n(R)$ has an inverse $A^{-1} \in M_n(R)$ satisfying $AA^{-1} = A^{-1}A = I_n$. This inverse A^{-1} is itself invertible (with inverse A), so $A^{-1} \in \text{GL}_n(R)$.

Thus $\text{GL}_n(R)$ satisfies all group axioms. $\square$

5.4.4. Matrix of an endomorphism.

Definition 5.4.17. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n with an ordered basis $\mathcal{E} = (e_1, \dots, e_n)$. Let $T : M \rightarrow M$ be an endomorphism (Definition 5.4.12) of M (i.e. an R -linear map).

1. **Column vector convention.** The *matrix of T with respect to $\mathcal{E}$ (column convention)* is the matrix $[T]_{\mathcal{E}} \in M_n(R)$ (Definition 5.1.5) whose j -th column is the coordinate column vector (Definition 5.4.10) of $T(e_j)$:

$$[T]_{\mathcal{E}} = \begin{pmatrix} [T(e_1)]_{\mathcal{E}} & \cdots & [T(e_n)]_{\mathcal{E}} \end{pmatrix}.$$

Equivalently, the entries $[T]_{\mathcal{E}}(i, j)$ are determined by

$$T(e_j) = \sum_{i=1}^n e_i [T]_{\mathcal{E}}(i, j)$$

if M is a right R -module, and by

$$T(e_j) = \sum_{i=1}^n [T]_{\mathcal{E}}(i, j) e_i$$

if M is a left R -module. For any $x \in M$, the coordinate column vectors transform as

$$[T(x)]_{\mathcal{E}} = [T]_{\mathcal{E}} [x]_{\mathcal{E}}$$

(Lemma 5.4.18).

2. **Row vector convention.** The *matrix of T with respect to $\mathcal{E}$ (row convention)* is the matrix $[T]_{\mathcal{E}}^r \in M_n(R)$ whose i -th row is the coordinate row vector (Definition 5.4.10) of $T(e_i)$:

$$[T]_{\mathcal{E}}^r = \begin{pmatrix} [T(e_1)]_{\mathcal{E}}^r \\ \vdots \\ [T(e_n)]_{\mathcal{E}}^r \end{pmatrix}.$$

For any $x \in M$, the coordinate row vectors transform as

$$[T(x)]_{\mathcal{E}}^r = [x]_{\mathcal{E}}^r [T]_{\mathcal{E}}^r.$$

The two conventions are related by transposition:

$$[T]_{\mathcal{E}}^r = [T]_{\mathcal{E}}^{\top}$$

(Definition 5.4.6).

Lemma 5.4.18. (♠ TODO: do a version of this for row vectors) Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n with an ordered basis $\mathcal{E} = (e_1, \dots, e_n)$. Let $T : M \rightarrow M$ be an R -linear endomorphism and let $[T]_{\mathcal{E}}$ be its matrix with respect to $\mathcal{E}$ (column convention) (Definition 5.4.17). Given $x \in M$, let $[x]_{\mathcal{E}}$ be the coordinate column vector (Definition 5.4.10) of x with respect to $\mathcal{E}$.

For any $x \in M$, the coordinate column vectors satisfy

$$[T(x)]_{\mathcal{E}} = [T]_{\mathcal{E}} [x]_{\mathcal{E}}.$$

Proof. Write $x = \sum_{j=1}^n e_j a_j$ if M is a right R -module, or $x = \sum_{j=1}^n a_j e_j$ if M is a left R -module, so that $[x]_{\mathcal{E}} = (a_1, \dots, a_n)^{\top}$.

1. Suppose M is a right R -module. By R -linearity,

$$\begin{aligned} T(x) &= T\left(\sum_{j=1}^n e_j a_j\right) = \sum_{j=1}^n T(e_j) a_j = \sum_{j=1}^n \left(\sum_{i=1}^n e_i [T]_{\mathcal{E}}(i, j)\right) a_j \\ &= \sum_{i=1}^n e_i \left(\sum_{j=1}^n [T]_{\mathcal{E}}(i, j) a_j\right). \end{aligned}$$

The coefficient of e_i in this expression is $\sum_{j=1}^n [T]_{\mathcal{E}}(i, j) a_j$, which is precisely the i -th entry of $[T]_{\mathcal{E}} [x]_{\mathcal{E}}$. Hence $[T(x)]_{\mathcal{E}} = [T]_{\mathcal{E}} [x]_{\mathcal{E}}$.

2. Suppose M is a left R -module. By R -linearity,

$$\begin{aligned} T(x) &= T\left(\sum_{j=1}^n a_j e_j\right) = \sum_{j=1}^n a_j T(e_j) = \sum_{j=1}^n a_j \left(\sum_{i=1}^n [T]_{\mathcal{E}}(i, j) e_i\right) \\ &= \sum_{i=1}^n \left(\sum_{j=1}^n a_j [T]_{\mathcal{E}}(i, j)\right) e_i. \end{aligned}$$

The coefficient of e_i in this expression is $\sum_{j=1}^n a_j [T]_{\mathcal{E}}(i, j)$, which is the i -th entry of $[T]_{\mathcal{E}} [x]_{\mathcal{E}}$ (since $[T]_{\mathcal{E}}(i, j)$ is the (i, j) -entry and a_j is the j -th entry of $[x]_{\mathcal{E}}$). Hence $[T(x)]_{\mathcal{E}} = [T]_{\mathcal{E}} [x]_{\mathcal{E}}$.

□

5.4.5. Coordinate row vectors.

Definition 5.4.19. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n with an ordered basis $\mathcal{E} = (e_1, \dots, e_n)$.

1. Suppose M is a right R -module. For any $x \in M$, there exist unique scalars $a_1, \dots, a_n \in R$ such that

$$x = \sum_{i=1}^n e_i a_i.$$

The *coordinate row vector of x with respect to $\mathcal{E}$* is the row vector

$$[x]_{\mathcal{E}}^r = (a_1 \ \cdots \ a_n) \in R^{1 \times n}.$$

2. Suppose M is a left R -module. For any $x \in M$, there exist unique scalars $a_1, \dots, a_n \in R$ such that

$$x = \sum_{i=1}^n a_i e_i.$$

The *coordinate row vector of x with respect to $\mathcal{E}$* is the row vector

$$[x]_{\mathcal{E}}^r = (a_1 \ \cdots \ a_n) \in R^{1 \times n}.$$

In both cases, the map $x \mapsto [x]_{\mathcal{E}}^r$ is an isomorphism of (left/right) R -modules $M \cong R^{1 \times n}$ (Definition 5.4.9). The coordinate row vector is related to the coordinate column vector (Definition 5.4.10) by transposition:

$$[x]_{\mathcal{E}}^r = [x]_{\mathcal{E}}^T$$

(Definition 5.4.6).

Lemma 5.4.20. Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n .

Let $\mathcal{E} = (e_1, \dots, e_n)$ and $\mathcal{F} = (f_1, \dots, f_n)$ be two ordered bases of M , and let $P \in \text{GL}_n(R)$ be the change-of-basis matrix from $\mathcal{F}$ to $\mathcal{E}$ (Definition 5.4.3). For any $x \in M$, denote by $[x]_{\mathcal{E}}^r, [x]_{\mathcal{F}}^r \in R^{1 \times n}$ its coordinate row vectors (Definition 5.4.19) with respect to $\mathcal{E}$ and $\mathcal{F}$, respectively.

Then for all $x \in M$,

$$[x]_{\mathcal{E}}^r = [x]_{\mathcal{F}}^r P^T,$$

where P^T denotes the transpose (Definition 5.4.6) of P .

Proof. By Lemma 5.4.11, the coordinate column vectors satisfy $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$ for both left and right modules. Transposing both sides gives

$$[x]_{\mathcal{E}}^T = ([x]_{\mathcal{F}})^T P^T.$$

Since $[x]_{\mathcal{E}}^r = [x]_{\mathcal{E}}^T$ and $[x]_{\mathcal{F}}^r = [x]_{\mathcal{F}}^T$ (Definition 5.4.19), we obtain $[x]_{\mathcal{E}}^r = [x]_{\mathcal{F}}^r P^T$ as claimed. $\square$

Lemma 5.4.21. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring (Definition 3.1.1) and let M be a free (left/right) R -module of finite rank n with an ordered basis $\mathcal{E} = (e_1, \dots, e_n)$. Let $T : M \rightarrow M$ be an R -linear endomorphism and let $[T]_{\mathcal{E}}^r$ (Definition 5.4.17) be its matrix with respect to $\mathcal{E}$ under the *row convention* (Definition 5.4.17). Given $x \in M$, let $[x]_{\mathcal{E}}^r$ (Definition 5.4.19) be the coordinate row vector of x with respect to $\mathcal{E}$.

For any $x \in M$, the coordinate row vectors satisfy

$$[T(x)]_{\mathcal{E}}^r = [x]_{\mathcal{E}}^r [T]_{\mathcal{E}}^r.$$

Proof. Write $x = \sum_{j=1}^n e_j a_j$ if M is a right R -module, or $x = \sum_{j=1}^n a_j e_j$ if M is a left R -module, so that $[x]_{\mathcal{E}}^r = (a_1, \dots, a_n)$.

1. Suppose that M is a right R -module. By R -linearity of T ,

$$T(x) = T\left(\sum_{j=1}^n e_j a_j\right) = \sum_{j=1}^n T(e_j) a_j.$$

By definition of the matrix under the row convention (Definition 5.4.17), we have $T(e_i) = \sum_{j=1}^n e_j [T]_{\mathcal{E}}^r(i, j)$ for each i . Substituting (renaming indices so that the outer sum runs over i),

$$\begin{aligned} T(x) &= \sum_{i=1}^n T(e_i) a_i = \sum_{i=1}^n \left(\sum_{j=1}^n e_j [T]_{\mathcal{E}}^r(i, j) \right) a_i \\ &= \sum_{j=1}^n e_j \left(\sum_{i=1}^n [T]_{\mathcal{E}}^r(i, j) a_i \right). \end{aligned}$$

The coefficient of e_j in this expression is $\sum_{i=1}^n [T]_{\mathcal{E}}^r(i, j) a_i$, which is precisely the j -th entry of the row vector $[x]_{\mathcal{E}}^r [T]_{\mathcal{E}}^r$. Hence, $[T(x)]_{\mathcal{E}}^r = [x]_{\mathcal{E}}^r [T]_{\mathcal{E}}^r$.

2. Suppose that M is a left R -module. By R -linearity of T ,

$$T(x) = T\left(\sum_{j=1}^n a_j e_j\right) = \sum_{j=1}^n a_j T(e_j).$$

By definition of the matrix under the row convention (Definition 5.4.17), we have $T(e_i) = \sum_{j=1}^n [T]_{\mathcal{E}}^r(i, j) e_j$ for each i . Substituting (renaming indices so that the outer sum runs over i),

$$\begin{aligned} T(x) &= \sum_{i=1}^n a_i T(e_i) = \sum_{i=1}^n a_i \left(\sum_{j=1}^n [T]_{\mathcal{E}}^r(i, j) e_j \right) \\ &= \sum_{j=1}^n \left(\sum_{i=1}^n a_i [T]_{\mathcal{E}}^r(i, j) \right) e_j. \end{aligned}$$

The coefficient of e_j in this expression is $\sum_{i=1}^n a_i [T]_{\mathcal{E}}^r(i, j)$, which is the j -th entry of the row vector $[x]_{\mathcal{E}}^r [T]_{\mathcal{E}}^r$. Hence, $[T(x)]_{\mathcal{E}}^r = [x]_{\mathcal{E}}^r [T]_{\mathcal{E}}^r$.

In both cases, the formula $[T(x)]_{\mathcal{E}}^r = [x]_{\mathcal{E}}^r [T]_{\mathcal{E}}^r$ holds. □

5.4.6. Hermitian forms and change of basis.

Lemma 5.4.22. Let (A, σ) be a ring with involution (Definition 5.11.2), where σ is denoted by $a \mapsto \bar{a}$. Let M be a free (left/right) A -module of finite rank n with an ε -Hermitian form (Definition 5.11.4) $h : M \times M \rightarrow A$ for some $\varepsilon \in Z(A)$ (Definition 3.1.7) with $\varepsilon \cdot \bar{\varepsilon} = 1$.

Let $\mathcal{E} = (e_1, \dots, e_n)$ and $\mathcal{F} = (f_1, \dots, f_n)$ be two ordered bases of M , and let $P \in \text{GL}_n(A)$ be the change-of-basis matrix from $\mathcal{F}$ to $\mathcal{E}$ (Definition 5.4.3). Denote by $H_{\mathcal{E}}$ and $H_{\mathcal{F}}$ the matrices of h with respect to $\mathcal{E}$ and $\mathcal{F}$, respectively.

1. If M is a right A -module (so that h is conjugate linear in the first variable and linear in the second (Definition 5.11.3)), then

$$H_{\mathcal{F}} = P^* H_{\mathcal{E}} P,$$

where P^* denotes the conjugate transpose of P .

2. If M is a left A -module (so that h is linear in the first variable and conjugate linear in the second (Definition 5.11.3)), then

$$H_{\mathcal{F}} = P^{\top} H_{\mathcal{E}} \overline{P},$$

where $P^{\top}$ denotes the transpose of P (Definition 5.4.6) and $\overline{P}$ denotes the entrywise conjugate of P .

In particular, $H_{\mathcal{E}}$ and $H_{\mathcal{F}}$ are $*$ -congruent when M is a right A -module. For left A -modules, the relation $H_{\mathcal{F}} = P^{\top} H_{\mathcal{E}} \overline{P}$ is not $*$ -congruence in general, but reduces to $*$ -congruence when the involution is trivial (e.g. when A is a commutative ring with the identity involution).

Proof. 1. Suppose M is a right A -module. For any $x, y \in M$, write $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$ and $[y]_{\mathcal{E}} = P[y]_{\mathcal{F}}$. By ??, we have

$$\begin{aligned} h(x, y) &= [x]_{\mathcal{E}}^* H_{\mathcal{E}} [y]_{\mathcal{E}} \\ &= (P[x]_{\mathcal{F}})^* H_{\mathcal{E}} (P[y]_{\mathcal{F}}) \\ &= [x]_{\mathcal{F}}^* (P^* H_{\mathcal{E}} P) [y]_{\mathcal{F}}. \end{aligned}$$

Since $h(x, y) = [x]_{\mathcal{F}}^* H_{\mathcal{F}} [y]_{\mathcal{F}}$ by the same lemma, and this holds for all $x, y \in M$, we conclude $H_{\mathcal{F}} = P^* H_{\mathcal{E}} P$.

2. Suppose M is a left A -module. For any $x, y \in M$, write $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$ and $[y]_{\mathcal{E}} = P[y]_{\mathcal{F}}$. By ??, we have

$$\begin{aligned} h(x, y) &= [x]_{\mathcal{E}}^{\top} H_{\mathcal{E}} \overline{[y]_{\mathcal{E}}} \\ &= (P[x]_{\mathcal{F}})^{\top} H_{\mathcal{E}} \overline{(P[y]_{\mathcal{F}})} \\ &= [x]_{\mathcal{F}}^{\top} (P^{\top} H_{\mathcal{E}} \overline{P}) \overline{[y]_{\mathcal{F}}}. \end{aligned}$$

Since $h(x, y) = [x]_{\mathcal{F}}^{\top} H_{\mathcal{F}} \overline{[y]_{\mathcal{F}}}$ by the same lemma, and this holds for all $x, y \in M$, we conclude $H_{\mathcal{F}} = P^{\top} H_{\mathcal{E}} \overline{P}$.

□

5.4.7. Matrix algebra as an involution algebra.

Proposition 5.4.23. Let (A, σ) be a ring with involution (Definition 5.11.2), where σ is denoted by $a \mapsto \bar{a}$. Let $M_n(A)$ be the matrix ring (Definition 5.1.5) over A . The map $X \mapsto X^* = \overline{X^{\top}}$ is an involution (Definition 5.11.2) on $M_n(A)$. Moreover, this involution along with the embedding $A \mapsto aI_n$ makes $M_n(A)$ into an involution algebra over A .

Proof. We verify the four involution axioms for the map $\cdot^* : M_n(A) \rightarrow M_n(A)$.

1. We show that $\cdot^*$ is additive. For $X, Y \in M_n(A)$,

$$(X + Y)^* = \overline{(X + Y)^\top} = \overline{X^\top + Y^\top} = \overline{X^\top} + \overline{Y^\top} = X^* + Y^*.$$

2. We show that $\cdot^*$ is anti-multiplicative. For $X, Y \in M_n(A)$,

$$(XY)^* = \overline{(XY)^\top} = \overline{Y^\top X^\top} = \overline{Y^\top} \overline{X^\top} = Y^* X^*,$$

where the second equality uses the fact that the transpose reverses the order of multiplication and the third uses that σ is a ring homomorphism applied entrywise.

3. We show that $\cdot^*$ preserves the identity. The identity matrix I_n has entries δ_{ij} . Its conjugate transpose is

$$(I_n)^* = \overline{I_n^\top} = \overline{I_n} = I_n,$$

since $\overline{1_A} = 1_A$ and $\overline{0_A} = 0_A$.

4. We show that $\cdot^*$ is an involution. For $X \in M_n(A)$,

$$(X^*)^* = (\overline{X^\top})^* = \overline{(\overline{X^\top})^\top} = \overline{\overline{X}} = X,$$

where the last equality uses $\overline{\overline{a}} = a$ for all $a \in A$.

Thus $*$ is an involution on $M_n(A)$.

It remains to show that the embedding $\iota : A \rightarrow M_n(A)$, $a \mapsto aI_n$, is a homomorphism of involution rings whose image lies in the center of $M_n(A)$. The map ι is a ring homomorphism because $(a + b)I_n = aI_n + bI_n$, $(ab)I_n = (aI_n)(bI_n)$, and $1_A I_n = I_n$. Its image lies in the center since aI_n commutes with every matrix: $(aI_n)X = aX = X(aI_n)$ for all $X \in M_n(A)$. Finally, ι respects the involutions:

$$\iota(\overline{a}) = \overline{a}I_n = \overline{aI_n} = \overline{\iota(a)}^\top = \iota(a)^*,$$

where $\overline{aI_n}$ denotes the entrywise conjugate of the scalar matrix aI_n , which coincides with its conjugate transpose because aI_n is symmetric. Hence ι is a homomorphism of involution rings, and $M_n(A)$ is an involution A -algebra. $\square$

5.5. Special endomorphisms.

Definition 5.5.1. Let F be a field (Definition 3.1.16), and let V be an F -vector space (Definition 5.1.1).

A linear endomorphism $T : V \rightarrow V$ is called *nilpotent* if there exists a positive integer k such that $T^k = 0$. The smallest such k is called the *index of nilpotency of T* .

An $n \times n$ matrix (Definition 5.1.5) A over F is called *nilpotent* if $A^k = 0$ for some positive integer k .

(♠ TODO: AI generated, verify)

Proposition 5.5.2. Let F be a field (Definition 3.1.16), and let V be a nonzero finite-dimensional F -vector space (Definition 5.1.1).

If $T : V \rightarrow V$ is a nilpotent endomorphism (Definition 5.5.1), then the only eigenvalue of T is 0. Consequently, its characteristic polynomial is $\chi_T(t) = t^{\dim_F(V)}$, and its determinant and trace are both 0. (♠ TODO: AI generated, verify)

Definition 5.5.3. Let F be a field (Definition 3.1.16), and let V be an F -vector space (Definition 5.1.1).

A linear endomorphism $P : V \rightarrow V$ is called *idempotent* (or a *projection operator*) if $P^2 = P$.

An idempotent linear map is precisely an Definition 3.2.1 in the ring $\text{End}_F(V)$. (♠ TODO: AI generated, verify)

Proposition 5.5.4. Let F be a field (Definition 3.1.16), and let V be an F -vector space (Definition 5.1.1).

If $P : V \rightarrow V$ is an idempotent linear map (Definition 5.5.3), then

$$V = \ker(P) \oplus \text{im}(P).$$

Moreover, P acts as the identity on $\text{im}(P)$ and as zero on $\ker(P)$. Conversely, if $V = U \oplus W$, there exists a unique idempotent linear map whose image is U and kernel is W . This decomposition is the vector-space special case of the direct-sum decomposition induced by an Definition 3.2.1 in a ring: $R = R(1 - e) \oplus Re$ for $e^2 = e$. (♠ TODO: AI generated, verify)

5.6. Invariant subspaces and operator decompositions.

Definition 5.6.1. Let F be a field (Definition 3.1.16), let V be an F -vector space (Definition 5.1.1), and let $T : V \rightarrow V$ be a linear endomorphism.

A subspace $W \subseteq V$ is called *T -invariant* (or an *invariant subspace for T*) if $T(W) \subseteq W$. A T -invariant subspace is precisely a submodule of V when V is viewed as an $F[t]$ -module via the action $t \cdot v = T(v)$. Thus invariant subspaces are the special case of Definition 3.5.2 for the polynomial-ring module structure induced by T . (♠ TODO: AI generated, verify)

Theorem 5.6.2. Let V be a nonzero finite-dimensional vector space over $\mathbb{C}$, and let $T : V \rightarrow V$ be a linear endomorphism.

Then T has a one-dimensional invariant subspace (Definition 5.6.1); equivalently, T has an eigenvector. This follows from the fundamental theorem of algebra applied to the characteristic polynomial, and is equivalent to the existence of a submodule of rank 1 in the $F[t]$ -module structure on V induced by T . (♠ TODO: AI generated, verify)

Definition 5.6.3. Let F be a field (Definition 3.1.16), let V be a finite-dimensional F -vector space (Definition 5.1.1), and let $T : V \rightarrow V$ be a linear endomorphism. Let $\lambda \in F$.

A nonzero vector $v \in V$ is called a *generalized eigenvector of T corresponding to λ* if $(T - \lambda I)^k v = 0$ for some positive integer k .

The *generalized eigenspace of T corresponding to λ* is the subspace

$$V_\lambda = \ker((T - \lambda I)^{\dim_F(V)}) = \{v \in V : (T - \lambda I)^k v = 0 \text{ for some } k \geq 1\}.$$

The generalized eigenspace V_λ is the primary component of the $F[t]$ -module structure on V induced by T , corresponding to the irreducible factor $(t - \lambda)$. This is the special case for

vector spaces of the primary decomposition of a finitely generated module over a principal ideal domain. (♠ TODO: AI generated, verify)

Theorem 5.6.4 (Primary Decomposition Theorem). Let F be a field (Definition 3.1.16), and let V be a finite-dimensional F -vector space (Definition 5.1.1). Let $T : V \rightarrow V$ be a linear endomorphism whose characteristic polynomial factors as

$$\chi_T(t) = p_1(t)^{e_1} \cdots p_r(t)^{e_r},$$

where $p_1, \dots, p_r$ are distinct monic irreducible polynomials in $F[t]$.

Then

$$V = V_{p_1} \oplus \cdots \oplus V_{p_r},$$

where $V_{p_i} = \ker(p_i(T)^{e_i})$ are T -invariant subspaces (Definition 5.6.1). Each restriction $T|_{V_{p_i}}$ has characteristic polynomial $p_i(t)^{\dim_F(V_{p_i})}$. This is the vector-space special case of the primary decomposition theorem for finitely generated modules over a principal ideal domain, applied to the $F[t]$ -module structure on V induced by T . The subspaces V_{p_i} are precisely the primary submodules of this module. (♠ TODO: AI generated, verify)

5.7. Cayley-Hamilton and Jordan form.

Theorem 5.7.1 (Cayley–Hamilton Theorem). Let F be a field (Definition 3.1.16), and let V be a finite-dimensional F -vector space (Definition 5.1.1). Let $T : V \rightarrow V$ be a linear endomorphism, and let $\chi_T(t) \in F[t]$ be its characteristic polynomial.

Then $\chi_T(T) = 0$ as an endomorphism of V . The Cayley–Hamilton theorem is a special case of the determinant-of-endomorphism formulation for finitely generated Definition 3.5.1s: if M is a finitely generated R -module and $\phi : M \rightarrow M$ satisfies $r_1 m_1 + \cdots + r_k m_k = 0$ for generators m_i , then $(r_1) \cdots (r_k) \phi = 0$ on M . (♠ TODO: AI generated, verify)

Proposition 5.7.2. Let F be a field (Definition 3.1.16), and let V be a finite-dimensional F -vector space (Definition 5.1.1). Let $T : V \rightarrow V$ be a linear endomorphism.

The minimal polynomial $m_T(t)$ divides the characteristic polynomial $\chi_T(t)$. Moreover, m_T and χ_T have the same irreducible factors (equivalently, the same roots in any extension field of F). The minimal polynomial characterizes the $F[t]$ -module structure of V under T : it is the monic generator of the annihilator ideal $\text{Ann}_{F[t]}(V) \subseteq F[t]$. The fact that m_T divides χ_T and shares the same irreducible factors reflects the structure theorem for finitely generated modules over a principal ideal domain. (♠ TODO: AI generated, verify)

Definition 5.7.3. Let F be a field (Definition 3.1.16), let $\lambda \in F$, and let $k \geq 1$.

The *Jordan block of size k with eigenvalue λ* is the $k \times k$ matrix

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}.$$

A square matrix over F is in *Jordan normal form* (or is a *Jordan matrix*) if it is block diagonal with Jordan blocks as its diagonal blocks:

$$J = \text{diag}(J_{k_1}(\lambda_1), \dots, J_{k_r}(\lambda_r)).$$

Jordan blocks correspond to the cyclic submodule decomposition of V as an $F[t]$ -module via T : each Jordan block $J_k(\lambda)$ represents the module $F[t]/(t - \lambda)^k$. The Jordan normal form is the special case of the invariant-factor decomposition for finitely generated modules over a principal ideal domain when all irreducible factors are linear. (♠ TODO: AI generated, verify)

Theorem 5.7.4 (Jordan Canonical Form). Let F be an algebraically closed field (Definition 4.1.12), and let V be a finite-dimensional nonzero F -vector space (Definition 5.1.1). Let $T : V \rightarrow V$ be a linear endomorphism.

There exists an ordered basis $\mathcal{B}$ of V such that the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is in Jordan normal form (Definition 5.7.3). Moreover, the Jordan blocks appearing in this matrix are uniquely determined by T up to their ordering. The Jordan canonical form is the vector-space special case of the invariant-factor decomposition for finitely generated torsion modules over a principal ideal domain, applied to the $F[t]$ -Definition 3.5.1 structure on V induced by T , with all irreducible factors linear (guaranteed by algebraic closure). (♠ TODO: AI generated, verify)

Proposition 5.7.5. Let F be a field (Definition 3.1.16), and let V be a finite-dimensional nonzero F -vector space (Definition 5.1.1). Let $T : V \rightarrow V$ be a linear endomorphism.

The following are equivalent:

1. T is diagonalizable.
2. The minimal polynomial $m_T(t)$ splits into distinct linear factors over F ; that is, $m_T(t) = (t - \lambda_1) \cdots (t - \lambda_r)$ with λ_i pairwise distinct.
3. The vector space V decomposes as the direct sum of the eigenspaces of T .

Diagonalizability is equivalent to the $F[t]$ -Definition 3.5.1 structure on V being a direct sum of cyclic modules $F[t]/(t - \lambda_i)$, i.e., the annihilator ideal is square-free. This is the special case of the structure theorem for finitely generated modules over a principal ideal domain when all invariant factors are linear and distinct. (♠ TODO: AI generated, verify)

5.8. Inner product spaces and the spectral theorem.

Definition 5.8.1. Let V be an inner product space over $\mathbb{R}$ or $\mathbb{C}$.

A set of vectors $\{v_1, \dots, v_n\} \subseteq V$ is called *orthogonal* if $\langle v_i, v_j \rangle = 0$ for all $i \neq j$. It is called *orthonormal* if in addition $\|v_i\| = 1$ for all i (equivalently, $\langle v_i, v_j \rangle = \delta_{ij}$).

An orthonormal set that is also a basis (Definition 5.1.2) of V is called an *orthonormal basis for V* . Orthogonality with respect to an inner product is the vector-space special case of orthogonality defined by a sesquilinear form on modules over a ring with Definition 5.11.2. (♠ TODO: AI generated, verify)

Theorem 5.8.2 (Gram–Schmidt Orthogonalization). Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $(v_1, \dots, v_n)$ be a linearly independent ordered set of vectors in V .

There exists an orthonormal (Definition 5.8.1) ordered set $(e_1, \dots, e_n)$ such that for each $k = 1, \dots, n$,

$$\text{span}\{e_1, \dots, e_k\} = \text{span}\{v_1, \dots, v_k\}.$$

The vectors are constructed by the recurrence: set $u_1 = v_1$, $e_1 = u_1/\|u_1\|$, and for $k \geq 2$,

$$u_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j, \quad e_k = \frac{u_k}{\|u_k\|}.$$

The Gram–Schmidt process generalizes to orthogonalization of bases for free modules over a ring with Definition 5.11.2, provided the necessary norms are invertible. (♠ TODO: AI generated, verify)

Definition 5.8.3. Let V be an inner product space, and let W be a subspace of V .

The *orthogonal complement of W* is the subspace

$$W^\perp = \{v \in V : \langle v, w \rangle = 0 \text{ for all } w \in W\}.$$

The orthogonal complement is the vector-space special case of the annihilator submodule defined by a sesquilinear form on modules over a ring with Definition 5.11.2. (♠ TODO: AI generated, verify)

Proposition 5.8.4. Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let W be a subspace of V .

Then

$$V = W \oplus W^\perp.$$

Moreover, $\dim_F(W) + \dim_F(W^\perp) = \dim_F(V)$, and $(W^\perp)^\perp = W$. This orthogonal decomposition is the vector-space special case of the direct-sum decomposition induced by a nondegenerate sesquilinear form on modules over a ring with Definition 5.11.2. The finite-dimensionality assumption ensures $W^{\perp\perp} = W$, which fails in infinite dimensions. (♠ TODO: AI generated, verify)

Definition 5.8.5. Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $T : V \rightarrow V$ be a linear endomorphism.

The unique linear endomorphism $T^* : V \rightarrow V$ satisfying

$$\langle T(v), w \rangle = \langle v, T^*(w) \rangle \quad \text{for all } v, w \in V$$

is called the *adjoint of T* .

If $\mathcal{B}$ is an orthonormal basis of V , then $[T^*]_{\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}*}$, where the right-hand side denotes the conjugate transpose of the matrix. The adjoint operator is the vector-space special case of the adjoint of a module endomorphism with respect to a sesquilinear form over a ring with Definition 5.11.2. (♠ TODO: AI generated, verify)

Definition 5.8.6. Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $T : V \rightarrow V$ be a linear endomorphism.

- T is called *self-adjoint* (or *Hermitian*, in the complex case) if $T^* = T$.
- T is called *skew-adjoint* (or *skew-Hermitian*) if $T^* = -T$.
- T is called *unitary* (resp. *orthogonal*, in the real case) if $T^*T = TT^* = I_V$.
- T is called *normal* if $T^*T = TT^*$.

In terms of an orthonormal basis, these conditions correspond to the matrix being equal to its conjugate transpose, negative conjugate transpose, having conjugate transpose as inverse, or commuting with its conjugate transpose, respectively. These classifications of operators are special cases of elements in a ring with Definition 5.11.2: self-adjoint ($x^* = x$), skew-adjoint ($x^* = -x$), unitary ($x^*x = xx^* = 1$), and normal ($x^*x = xx^*$). (♠ TODO: AI generated, verify)

Theorem 5.8.7 (Spectral Theorem for Normal Operators). Let V be a finite-dimensional nonzero inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be a linear endomorphism.

The following are equivalent:

1. T is normal (Definition 5.8.6).
2. There exists an orthonormal basis (Definition 5.8.1) of V consisting entirely of eigenvectors of T .
3. The matrix of T with respect to some orthonormal basis is diagonal.

If moreover V is a real inner product space and T is self-adjoint (Definition 5.8.6), then the equivalent conditions hold and all eigenvalues of T are real. The spectral theorem for normal operators is the vector-space special case of the diagonalization theorem for normal elements in a C^* -algebra (a complete normed algebra with Definition 5.11.2). The finite-dimensional hypothesis ensures the existence of eigenvalues. (♠ TODO: AI generated, verify)

Proposition 5.8.8. Let V be a finite-dimensional nonzero inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be a self-adjoint (Definition 5.8.6) linear endomorphism.

The following are equivalent:

1. $\langle T(v), v \rangle > 0$ for all nonzero $v \in V$ (T is *positive definite*).
2. All eigenvalues of T are strictly positive.
3. There exists an invertible linear map $S : V \rightarrow V$ such that $T = S^*S$.

If one replaces " > 0 " with " ≥ 0 " and "strictly positive" with "nonnegative", the same equivalences hold for *positive semidefinite* operators. Positive definiteness of self-adjoint operators is the vector-space special case of positive elements in a ring with Definition 5.11.2. The factorization $T = S^*S$ generalizes to the Cholesky decomposition for matrices over such rings. (♠ TODO: AI generated, verify)

5.9. Bilinear forms.

Definition 5.9.1. Let F be a field (Definition 3.1.16), and let V be an F -vector space (Definition 5.1.1).

A map $B : V \times V \rightarrow F$ is called a **bilinear form on V** if it is linear in each argument separately; that is, for all $u, v, w \in V$ and $\lambda \in F$,

$$\begin{aligned} B(u + v, w) &= B(u, w) + B(v, w), & B(\lambda u, w) &= \lambda B(u, w), \\ B(u, v + w) &= B(u, v) + B(u, w), & B(u, \lambda v) &= \lambda B(u, v). \end{aligned}$$

This is a special case of a multilinear map (Definition 5.10.1) when V is viewed as an F -module (Definition 3.5.1).

A bilinear form on V corresponds to an F -linear map $V \rightarrow V^*$ via $v \mapsto B(v, -)$, which is a Definition 3.5.4 from V to its dual module. (♠ TODO: AI generated, verify)

Definition 5.9.2. Let F be a field (Definition 3.1.16), and let B be a bilinear form (Definition 5.9.1) on an F -vector space V .

- B is called **symmetric** if $B(v, w) = B(w, v)$ for all $v, w \in V$.
- B is called **skew-symmetric** (or **alternating**) if $B(v, w) = -B(w, v)$ for all $v, w \in V$. Equivalently, $B(v, v) = 0$ for all $v \in V$ (assuming $\text{char}(F) \neq 2$).

A symmetric bilinear form is a special case of a quadratic form (Definition 5.12.1) via the correspondence $Q(v) = B(v, v)$ (when $\text{char}(F) \neq 2$).

Symmetric and skew-symmetric bilinear forms on vector spaces are special cases of symmetric and skew-symmetric sesquilinear forms on modules over a ring with involution, relating to Definition 5.12.1 via the polarization identity. (♠ TODO: AI generated, verify)

Proposition 5.9.3. Let F be a field (Definition 3.1.16), and let V be an n -dimensional F -vector space (Definition 5.1.1). Let B be a bilinear form (Definition 5.9.1) on V , and let $\mathcal{B} = (v_1, \dots, v_n)$ be an ordered basis.

There exists a unique $n \times n$ matrix (Definition 5.1.5) M over F , with entries $M(i, j) = B(v_i, v_j)$, such that for all $x, y \in V$,

$$B(x, y) = [x]_{\mathcal{B}}^{\top} M [y]_{\mathcal{B}},$$

where $[x]_{\mathcal{B}}$ denotes the coordinate column vector of x in basis $\mathcal{B}$.

If $\mathcal{B}'$ is another ordered basis with change-of-basis matrix $P = [I]_{\mathcal{B}'}^{\mathcal{B}}$, then the matrix of B in $\mathcal{B}'$ is

$$M' = P^{\top} M P.$$

Moreover, B is symmetric (Definition 5.9.2) (resp. skew-symmetric) if and only if its matrix M is symmetric (resp. skew-symmetric) in every basis. The matrix representation of a bilinear form and its change-of-basis transformation $M' = P^{\top} M P$ is the special case for free F -modules of the matrix representation of a bilinear form on Definition 3.5.4s with chosen bases. (♠ TODO: AI generated, verify)

5.10. Multilinear maps.

Definition 5.10.1. 1. Let $R_0, \dots, R_k$ be (not necessarily commutative) rings (Definition 3.1.1). Let M_i be a $R_{i-1} - R_i$ -bimodule (Definition 3.5.1) for $i = 1, \dots, k$, and let N be an $R_0 - R_k$ -bimodule. A function $\Phi : M_1 \times \dots \times M_k \rightarrow N$ is called a **multilinear map** (or $R_0 - R_k$ -multilinear) if

- for each $j = 1, \dots, k$ and fixed $m_i \in M_i$ for $i \neq j$, the map $M_j \rightarrow N$ given by $m_j \mapsto \Phi(m_1, \dots, m_j, \dots, m_k)$ is a group homomorphism (Definition 2.1.17) and
- for all $m_i \in M_i$ for $i = 1, \dots, k$ and $r_j \in R_j$ where $j \in \{1, \dots, k-1\}$, we have

$$\Phi(m_1, \dots, m_j r_j, m_{j+1}, \dots, m_k) = \Phi(m_1, \dots, m_j r_j, r_j m_{j+1}, \dots, m_k).$$

- Φ is left R_0 -linear (Definition 3.5.4) in the first argument and right R_k -linear in the k th argument, i.e. for all $r_0 \in R_0$ and $r_k \in R_k$ we have

$$\Phi(r_0 m_1, m_2, \dots, m_{k-1}, m_k r_k) = r_0 \cdot \Phi(m_1, \dots, m_k) \cdot r_k.$$

2. Let R be a (not necessarily commutative) ring and let M be a two-sided R -module (Definition 3.5.4). A **multilinear form** is a multilinear map $M^r \rightarrow R$ (where M^r here is the set theoretic Cartesian product (Definition A.1.4), rather than a product of groups or modules) for some $r \geq 0$.

In particular, when R be a commutative ring (Definition 3.1.6), and M_i for $i = 1, \dots, k$ and N are R -modules, we may speak of a multilinear map $\Phi : M_1 \times \dots \times M_k \rightarrow N$. We may thus also speak of multilinear maps $M^r \rightarrow R$ for $r \geq 0$

Additionally, we may speak of **bilinear maps/forms**, **trilinear maps/forms**, etc.

Definition 5.10.2. Let R be a (not necessarily commutative) ring (Definition 3.1.1), M an two-sided R -module (Definition 3.5.4), $k \in \mathbb{N}$ and $\Phi : M^k \rightarrow R$ a multilinear form (Definition 5.10.1)

1. Φ is **symmetric** if for every permutation (Definition 2.1.48) $\sigma \in S_k$ and all $x_1, \dots, x_k \in M$,

$$\Phi(x_1, \dots, x_k) = \Phi(x_{\sigma(1)}, \dots, x_{\sigma(k)}).$$

2. Φ is **antisymmetric** if for all $\sigma \in S_k$ and all $x_1, \dots, x_k \in M$,

$$\Phi(x_{\sigma(1)}, \dots, x_{\sigma(k)}) = \text{sgn}(\sigma) \Phi(x_1, \dots, x_k),$$

where $\text{sgn}(\sigma)$ is the sign (Definition 2.1.49) of the permutation σ .

3. Φ is **alternating** if whenever $x_i = x_j$ for some $i \neq j$, $\Phi(x_1, \dots, x_k) = 0$. Every alternating form is antisymmetric, since if $x_i = x_j$ and we swap coordinates, both terms are zero.

For $k = 2$, these definitions specialize to **bilinear forms**; in particular:

- Φ is symmetric if $\Phi(x, y) = \Phi(y, x)$.
- Φ is antisymmetric if $\Phi(x, y) = -\Phi(y, x)$.
- Φ is alternating if $\Phi(x, x) = 0$ for all x .

Definition 5.10.3. Let $n \in \mathbb{N}$. The symmetric group (Definition 2.1.48) on n letters is denoted by $\mathfrak{S}_n$. For a two-sided module (Definition 3.5.1) M over a (not necessarily commutative) ring (Definition 3.1.1), this group acts (on the right) (Definition 2.1.55) on the tensor

power $M^{\otimes R^n}$ (Definition 3.5.62) by permutation of the tensor factors: on the pure tensors, the action is given by

$$(x_1 \otimes \cdots \otimes x_n) \cdot \sigma := x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}, \quad \sigma \in \mathfrak{S}_n, x_i \in M$$

and the action is extended linearly.

Definition 5.10.4. Let R be a (not necessarily commutative) ring (Definition 3.1.1), and let M an two-sided R -module (Definition 3.5.1).

1. The *symmetric power of M of degree n* , denoted by $S_R^n(M)$ or $\text{Sym}^n(M) = \text{Sym}_R^n(M)$, is the quotient two-sided module (Definition 3.5.24)

$$S_R^n(M) := M^{\otimes R^n} / I_{\text{sym}},$$

(Definition 3.5.62) where I_{sym} is the two-sided (Definition 3.5.1) submodule (Definition 3.5.2) of $M^{\otimes R^n}$ generated by (Definition 3.5.12) all elements of the form

$$x_1 \otimes \cdots \otimes x_n - (x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}), \quad x_i \in M, \sigma \in \mathfrak{S}_n.$$

(Definition 2.1.48) (Definition 5.10.3)

2. The *exterior power of M of degree n* , denoted by $\Lambda_R^n(M)$, is the quotient two-sided module (Definition 3.5.24)

$$\Lambda_R^n(M) := M^{\otimes R^n} / I_{\text{alt}},$$

where I_{alt} is two-sided submodule (Definition 3.5.2) of $M^{\otimes R^n}$ generated by (Definition 3.5.12) all elements of the form

$$x_1 \otimes \cdots \otimes x_n - \text{sgn}(\sigma) (x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}), \quad x_i \in M, \sigma \in \mathfrak{S}_n.$$

(Definition 2.1.48) (Definition 2.1.49)

In particular, we often speak of symmetric powers of exterior powers of modules over commutative rings (Definition 3.1.6) and even vector spaces (Definition 5.1.1) over fields (Definition 3.1.16).

5.11. Sesquilinear and Hermitian forms.

Definition 5.11.1. Let $\mathcal{C}$ be a category (Definition A.4.1). An *involution on an object $X \in \text{Ob}(\mathcal{C})$* is an automorphism $i : X \rightarrow X$ such that $i \circ i = \text{id}_X$.

Definition 5.11.2. Let R be a unital associative ring (Definition 3.1.1). An *involution on R* is a anti-ring homomorphism of order 2, i.e. a map $\sigma : R \rightarrow R$ satisfying the following conditions for all $a, b \in R$:

1. $\sigma(a + b) = \sigma(a) + \sigma(b)$;
2. $\sigma(ab) = \sigma(b)\sigma(a)$ (anti-homomorphism property);
3. $\sigma(1) = 1$;
4. $\sigma(\sigma(a)) = a$.

In many contexts, an involution may simply be notated by $\cdot \mapsto \bar{\cdot}$. A *ring with involution* is a ring equipped with an involution.

Equivalently, a ring with involution is a dagger object in the category of rings.

Definition 5.11.3. Let (A, σ) be a ring with involution (Definition 5.11.2).

1. Let M be a right A -module (Definition 3.5.1). A **σ -sesquilinear form** (or simply **sesquilinear form**) on M is a map $b : M \times M \rightarrow A$ such that

(a) b is bi-additive, i.e.

$$b(u + v, w) = b(u, w) + b(v, w) \text{ and } b(u, v + w) = b(u, v) + b(u, w)$$

for all $u, w \in M$, and

(b) b is conjugate-linear in the first variable and linear in the second variable, i.e.

$$b(ua_1, va_2) = \sigma(a_1) \cdot b(u, v) \cdot a_2$$

for all $u, v \in M$ and $a_1, a_2 \in A$.

2. Let M be a left A -module (Definition 3.5.1). A **σ -sesquilinear form** (or simply **sesquilinear form**) on M is a map $b : M \times M \rightarrow A$ such that

(a) b is bi-additive, i.e.

$$b(u + v, w) = b(u, w) + b(v, w) \text{ and } b(u, v + w) = b(u, v) + b(u, w)$$

for all $u, w \in M$, and

(b) b is linear in the first variable and conjugate-linear in the second variable, i.e.

$$b(a_1u, a_2v) = a_1 \cdot b(u, v) \cdot \sigma(a_2)$$

for all $u, v \in M$ and $a_1, a_2 \in A$.

Note that the difference between sesquilinear forms on right A -modules vs. left A -modules is which variable is linear and which variable is conjugate-linear. In case the module is both a left and a right A -module, one should more directly emphasize which variable is linear and which is conjugate-linear. [Knu91] discusses sesquilinear forms on right A -modules. We usually adopt this same convention by default.

We may denote the set of sesquilinear forms on M by $\text{Sesq}_A(M)$. It is a module over the center of A (Definition 3.1.7).

Definition 5.11.4. Let (A, σ) be a ring with involution (Definition 5.11.2) where σ is also denoted by $\bar{\cdot}$, let $\varepsilon \in Z(A)$ be an element such that $\varepsilon \cdot \bar{\varepsilon} = 1$, and let M be a (left or right) A -module (Definition 3.5.1). A **ε -hermitian form on M** (or simply **hermitian form on M** , especially when ε is understood or is just 1) is a σ -sesquilinear form (Definition 5.11.3) $\phi : M \times M \rightarrow A$ satisfying the symmetry condition:

$$\phi(y, x) = \varepsilon \cdot \overline{\phi(x, y)}$$

for all $x, y \in M$. The pair (M, ϕ) is often called a **hermitian module**.

1-hermitian forms are may simply be called **hermitian forms**, and (-1) -hermitian forms are called **skew-hermitian forms**.

As with sesquilinear forms (Definition 5.11.3) more generally, ε -Hermitian forms are either conjugate linear on the first variable and linear on the second variable or linear on the first variable and conjugate linear on the second variable, depending on whether M is regarded as a right A -module or a left A -module.

Definition 5.11.5. Let (A, σ) be a ring with involution (Definition 5.11.2). Let N be a one-sided A -module (Definition 3.5.1). The *twisted dual module*, denoted by notations such as N^* , (or sometimes $N^\vee$; not to be confused with the standard dual (Definition 5.1.4) $N^\vee$) or $\text{Hom}_A(N, A)_\sigma$, is the following:

1. In the case that N is a right A -module, it is the set of all right A -linear maps (Definition 3.5.4) $f : N \rightarrow A$, equipped with the right A -module structure given by

$$(f \cdot r)(n) = \sigma(r) \cdot f(n)$$

for all $r \in A$, $f \in N^\vee$, $n \in N$. Equivalently, it may be constructed as $\overline{N^\vee}$, where by $\bar{\cdot}$, we mean the opposite module and by $\cdot^\vee$, we mean the standard dual (Definition 5.1.4).

2. In the case that N is a left A -module, it is the set of all left A -linear maps (Definition 3.5.4) $f : N \rightarrow A$, equipped with the left A -module structure given by

$$(r \cdot f)(n) = f(n) \cdot \sigma(r)$$

for all $r \in A$, $f \in N^\vee$, $n \in N$. Equivalently, it may be constructed as $\overline{N^\vee}$, where by $\bar{\cdot}$, we mean the opposite module and by $\cdot^\vee$, we mean the standard dual (Definition 5.1.4).

Twisted dual module constructions thus yield functors

$$\begin{aligned} \text{Mod}_A^{\text{op}} &\rightarrow \text{Mod}_A \\ {}_A \text{Mod}^{\text{op}} &\rightarrow {}_A \text{Mod}. \end{aligned}$$

(Definition A.4.12) (Definition 3.5.6)

Given a morphism $h : M \rightarrow N$ of right or left A -modules, we may write h^* for the morphism $h^* : N^* \rightarrow M^*$ in functorial (Definition A.4.5) fashion.

The construction $N^* = \overline{N^\vee}$ is naturally isomorphic (Definition A.4.6) to $(\overline{N})^\vee$.

Definition 5.11.6. Let (A, σ) be a ring with involution (Definition 5.11.2) and M a right A -module (Definition 3.5.1) equipped with a sesquilinear form (Definition 5.11.3) $\phi : M \times M \rightarrow A$. Let

$$\hat{\phi} : M \rightarrow M^*, \quad x \mapsto \phi(x, \cdot)$$

be the induced (left) adjoint map (Definition 5.11.7).

1. The form ϕ is said to be *nondegenerate* if $\hat{\phi}$ is injective.

In the case where ϕ is an ε -hermitian form (Definition 5.11.4), this condition is equivalent to requiring that $\phi(x, y) = 0$ for all $y \in M$ implies $x = 0$.

2. The form ϕ is said to be *nonsingular* or *regular* if $\hat{\phi}$ is an isomorphism.

Definition 5.11.7 ([Knu91, Chapter I, 2.2]). Let (A, σ) be a ring with involution (Definition 5.11.2) and M a right/left A -module (Definition 3.5.1) equipped with a sesquilinear form (Definition 5.11.3) $b : M \times M \rightarrow A$.

1. The *left adjoint map* is the assignment

$$\hat{b}_l : u \mapsto (v \mapsto b(u, v)).$$

In the case that M is a right A -module, it is a homomorphism morphism of right A -modules:

$$\hat{b}_l : M \rightarrow M^*, \quad u \mapsto (v \mapsto b(u, v))$$

where $M^* = \text{Hom}_A(M, A)_\sigma$ is the twisted dual module (Definition 5.11.5).

2. The **right adjoint map** is the assignment

$$\hat{b}_r : v \mapsto (u \mapsto b(u, v)).$$

In the case that M is a left A -module, it is a homomorphism morphism of left A -modules:

$$\hat{b}_r : M \rightarrow M^*, \quad v \mapsto (u \mapsto \sigma(b(u, v)))$$

When we simply talk about the **adjoint map** of b on the right/left A -module M , we will mean the left/right adjoint respectively and denote it by $\hat{b}$ or by h_b .

5.12. Quadratic forms.

Definition 5.12.1 (Quadratic form over a ring with involution). Let R be a ring with involution (Definition 5.11.2).

1. A **quadratic form on a right R -module M** is a map $q : M \rightarrow R$ such that:
 - (a) $q(rm) = rq(m)\bar{r}$ for all $r \in R, m \in M$;
 - (b) The map $B_q : M \times M \rightarrow R$ defined by $B_q(x, y) = q(x + y) - q(x) - q(y)$ is Hermitian.
2. A **quadratic form on a left R -module N** is a map $q : N \rightarrow R$ such that:
 - (a) $q(rn) = \bar{r}q(n)r$ for all $r \in R, n \in N$;
 - (b) The map $B_q : N \times N \rightarrow R$ defined by $B_q(x, y) = q(x + y) - q(x) - q(y)$ is Hermitian.

In either case, the form B_q is called the **polarization of q** or the **associated Hermitian form**, which may also be denoted by $\langle x, y \rangle_R = B_q(x, y)$.

A **quadratic module** refers to a module equipped with a quadratic form.

Definition 5.12.2 (Isometry of quadratic modules). Let R be a ring with involution (Definition 5.11.2) and $(M, q), (M', q')$ quadratic modules (Definition 5.12.1) over R (right or left modules consistently).

An **isometry** $\phi : M \rightarrow M'$ is an R -module isomorphism (Definition 3.5.4) such that $q'(\phi(m)) = q(m)$ for all $m \in M$.

Two quadratic modules (M, q) and (M', q') are **isometric**, written $(M, q) \cong (M', q')$, if there exists an isometry $\phi : M \rightarrow M'$.

Definition 5.12.3 (Non-degenerate quadratic form). Let R be a ring with involution (Definition 5.11.2). Let $q : M \rightarrow R$ be a quadratic form (Definition 5.12.1). The quadratic form $q : M \rightarrow R$ is non-degenerate if for every $x \in M$, the map $y \mapsto \langle x, y \rangle_R$ is zero only if $x = 0$.

Definition 5.12.4 (Quadratic form on a module over a commutative ring). Let R be a commutative ring (Definition 3.1.6). A **quadratic form on a R -module M** is a map $q : M \rightarrow R$ satisfying:

1. $q(rm) = r^2q(m)$ for all $r \in R, m \in M$;
2. The map $b_q : M \times M \rightarrow R$ given by $b_q(x, y) = q(x + y) - q(x) - q(y)$ is R -bilinear (Definition 5.10.1).

The associated bilinear form b_q is called the **polarization of q** . It is also often written as B_q . One also often write $\langle x, y \rangle = B_q(x, y)$. Equivalently, a quadratic form on an R -module M for a commutative (unital) ring R is equivalent to a quadratic form (Definition 5.12.1) on M where R is considered as an involution ring (Definition 5.11.2) under the identity map.

A **quadratic module** refers to a module equipped with a quadratic form.

Definition 5.12.5. Let R be a commutative ring (Definition 3.1.6), and let M be a quadratic module (Definition 5.12.4).

An *isometry* $\phi : M \rightarrow M'$ is an R -module isomorphism such that $q'(\phi(m)) = q(m)$ for all $m \in M$.

Two quadratic modules (M, q) and (M', q') are *isometric*, written $(M, q) \cong (M', q')$, if there exists an isometry $\phi : M \rightarrow M'$.

Equivalently, these notions can be defined via Definition 5.12.2 where R is considered as an involution ring with the identity morphism.

Definition 5.12.6 (Isotropic and hyperbolic). Let R be a commutative ring (Definition 3.1.6). Let $q : M \rightarrow R$ be a quadratic form (Definition 5.12.4) on a module (Definition 3.5.1) over R .

1. An element $x \in M$ is **isotropic** if $q(x) = 0$.
2. A subspace $L \subset M$ is **isotropic** if $q(L) = 0$.
3. The quadratic form q is **anisotropic** if it has no non-zero isotropic vectors, i.e., $q(x) = 0$ implies $x = 0$.

Definition 5.12.7. Let R be a commutative ring (Definition 3.1.6). Let $q : M \rightarrow R$ be a quadratic form (Definition 5.12.4) on a module (Definition 3.5.1) over R .

The quadratic form q is called **non-degenerate** if $B_q(x, -) = 0$ implies $x = 0$. Equivalently, q is nondegenerate if it is nondegenerate (Definition 5.12.3) when considered as a quadratic form (Definition 5.12.1) on the module M where R is considered as an involution ring with the identity map.

Definition 5.12.8. Let R be a commutative ring (Definition 3.1.6). A **hyperbolic plane** is a quadratic module (Definition 5.12.4) isometric (Definition 5.12.5) to R^2 (Definition 3.5.22) equipped with the quadratic form given by $q(x, y) = xy$.

Lemma 5.12.9 (Properties of the hyperbolic plane). Let (H, q_H) be a hyperbolic plane (Definition 5.12.8) over a commutative ring (Definition 3.1.6) R . Then:

1. The polarization (Definition 5.12.4) is $B_{q_H}((x_1, y_1), (x_2, y_2)) = x_1y_2 + y_1x_2$.
2. The vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ satisfy $q_H(e_1) = q_H(e_2) = 0$ and $B_{q_H}(e_1, e_2) = 1$.
♠ TODO: symplectic basis
3. H admits a symplectic basis $\{e_1, e_2\}$ where $B_{q_H}(e_i, e_j) = \delta_{i,3-j}$.

Proposition 5.12.10 (Isomorphism criterion). Let R be a commutative ring (Definition 3.1.6), M a module (Definition 3.5.1) of R , $H \cong R^2$ a submodule (Definition 3.5.2) of M , and $q : M \rightarrow R$ a quadratic form (Definition 5.12.4). Then $(H, q|_H)$ is a hyperbolic plane (Definition 5.12.8) if and only if $q|_H \cong q_H$ as quadratic forms via some basis of H .

Proposition 5.12.11 (Witt cancellation). Let R be a field (Definition 3.1.16) and $q : M \rightarrow R$, $q' : M' \rightarrow R$ quadratic forms (Definition 5.12.4) such that $(M, q) \cong (M', q') \oplus (H, q_H)$ as quadratic modules, where (H, q_H) is a hyperbolic plane (Definition 5.12.8). Then $(M, q) \cong (M', q')$ (Definition 5.12.5).

Theorem 5.12.12 (Witt decomposition theorem). Let R be a field (Definition 3.1.16) and $q : M \rightarrow R$ a non-degenerate (Definition 5.12.7) quadratic form (Definition 5.12.4). Then there exists a unique (up to isomorphism) decomposition $M \cong M_a \oplus H^k$ where M_a is anisotropic (Definition 5.12.6) and H^k is a direct sum of k hyperbolic planes (Definition 5.12.8). ♠ TODO: direct sum of quadratic modules

The integer k is called the **Witt index of q** .

Corollary 5.12.13 (Dimension formula). Let R be a field (Definition 3.1.16) and $q : M \rightarrow R$ a non-degenerate (Definition 5.12.7) quadratic form (Definition 5.12.4). We have $\dim M \equiv \dim M_a \pmod{2}$.

6. MORE GROUP THEORY

6.1. Topological groups.

Definition 6.1.1 (Topological groups). A **topological group** is a group (Definition 2.1.1) $(G, \cdot)$ together with a topology (Definition A.3.1) $\mathcal{T}$ on G such that the maps

$$\begin{aligned} \mu : G \times G &\rightarrow G, & (g, h) &\mapsto g \cdot h, \\ \iota : G &\rightarrow G, & g &\mapsto g^{-1}, \end{aligned}$$

are continuous (Definition A.3.2) with respect to the product topology on $G \times G$ and the topology $\mathcal{T}$ on G .

A **abelian topological group** or a **commutative topological group** is a topological group that is abelian as a group (Definition 2.1.1).

Definition 6.1.2 (Topological rings). ♠ TODO: Product topology A **topological ring** is a (not necessarily commutative) ring (Definition 3.1.1) $(R, +, \cdot)$ together with a topology

(Definition A.3.1) $\mathcal{T}$ on R such that the maps

$$\begin{aligned} + : R \times R &\rightarrow R & (a, b) &\mapsto a + b \\ \cdot : R \times R &\rightarrow R & (a, b) &\mapsto a \cdot b \\ - : R &\rightarrow R & a &\mapsto -a \end{aligned}$$

are continuous (Definition A.3.2) with respect to the product topology on $R \times R$ and the topology $\mathcal{T}$ on R . Equivalently, a topological ring is a ring whose addition structure forms a topological group (Definition 6.1.1) and whose multiplication structure is also continuous.

Definition 6.1.3 (Topological fields). A *topological field* is a field (Definition 3.1.16) $(F, +, \cdot)$ together with a topology (Definition A.3.1) $\mathcal{T}$ on F such that the maps

$$\begin{aligned} + : F \times F &\rightarrow F & (a, b) &\mapsto a + b \\ \cdot : F \times F &\rightarrow F & (a, b) &\mapsto a \cdot b \\ - : F &\rightarrow F & a &\mapsto -a \\ \cdot^{-1} : F^\times &\rightarrow F^\times & a &\mapsto a^{-1} \end{aligned}$$

are continuous (Definition A.3.2) with respect to the product topology on $F \times F$, the topology $\mathcal{T}$ on F , and the subset topology on the unit group $F^\times = F - \{0\}$ (Definition 3.1.11). Equivalently, a topological field is a field with a topology which becomes a topological ring (Definition 6.1.2) under the ring structure and whose multiplicative inverse is also continuous.

Definition 6.1.4 (Topological fields). (♠ TODO: Product topology) Let $(F, +, \cdot)$ be a topological field (Definition 6.1.3). A *topological vector space over F* or an *F -topological vector space* is an F -vector space $(V, +, \cdot)$ together with a topology (Definition A.3.1) $\mathcal{T}_V$ on V such that the maps

$$\begin{aligned} + : V \times V &\rightarrow V & (v, w) &\mapsto v + w \\ \cdot : F \times V &\rightarrow V & (a, v) &\mapsto a \cdot v \\ - : V &\rightarrow V & v &\mapsto -v \end{aligned}$$

are continuous (Definition A.3.2) with respect to the product topologies on $V \times V$ and $F \times V$ and the topology $\mathcal{T}_V$ on V . Equivalently, a topological vector space is a vector space over a topological field with a topology which becomes a topological group (Definition 6.1.1) under the additive structure and whose scalar multiplication map is continuous.

6.2. Profinite groups.

Definition 6.2.1 (Profinite groups). A *profinite group* is a topological group (Definition 6.1.1) that is compact (Definition A.3.3), Hausdorff, and totally disconnected.

Equivalently, a profinite group is the inverse limit (Definition A.4.19) of an inverse system (Definition A.4.18) of finite groups equipped with the discrete topology. That is,

$$G \cong \varprojlim_{i \in I} G_i,$$

where each G_i is a finite group and the connecting maps are group homomorphisms. The topology on G is called the **profinite topology** and is the roughest topology making the projection maps $G \rightarrow G_i$ continuous (Definition A.3.2); equivalently, the preimages of singletons under the projection maps $G \rightarrow G_i$ form a basis of open subsets (and in fact a basis of clopen subsets). Also equivalently, the profinite topology of G is equivalent to the profinite topology of G considered as a profinite set.

Definition 6.2.2. The **category of profinite groups** refers to the category whose objects are profinite groups (Definition 6.2.1) and whose morphisms are continuous homomorphisms (Definition 2.1.17) between them.

Definition 6.2.3. Let G be a group (Definition 2.1.1). The **profinite completion of G** , denoted by $\widehat{G}$, is the inverse limit (Definition A.4.20) of the inverse system (Definition A.4.18) consisting of the quotient groups (Definition 2.1.40) G/N where N runs over all normal subgroups (Definition 2.1.36) of finite index (Definition 2.1.27) in G , ordered by inclusion. In other words,

$$\widehat{G} = \varprojlim_{N \in \mathcal{N}} G/N$$

where $\mathcal{N} = \{N \trianglelefteq G : [G : N] < \infty\}$.

There exists a canonical homomorphism $\iota : G \rightarrow \widehat{G}$ defined by $g \mapsto (gN)_{N \in \mathcal{N}}$, where $\mathcal{N}$ is the set of normal subgroups of finite index in G .

Proposition 6.2.4. Let G be a group (Definition 2.1.1) and $\widehat{G}$ its profinite completion (Definition 6.2.3). The group $\widehat{G}$, equipped with the profinite topology (where each G/N is discrete), is a profinite group (Definition 6.2.1).

Theorem 6.2.5. Let G be a group (Definition 2.1.1) and $\widehat{G}$ its profinite completion (Definition 6.2.3) with the canonical map $\iota : G \rightarrow \widehat{G}$. For every profinite group (Definition 6.2.1) H and every continuous homomorphism (Definition 2.1.17) $f : G \rightarrow H$ (where G is given the discrete topology), there exists a unique continuous homomorphism $\hat{f} : \widehat{G} \rightarrow H$ such that $\hat{f} \circ \iota = f$.

In other words, the functor (Definition A.4.5) $G \mapsto \widehat{G}$ from the category of groups (Definition 2.1.18) to the category of profinite groups (Definition 6.2.2) is left adjoint (Definition A.4.11) to the inclusion functor $\widehat{G} \hookrightarrow G$.

Proposition 6.2.6. (♠ TODO:) Let G be a group and $\iota : G \rightarrow \widehat{G}$ be the canonical map to its profinite completion. The map ι is injective if and only if G is residually finite.

Definition 6.2.7. Let G be a profinite group. Let $S \subseteq G$ be a subset. The **subgroup of G topologically/profinutely generated by S** refers to the closure in G of the subgroup algebraically generated by S (Definition 2.1.22).

Definition 6.2.8. (♠ TODO: dense) Let G be a profinite group (Definition 6.2.1). It is said to be **topologically finite generated** if there exists a finite subset $S \subseteq G$ that topologically generates G , i.e. $\overline{\langle S \rangle} = G$ or equivalently such that the subgroup algebraically generated by S (Definition 2.1.22) is dense in G .

Definition 6.2.9. A profinite group (Definition 6.2.1) G is said to be *strongly complete* if the following equivalent conditions hold:

1. Every subgroup (Definition 2.1.19) of finite index (Definition 2.1.27) in G is open (Definition A.3.1).
2. G equals its own profinite completion (Definition 6.2.3).
3. Every group homomorphism (Definition 2.1.17) from G to any profinite group is continuous.

Theorem 6.2.10 ([NS07, Theorem 1.1.]). Every topologically finitely generated (Definition 6.2.8) profinite group (Definition 6.2.1) is strongly complete (Definition 6.2.9).

APPENDIX A. MISCELLANEOUS DEFINITIONS

A.1. Set Theory.

Definition A.1.1 (Small Set). A *small set* refers to a set; the adjective of *small* is used to emphasize that the set is a set, rather than a proper class.

Alternatively, a *small set* is any object that belongs to a fixed universe of sets $\mathcal{U}$, called the *universe of small sets*. In other words, a set A is said to be *small* if and only if $A \in \mathcal{U}$. Elements of $\mathcal{U}$ may themselves be sets, but $\mathcal{U}$ is not assumed to be a set; it may be a proper class.

Definition A.1.2 (Subset). Let A and B be sets. The set A is said to be a *subset of B* , written as $A \subseteq B$ or $A \subset B$, if every element of A is also an element of B , that is,

$$A \subseteq B \iff \forall x (x \in A \Rightarrow x \in B).$$

If $A \subseteq B$ and $A \neq B$, then A is called a *proper subset of B* ; this is commonly denoted by $A \subsetneq B$.

Definition A.1.3. Let X and Y be sets and let $f : X \rightarrow Y$ be a function.

- The function f is said to be *injective* (or *one-to-one*) if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- The function f is said to be *surjective* (or *onto*) if for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- The map f is *bijective* if it is both injective and surjective. In this case, there exists a unique *inverse map* $f^{-1} : Y \rightarrow X$ such that for all $x \in X$ and $y \in Y$,

$$f^{-1}(f(x)) = x \text{ and } f(f^{-1}(y)) = y.$$

By the definition of cardinality (Definition A.1.13),

Definition A.1.4 (Product of Sets). Let I be a (possibly infinite (Definition A.1.5) but small (Definition A.1.1)) index set and let $\{A_i\}_{i \in I}$ be a family of sets indexed by I . The *Cartesian product of the family $\{A_i\}_{i \in I}$* , denoted by $\prod_{i \in I} A_i$, is defined as the set of all tuples/functions (Definition A.1.6)

$$\prod_{i \in I} A_i := \{(a_i)_{i \in I} \mid a_i \in A_i \text{ for all } i \in I\},$$

where $(a_i)_{i \in I}$ denotes a function from I to $\bigcup_{i \in I} A_i$ such that $(a_i)_{i \in I}(i) = a_i \in A_i$ for each $i \in I$.

The self product of a set A indexed by I is often denoted by A^I . Note that elements of A^I can be identified with functions (Definition A.1.6) $I \rightarrow A$. The finite self product of A taken n times is often denoted by A^n . For finitely many sets $A_1, \dots, A_n$, their Cartesian product is denoted by $A_1 \times \dots \times A_n$. Elements of such a finite product may be written as $(a_1, \dots, a_n)$.

Definition A.1.5. Let A be a set.

- The set A is said to be **countably infinite** (or simply **countable**) if there exists a bijection $f : \mathbb{N} \rightarrow A$.
- The set A is said to be **finite** if there exists some $n \in \mathbb{N}$ and a bijection $g : \{1, 2, \dots, n\} \rightarrow A$.
- The set A is said to be **at most countable** if it is either finite or countably infinite.
- The set A is said to be **uncountable** if it is not at most countable.

Definition A.1.6. Let X and Y be sets. A **map** (or **function**) from X to Y is a rule f assigning to each element $x \in X$ exactly one element $f(x) \in Y$. We write $f : X \rightarrow Y$.

One also says that a function f is an “assignment between sets” that is **well-defined**. In practice, to prove the **well-definedness of a function** is to show that that an assignment is indeed a function, i.e. that f associates to x one and only one element of Y .

The element $f(x)$ of Y is called the **image of x under f** .

Equivalently, a function $f : X \rightarrow Y$ is identified with a binary relation (Definition A.1.8) $f \subseteq X \times Y$ such that for every element $x \in X$, there is exactly one element $y_x \in Y$ such that $(x, y_x) \in f$.

We say that X is the **domain** and that Y is the **codomain of f** . We say that f **maps into Y** .

Definition A.1.7. Let X be a set. The **identity function on X** , denoted by id_X , is the function (Definition A.1.6) $\text{id}_X : X \rightarrow X$ defined by

$$\text{id}_X(x) = x \quad \text{for all } x \in X.$$

It is the unique function on X satisfying $f \circ \text{id}_X = f = \text{id}_X \circ f$ for every function $f : X \rightarrow Y$ and every function $f : Y \rightarrow X$.

Definition A.1.8. Let n be a positive integer. Let $X_1, X_2, \dots, X_n$ be sets. An **n -ary relation** on these sets is a subset $R \subseteq X_1 \times X_2 \times \dots \times X_n$. The integer n is called the **arity** (or **degree**) of the relation. The sets $X_1, \dots, X_n$ are called the **domains of the relation**.

If $(x_1, x_2, \dots, x_n) \in R$, we say that the elements $x_1, \dots, x_n$ are **related** by R .

In the special case where $X_1 = X_2 = \dots = X_n = X$, we say that R is an **n -ary relation on the set X** . In this case, $R \subseteq X^n$.

Specific arities have standard names:

- A **unary relation** on X is a subset of X (arity $n = 1$).

- A **binary relation** from X to Y is a subset of $X \times Y$ (arity $n = 2$).
- A **ternary relation** is a subset of $X \times Y \times Z$ (arity $n = 3$).

Definition A.1.9. An **equivalence relation on a set X** is a binary relation (Definition A.1.8) on X that is reflexive, symmetric, and transitive.

Definition A.1.10. If $\sim$ is an equivalence relation on a set X and $x \in X$, the **equivalence class of x** , denoted by $[x]$ or $[x]_\sim$, is the set defined by

$$[x] = \{y \in X \mid x \sim y\}.$$

The set of all equivalence classes is called the **quotient set of X by $\sim$** , denoted by $X/\sim$.

Definition A.1.11. A **partition of a set X** is a collection $\mathcal{P}$ of nonempty subsets of X such that:

1. The union of all subsets in $\mathcal{P}$ is X : $\bigcup_{A \in \mathcal{P}} A = X$.
2. Any two distinct subsets in $\mathcal{P}$ are disjoint: if $A, B \in \mathcal{P}$ with $A \neq B$, then $A \cap B = \emptyset$.

The subsets in $\mathcal{P}$ are called the **blocks** (or **parts** or **cells**) of the partition.

Lemma A.1.12. Let X be a set.

1. If $\sim$ is an equivalence relation (Definition A.1.9) on X , then the quotient set (Definition A.1.10) $X/\sim$ is a partition (Definition A.1.11) of X .
2. If $\mathcal{P}$ is a partition (Definition A.1.11) of X , then the relation $\sim_{\mathcal{P}}$ on X defined by $x \sim_{\mathcal{P}} y \iff \exists A \in \mathcal{P}$ such that $x, y \in A$ is an equivalence relation (Definition A.1.9) on X , and its quotient set $X/\sim_{\mathcal{P}}$ is canonically identified with $\mathcal{P}$.

These constructions are mutually inverse: starting from an equivalence relation, forming its partition, and then forming the equivalence relation from that partition recovers the original equivalence relation; and starting from a partition, forming its equivalence relation, and then forming the partition from that equivalence relation recovers the original partition.

Definition A.1.13. An ordinal number κ is a **cardinal number** (or simply a **cardinal**) if for every ordinal $\alpha < \kappa$, there is no bijection (Definition A.1.3) between α and κ . Equivalently, a cardinal is an initial ordinal—an ordinal that is not equinumerous with any smaller ordinal.

The **cardinality of an arbitrary set X** , denoted by $|X|$, $\text{card}(X)$, or $\#X$, is the unique cardinal number κ such that there exists a bijection (Definition A.1.3) between X and κ . (The existence of such a κ for every set requires the Axiom of Choice).

A.2. Algebraic definitions.

Definition A.2.1 (Semigroup). A **semigroup** is a pair $(M, \cdot)$ where M is a set and $\cdot : M \times M \rightarrow M$ is an associative (Definition 1.0.4) binary operation (Definition 1.0.1), i.e. $\cdot$ satisfies the following:

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in M.$$

A semigroup is said to be **commutative** or **abelian** if $\cdot$ is also commutative (Definition A.2.3), i.e. $\cdot$ satisfies the following:

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in M.$$

Definition A.2.2 (Monoid). A **monoid** is a semigroup (Definition A.2.1) $(M, \cdot)$ such that there exists an identity element (Definition 1.0.2) $e \in M$ for $\cdot$, i.e. e has the property:

$$e \cdot a = a \cdot e = a \quad \text{for all } a \in M.$$

Definition A.2.3. Let $\cdot : S \times S \rightarrow S$ be a binary operation (Definition 1.0.1) on a set. It is said to be **commutative** if

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in S.$$

A.3. Topological definitions.

Definition A.3.1 (Topology). Let X be a set. A **topology on X** is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. For any collection $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ (with I arbitrary), the union $\bigcup_{i \in I} U_i \in \mathcal{T}$,
3. For any finite collection $\{U_1, \dots, U_n\} \subseteq \mathcal{T}$, the intersection $U_1 \cap \dots \cap U_n$ is in $\mathcal{T}$.
Equivalently, for any $U_1, U_2 \in \mathcal{T}$, the intersection $U_1 \cap U_2$ is in $\mathcal{T}$

If $\mathcal{T}$ is a topology on X , the pair $(X, \mathcal{T})$ is called a **topological space**. Members of $\mathcal{T}$ are called **open sets**.

A subset $C \subseteq X$ is **closed** (also called a **closed subset**) if its complement $X \setminus C$ is an open set in $\mathcal{T}$.

One very often refers to X as a topological space, omitting the notation of the topology $\mathcal{T}$.

The collection of all topologies on a set X may be denoted by notations such as **Top(X)**, **Top(X)**, or **Top(X)**.

Definition A.3.2. Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open subsets, and let $k \in \mathbb{N}_0 \cup \{\infty\}$. A **continuous morphism** (or a **continuous map** or simply a **map between topological spaces**) from U to V is a function $f : U \rightarrow V$ that satisfies one of the following equivalent characterizations:

1. f is a continuous map from U to V as topological spaces.
2. for every point $x \in U$, for every open neighborhood W of $f(x)$ in V , there exists an open neighborhood O of x in U satisfying

$$f(O) \subseteq W.$$

We write **$C(U, V) = C^0(U, V)$** for the set of continuous maps $U \rightarrow V$.

Definition A.3.3 (Compact topological space). A topological space $(X, \mathcal{T})$ is **compact** if every open cover of X admits a finite subcover; that is, for every collection $\{U_i\}_{i \in I}$ of open sets in $\mathcal{T}$ such that $X = \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ such that $X = \bigcup_{j=1}^n U_{i_j}$.

Some mathematicians, e.g. algebraic geometers, would refer to this property as **quasi-compactness**.

A.4. Categorical definitions.

Definition A.4.1 (Category). A *category* $\mathcal{C}$ consists of the following data:

- A class of *objects* denoted $\text{Ob}(\mathcal{C})$.
- For each pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, a class

$$\text{Hom}_{\mathcal{C}}(X, Y)$$

of *morphisms* (also called *arrows* or *homs*). If the category $\mathcal{C}$ is clear, then this *hom-class* is also denoted by $\text{Hom}(X, Y)$. It may also be denoted by $\text{hom}_{\mathcal{C}}(X, Y)$ or $\text{hom}(X, Y)$, especially to distinguish from other types of hom's (e.g. internal hom's)

- For each triple of objects X, Y, Z , a composition law

$$\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z),$$

denoted $(g, f) \mapsto g \circ f$.

- For each object X , an *identity morphism*

$$\text{id}_X \in \text{Hom}_{\mathcal{C}}(X, X).$$

These data satisfy the following axioms:

- (Associativity) For all morphisms $f \in \text{Hom}_{\mathcal{C}}(X, Y)$, $g \in \text{Hom}_{\mathcal{C}}(Y, Z)$, and $h \in \text{Hom}_{\mathcal{C}}(Z, W)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- (Identity) For all $f \in \text{Hom}_{\mathcal{C}}(X, Y)$,

$$\text{id}_Y \circ f = f = f \circ \text{id}_X.$$

One often writes $X \in \mathcal{C}$ synonymously with $X \in \text{Ob}(\mathcal{C})$, i.e. to denote that X is an object of $\mathcal{C}$.

We may call a category as above an *ordinary category* to distinguish this notion from the notions of *categories enriched in monoidal categories* or higher/ n -categories. (🔥 TODO: define n -categories)

A category as defined above may be called called a *large category* or a *class category* to emphasize that the hom-classes may be proper classes rather than sets (note, however, that the possibility that hom-classes are sets is not excluded for large categories). Accordingly, a *category* may often refer to a locally small category (Definition A.4.3), which is a category whose hom-classes are all sets.

Definition A.4.2. Let $\mathcal{C}$ be a category (Definition A.4.1) and X be an object of $\mathcal{C}$.

1. An *automorphism of X* is a morphism $f : X \rightarrow X$ that is an isomorphism. That is, there exists a morphism $g : X \rightarrow X$ such that $f \circ g = \text{id}_X$ and $g \circ f = \text{id}_X$.
2. Assume that $\mathcal{C}$ is a locally small category (Definition A.4.3). The *automorphism group of X* , denoted $\text{Aut}_{\mathcal{C}}(X)$, is the set of all automorphisms of X . It is indeed a group (Definition 2.1.1) (Proposition 2.5.2) and the group operation is the composition of morphisms in $\mathcal{C}$.

Definition A.4.3 (Locally small category). A (large) category (Definition A.4.1) $\mathcal{C}$ is called a *locally small category* if for every pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, the collection $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a (small (Definition A.1.1)) *set* (as opposed to a proper class). In other words, each hom-class is a set and may even be called a *hom-set*.

In some contexts, a locally small category may simply be called a *category*, especially when genuinely large categories are not considered.

A category $\mathcal{C}$ is called a *small category* if it is a locally small category and the class $\text{Ob}(\mathcal{C})$ of objects is a set.

Remark A.4.4. Many “concrete” categories considered in “classical mathematics” or outside of more “abstract” category theory tend to be locally small. For example, the categories of sets, groups, R -modules, vector spaces, topological spaces, schemes, manifolds, sheaves on “small enough” sites are all locally small.

Definition A.4.5. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition A.4.1).

1. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ (from $\mathcal{C}$ to $\mathcal{D}$) consists of :
 - For each object X in $\mathcal{C}$, an object $F(X)$ in $\mathcal{D}$.
 - For each morphism $f : X \rightarrow Y$ in $\mathcal{C}$, a morphism $F(f) : F(X) \rightarrow F(Y)$ in $\mathcal{D}$,
such that:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for all objects } X \text{ in } \mathcal{C},$$

$$F(g \circ f) = F(g) \circ F(f) \quad \text{for all } X, Y, Z \in \text{Ob}(\mathcal{C}) \text{ and all } f : X \rightarrow Y, g : Y \rightarrow Z \text{ in } \mathcal{C}.$$

Functors as defined above are also referred to as *covariant functors* to distinguish them from contravariant functors

2. A *contravariant functor from $\mathcal{C}$ to $\mathcal{D}$* refers to a covariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. Equivalently, such a functor consists of
 - For each object X in $\mathcal{C}$, an object $F(X)$ in $\mathcal{D}$.
 - For each morphism $f : X \rightarrow Y$ in $\mathcal{C}$, a morphism $F(f) : F(Y) \rightarrow F(X)$ in $\mathcal{D}$,
such that:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for all objects } X \text{ in } \mathcal{C},$$

$$F(g \circ f) = F(f) \circ F(g) \quad \text{for all } X, Y, Z \in \text{Ob}(\mathcal{C}) \text{ and all } f : X \rightarrow Y, g : Y \rightarrow Z \text{ in } \mathcal{C}.$$

Note that declarations such as “Let $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ be a contravariant functor” can be common; such declarations usually mean “Let F be a contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ ” as opposed to “Let F be a contravariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$ ”. further note that a contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is equivalent to a covariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$.

Definition A.4.6. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition A.4.1). Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors (Definition A.4.5).

A *natural transformation η between F and G* is a family of morphisms $\eta_X : F(X) \rightarrow G(X)$ in $\mathcal{D}$, one for each object X in $\mathcal{C}$, such that for every morphism $f : X \rightarrow Y$ in $\mathcal{C}$,

$$G(f) \circ \eta_X = \eta_Y \circ F(f)$$

in $\mathcal{D}$. In other words, the following diagram commutes:

$$\begin{array}{ccc}
F(X) & \xrightarrow{F(f)} & F(Y) \\
\eta_X \downarrow & & \downarrow \eta_Y \\
G(X) & \xrightarrow{G(f)} & G(Y)
\end{array}$$

We write such a natural transformation by $\eta : F \Rightarrow G$. We might also write $\eta : F \Rightarrow G : \mathcal{C} \rightarrow \mathcal{D}$ to fully specify that η is a natural transformation from F to G , which are in turn functors from $\mathcal{C}$ to $\mathcal{D}$.

If η_X is an isomorphism for all objects X of $\mathcal{C}$, then η is said to be a *natural isomorphism*.

Definition A.4.7. An *equivalence of categories* between two (large) categories (Definition A.4.1) $\mathcal{C}$ and $\mathcal{D}$ consists of a pair of functors (Definition A.4.5)

$$F : \mathcal{C} \rightarrow \mathcal{D} \quad \text{and} \quad G : \mathcal{D} \rightarrow \mathcal{C}$$

together with natural isomorphisms (Definition A.4.6)

$$\eta : \text{Id}_{\mathcal{C}} \xrightarrow{\sim} G \circ F \quad \text{and} \quad \epsilon : F \circ G \xrightarrow{\sim} \text{Id}_{\mathcal{D}}.$$

Such functors F and G may be called *(natural) inverses of each other*.

When $\mathcal{C}$ and $\mathcal{D}$ are locally small categories (Definition A.4.3), F is an equivalence of categories if and only if F is fully faithful (Definition A.4.26) and essentially surjective

Definition A.4.8 (Category of objects over a fixed object). Let $\mathcal{C}$ be a category (Definition A.4.1) and let $X \in \text{Ob}(\mathcal{C})$ be a fixed object.

1. The *category of objects over X* (or synonymously the *slice category of X in $\mathcal{C}$* or the *over category of X in $\mathcal{C}$*), commonly denoted $\mathcal{C}/X$, $\mathcal{C}_{/X}$, or $(\mathcal{C} \downarrow X)$ is the category defined as follows:
 - An object of $\mathcal{C}/X$ is a morphism $f : A \rightarrow X$ in $\mathcal{C}$, where $A \in \text{Ob}(\mathcal{C})$.
 - A morphism from $f : A \rightarrow X$ to $g : B \rightarrow X$ in $\mathcal{C}/X$ is a morphism $h : A \rightarrow B$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{array}{ccc}
A & \xrightarrow{h} & B \\
& \searrow f & \downarrow g \\
& & X
\end{array}$$

i.e. such that $g \circ h = f$.

- The identity morphisms and composition in $\mathcal{C}/X$ are inherited from $\mathcal{C}$.
2. The *category of objects under X* (or synonymously the *coslice category of X in $\mathcal{C}$* or the *under category of X in $\mathcal{C}$*), commonly denoted $X/\mathcal{C}$, $X \backslash \mathcal{C}$, $\mathcal{C}_{X/}$, or $(X \downarrow \mathcal{C})$, is the category defined as follows:
 - An object of $X/\mathcal{C}$ is a morphism $f : X \rightarrow A$ in $\mathcal{C}$, where $A \in \text{Ob}(\mathcal{C})$.

- A morphism from $f: X \rightarrow A$ to $g: X \rightarrow B$ in $X/\mathcal{C}$ is a morphism $h: A \rightarrow B$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & A \\ & \searrow g & \downarrow h \\ & & B \end{array}$$

i.e. such that $h \circ f = g$.

- The identity morphisms and composition in $X/\mathcal{C}$ are inherited from $\mathcal{C}$.

Definition A.4.9. Let $\mathcal{C}$ be a (large) category (Definition A.4.1).

1. An object $I \in \mathcal{C}$ is called an **initial object** if for every object $X \in \mathcal{C}$ there exists a unique morphism

$$I \rightarrow X.$$

Equivalently, an initial object is a limit (Definition A.4.19) of the empty diagram, if such a limit exists.

2. An object $F \in \mathcal{C}$ is called a **final object** (or **terminal object**) if for every object $X \in \mathcal{C}$ there exists a unique morphism

$$X \rightarrow F.$$

Equivalently, a final object is a colimit (Definition A.4.19) of the empty diagram, if such a colimit exists.

3. An object $Z \in \mathcal{C}$ is called a **zero object** if Z is both initial and final in $\mathcal{C}$. In particular, for every object $X \in \mathcal{C}$ there exist unique morphisms

$$Z \rightarrow X \quad \text{and} \quad X \rightarrow Z.$$

In particular, if initial/final/zero objects exist in a category, then they are unique up to unique isomorphism.

Definition A.4.10 (Full subcategory). Let $\mathcal{C}$ be a (large) category (Definition A.4.1). A **full subcategory** $\mathcal{D}$ of $\mathcal{C}$ is a subcategory such that for every pair of objects $X, Y \in \text{Ob}(\mathcal{D})$, the morphism classes coincide:

$$\text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y).$$

In other words, a full subcategory includes all morphisms between its objects that exist in the ambient category $\mathcal{C}$.

Definition A.4.11. Let $\mathcal{C}$ and $\mathcal{D}$ be categories (Definition A.4.1). Let $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{C}$ be functors.

An **adjunction between F and G** consists of two natural transformations (Definition A.4.6): $\eta: \text{Id}_{\mathcal{C}} \Rightarrow GF$ (the **unit**), and $\varepsilon: FG \Rightarrow \text{Id}_{\mathcal{D}}$ (the **counit**)

These must satisfy the triangle identities: For every object $X \in \mathcal{C}$ and $Y \in \mathcal{D}$,

$$\varepsilon_{FX} \circ F(\eta_X) = \text{id}_{FX}$$

$$G(\varepsilon_Y) \circ \eta_{GY} = \text{id}_{GY}.$$

In diagrammatic form, the triangle identities assert that the following are commutative diagrams:

$$\begin{array}{ccc}
 F(X) & \xrightarrow{F(\eta_X)} & FGF(X) \\
 \searrow \text{id}_{F(X)} & & \downarrow \varepsilon_{F(X)} \\
 & & F(X)
 \end{array}
 \quad
 \begin{array}{ccc}
 G(Y) & \xrightarrow{\eta_{G(Y)}} & GFG(Y) \\
 \searrow \text{id}_{G(Y)} & & \downarrow G(\varepsilon_Y) \\
 & & G(Y)
 \end{array}$$

We say that F is a **left adjoint to G** and G is a **right adjoint to F** (written $F \dashv G$).

In the case that $\mathcal{C}$ and $\mathcal{D}$ are locally small (Definition A.4.3) categories (or U -locally small categories if a universe U is available), we have an adjunction $F \dashv G$ if and only if for every object X in $\mathcal{C}$ and Y in $\mathcal{D}$ there is a natural isomorphism (Definition A.4.6)

$$\text{Hom}_{\mathcal{D}}(F(X), Y) \cong \text{Hom}_{\mathcal{C}}(X, G(Y))$$

that is natural in both X and Y . More precisely, there are two functors

$$\mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \mathbf{Sets}$$

given by

$$\begin{aligned}
 (X, Y) &\mapsto \text{Hom}_{\mathcal{D}}(F(X), Y) \\
 (X, Y) &\mapsto \text{Hom}_{\mathcal{C}}(X, G(Y))
 \end{aligned}$$

respectively, and there is a natural isomorphism between them.

In this case, the **unit of the adjunction** is the natural transformation $\eta : \text{Id}_{\mathcal{C}} \Rightarrow GF$ such that,

1. for every $X \in \mathcal{C}$, the morphism $\eta_X : X \rightarrow GF(X)$ (each called a **unit morphism**) in $\mathcal{C}$ is obtained as the image of $\text{id}_{F(X)}$ via the adjoint isomorphism

$$\text{Hom}_{\mathcal{D}}(F(X), F(X)) \cong \text{Hom}_{\mathcal{C}}(X, GF(X)).$$

2. for every $Y \in \mathcal{D}$, the morphism $\epsilon_Y : FG(Y) \rightarrow Y$ (each called a **counit morphism**) in $\mathcal{D}$ is obtained as the image of $\text{id}_{G(Y)}$ via the adjoint isomorphism

$$\text{Hom}_{\mathcal{C}}(G(Y), G(Y)) \cong \text{Hom}_{\mathcal{D}}(FG(Y), Y).$$

Definition A.4.12 (Opposite category). Let $\mathcal{C}$ be a (large) category (Definition A.4.1). The **opposite category** of $\mathcal{C}$, denoted $\mathcal{C}^{\text{op}}$, is defined as follows:

- The objects of $\mathcal{C}^{\text{op}}$ are the same as those of $\mathcal{C}$.
- For any pair of objects $X, Y \in \mathcal{C}$, the morphisms from X to Y in $\mathcal{C}^{\text{op}}$ are given by the morphisms from Y to X in $\mathcal{C}$:

$$\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) := \text{Hom}_{\mathcal{C}}(Y, X).$$

- Composition in $\mathcal{C}^{\text{op}}$ is defined by reversing the order of composition in $\mathcal{C}$. That is, for morphisms $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y)$ and $g \in \text{Hom}_{\mathcal{C}^{\text{op}}}(Y, Z)$, their composition is

$$g \circ_{\mathcal{C}^{\text{op}}} f := f \circ_{\mathcal{C}} g.$$

Intuitively, the category $\mathcal{C}^{\text{op}}$ thus "reverses" the direction of all morphisms in $\mathcal{C}$.

Definition A.4.13. Let $\mathcal{C}$ be a (large) category (Definition A.4.1), and let $c_1, c_2, \dots, c_n \in \text{Ob}(\mathcal{C})$ be objects.

A *product of $c_1, c_2, \dots, c_n$* in $\mathcal{C}$ is a tuple

$$(c_1 \times c_2 \times \cdots \times c_n, p_1 : c_1 \times \cdots \times c_n \rightarrow c_1, \dots, p_n : c_1 \times \cdots \times c_n \rightarrow c_n)$$

such that for every object $d \in \text{Ob}(\mathcal{C})$ and every family of morphisms

$$(f_1 : d \rightarrow c_1, f_2 : d \rightarrow c_2, \dots, f_n : d \rightarrow c_n),$$

there exists a unique morphism

$$\langle f_1, f_2, \dots, f_n \rangle : d \rightarrow c_1 \times c_2 \times \cdots \times c_n$$

satisfying

$$p_i \circ \langle f_1, f_2, \dots, f_n \rangle = f_i \quad \text{for all } 1 \leq i \leq n.$$

The morphisms p_i are called the *projection morphisms*, and the morphism $\langle f_1, f_2, \dots, f_n \rangle$ is called the *product morphism*. The product $c_1 \times \cdots \times c_n$ is also denoted by $\prod_{i=1}^n c_i$, $c_1 \oplus \cdots \oplus c_n$, or $\bigoplus_{i=1}^n c_i$.

When $n = 0$, the product is defined to be a terminal object (Definition A.4.9) of $\mathcal{C}$, if one exists. When $n = 2$, we simply speak of the *binary product* or *biproduct* of c_1 and c_2 .

Definition A.4.14. Let $\mathcal{C}$ be a (large) (Definition A.4.1) pointed category, i.e. a category with a zero object (Definition A.4.9) 0 . Let $X, Y \in \text{Ob}(\mathcal{C})$ be an object and let $f : X \rightarrow Y$ be a morphism.

1. A morphism $i : K \rightarrow X$ is called the *kernel of f* if:
 - (a) $f \circ i = 0$, where 0 is the zero morphism $K \rightarrow Y$,
 - (b) for any morphism $g : Z \rightarrow X$ such that $f \circ g = 0$, there exists a unique morphism $u : Z \rightarrow K$ such that $g = i \circ u$.

The kernel, if it exists, is unique up to unique isomorphism. $\ker(f)$ denotes the object K determined (up to isomorphism) by a kernel of f .

2. a morphism $p : Y \rightarrow Q$ is called the *cokernel of f* if:
 - (a) $p \circ f = 0$, where 0 is the zero morphism (Definition A.4.9) $X \rightarrow Q$,
 - (b) for any morphism $g : Y \rightarrow Z$ such that $g \circ f = 0$, there exists a unique morphism $v : Q \rightarrow Z$ such that $g = v \circ p$.

The cokernel, if it exists, is unique up to unique isomorphism. $\text{coker}(f)$ denotes the object Q determined (up to isomorphism) by a cokernel of f .

Definition A.4.15. Let $\mathcal{C}$ be a category (Definition A.4.1), and let $f : A \rightarrow B$ be a morphism in $\mathcal{C}$.

1. An *image of f* consists of an object $I \in \text{Ob}(\mathcal{C})$ together with a factorization of f into two morphisms

$$A \xrightarrow{e} I \xrightarrow{m} B,$$

where e is an epimorphism (Definition A.4.23) and m is a monomorphism (Definition A.4.23), such that for any other factorization

$$A \xrightarrow{e'} I' \xrightarrow{m'} B$$

with e' epi and m' mono, there exists a unique isomorphism $\varphi : I \simeq I'$ satisfying $m = m'\varphi$ and $\varphi e = e'$.

$$\begin{array}{ccc} & I & \\ e \nearrow & \downarrow \varphi & \nwarrow m \\ A & & B \\ e' \searrow & \downarrow & \swarrow m' \\ & I' & \end{array}$$

The monomorphism $m : I \rightarrow B$ (or equivalently its subobject class) is called the **image of f in $\mathcal{C}$** . The morphisms e and m may be suppressed from notation and the image of f may simply be denoted as **$\text{im}(f)$** or **$f(A)$** .

- Let $\mathcal{C}$ be a category (Definition A.4.1), and let $f : A \rightarrow B$ be a morphism in $\mathcal{C}$. A **coimage of f** consists of an object $C \in \text{Ob}(\mathcal{C})$ together with a factorization of f into two morphisms

$$A \xrightarrow{e} C \xrightarrow{m} B,$$

where e is an epimorphism and m is a monomorphism, such that for any other factorization

$$A \xrightarrow{e'} C' \xrightarrow{m'} B$$

with e' epi and m' mono, there exists a unique isomorphism $\varphi : C \simeq C'$ satisfying $m = m'\varphi$ and $\varphi e = e'$.

$$\begin{array}{ccc} & C & \\ e \nearrow & \downarrow \varphi & \nwarrow m \\ A & & B \\ e' \searrow & \downarrow & \swarrow m' \\ & C' & \end{array}$$

The epimorphism $e : A \rightarrow C$ (or equivalently its quotient class) is called the **coimage of f in $\mathcal{C}$** . The morphisms e and m may be suppressed from notation and the coimage of f may simply be denoted as **$\text{coim}(f)$** .

Definition A.4.16 (Complete and Cocomplete Category). Let $\mathcal{C}$ be a category (Definition A.4.1).

- The category $\mathcal{C}$ is called **complete** (resp. **finitely complete**) if all small limits (Definition A.4.22) (resp. finite limits) exist in $\mathcal{C}$; that is, for every small diagram $D : J \rightarrow \mathcal{C}$ (with J a small category), the limit $\lim D$ exists and is an object of $\mathcal{C}$.
- The category $\mathcal{C}$ is called **cocomplete** (resp. **finitely cocomplete**) if all small colimits (Definition A.4.22) (resp. finite colimits) exist in $\mathcal{C}$; that is, for every small diagram $D : J \rightarrow \mathcal{C}$, the colimit $\text{colim } D$ exists and is an object of $\mathcal{C}$.

Definition A.4.17 (Filtered category). 1. A *filtered category* is a (nonempty, large) category $\mathcal{I}$ satisfying the following conditions:

- For every finite collection of objects $i_1, i_2, \dots, i_n$ in $\mathcal{I}$, there exists an object j and morphisms

$$\phi_k : i_k \rightarrow j, \quad \text{for each } k = 1, \dots, n.$$

- For every pair of morphisms $f, g : i \rightarrow j$ in $\mathcal{I}$, there exists an object k and a morphism

$$h : j \rightarrow k$$

that satisfies

$$h \circ f = h \circ g.$$

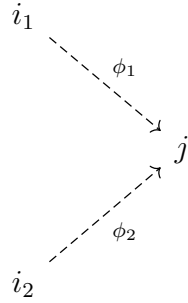


FIGURE 1. *

Condition 1: Upper Bound

$$i \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} j \dashrightarrow^h k$$

FIGURE 2. *

Condition 2: Coequalizing map

In other words, $\mathcal{I}$ is nonempty, any finite diagram of objects admits a cocone (Definition A.4.19), and any pair of parallel morphisms become equal after post-composition with an appropriate morphism.

2. Dually, a *Cofiltered category* is a category whose opposite category (Definition A.4.12) is filtered. More explicitly, A cofiltered category is a (nonempty, large) category $\mathcal{I}$ satisfying the following conditions:

- For every finite collection of objects $i_1, i_2, \dots, i_n$ in $\mathcal{I}$, there exists an object j and morphisms

$$\phi_k : j \rightarrow i_k, \quad \text{for each } k = 1, \dots, n.$$

- For every pair of morphisms $f, g : j \rightarrow i$ in $\mathcal{I}$, there exists an object k and a morphism

$$h : k \rightarrow j$$

that satisfies

$$f \circ h = g \circ h.$$

In other words, $\mathcal{I}$ is nonempty, any finite diagram of objects admits a cone, and any pair of parallel morphisms become equal after pre-composition with an appropriate morphism.

Definition A.4.18 (Systems in a category). Let $\mathcal{C}$ be a (large) category. Let I be a (large) category.

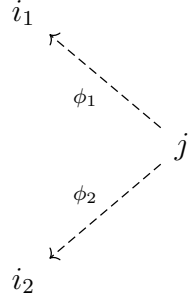


FIGURE 3. *

Condition 1: Lower Bound

$$k \dashrightarrow^h j \begin{matrix} \xrightarrow{f} \\ \xrightarrow{g} \end{matrix} i$$

FIGURE 4. *

Condition 2: Equalizing map

1. A diagram/system $I \rightarrow C$ is called **filtered** (resp. **cofiltered**) if I is a filtered (Definition A.4.17) (resp. cofiltered (Definition A.4.17)) category.
2. A diagram/system $I \rightarrow C$ is called **directed** (resp. **codirected**) if I is small and thing, i.e. is regardable/comes from a directed (resp. codirected) partially ordered set. A **direct system** or **inductive system** is synonymous for a directed system and a **inverse system** or **projective system** is synonymous for a codirected system.

One might also speak of a **filtered direct/inductive system** synonymously for a filtered system to emphasize that the indexing category is a general filtered category, rather than a directed poset.

Definition A.4.19 (Cones, limits and colimits in a category). Let $\mathcal{C}$ be a (large) category (Definition A.4.1), let I be a (large) category, and let $D : I \rightarrow \mathcal{C}$ be a diagram.

1. A **cone to the diagram D** is an object $L \in \mathcal{C}$ together with a family of morphisms

$$\{\pi_i : L \rightarrow D(i)\}_{i \in I}$$

such that for every morphism $f : i \rightarrow j$ in I , the diagram

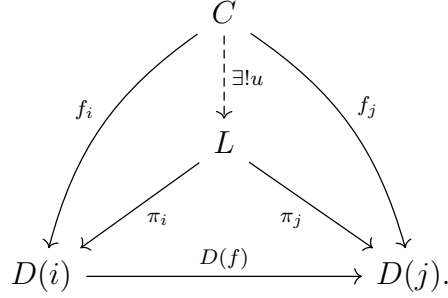
$$\begin{array}{ccc} & L & \\ \pi_i \swarrow & & \searrow \pi_j \\ D(i) & \xrightarrow{D(f)} & D(j) \end{array}$$

commutes, i.e. $D(f) \circ \pi_i = \pi_j$.

2. A cone $(L, \{\pi_i\})$ is called a **limit of D** if it satisfies the following “universal property”: for any cone $(C, \{f_i\})$ over D , there exists a *unique* morphism $u : C \rightarrow L$ such that

$$\pi_i \circ u = f_i \quad \text{for all } i \in I.$$

Visually, the following diagrams commute every morphism $f : i \rightarrow j$ in I :

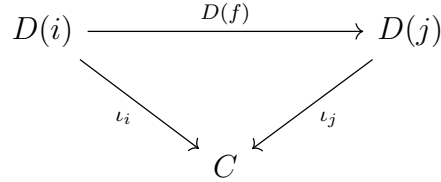


If such a cone exists, then the object L is necessarily unique up to unique isomorphism by the universal property. In this case, L is denoted by $\lim_{i \in I} D$ or $\lim D$.

3. A **cocone from the diagram D** is an object $C \in \mathcal{C}$ together with a family of morphisms

$$\{\iota_i : D(i) \rightarrow C\}_{i \in I}$$

such that for every morphism $f : i \rightarrow j$ in I , the diagram

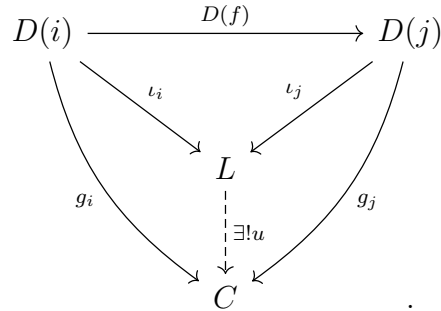


commutes, i.e. $\iota_j \circ D(f) = \iota_i$.

4. A cocone $(L, \{\iota_i\})$ is called a **colimit of D** if it satisfies the following “universal property”: for any cocone $(C, \{g_i\})$ under D , there exists a *unique* morphism $u : L \rightarrow C$ such that

$$u \circ \iota_i = g_i \quad \text{for all } i \in I.$$

Visually, the following diagrams commute every morphism $f : i \rightarrow j$ in I :



If such a cocone exists, then the object L is necessarily unique up to unique isomorphism by the universal property. In this case, L is denoted by $\text{colim}_{i \in I} D$ or $\text{colim } D$.

A limit/colimit is called **finite** (resp. **small**) if the index category I is finite (resp. small).

Some authors use the terms **projective limit** or **inverse limit** to refer to what is defined here as a limit. Similarly, the terms **inductive limit** or **direct limit** are sometimes used to mean a colimit. However, these phrases can have more specific meanings to other authors: a *projective* or *inverse limit* may refer to a limit over a diagram indexed by a codirected poset. Likewise, an *inductive* or *direct limit* may refer to a colimit over a directed poset (see Definition A.4.20) .

Thus, while the terms are sometimes used interchangeably with “limit” and “colimit,” they may also emphasize particular indexing shapes and directions, distinguishing them from general limits and colimits taken over arbitrary small categories.

Definition A.4.20 (Special cases of limits). Let $\mathcal{C}$ be a (large) category. Let I be a (large) category. Let $I \rightarrow \mathcal{C}$ be a diagram/system.

- Suppose that the system is a cofiltered system (Definition A.4.18), i.e. I is a cofiltered category. A limit (Definition A.4.19) of this diagram is often denoted by

$$\varprojlim_{i \in I} D(i)$$

and may be called a *cofiltered (inverse/projective) limit*. In case that the system is more specifically an inverse/projective system (Definition A.4.18), i.e. I is a cofiltered poset, the preferred term for such a limit is *inverse/projective limit*.

- Suppose that the system is a filtered system, i.e. I is a filtered category. A colimit of this diagram is often denoted by

$$\varinjlim_{i \in I} D(i)$$

and may be called a *filtered colimit* or a *direct/inductive/injective limit*. In case that the system is more specifically a direct/inductive system, i.e. I is a filtered poset, the preferred term for such a limit is *direct/inductive limit*.

Definition A.4.21 (Product in a category). Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$ indexed by a class I .

1. A *product of the family $\{X_i\}$* is an object P of $\mathcal{C}$ together with a “universal” family of morphisms

$$\pi_i : P \rightarrow X_i, \quad \text{for each } i \in I.$$

More precisely, for any object Y and any family of morphisms $\{f_i : Y \rightarrow X_i\}_{i \in I}$, there exists a unique morphism

$$f : Y \rightarrow P$$

making the following diagram commute for all $i \in I$, i.e. $\pi_i \circ f = f_i$:

$$\begin{array}{ccc} Y & & \\ \downarrow \exists! f & \searrow f_i & \\ \prod X_i & \xrightarrow{\pi_i} & X_i \end{array}$$

Such a product is often denoted by $\prod_{i \in I} X_i$. If $\prod_{i \in I} X_i$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism by the universal property described above.

Equivalently, the product $\prod_{i \in I} X_i$ is the limit (Definition A.4.19) of the diagram $I \rightarrow \mathcal{C}, i \mapsto X_i$, where I is made into a category whose objects are the members of I and whose morphisms are just the identity morphisms.

2. A **coproduct** (or synonymously **direct sum**) of the family $\{X_i\}$ is an object C of $\mathcal{C}$ together with a “universal” family of morphisms

$$\iota_i : X_i \rightarrow C, \quad \text{for each } i \in I.$$

More precisely, for any object Y and any family of morphisms $\{g_i : X_i \rightarrow Y\}_{i \in I}$, there exists a unique morphism

$$g : C \rightarrow Y$$

making the following diagram commute for all $i \in I$, i.e. $g \circ \iota_i = g_i$:

$$\begin{array}{ccc} X_i & \xrightarrow{\iota_i} & \coprod X_i \\ & \searrow g_i & \downarrow \exists! g \\ & & Y \end{array}$$

Such a coproduct is often denoted by $\coprod_{i \in I} X_i$ or $\oplus_{i \in I} X_i$. If $\coprod_{i \in I} X_i$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism by the universal property described above.

Equivalently, the coproduct $\coprod_{i \in I} X_i$ is the colimit (Definition A.4.19) of the diagram $I \rightarrow \mathcal{C}, i \mapsto X_i$, where I is made into a category whose objects are the members of I and whose morphisms are just the identity morphisms.

Definition A.4.22. Let $\mathcal{C}$ be a (large) category (Definition A.4.1), let I be a small category (Definition A.4.3), and let $D : I \rightarrow \mathcal{C}$ be a diagram.

A limit or colimit (Definition A.4.19) is called **finite** (resp. **small**) if the indexing category I has finitely many objects and morphisms (resp. if I is a small category (Definition A.4.3)).

Definition A.4.23 (Monomorphism and Epimorphism in Categories). Let $\mathcal{C}$ be a category (Definition A.4.1). For objects $A, B \in \mathcal{C}$, let $f : A \rightarrow B$ be a morphism in $\mathcal{C}$.

- The morphism f is called a **monomorphism** (or a **monic morphism**) if for every object X and every pair of morphisms $g_1, g_2 : X \rightarrow A$, the equality $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$.
- The morphism f is called an **epimorphism** (or an **epic morphism**) if for every object Y and every pair of morphisms $h_1, h_2 : B \rightarrow Y$, the equality $h_1 \circ f = h_2 \circ f$ implies $h_1 = h_2$.

Definition A.4.24. Let $\mathcal{C}$ be a category (Definition A.4.1). Let $X \in \text{Ob}(\mathcal{C})$ be an object. A **subobject of X** refers to a monomorphism (Definition A.4.23) $i : Y \hookrightarrow X$ in $\mathcal{C}$. We regard two subobjects (Y, i) and (Y', i') of X as isomorphic if there exists an isomorphism $f : Y \rightarrow Y'$ such that $i = i' \circ f$. One often leaves the monomorphism i implicit, suppressing it from the notation. One also often write $Y \subseteq X$ to denote that Y is a subobject of X .

Definition A.4.25. Let $\mathcal{C}$ be an abelian category (Definition 3.5.18). Let $X \in \text{Ob}(\mathcal{C})$ be an object. Let $i : A \hookrightarrow X$ be a subobject (Definition A.4.24). The cokernel (Definition A.4.14) $\pi : X \rightarrow X/A := \text{coker}(i)$ is called the **quotient object of X by A** . The object X/A is determined up to canonical isomorphism.

Definition A.4.26. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition A.4.1). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition A.4.5).

1. F is called **full** if for every pair of objects $x, y \in \text{Ob}(\mathcal{C})$, the induced rule/assignment/class function

$$F_{x,y} : \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is “surjective”, i.e. for all morphisms $g : F(x) \rightarrow F(y)$, there exists some morphism $f : x \rightarrow y$ such that $F(f) = g$.

2. F is called **faithful** if for every pair of objects $x, y \in \text{Ob}(\mathcal{C})$, the induced class function (assignment)

$$F_{x,y} : \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is “injective”, i.e., for any morphisms $f_1, f_2 \in \text{Hom}_{\mathcal{C}}(x, y)$, if $F(f_1) = F(f_2)$ in $\text{Hom}_{\mathcal{D}}(F(x), F(y))$, then $f_1 = f_2$.

3. F is called **fully faithful** if it is both full and faithful.

Definition A.4.27 (Reflecting a type of morphism). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between (large) categories (Definition A.4.5), and let $\mathcal{P}$ be a property of morphisms (or more generally a property of sequences or families of morphisms) that is stable under isomorphism (e.g. monomorphism, epimorphism (Definition A.4.23), isomorphism, etc.). We say that F **reflects $\mathcal{P}$ -morphisms** if for every morphism $f : x \rightarrow y$ in $\mathcal{C}$, whenever $F(f)$ has property $\mathcal{P}$ in $\mathcal{D}$, it follows that f has property $\mathcal{P}$ in $\mathcal{C}$.

One can speak of more general notions of reflection (e.g. reflection of limits (Definition A.4.19), of colimits (Definition A.4.19), of “exact sequences”, say for abelian categories, etc.)

Lemma A.4.28. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.5.17) between additive categories (Definition 3.5.16). The functor F is faithful (Definition A.4.26) if and only if for any morphism f in $\mathcal{A}$, we have $F(f) = 0$ exactly when $f = 0$.

Proof. To say that F is faithful means that for every pair of objects X and Y in $\mathcal{A}$, the induced abelian group (Definition 2.1.1) homomorphism (Definition 2.1.17)

$$F_{X,Y} : \text{Hom}_{\mathcal{A}}(X, Y) \rightarrow \text{Hom}_{\mathcal{B}}(F(X), F(Y))$$

on the Hom-sets is injective (Definition A.1.3). Equivalently, this means that for every morphism $f : X \rightarrow Y$ in $\mathcal{A}$, we have $F(f) = 0$ if and only if $f = 0$. $\square$

Proposition A.4.29 (Reflection of morphism properties by full or faithful functors). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition A.4.5) between locally small categories (Definition A.4.3).

(1) If F is faithful (Definition A.4.26), then F reflects (Definition A.4.27) monomorphisms and epimorphisms (Definition A.4.23). That is, if $f : x \rightarrow y$ in $\mathcal{C}$ is such that $F(f)$ is a monomorphism (resp. epimorphism) in $\mathcal{D}$, then f is a monomorphism (resp. epimorphism) in $\mathcal{C}$.

(♠ TODO: define split monos and epis) (2) If F is fully faithful, then F reflects isomorphisms, split monomorphisms, and split epimorphisms. That is, if $f : x \rightarrow y$ in $\mathcal{C}$ is such that $F(f)$ is an isomorphism in $\mathcal{D}$, then f is an isomorphism in $\mathcal{C}$.

Proposition A.4.30. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor (Definition 3.5.29) between abelian categories (Definition 3.5.18). The functor F is faithful (Definition A.4.26) if and only if it reflects (Definition A.4.27) short exact sequences (Definition 3.5.28).

Proof. Suppose F is faithful. Let

$$A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3$$

be a sequence in $\mathcal{A}$ such that

$$F(A_1) \xrightarrow{F(f)} F(A_2) \xrightarrow{F(g)} F(A_3)$$

is exact in $\mathcal{B}$. Because F is exact, it preserves kernels and images, so

$$F(\ker(g)) = \ker(F(g)), \quad F(\operatorname{im}(f)) = \operatorname{im}(F(f)).$$

Exactness in $\mathcal{B}$ gives

$$\ker(F(g)) = \operatorname{im}(F(f)),$$

so

$$F(\ker(g)) = F(\operatorname{im}(f)).$$

Since F is faithful, it reflects (Definition A.4.27) monomorphisms and epimorphisms (Definition A.4.23) (??), and morphisms that become equal via F must be equal in $\mathcal{A}$. Hence,

$$\ker(g) = \operatorname{im}(f).$$

Therefore, the sequence

$$A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3$$

is exact in $\mathcal{A}$, i.e., F reflects short exact sequences.

Conversely, suppose F reflects short exact sequences. Let $f : A \rightarrow B$ be a morphism in $\mathcal{A}$ such that

$$F(f) = 0.$$

Consider the sequence

$$0 \rightarrow A \xrightarrow{f} B.$$

Since $F(f) = 0$, we have the sequence

$$0 \rightarrow F(A) \xrightarrow{F(f)} F(B)$$

is exact at $F(A)$. Because F reflects short exact sequences, the original sequence must be exact at A , so f is a monomorphism with zero image, hence $f = 0$. Thus F is faithful by Lemma A.4.28. $\square$

A.5. Homological algebra.

Definition A.5.1 (n-ary Additive Functor). Let I be a finite set with $|I| = n$. Let $\{\mathcal{A}_i\}_{i \in I}$ be additive categories (Definition 3.5.16) and let $\mathcal{B}$ be an additive category. An *n-ary additive functor* (or *multilinear functor*)

$$F : \prod_{i \in I} \mathcal{A}_i \rightarrow \mathcal{B}$$

is a functor such that for each fixed collection of all but one variable, the resulting functor in the remaining variable is additive (Definition 3.5.17). Equivalently, for every $j \in I$ and objects $(A_i)_{i \in I}$ and morphisms $f_1, f_2 : A_j \rightarrow A'_j$ in $\mathcal{A}_j$, we have

$$\begin{aligned} & F(A_1, \dots, A_{j-1}, f_1 + f_2, A_{j+1}, \dots, A_n) \\ &= F(A_1, \dots, A_{j-1}, f_1, A_{j+1}, \dots, A_n) \\ &+ F(A_1, \dots, A_{j-1}, f_2, A_{j+1}, \dots, A_n), \end{aligned}$$

and F preserves zero morphisms componentwise:

$$F(A_1, \dots, 0_{A_j, A'_j}, \dots, A_n) = 0_{F(A_1, \dots), F(A'_1, \dots)}.$$

A bifunctor that satisfies this property for $n = 2$ is simply called a **biadditive functor**.

Definition A.5.2 (Flat object with respect to a tensor (monoidal) product). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.5.18), and let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) (in practice, the following definitions are usually considered when F is some kind of “tensor product” $\otimes$ and is right exact (Definition 3.5.29) in each variable).

1. An object $X \in \mathcal{A}$ is called **flat (with respect to F on the left)** if the functor $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ is exact (Definition 3.5.29).
2. An object $Y \in \mathcal{B}$ is called **flat (with respect to F on the right)** if the functor $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ is exact (Definition 3.5.29).

If $\mathcal{A} = \mathcal{B} = \mathcal{C}$ and F makes $\mathcal{A}$ into a symmetric monoidal category, then certainly $F(X, -)$ is exact if and only if $F(-, Y)$ is exact, i.e. flatness is equivalent on the two sides of $\otimes$.

Remark A.5.3. The above notion of flatness is non-standard terminology, unless the bifunctor F is some kind of tensor product.

Lemma A.5.4. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.5.18), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) that is right exact (Definition 3.5.29) in each variable.

1. Say that $\mathcal{B}$ has enough projectives and let $\text{Tor}_n(A, B)$ be the Tor object computed as the left derived functor of $A \otimes -$ applied to B . If A is flat (Definition A.5.2), then $\text{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all B .
2. Say that $\mathcal{A}$ has enough projectives and let $\text{Tor}_n(A, B)$ be the Tor object computed as the left derived functor of $- \otimes B$ applied to A . If B is flat (Definition A.5.2), then $\text{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all A .
3. Say that $\mathcal{B}$ has enough flats and let $\text{Tor}_n(A, B)$ be the Tor object computed as $H_n(A \otimes F_\bullet)$ for a flat resolution $F_\bullet \rightarrow B$. If A is flat (Definition A.5.2), then $\text{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all B .
4. Say that $\mathcal{A}$ has enough flats and let $\text{Tor}_n(A, B)$ be the Tor object computed as $H_n(F_\bullet \otimes B)$ for a flat resolution $F_\bullet \rightarrow A$. If B is flat (Definition A.5.2), then $\text{Tor}_n(A, B) = 0$ for all $n \neq 0$ and all A .

Proof. We prove the first part. The other parts hold similarly.

If A is flat, then for any $B \in \mathcal{B}$, letting $P_\bullet \rightarrow B$ be some projective resolution, the exactness of $A \otimes -$ implies that $\text{Tor}_n(A, B) = H_n(A \otimes P_\bullet) = 0$ for all $n \neq 0$. $\square$

Lemma A.5.5 (Dimension shifting, cf. [Wei94, Exercise 2.4.3]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.5.17) between abelian categories (Definition 3.5.18). Let A be an object of $\mathcal{A}$.

1. Suppose that F is right exact (Definition 3.5.29).
 - (a) Suppose that $0 \rightarrow M \rightarrow C \rightarrow A \rightarrow 0$ is an exact sequence in $\mathcal{A}$ where C is F -acyclic, and that A and M have projective resolutions. We have $L_i F(A) \cong L_{i-1} F(M)$ for $i \geq 2$ and $L_1 F(A) \cong \ker(F(M) \rightarrow F(C))$.
 - (b) Suppose that $\mathcal{A}$ has enough projectives. Let

$$0 \rightarrow M_m \rightarrow C_m \rightarrow C_{m-1} \rightarrow \cdots \rightarrow C_0 \rightarrow A \rightarrow 0$$

be an acyclic complex with the C_i F -acyclic. We have

- (i) $L_i F(A) \cong L_{i-m-1} F(M_m)$ for $i \geq m+2$ and
- (ii) $L_{m+1} F(A) \cong \ker(F(M_m) \rightarrow F(C_m))$.

2. Suppose that $G : \mathcal{A} \rightarrow \mathcal{B}$ is left exact (Definition 3.5.29).
 - (a) Suppose that $0 \rightarrow A \rightarrow I \rightarrow N \rightarrow 0$ is a short exact sequence in $\mathcal{A}$ where I is G -acyclic, and that A and N have injective resolutions. We have $R^i G(A) \cong R^{i-1} G(N)$ for $i \geq 2$ and $R^1 G(A) \cong \text{coker}(G(I) \rightarrow G(N))$.
 - (b) Suppose that $\mathcal{A}$ has enough injectives. Let

$$0 \rightarrow A \rightarrow I_0 \rightarrow I_1 \rightarrow \cdots \rightarrow I_m \rightarrow N_m \rightarrow 0$$

be an acyclic complex with the I_j G -acyclic. We have

- (i) $R^i G(A) \cong R^{i-m-1} G(N_m)$ for $i \geq m+2$ and
- (ii) $R^{m+1} G(A) \cong \text{coker}(G(I_m) \rightarrow G(N_m))$.

Proof. 1. Suppose that F is right exact

- (a) Since C is F -acyclic, the long exact sequence of derived functors associated to $0 \rightarrow M \rightarrow C \rightarrow A \rightarrow 0$ yields an exact sequence

$$0 \rightarrow L_1 F(A) \rightarrow F(M) \rightarrow F(C) \rightarrow F(A) \rightarrow 0$$

along with isomorphisms $L_i F(A) \cong L_{i-1} F(M)$ for $i \geq 2$. In particular, $L_1 F(A) = \ker(F(M) \rightarrow F(C))$.

- (b) We now proceed by induction. The above proves the base case of $m = 0$. Associated to the acyclic complex

$$0 \rightarrow M_m \rightarrow C_m \rightarrow C_{m-1} \rightarrow \cdots \rightarrow C_0 \rightarrow A \rightarrow 0$$

is a short exact sequence

$$0 \rightarrow M_m \rightarrow C_m \rightarrow M_{m-1} \rightarrow 0$$

where $M_{m-1} = \text{coker}(M_m \rightarrow C_m)$ and an acyclic complex

$$0 \rightarrow M_{m-1} \rightarrow C_{m-1} \rightarrow \cdots \rightarrow C_0 \rightarrow A \rightarrow 0.$$

The derived functor long exact sequence of the short exact sequence yields isomorphisms

$$L_i F(M_{m-1}) \cong L_{i-1} F(M_m)$$

for all $i \geq 1$. By induction, we also have $L_i F(A) \cong L_{i-(m-1)-1} F(M_{m-1}) = L_{i-m} F(M_{m-1})$ for $i \geq m+1$, so we in fact have

$$L_i F(A) \cong L_{i-m} F(M_{m-1}) \cong L_{i-m-1} F(M_m)$$

for all $i \geq m+1$. Moreover, the derived functor long exact sequence also includes

$$0 \rightarrow L_1 F(M_{m-1}) \rightarrow F(M_m) \rightarrow F(C_m) \rightarrow F(M_{m-1}) \rightarrow 0,$$

so $L_1 F(M_{m-1}) \cong \ker(F(M_m) \rightarrow F(C_m))$. We found that $L_1 F(M_{m-1}) \cong L_{m+1} F(A)$, so $L_{m+1} F(A)$ is the kernel of $F(M_m) \rightarrow F(C_m)$ as desired.

2. These hold dually.

□

Corollary A.5.6. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.5.18), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) that is right exact (Definition 3.5.29) in each variable. Suppose that $\mathcal{A}$ and $\mathcal{B}$ have enough projectives and flats. Further suppose that small (Definition A.4.22) filtered colimits (Definition A.4.20) which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 3.5.19)).

For any objects $A \in \mathcal{A}$ and $B \in \mathcal{B}$, all notions of $\text{Tor}_n(A, B)$ as defined in Definition A.5.14 naturally agree with one another.

Proof. This follows from Theorem A.5.13 and Lemma A.5.7. □

Lemma A.5.7 (cf. [Wei94, Flat Resolution Lemma 3.2.8]). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.5.18), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) that is right exact (Definition 3.5.29) in each variable. Suppose that $\mathcal{A}$ and $\mathcal{B}$ have enough projective objects.

Let $A \in \mathcal{A}$ and $B \in \mathcal{B}$ be objects.

1. Suppose that A has some flat resolution. The Tor objects (Definition A.5.14)

$$\text{Tor}_n(A, B) \in \mathcal{C}$$

obtained via a projective resolution of A and via a flat resolution of A are naturally isomorphic. In particular, either notion of Tor_n is well defined.

2. Suppose that B has some flat resolution. The Tor objects

$$\text{Tor}_n(A, B) \in \mathcal{C}$$

obtained via a projective resolution of B and via a flat resolution of B are naturally isomorphic. In particular, either notion of Tor_n is well defined.

Proof. We show 1. Write $\text{Tor}_n(A, B)$ for the tor object obtained via a projective resolution of A . Write $H_n(F_\bullet \otimes B)$ for the tor object obtained via a flat resolution $F_\bullet \rightarrow A$ of A . We argue by induction. For $n = 0$, we have $\text{Tor}_0(A, B) \cong H_0(F_\bullet \otimes B)$ because $- \otimes B$ is right exact by assumption.

Let K be such that $0 \rightarrow K \rightarrow F_0 \rightarrow A \rightarrow 0$ is exact. write $E_\bullet = (\cdots \rightarrow F_2 \rightarrow F_1)$ so that $E_\bullet \rightarrow K$ is a resolution of K by flat objects (??). Since $\mathcal{B}$ has enough projectives, further note that all flat objects of $\mathcal{A}$ are acyclic for the functor $- \otimes B$ by Lemma A.5.4.

Since $\mathcal{A}$ has enough projectives, Lemma A.5.5 thus applies to yield

$$\mathrm{Tor}_1(A, B) = \ker(K \otimes B \rightarrow F_0 \otimes B) = \ker \left\{ \frac{F_1 \otimes B}{\mathrm{im}(F_2 \otimes B) \rightarrow F_0 \otimes B} = H_1(F \otimes B) \right\},$$

thus establishing the desired isomorphism for $n = 1$. For $n \geq 2$, use induction to see that

$$\mathrm{Tor}_n(A, B) \cong \mathrm{Tor}_{n-1}(K, B) \cong H_{n-1}(E_\bullet \otimes B) = H_n(F \otimes B).$$

□

Lemma A.5.8. Let $\mathcal{A}$ be a preadditive category (Definition 3.5.16). Finite products (Definition A.4.21) in $\mathcal{A}$ coincide with finite coproducts (Definition A.4.21). More precisely, if $\{A_i\}_{i=1}^n$ is a finite collection of objects of $\mathcal{A}$, then

1. if $\prod_{i=1}^n A_i$ exists, then so does $\coprod_{i=1}^n A_i$ and these are naturally isomorphic.
2. if $\coprod_{i=1}^n A_i$, then so does $\prod_{i=1}^n A_i$ and these are naturally isomorphic.

Proof. (♠ TODO:)

□

Lemma A.5.9. Let C be a double complex of objects in an additive category (Definition 3.5.16) $\mathcal{A}$. If C is locally bounded along diagonals, then the complexes $\mathrm{Tot}^\oplus(A)$ and $\mathrm{Tot}^\Pi(A)$ are naturally isomorphic.

Proof. Since C is locally bounded along diagonals, the degree n components

$$\begin{aligned} (\mathrm{Tot}^\oplus)^n(A) &= \bigoplus_{p+q=n} A^{p,q}, \\ (\mathrm{Tot}^\Pi)^n(A) &= \prod_{p+q=n} A^{p,q}. \end{aligned}$$

are finite direct sums and finite products respectively and hence are naturally isomorphic (Lemma A.5.8). The differential maps of the two total complexes also naturally coincide. □

Lemma A.5.10 (cf. [Wei94, Acyclic Assembly Lemma 2.7.3]). (♠ TODO: It may be the case that this is generalizable beyond first quadrant double complexes, but I don't have a slick way to show this. See the commented out code for the statements; also, it may be necessary to assume something like AB4* for such statements) Let $\mathcal{A}$ be an abelian category (Definition 3.5.18) for which (small) filtered colimits (Definition A.4.20) which exist are exact (e.g. which holds if $\mathcal{A}$ satisfies Ab5 (Definition 3.5.19)). Let C be a double complex in $\mathcal{A}$.

If C has exact columns or has exact rows and C is a bounded below or bounded above double complex, then $\mathrm{Tot}^\Pi(C)$ is an acyclic chain complex.

Proof. We show that if C has exact columns and C is bounded below, then $\mathrm{Tot}^\Pi(C)$ is an acyclic chain complex; it can then be argued symmetrically that if C has exact columns

and C is bounded above, then $\mathrm{Tot}^\Pi(C)$ is acyclic. Moreover, the case of exact rows can be deduced by reflecting the rows and columns of double complexes.

Note that since C is assumed to be bounded below and hence is locally bounded along diagonals, $\mathrm{Tot}^\Pi(C)$ and $\mathrm{Tot}^\oplus(C)$ exist, are constructed by finite products (which are also finite coproducts), and are naturally isomorphic by Lemma A.5.9. Further recall that finite coproducts in an abelian category are exact.

Define the sub-double complexes $F^k C$ of C by

$$(F^k C)^{p,q} = \begin{cases} C^{p,q} & \text{if } p \leq k \\ 0 & \text{otherwise} \end{cases}.$$

This yields a filtration

$$\dots \subseteq F^{k-1} C \subseteq F^k C \subseteq F^{k+1} C \subseteq \dots \subseteq C.$$

Moreover, for each n ,

$$(\mathrm{Tot}^\Pi(F^k C))^n = \prod_{p \leq k, p+q=n} C^{p,q}.$$

For each n , the above stabilizes as $k \rightarrow \infty$ to $\mathrm{Tot}^\Pi(C)^n$. Now let

$$D^k = F^k C / F^{k-1} C = \begin{cases} C^{p,q} & \text{if } p = k \\ 0 & \text{otherwise.} \end{cases}$$

Since each column $C^{k,*}$ is exact by assumption, the total complex $\mathrm{Tot}^\Pi(D^k)$ is acyclic. Note that we have short exact sequences

$$0 \rightarrow F^{k-1} C \rightarrow F^k C \rightarrow D^k \rightarrow 0$$

of double complexes. The totalization functor $\mathrm{Tot}^\Pi(-)$ in this case is exact because all of the double complexes are locally bounded along diagonals. We hence have a short exact sequence

$$0 \rightarrow \mathrm{Tot}^\Pi(F^{k-1} C) \rightarrow \mathrm{Tot}^\Pi(F^k C) \rightarrow \mathrm{Tot}^\Pi(D^k) \rightarrow 0.$$

Since $\mathrm{Tot}^\Pi(D^k)$ is acyclic, the long exact cohomology sequences yield isomorphisms

$$H^n(\mathrm{Tot}^\Pi(F^{k-1} C)) \cong H^n(\mathrm{Tot}^\Pi(F^k C)).$$

Since C is assumed to be bounded, the subcomplex $F^k C$ is zero for sufficiently negative k , in which case $\mathrm{Tot}^\Pi(F^k C)$ is acyclic. By induction on k , $\mathrm{Tot}^\Pi(F^k C)$ remains acyclic for all k . Moreover, the filtered colimit $\varinjlim_k \mathrm{Tot}^\Pi(F^k C)$ is $\mathrm{Tot}^\Pi(C)$. The assumed exactness of filtered colimits in $\mathcal{A}$ concludes that $\mathrm{Tot}^\Pi(C)$ is acyclic.

By symmetry, if C instead has exact rows, then $\mathrm{Tot}^\Pi(C)$ is an acyclic chain complex. $\square$

Lemma A.5.11. Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) of abelian categories (Definition 3.5.18). Assume that (small) filtered colimits which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 3.5.19)).

Let $A \in \mathcal{A}$ and $B \in \mathcal{B}$ be objects.

1. Suppose that left resolutions $P_{A,\bullet} \rightarrow A$ and $P_{B,\bullet} \rightarrow B$ exist such that $P_{A,i}$ and $P_{B,i}$ are flat (Definition A.5.2) with respect to F on the left and right respectively, i.e. $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i .

The complexes $F(P_{A,\bullet}, B)$ and $F(A, P_{B,\bullet})$ are quasi-isomorphic to the complex $\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))$.

2. Suppose that right resolutions $A \rightarrow I^{A,\bullet}$ and $B \rightarrow I^{B,\bullet}$ exist such that $I^{A,i}$ and $I^{B,i}$ are flat (Definition A.5.2) with respect to F on the left and right respectively, $F(I^{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, I^{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i .

The complexes $F(I^{A,\bullet}, B)$ and $F(A, I^{B,\bullet})$ are quasi-isomorphic to the complex $\text{Tot}(F(I^{A,\bullet}, I^{B,\bullet}))$.

Proof. We prove 1. The other part is the dual statement.

Choose resolutions $P_{A,\bullet} \xrightarrow{\varepsilon} A$ and $P_{B,\bullet} \xrightarrow{\eta} B$ such that $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i . Identifying A and B with complexes concentrated in degree 0, we can form the three double complexes $F(P_{A,\bullet}, P_{B,\bullet})$, $F(A, P_{B,\bullet})$, and $F(P_{A,\bullet}, B)$. Note that the augmentation morphisms ε and η induce morphisms $\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet})) \rightarrow \text{Tot}(F(A \otimes P_{B,\bullet}))$, $\text{Tot}(F(P_{A,\bullet}, B))$.

Let C be the double complex of objects in $\mathcal{C}$ obtained from $F(P_{A,\bullet}, P_{B,\bullet})$ by adding $F(A, P_{B,\bullet}[-1])$ in the column $p = -1$. One can show that the translate $\text{Tot}(C)[1]$ is the mapping cone of the map

$$\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet})) \xrightarrow{\varepsilon \otimes \text{id}} \text{Tot}(F(A, P_{B,\bullet})) = F(A, P_{B,\bullet}).$$

Moreover, since each $F(-, P_{B,i})$ is an exact functor, every row of C is exact, so $\text{Tot}(C)$ is exact by Lemma A.5.10. Therefore, $F(\varepsilon, \text{id})$ is a quasi-isomorphism and hence

$$H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \xrightarrow{H_*(F(\varepsilon, P_{B,\bullet}))} H_*(F(A, P_{B,\bullet}))$$

is a natural isomorphism.

By symmetry, there is a natural isomorphism $H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \rightarrow H_*(F(P_{A,\bullet}, B))$. $\square$

Theorem A.5.12 (Balancing generalized derived functors of a biadditive functor of abelian categories computed via flat resolutions). Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) of abelian categories (Definition 3.5.18). Assume that (small) filtered colimits which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 3.5.19)).

Let $A \in \mathcal{A}$ and $B \in \mathcal{B}$ be objects.

1. Suppose that left resolutions $P_{A,\bullet} \rightarrow A$ and $P_{B,\bullet} \rightarrow B$ exist such that $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i , i.e. $P_{A,\bullet}$ and $P_{B,\bullet}$ are flat resolutions of A and B respectively.
 - (a) The objects $L_n^I F(A, B)$ and $L_n^{II} F(A, B)$ are naturally isomorphic.
 - (b) The objects $L_n^I F(A, B)$ and $L_n^{II} F(A, B)$ are well defined (up to natural isomorphism), i.e. do not depend on the choice of left resolutions of A and B respectively.
2. Suppose that right resolutions $A \rightarrow I^{A,\bullet}$ and $B \rightarrow I^{B,\bullet}$ exist such that $F(I^{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, I^{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i .
 - (a) The objects $R_I^n F(A, B)$ and $R_{II}^n F(A, B)$ are naturally isomorphic.

- (b) The objects $R_I^n F(A, B)$ and $R_{II}^n F(A, B)$ are well defined (up to natural isomorphism), i.e. do not depend on the choice of left resolutions of A and B respectively.

Proof. We prove 1. The other part is the dual statement.

Choose resolutions $P_{A,\bullet} \xrightarrow{\varepsilon} A$ and $P_{B,\bullet} \xrightarrow{\eta} B$ such that $F(P_{A,i}, -) : \mathcal{B} \rightarrow \mathcal{C}$ and $F(-, P_{B,i}) : \mathcal{A} \rightarrow \mathcal{C}$ are exact for all i . As per Lemma A.5.11, $F(\varepsilon, \text{id})$ is a quasi-isomorphism and hence

$$H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \xrightarrow{H_*(F(\varepsilon, P_{B,\bullet}))} H_*(F(A, P_{B,\bullet}))$$

is a natural isomorphism. By symmetry, there is a natural isomorphism $H_*(\text{Tot}(F(P_{A,\bullet}, P_{B,\bullet}))) \rightarrow H_*(F(P_{A,\bullet}, B))$. Therefore, $L_n^I(A, B)$ and $L_n^{II}(A, B)$ are naturally isomorphic as claimed. In particular, $L_n^I(A, B)$ and $L_n^{II}(A, B)$ are independent of the choice of resolution of A and B respectively. □

Theorem A.5.13 (Balancing of Tor, cf. [Wei94, Theorem 2.7.2]). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.5.18), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) that is right exact (Definition 3.5.29) in each variable. Assume that (small) filtered colimits which exist in $\mathcal{C}$ are exact (e.g. which holds if $\mathcal{C}$ satisfies Ab5 (Definition 3.5.19)).

Given $A \in \mathcal{A}$ and $B \in \mathcal{B}$ for which flat resolutions exist, let $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ respectively be the Tor objects $\text{Tor}_n^A(A, B)$ (Definition A.5.14) computed via flat resolutions of A in $\mathcal{A}$ and of B in $\mathcal{B}$.

1. $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ are naturally isomorphic.
2. $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ are independent of the choice of flat resolution of A and B respectively.

In particular, we may identify the objects $\text{Tor}_n^I(A, B)$ and $\text{Tor}_n^{II}(A, B)$ and simply write $\text{Tor}_n(A, B)$ for either.

Proof. This follows from Theorem A.5.12. □

Definition A.5.14 (General circumstances for the existence of Tor functors). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.5.18), and let $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a biadditive functor (Definition A.5.1) that is right exact (Definition 3.5.29) in each variable.

(♠ TODO: Do notation for tor in two categories, one category, modules over rings)

For objects $A \in \mathcal{A}$ and $B \in \mathcal{B}$, the *Tor objects*

$$\text{Tor}_n(A, B) = \text{Tor}_n^{A, B}(A, B)$$

may be defined in one of several not-necessarily-equivalent ways:

1. as the left derived functors of $A \otimes -$ computed via projective resolutions of B in $\mathcal{B}$, assuming that such a projective resolution exists.

2. as the left derived functors of $- \otimes B$ computed via projective resolutions of B in $\mathcal{A}$, assuming that such a projective resolution exists.
3. as the “left derived functors” of $A \otimes -$ computed via flat resolutions of B in $\mathcal{B}$, assuming that such a flat resolution exists. More precisely, $\mathrm{Tor}_n(A, B) = H_n(A \otimes F_\bullet)$ in this case where $F_\bullet \rightarrow B$ is a flat resolution. A priori, this might depend on the choice of flat resolution.
4. as the “left derived functors” of $- \otimes B$ computed via flat resolutions of A in $\mathcal{A}$. More precisely, $\mathrm{Tor}_n(A, B) = H_n(F_\bullet \otimes B)$ in this case where $F_\bullet \rightarrow A$ is a flat resolution. A priori, this might depend on the choice of flat resolution.

See Theorem A.5.13 and Lemma A.5.7, which describe sufficient conditions under which the Tor functors defined via flat resolutions are independent of the choice of flat resolution. In particular Corollary A.5.6 shows that all of the above notions are in natural agreement if $\mathcal{A}$ and $\mathcal{B}$ have enough projectives and have enough flats and filtered direct limits that exist in $\mathcal{C}$ are exact.

For any of the above notions of Tor_n , note that

1. for fixed $A \in \mathcal{A}$, if $\mathrm{Tor}_n(A, B)$ exists and is well defined for any $B \in \mathcal{B}$, then $\mathrm{Tor}_n(A, -)$ is an additive functor $\mathcal{B} \rightarrow \mathcal{C}$.
2. for fixed $B \in \mathcal{B}$, if $\mathrm{Tor}_n(A, B)$ exists and is well defined for any $A \in \mathcal{A}$, then $\mathrm{Tor}_n(A, -)$ is an additive functor $\mathcal{A} \rightarrow \mathcal{C}$.
3. if $\mathrm{Tor}_n(A, B)$ exists and is well defined for any $A \in \mathcal{A}$ and $B \in \mathcal{B}$, then $\mathrm{Tor}_n(-, -)$ is an biadditive $\mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$.

In the case that $\mathcal{A}$ is the category of $R - S$ -modules, $\mathcal{B}$ is the category of $S - T$ -modules, $\mathcal{C}$ is the category of $R - T$ -modules for some (not necessarily commutative) rings R, S, T (Definition 3.1.1), and $\otimes$ is the usual tensor product between $R - S$ -modules and $S - T$ -modules producing $R - T$ -modules, then the Tor functors may be denoted by

$$\mathrm{Tor}_n^S(A, B)$$

for $A \in \mathcal{A}$ and $B \in \mathcal{B}$.

APPENDIX B. SESQUILINEAR AND HERMITIAN FORMS ON SHEAVES OF RING MODULES OVER SHEAVES OF RINGS

Definition B.0.1. Let $(\mathcal{S}, \tau)$ be a site. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{S}$. An *involution on $\mathcal{O}$* is a morphism of sheaves of rings $\sigma : \mathcal{O} \rightarrow \mathcal{O}^{\mathrm{op}}$ such that $\sigma \circ \sigma = \mathrm{id}_{\mathcal{O}}$, i.e. an involution (Definition 5.11.1) in the category of sheaves of rings on $\mathcal{S}$.

Proposition B.0.2. Let (R, σ) be a ring with involution (Definition 5.11.2) and $\phi : M \times M \rightarrow R$ a sesquilinear form (Definition 5.11.3). If M is a finitely generated (Definition 3.5.15) projective (Definition 3.5.77) R -module, then the adjoint map $\hat{\phi} : M \rightarrow \mathrm{Hom}_R(M, R)_\sigma$ is an isomorphism if and only if ϕ is non-degenerate (Definition 5.11.6).

Definition B.0.3. Let $(\mathcal{S}, \tau)$ be a site. Let $(\mathcal{O}, \sigma)$ be a sheaf of rings on $\mathcal{S}$ with involution (Definition B.0.1) $\sigma : \mathcal{O} \rightarrow \mathcal{O}$. Let $\mathcal{E}$ be a sheaf of left $\mathcal{O}$ -modules on $\mathcal{S}$.

A **sesquilinear form on $\mathcal{E}$** is a morphism of sheaves of abelian groups

$$\phi : \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{O}$$

such that for every object $U \in \mathcal{S}$, the induced map on sections

$$\phi_U : \mathcal{E}(U) \times \mathcal{E}(U) \rightarrow \mathcal{O}(U)$$

is a σ_U -sesquilinear form (Definition 5.11.3) on the $\mathcal{O}(U)$ -module $\mathcal{E}(U)$, where $\sigma_U : \mathcal{O}(U) \rightarrow \mathcal{O}(U)$ is the involution (Definition 5.11.2) on global sections.

Definition B.0.4. Let $(\mathcal{O}, \sigma)$ be a sheaf of rings with involution (Definition B.0.1) on a site $\mathcal{S}$. Let $\mathcal{E}$ be a sheaf of left $\mathcal{O}$ -modules.

A **hermitian form on $\mathcal{E}$** is a morphism of sheaves of sets

$$\phi : \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{O}$$

such that for every object U in $\mathcal{S}$, the map on sections

$$\phi_U : \mathcal{E}(U) \times \mathcal{E}(U) \rightarrow \mathcal{O}(U)$$

is a hermitian form (Definition 5.11.4) on the $\mathcal{O}(U)$ -module $\mathcal{E}(U)$ with respect to the involution (Definition 5.11.2) $\sigma_U : \mathcal{O}(U) \rightarrow \mathcal{O}(U)$.

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